



nuclear magnetic resonance spectroscopy

netic moment are placed in a constant magnetic field of high intensity and are simultaneously exposed to electromagnetic radiation of an appropriate frequency in the radiofrequency region, whereby the nuclei are caused to change from a low-energy to a high-energy state with the absorption of energy. Nuclei that display this phenomenon include those of the nuclides protium, tritium, carbon-13, nitrogen-15, fluorine-19, and phosphorus-31; all these nuclei have nonzero spins and, therefore, magnetic moments. *See also* **nuclear magnetic resonance spectroscopy**.

nuclear magnetic resonance spectroscopy *abbr.*: NMR spectroscopy *or* nmr spectroscopy; absorption spectroscopy in the radiofrequency region based on the phenomenon of **nuclear magnetic resonance**. Unless otherwise qualified the term usually denotes nuclear magnetic resonance spectroscopy of protium atoms, also known as proton magnetic resonance spectroscopy (*abbr.*: pmr spectroscopy). Nuclei of the same nuclide in different chemical environments give rise to distinguishable spectral lines, the positions, intensities, widths, and multiplicities of which give considerable information about the structure of complicated molecules containing such nuclei. The technique is also useful for determination of the concentrations of particular substances in complex environments, including that of an intact mammal. *Compare* **electron spin resonance spectroscopy**, **magnetic resonance imaging**.

nuclear magneton *symbol*: μ_N ; a quantity of magnetic moment, given by $\mu_N = eh/4\pi m_p = (m_e/m_p)\mu_B$, where e is the elementary charge, h is the Planck constant, m_e and m_p are the rest masses of an electron and a proton respectively, and μ_B is the **Bohr magneton**. It is a fundamental physical constant, of value $5.050\,786\,6(17) \times 10^{-27} \text{ J T}^{-1}$.

nuclear matrix *see* **matrix**.

nuclear medicine the branch of medicine concerned with the use of radioactive nuclides for diagnosis or treatment of disease, or for medical research.

nuclear membrane *an alternative name for* **nuclear envelope**.

nuclear mitotic apparatus protein *abbr.*: NuMA; one of a family of abundant coiled-coil nuclear proteins located in the nucleus during interphase, but which, when the nuclear envelope disassembles in prometaphase, rapidly redistribute to the developing spindle poles. Example from human: database code S33414, 45 amino acids (4.73 kDa) (one of multiple forms).

nuclear Overhauser effect *or* **nuclear Overhauser enhancement** *abbr.*: NOE; in nuclear magnetic resonance spectroscopy, a phenomenon in which saturation of an electron resonance by the simultaneous application of a radiofrequency field causes, under appropriate conditions, a considerable increase in the polarization of the interacting nuclei and hence in the NMR signal produced.

nuclear Overhauser effect spectroscopy *abbr.*: NOESY; in **nuclear magnetic resonance spectroscopy** advantage can be taken of the fact that the energy states of the spins of two nuclei subject to the **nuclear Overhauser effect** (NOE) are slightly perturbed by the mutual dipolar coupling (the effects of J-coupling are ignored). A perturbation of the populations of the states of one of the spins will lead to a transfer of the populations between the upper and lower energy states of the second spin, thereby modifying the resonance intensity. The NOE can be either positive or negative depending on the nuclei involved and the rapidity of the relative motions of the two spins. The strength of the dipolar coupling is distance dependent. Thus measurements of the NOE between two nuclei can lead to the estimation of the distance between them. Conversely, if the distance between the two nuclei is fixed (e.g. in a bond), then the NOE measurement can yield an indication of the dynamic motions in the molecule. Two-dimensional spectroscopy, known as 2D NOESY, is used to elucidate dipolar coupling networks present in complex biomolecules, using the NOEs between individual ^1H nuclei in the molecule. The network of NOEs obtained provides a basis for the determination

of the three-dimensional structure in solution. Applications include the structures of small proteins and nucleic acids as well as protein-drug and protein-nucleic acid complexes.

nuclear pore any of the numerous similar discrete openings in the **nuclear envelope** of a eukaryotic cell, where the inner and outer nuclear membranes are joined. The nuclear pores are the central feature of a complex which has a mass of $\approx 125 \text{ MDa}$ in higher eukaryotes and are estimated to contain about 100 different polypeptides. The complex constitutes a diffusion channel about 9 nm in diameter through which proteins enter the nucleus and ribonucleoproteins leave. Small proteins enter freely but the entry of larger proteins is controlled by a mechanism of active transport. Such proteins contain within their primary structure one or more clusters of basic amino acids which comprise the nuclear localization sequence (*abbr.*: NLS). Two steps are involved: binding to the nuclear envelope, followed by energy-dependent transit through the complex. Four soluble proteins have been shown to assist in the process: karyopherin α , *other name*: importin α (the NLS receptor), karyopherin β , *other name*: importin β (which mediates binding to the nuclear pore complex), a small guanosine triphosphatase (*abbr.*: GTPase), Ran, and p10/NTF2 (a protein of hitherto ill-defined function but now claimed to coordinate the Ran-dependent association and disassociation reaction). Ran cycles between a GDP- and GTP-bound form and is thereby involved in the consumption of energy for the protein import. The exchange factor for Ran (RCC1) is located inside the nucleus whereas its GTPase-activating protein (Rab-GAP1 or RAN1) is cytoplasmic. Much less is known about the mechanism whereby the export of ribonucleoprotein is controlled. RNA itself does not seem to leave the nucleus.

nuclear rest *see* **nucleoid** (def. 4).

nuclear transformation transformation of a nuclide; the change of one nuclide into another.

nuclease any enzyme within subclass EC3.1, that catalyses the hydrolysis of ester linkages in nucleic acids. Nucleases with sugar specificity are referred to as ribonucleases or deoxyribonucleases; **endonucleases** create internal breaks, while 5'- and 3'-**exonucleases** remove nucleotide residues from the indicated end. Some nucleases are specific for double- or single-stranded nucleic acids.

nucleate **1** or **nucleated** possessing a nucleus. **2** to form, to act as, or to provide with, a nucleus or nuclei. **3** any salt of a nucleic acid.

nucleation **1** the act or process of nucleating or forming a nucleus. **2** (*in chemistry*) **a** the aggregation of molecules to form a polymer or a new phase within a medium. **b** (*also known as seeding*) the addition of a small quantity of finely divided crystalline material to a supersaturated solution in order to promote crystallization.

nuclei the plural of **nucleus**.

nucleic acid *abbr.*: NA; any single- or double-stranded polynucleotide of molar mass in the range 20 kDa to 40 GDa or more. Nucleic acids are either **deoxyribonucleic acids** (DNA) or **ribonucleic acids** (RNA). The phosphoric residue linking any constituent mononucleotide residue to the next bears one free hydroxyl group, which is weakly acidic. They are universal constituents of living matter and are concerned with the storage, transmission, and transfer of genetic information.

nucleics *an informal collective term sometimes used for* nucleotides, nucleosides, and nucleobases, whether constituents of nucleic acids or compounds of low M_r , and including their metabolic relatives and synthetic analogues.

nuclein the name given by J. F. Miescher (1844–95) to an unusual phosphorus-containing material that he was able to isolate from the nuclei of pus cells, now known to have been **nucleoprotein**.

nucleinase *a name once proposed to designate the enzymes now known as* **nucleases**.

nucleo+ *or* (*before a vowel*) **nucle+** *comb. form* denoting **1** nucleus *or* nuclear. **2** nucleic acid.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



nucleobase

nucleobase any nitrogenous base that is a constituent of a nucleoside, nucleotide, or nucleic acid.

nucleobindin a DNA-binding protein containing a signal peptide, leucine zipper, and basic amino acid-rich regions. It was derived from a cell line from the lupus erythematosus-prone mouse MRL/T. Example (precursor) from human: database code NUBN_HUMAN, 460 amino acids (53.67 kDa).

nucleocapsid the structure within a virus that comprises the proteinaceous capsid and the genomic nucleic acid. In enveloped viruses the nucleocapsid is surrounded by an envelope.

nucleocidin 4'-C-fluoroadenosine 5'-sulfamate; an analogue of adenylyl produced by *Streptomyces calvus*, and the first naturally occurring derivative of a fluorosugar to be described. It has antibiotic activity against trypanosomes and inhibits translation in both eukaryotic and prokaryotic systems.

nucleocytoplasmic (of a structure or process) occurring between the nucleus and the cytoplasm; (of a property) relating the nucleus to the cytoplasm.

nucleodisome a structure consisting of two nucleosomes and isolable under certain conditions from some forms of chromatin.

nucleohistone any nucleoprotein consisting of DNA and histone(s).

nucleoid 1 resembling a nucleus. 2 the part or parts of a prokaryotic cell containing the genome and functionally equivalent to the nucleus of a eukaryotic cell. The meaning is sometimes extended to include the electron-dense inner zone observable in certain viruses. *Other names:* bacterial nucleus; chromatinic body; karyoid; nuclear area; nuclear body; nucleoid body; prokaryon. 3 the nucleus-like structure liberated from a eukaryotic cell by mild lysis. It contains histone-free supercoiled DNA within a flexible cage of RNA and protein. 4 or **nuclear rest** the nuclear residue sometimes observable in an erythrocyte. 5 the very dense core of a **microbody**.

nucleolar DNA or **ribosomal DNA** the DNA comprising the **nucleolar organizer(s)**.

nucleolar organizer or **nucleolus organizer** the portion of the genome of a eukaryotic cell that is associated with the **nucleolus** and contains the genes from which 45S ribosomal-precursor RNA is transcribed.

nucleolar phosphoprotein B23 or **nucleophosmin** or **numatrin** a monomeric, or under some conditions disulfide-linked dimeric, nucleolar protein that binds to single-stranded nucleic acids and may be bound to rRNA in the nucleolus. Example (precursor) from human: database code B232_HUMAN, 294 amino acids (32.54 kDa).

nucleolin or **protein C23** the major nucleolar protein of growing eukaryotic cells believed to function in pre-rRNA processing and ribosome assembly. It is a phosphoprotein found associated with intranucleolar chromatin and periribosomal particles. Example from human: database code NUCL_HUMAN, 706 amino acids (76.13 kDa).

nucleolus (*pl.* **nucleoli**) a small, dense body one or more of which are present in the nucleus of eukaryotic cells. It is rich in RNA and protein, is not bounded by a limiting membrane, and is not seen during mitosis. Its prime function is the transcription of the nucleolar DNA into 45S ribosomal-precursor RNA, the processing of this RNA into 5.8S, 18S, and 28S components of ribosomal RNA, and the association of these components with 5S RNA and proteins synthesized outside the nucleolus. This association results in the formation of ribonucleoprotein precursors; these pass into the cytoplasm and mature into the 40S and 60S subunits of the ribosome. —**nucleolar adj.**

nucleolytic capable of causing hydrolysis of nucleic acid.

nucleon a neutron or a proton.

nucleonics the branch of physics concerned with the practical applications of nuclear phenomena.

nucleon number or **mass number** *symbol:* *A*; a dimensionless physical quantity, having an integral value for any given nuclide, equal to the total number of nucleons (i.e. protons and

neutrons) contained in the nucleus of an atom of the nuclide; it also is the integer nearest to the numerical value of the relative atomic mass of that nuclide and is equal to the arithmetical sum of the **neutron number**, *N*, of the nuclide and the **proton number**, *Z*, of the relevant chemical element: i.e. $A = N + Z$. The nucleon number may be attached to the symbol for the chemical element as a left superscript, or to the name of the element as a suffix; e.g. ^{14}C , or carbon-14.

nucleophile or **nucleophilic reagent** any reagent that is preferentially attracted to a region of low electron density in a chemical reaction. *Compare* **electrophile**.

nucleophilic 1 of, pertaining to, or being a **nucleophile**; having or involving an affinity for regions of low electron density in a chemical reactant. 2 describing a chemical reaction in which a nucleophilic reagent participates.

nucleophilic catalysis catalysis by a **Lewis base**, which donates an electron pair to the reactant.

nucleophilic displacement *an alternative term for nucleophilic substitution reaction.*

nucleophilicity the relative reactivity of a nucleophile, measured in terms of the relative rate constants of different nucleophiles towards a common reactant.

nucleophilic reagent *an alternative term for nucleophile.*

nucleophilic substitution reaction or **nucleophilic displacement** a chemical reaction in which a **nucleophile** effects heterolytic substitution in another reactant, both bonding electrons being supplied by the nucleophile.

nucleophosmin *see* **nucleolar phosphoprotein B23**.

nucleoplasm all the protoplasm contained within the nuclear envelope of a living cell. —**nucleoplasmic adj.**

nucleoplasmin a heat-stable acidic protein present in the nucleoplasm of many cell types. It forms complexes with histones and functions in the assembly of nucleosomes in the formation of chromatin. Example from *Xenopus laevis*: database code NUPL_XENLA, 200 amino acids (22.02 kDa).

nucleoprotamine any nucleoprotein consisting of DNA and protamine(s).

nucleoprotein any complex of a protein with a deoxyribonucleic acid (**deoxyribonucleoprotein**) or a ribonucleic acid (**ribonucleoprotein**).

nucleosidase any enzyme that hydrolyses the *N*-glycosidic bond in a **nucleoside** (or in a nucleotide or some other derivative of a nucleoside). Such enzymes are classified within the sub-subclass EC 3.2.2, i.e. enzymes that hydrolyse *N*-glycosyl compounds.

nucleoside 1 *symbol* (for a residue of an undetermined, unknown, or unspecified nucleoside): *N* or *Nuc*; any **glycosylamine** that is a component of a nucleic acid and that consists of a nitrogenous base linked either to β -D-ribofuranose (forming a ribonucleoside, as in RNA) or to 2-deoxy- β -D-ribofuranose (forming a deoxyribonucleoside, as in DNA). The base is either a purine (linked at N-9) or a pyrimidine (linked at N-1, or in the case of pseudouridine at C-5). 2 any compound that is a glycosylated heterocyclic nitrogenous base; the glycoside moiety commonly is ribose, but is sometimes a different sugar or a modified sugar; e.g. cytarabine (1- β -D-arabinofuranosylcytosine).

nucleoside antibiotic any of a group of **nucleosides** (def. 2) that are not components of nucleic acids but are produced by various microorganisms and fungi (especially *Streptomyces* spp.), and have antibiotic activity. They contain unusual nitrogenous bases and/or unusual sugars (or sugar derivatives), and may also contain one or more additional components. Examples include **cordycepin**, **nucleocidin**, **puromycin**, and **showdomycin**.

nucleoside cyclic phosphate or **cyclic nucleotide** any nucleotide that consists of a nucleoside doubly esterified with one [ortho]phosphate molecule on its glycoside moiety (usually at the 3' and 5' positions, but sometimes at the 2'- and 3'-positions) e.g. cyclic AMP.

nucleoside diphosphatase EC 3.6.1.6; *systematic name:* nucleoside diphosphatase



leoside diphosphatase phosphohydrolase; an unspecific enzyme that hydrolyses a number of nucleoside diphosphates to a nucleoside monophosphate and orthophosphate. Substrates include IDP, GDP, UDP, and D-ribose 5-phosphate. There is also the more specific **guanosine-diphosphatase**, EC 3.6.1.42; *systematic name*: GDP phosphohydrolase, which hydrolyses guanosine diphosphate to guanylic acid and orthophosphate; example from yeast: database code GDA1_YEAST, 518 amino acids (56.76 kDa).

nucleoside diphosphate any nucleotide that consists of a nucleoside esterified with diphosphate on its glycoside moiety (usually at the 5' position, when the symbol is NDP).

nucleoside diphosphate reductase EC 1.17.4.1; *recommended name*: ribonucleoside-diphosphate reductase; *systematic name*: 2'-deoxyribonucleoside-diphosphate:oxidized thioredoxin 2'-oxidoreductase; *other name*: ribonucleotide reductase. A nucleotide reductase enzyme, present in most organisms, that catalyses the formation of a 2'-deoxyribonucleoside diphosphate, oxidized thioredoxin, and H₂O from a ribonucleoside diphosphate and reduced thioredoxin. It requires iron and ATP. The enzyme from *Escherichia coli* is formed from B2 protein (dimer of β chains; Fe-binding subunit, 3-D structure known) and B1 (dimer of α chains), which contains a disulfide/sulfhydryl redox system; it is the enzyme responsible for the biosynthesis of deoxyribonucleotides from ribonucleotides, e.g. in *de novo* biosynthesis. The B2 dimer contains two Fe(III) ions (one on each monomer) that form a Fe³⁺-O-Fe³⁺ group. This forms a free radical centre (involving an enzyme tyrosyl residue) that abstracts a hydrogen from the ribose (ultimately from the 2' position, but the mechanism is complex). This leads to loss of the 2'-OH. Hydride ions are lost from enzyme sulfhydryl groups to react with these OH[•] groups to form water, resulting in formation of disulfide groups. The hydrogen abstracted by the free radical centre is returned to what now becomes 2'-deoxyribose. The enzyme's disulfide groups are reduced in turn by thioredoxin. There is an allosteric site on the B1 subunit dimer; ATP acts at this site as an activator and dATP as an inhibitor, in a complex interaction with a second site that determines the specificity of the enzyme towards different nucleotides. Example from *E. coli*, α chain: database code RIR1_EC01, 761 amino acids (85.64 kDa); β chain: database code RIR2_EC01, 375 amino acids (43.34 kDa). A **nucleoside triphosphate reductase** is produced when the organisms are growing anaerobically.

nucleosidediphosphosugar or **nucleoside diphosphate sugar** or **nucleotide sugar** any **nucleotide** (def. 2) in which the distal phosphoric residue of a nucleoside 5'-diphosphate is in glycosidic linkage with a monosaccharide (or a derivative of a monosaccharide).

nucleoside monophosphate *see* **nucleoside phosphate**.

nucleoside phosphate or **nucleoside monophosphate** any nucleotide that consists of a nucleoside esterified with [ortho]phosphate on its glycoside moiety (usually at the 5'-position, when the symbol is NMP).

nucleoside-triphosphatase *abbr.*: NTPase; EC 3.6.1.15; an unspecific diphosphate phosphohydrolase enzyme that catalyses the hydrolysis of the distal diphosphate bond in various nucleoside triphosphates, with formation of [ortho]phosphate and the corresponding nucleoside diphosphate. It acts also on certain other diphospho-compounds.

nucleoside triphosphate any nucleotide that consists of a nucleoside esterified with triphosphate on its glycoside moiety (usually at the 5'-position, when the symbol is NTP).

nucleoside triphosphate reductase EC 1.17.4.2; *recommended name*: ribonucleoside-triphosphate reductase; *systematic name*: 2'-deoxyribonucleoside-triphosphate:oxidized thioredoxin 2'-oxidoreductase. A nucleotide reductase enzyme found in *Lactobacillus leichmannii* (and presumably related organisms). It catalyses the formation of a 2'-deoxyribonucleoside triphosphate, oxidized thioredoxin, and H₂O from a ribonucleoside triphosphate and reduced thioredoxin. It re-

quires cobalt and ATP. Example from *L. leichmannii*: database code LBARTR, 739 amino acids (81.85 kDa).

nucleosome a particle that forms the primary packing unit of DNA in chromatin. Nucleosomes give electron micrographs of decondensed chromatin a 'beads-on-a-string' appearance, and are released on mild digestion of eukaryotic nuclei with micrococcal endonuclease. Each consists of a segment of duplex DNA, 160–240 base pairs long, with associated **histone**; about 146 base pairs of the DNA comprise a **core particle** and the remainder form the linker DNA. The core particle is a disklike structure, 11 nm in diameter and 5.7 nm thick, consisting of two molecules each of histones H2A, H2B, H3, and H4 with approximately two superhelical turns of double-stranded DNA wound round it. In the intact nucleosome, an average of one molecule of histone H1 is on the exterior of each core particle and appears to mediate the packing together of adjacent nucleosomes in condensed chromatin. *See also* **nu body**.

nucleotidase any phosphoric monoester hydrolase enzyme that catalyses the hydrolysis of a **nucleotide** (def. 1, 2) to yield a nucleoside and orthophosphate. The principal ones are classified under three EC numbers (1) EC 3.1.3.31; *recommended name*: nucleotidase; enzymes having a wide specificity for 2', 3', and 5'-nucleotides, and that also hydrolyse glycerol phosphate and 4-nitrophenyl phosphate; (2) EC 3.1.3.6; *recommended name*: 3'-nucleotidase; an enzyme having a wide specificity for 3'-ribonucleotides; these enzymes hydrolyse a 3'-ribonucleotide to a ribonucleoside and orthophosphate. Some such enzymes are not sugar specific. Example, nuclease PA3 (deoxyribonuclease PA3) from *Penicillium* sp: database code NUP3_PENSQ, 270 amino acids (29.18 kDa). (3) EC 3.1.3.5; *recommended name*: 5'-nucleotidase; an enzyme having a wide specificity for 5'-ribonucleotides. These enzymes hydrolyse a 5'-ribonucleotide to a ribonucleoside and orthophosphate. Example, (precursor) human ecto-nucleotidase (CD73): database code 5NTD_HUMAN, 574 amino acids (63.29 kDa).

nucleotide 1 or **mononucleotide** or **nucleoside [mono]phosphate** any compound that consists of a **nucleoside** (def. 1) esterified with [ortho]phosphate at either the 3'- or the 5'-hydroxyl group of its glycoside moiety (ribonucleosides giving **ribonucleotides**, and deoxyribonucleosides giving **deoxyribonucleotides**). Nucleotides are the constitutional units into which nucleic acids are broken down by partial hydrolysis and from which they are considered to be built up. **2** any compound consisting of a **nucleoside** (def. 2) that is esterified with [ortho]phosphate or an oligophosphate at any hydroxyl group on its glycoside moiety. In this sense the term includes **nucleoside cyclic phosphate**, **nucleoside diphosphate**, **nucleoside diphosphosugar**, **nucleoside triphosphate**, and pyridine nucleotide. **3** any compound containing a moiety of a nucleotide (def. 1, 2). Included are: any carrying an additional phosphoric group at another position on the glycoside moiety; any formed (actually or conceptually) from two or more such nucleotides, whether the same or not, joined together in phosphoric-ester linkage (to form an **oligonucleotide** or a **polynucleotide**); and, in certain instances, any formed from such a nucleotide joined through a phosphoric anhydride link either to a second nucleotide or to some other phosphate ester. Exceptionally, the term includes also certain analogous compounds (especially flavin mononucleotide, and hence also flavin-adenine dinucleotide) that, being derivatives of ribitol, are not true nucleoside phosphates or else are not formed exclusively from them.

nucleotide-binding fold or **nucleotide-binding pocket** a common structural feature that is possessed by numerous dehydrogenase enzymes utilizing either of the nucleotide coenzymes NAD or NADP as the acceptor. X-ray crystallographic analysis shows that the coenzyme fits in a pocket made by the folding of the polypeptide chain. Many such enzymes combine specifically with **blue agarose** owing to putative structural similarity of the bound dye (Reactive Blue 2) to the coenzyme, a property that can be exploited in the separation and purification of these enzymes by affinity chromatography.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



nucleotide coenzyme

A similar property is shown by some other nucleotide-binding enzymes (and also by certain other proteins that do not bind nucleotides).

nucleotide coenzyme any **nucleotide** (def. 2, 3) that is either (1) a coenzyme (e.g. nicotinamide-adenine dinucleotide), or (2) a substrate-carrier in an anabolic reaction (e.g. uridine diphosphate glucose).

nucleotide-gated channel *see* **ion channel**.

nucleotide reductase *see* **nucleoside diphosphate reductase**, **nucleoside triphosphate reductase**.

nucleotide sugar an alternative name for **nucleoside diphosphate sugar**.

nucleotide unit the repeating unit of a polynucleotide chain, defined typically by the sequence of atoms from the phosphorus atom at the 5' end of any one of its component glucose residues to the oxygen atom at the 3' end of the same residue and including all the atoms of the glucose residue and its attached phosphoric residue and nucleobase; atypically, the nucleotide unit at the 5' end of a chain may lack a phosphate group. Nucleotide units in a given chain are numbered sequentially from the chain's 5' end irrespective of the presence or absence of a phosphate group on the 5'-terminal unit.

nucleotidyl the acyl group formed from a nucleoside [mono]phosphate by removal of a hydroxyl group from its phosphoric group.

nucleotidyltransferase an enzyme of subclass EC 2.7.7; such enzymes transfer a nucleotidyl group to a reactant. Possession of such activity is a common mechanism of drug resistance (especially to **aminoglycoside antibiotics**) in bacteria. An example is gentamicin 2''-nucleotidyltransferase (EC 2.7.7.46); *systematic name*: NTP:gentamicin 2''-nucleotidyltransferase; *other name*: 2''-aminoglycoside nucleotidyltransferase. It catalyses a reaction between NTP and gentamicin to form pyrophosphate and 2''-nucleotidylgentamicin. This enzyme can be isolated from *Escherichia coli* and *Klebsiella pneumoniae* plasmids, and confers resistance to kanamycin, gentamicin, dibekacin, sisomicin, neomycin, and tobramycin by adenylating the 2''-OH group of these antibiotics. Example from *E. coli*: database code AADB_ECOLI, 177 amino acids (19.85 kDa).

nucleus (pl. **nuclei**) **1** (in biology) the most conspicuous organelle of a eukaryotic cell; it contains the chromosomes and (except for some activity in mitochondria) is the sole site of DNA replication and of RNA synthesis in the cell. Usually a spheroidal body, it is separated from the cytoplasm by the nuclear envelope. **2** (in physics) **atomic nucleus** the positively charged central part of an atom, with which is associated almost the whole of the mass of the atom but only a very small part of its volume. It is composed of nucleons, i.e. protons and neutrons. **3** (in organic chemistry) a characteristic arrangement of atoms, especially a ring structure, that occurs in a series of related organic compounds; e.g. the benzene nucleus. **4** a particle on which a crystal, droplet, or bubble forms in a fluid.

nuclide any individual species of atom, whether stable, metastable, or unstable, that is characterized by the constitution of its nucleus and by its **nucleon number**, **proton number**, and nuclear energy state; its mean life must be long enough for the species to be observable. Nuclides having the same proton number but different nucleon numbers or different nuclear energy states are termed isotopic nuclides, or **isotopes**. Nuclides having the same nucleon number but different proton numbers are termed isobaric nuclides, or **isobars**. Nuclides having the same neutron number but different proton numbers are termed isotonic nuclides, or **isotones**. A given nuclide may be specified by attaching its nucleon number either to the name of the relevant chemical element as a suffix, or to the symbol for the chemical element as a left superscript; the proton number may be attached to the symbol as a left subscript; e.g. carbon-14, ^{14}C , or $^{14}_{6}\text{C}$. A metastable nuclide (i.e. one with an elevated nuclear energy state) may be differentiated from its counterpart with the same nucleon number by attaching the

suffix m to the nucleon number; e.g. technetium-99m or $^{99\text{m}}\text{Tc}$. (Absence of a nucleon number attached to its name or symbol implies the presence of all isotopes of the element in natural abundance.) —**nuclidic** *adj.*

nude a mutant strain of mouse or rat that lacks a **thymus gland** and hence is devoid of **T lymphocytes**. A linked locus also produces hairlessness, hence the name.

NUF1 *abbr.* for nuclear filament-related protein.

Nujol a proprietary name for a high-boiling-point fraction of liquid paraffin widely used for making mulls for infrared spectroscopy. It is also used medicinally as a laxative, and was referred to by the poet e. e. cummings in *Poem, or Beauty Hurts Mr. Vinal*: '...little liver pill-/hearted-Nujol-needing-There's-A-Reason...'.

null a minimum or zero value, especially of an electric current.

null cell an alternative term for **null lymphocyte**.

nullisomic **1** describing a cell, organism, strain, etc. that lacks one or more pairs of homologous chromosomes. **2** such a cell, organism, strain, etc.

null lymphocyte or **null cell** a **lymphocyte** that is not recognizable as either a B lymphocyte or a T lymphocyte.

null method or **zero method** a method of measurement in which a zero reading in a sensitive, but not necessarily calibrated instrument, e.g. a galvanometer, enables an unknown value of some property to be deduced by comparison with reference values.

NuMA *abbr.* for nuclear mitotic apparatus protein.

numatrin another name for **nucleolar phosphoprotein B23**.

number **1** a concept or expression of quantity reckoned in ones or units; a particular value of such a quantity (which may be integral or zero). **2** the symbol or group of symbols used to represent a number (def. 1); numeral; figure. **3** a numeral or string of numerals used for identification (of something, especially a member of a sequence). **4** to assign numbers to (members of a series); to label with a number or numbers.

number-average molar mass *see* **average molar mass**.

number-average relative molecular mass or (formerly) **number-average molecular weight** *see* **average relative molecular mass**.

number concentration *symbol*: C ; *see* **concentration** (def. 1d).

number fraction *symbol*: δ ; the number of defined particles or elementary entities of a specified component of a system divided by the total number of defined particles in the system.

numerical aperture (in optics) *abbr.*: NA; the arithmetic product of the refractive index of the medium in which the objective (lens) of a microscope is situated and the sine of half the angle of view of the objective. The resolving power of the lens is proportional to its numerical aperture.

nurse cell a cell that is connected by cytoplasmic bridges to an oocyte and thereby conveys macromolecules to the growing oocyte.

nus any of three *Escherichia coli* genes that are involved in transcriptional termination. Their gene products are as follows. NusA is a protein that binds to RNA polymerase, RNA, rho, and the antitermination protein pN, the product of the N gene. It may also bind to NusB. It participates in both the termination and antitermination of transcription. Database code NUSA_ECOLI, 495 amino acids (54.81 kDa). NusB is involved in the transcription termination process at certain sites during normal bacterial growth and is essential for the function of pN. Database code NUSB_ECOLI, 139 amino acids (15.67 kDa). NusG is involved in transcription antitermination. Database code NUSG_ECOLI, 181 amino acids (20.51 kDa).

NutraSweet a proprietary name for **aspartame**.

nutrient **1** any chemical substance that can be used by an organism to sustain its metabolic activities. **2** serving as or conveying nourishment. **3** (in microbiology) describing a solid or liquid medium that is able to support the growth of a range of nutritionally undemanding chemoorganotrophs.

nutrition **1** the supply of substances (i.e. nutrients) required by



nutritional mutant

an organism, directly or indirectly, for its metabolic activities, i.e. for the provision of energy, for growth, and/or for the renewal of degraded components. **2** the act or process of supplying or receiving nutrients. **3** the scientific study of the nutrient requirements of particular organisms and of the supply of those nutrients. —**nutritional adj.**

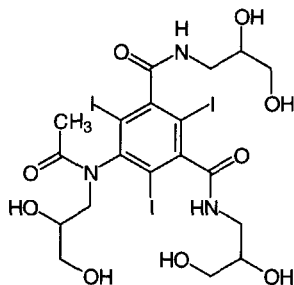
nutritional mutant a mutant strain of a microorganism (alga, bacterium, or fungus) that requires a supply of one or more growth factors in addition to those required by the wild type, i.e. a mutation in a **prototroph** to form an **autotroph**.

nutritive 1 containing nutrient(s). **2** of, concerned in, or promoting nutrition.

nux vomica the dried seeds of the Indian tree, *Strychnos nuxvomica*. They are a source of **strychnine**, **brucine**, and other poisonous alkaloids.

Nva symbol for the α -amino acid **norvaline** (now known as 2-aminovaleric acid, symbol: Avl).

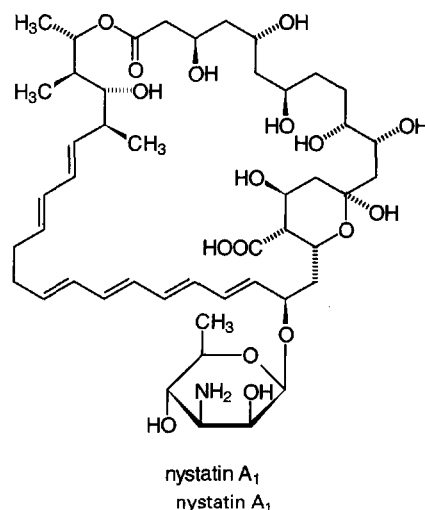
Nycodenz the proprietary name for *N,N'*-bis(2,3-dihydroxypropyl)-5-[*N*-(2,3-dihydroxypropyl)acetamido]-2,4,6-triiodoisophthalamide; a dense, synthetic, nonionic substance, highly soluble in water, that is capable of forming (autoclavable) aqueous solutions of densities up to 2.1 g mL⁻¹. It is one of several synthetic water-soluble triiodobenzene derivatives designed for use as intensifiers in radiology that have found use in isopycnic gradient fractionation of biological particles.



nyctalopia the medical term for **night blindness**.

nylon any of a class of synthetic thermoplastic substances, characterized chemically as long-chain polyamides and formed by condensation polymerization of dicarboxylic acids with diamines. They have high melting points, are insoluble, tough, resistant to fracture, and exhibit low friction. Hence they are useful in the manufacture of laboratory apparatus for which these properties are advantageous. Nylon can be formed into fibres or sheets and used for coatings, fabrics, and many other products. It is also used as an alternative to nitrocellulose in hybridization membranes.

nystatin a **mycosamine**-containing polyene antifungal antibiotic complex (three components) produced by *Streptomyces noursei* and other *Streptomyces* spp. It is very similar to **amphotericin B** in its properties and mechanism. Two proprietary names are Fungicidin and Mycostatin.



NZO mouse abbr. for New Zealand obese mouse.

Oo

o+ prefix (in peptide hormone nomenclature) denoting ovine.

o- prefix (in chemical nomenclature) denoting **ortho-**. Compare **m-**, **p-**.

O symbol for 1 oxygen. **2** a residue of the ribonucleoside orotidine (alternative to Ord).

O- prefix (in chemical nomenclature) indicating substitution on oxygen; a particular oxygen atom in a molecule may be specified by suffixing an appropriate superscript.

OA abbr. for ovalbumin.

OAA abbr. for oxaloacetic acid, or oxaloacetate.

OAG abbr. for 1-oleoyl-2-acetyl-glycerol.

Oakley-Fulthorpe technique a simple double immunodiffusion method that can be used for quantitative determination of antigen concentrations; it is essentially a development of the **Oudin technique**. As originally described, the antiserum in 1% agar is placed at the bottom of a cylindrical tube above which is placed a layer of 1% agar gel in 1% saline, and 0.5% *o*-cresol (as a preservative). Subsequently a bacterial filtrate (containing antigen) is layered onto the agar-saline gel and the tube then incubated at 37 °C. Disks or lines of flocculation of antigen-antibody complexes are produced in the agar-saline gel, from whose positions the concentration of antigen(s) may be determined. [After C. L. Oakley and A. J. Fulthorpe, who described the technique in 1953.]

O antigen **1** any of the lipopolysaccharide-protein somatic antigens of the cell walls of Gram-negative bacteria, especially those of the Enterobacteriaceae. They are useful in the serological classification of *Salmonella*, *Shigella*, and other bacteria. **2** (sometimes) an alternative name for H antigen (present on erythrocytes of blood group O); see **blood-group substance**.

ob see **ob/ob**.

obelin a luminescent protein that is obtained from the jellyfish *Obelia geniculata*. It is similar to **aequorin** in properties and use.

obese having an excessive amount of body fat. —**obesity** *n*.

objective (in optics) the lens or system of lenses in an optical instrument (especially a compound microscope) that is positioned nearest to the object to be viewed.

oblate **1** having an equatorial diameter longer than the polar diameter. **2** describing an **ellipsoid of rotation** (including a spheroid) generated by rotation of an ellipse about its minor axis. **3** describing any protein of **oblate** (def. 1) dimensions as observed in hydrodynamic measurements. Compare **prolate**.

obligate by necessity, or without option; used especially of the nature of the environment or of the mode of life of an organism; e.g. obligate aerobe (see **aerobe**), obligate anaerobe (see **anaerobe**). Compare **facultative**.

O blood-group substance see **ABO system**.

ob/ob the genotype symbol for any of a number of strains of mouse with an inherited form of severe obesity, due to a homozygous recessive autosomal mutation on chromosome 6. Such animals display marked hyperglycemia and insulin resistance. A protein named **leptin**, or OB protein, has been implicated in the control of fat mass and is a product of the *ob* mutant gene, being deficient in *ob/ob* mice.

obtuse angle any plane angle greater than 0.5 π rad or 90° and less than π rad or 180°.

Oc symbol for the octyl group.

Occam's razor a variant spelling of **Ockham's razor**.

occlude **1** to block or obstruct (a passage, tube, opening, etc.). **2** (in chemistry) to take up and retain something in the interstices of a solid; e.g. to absorb a gas into a metal or to entrap liquid within crystals during their formation. —**occlusion** *n*.

occludin an integral membrane protein that localizes in tight junctions in chick liver. There are four putative membrane-

spanning segments within the N-terminal half. Example from chick: database code A49467, 504 amino acids (55.80 kDa).

occult concealed or hidden; not apparent to the unaided eye.

occult blood blood that is present in a sample in such small quantities, e.g. in feces, that it can only be detected by chemical testing or by microscopy.

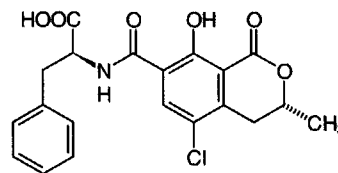
occult virus a virus whose presence in a host is inferred but whose isolation or detection cannot be achieved.

occupancy **1** the state or condition of being filled or occupied. **2** the number of similar sites occupied, usually expressed as a fraction or percentage of the total number available. It can refer, e.g., to the number of receptors on a cell surface occupied by a particular agonist or to the number of atomic orbitals of a particular subshell filled by electrons.

occupation theory of agonist action a theory stating that the magnitude of the response to an agonist is directly proportional to the fraction of specific receptors occupied by molecules of the agonist; it assumes that the binding of agonist to receptor is reversible. According to this theory the response will increase to a maximal value when all the receptors are occupied. Compare **rate theory of agonist action**.

Ochoa, Severo (1905–93), Spanish-born US biochemist; Nobel Laureate in Physiology or Medicine (1959) jointly with A. Kornberg 'for their discovery of the mechanisms in the biological synthesis of ribonucleic acid and deoxyribonucleic [sic] acid'.

ochratoxin any one of several **mycotoxins** produced by the fungus *Aspergillus ochraceus*, other *Aspergillus* spp., and *Penicillium* spp.; the major component, ochratoxin A, is a complex, chlorine-containing derivative of L-phenylalanine that inhibits phosphoenolpyruvate carboxykinase (EC 4.1.1.49). These toxins may occur on contaminated foodstuffs such as corn (maize), peanuts, storage grains, etc.



ochratoxin A

ochre codon or **ochre triplet** symbol: UAA; one of the three terminator codons or **nonsense codons** found in messenger RNA.

ochre mutant any cell or virus carrying an **ochre mutation**.

ochre mutation a mutation that gives rise to an **ochre codon** at an abnormal position in a messenger RNA molecule.

ochre suppressor any of a number of mutations, occurring in *Escherichia coli*, that suppress an **ochre codon** in messenger RNA thereby allowing insertion of one of several alternative amino acids into a polypeptide chain at that site. The mutation results in changed anticodons of the transfer RNA molecules, consequent on single-base substitution in their DNA coding sequences. Ochre suppressors also suppress **amber codons**.

ochronosis the presence of brown-black pigment in the skin, cartilage, and other tissues due to **alcaptonuria**.

Ockham's razor or **Occam's razor** a maxim stating that when explaining something there should be a minimal number of assumptions. *Entia non sunt multiplicanda praeter necessitatem* (entities should not be multiplied except from necessity). [After William of Ockham (1285–1349), English philosopher.]

Oco symbol for the octanoyl group.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

OCS the proprietary name for a commercial xylene-fluor mixture suited to the liquid-scintillation counting of a wide range of toluene- and xylene-soluble organic samples, especially specimens solubilized in **NCS**.

oct+ a variant form of **octa+** (sometimes before a vowel).

OCT (in clinical chemistry) *abbr.* for ornithine carbamoyltransferase.

OCT a family of genes encoding transcription factors for eukaryotic RNA polymerase II promoters. The proteins contain a **POU domain** and contribute to the naming of that domain. They are leucine zipper proteins and bind to **octamer** sequences. Examples from human: Oct-1, a factor for snRNA and histones; database code OCT1_HUMAN, 743 amino acids (76.40 kDa); Oct 2, a factor for immunoglobulin promoters; database code OCT2_HUMAN, 478 amino acids (51.02 kDa).

octa+ or **octo+** or (sometimes before a vowel) **oct+** *comb. form* denoting eight, eightfold, eight times.

octacosa+ *comb. form* denoting twenty-eight, twenty-eightfold, twenty-eight times.

octacosanoate *see montanate*.

octacosanoyl *see montanoyl*.

octadeca+ *comb. form* denoting eighteen, eighteenfold, eighteen times.

octadecadienoic acid any straight-chain fatty acid having eighteen carbon atoms and two double bonds per molecule. Linoleic acid (*see linoleate*) is the *all-Z*-(9,12)-isomer and is a constituent of most vegetable oils and animal fats. The *all-E*-isomer of this acid is linolelaidic acid, a constituent of the seeds of *Chilopsis linearis*, which also contain the *all-E*-(10,12)-isomer. A number of other isomers have been synthesized chemically.

octadecanoic acid the systematic name for stearic acid; *see stearate*.

octadecapentaenoic acid any straight-chain fatty acid having eighteen carbon atoms and five double bonds per molecule. The *all-Z*-(3,6,9,12,15)-isomer occurs naturally in dinoflagellates.

octadecatetraenoic acid any straight-chain fatty acid having eighteen carbon atoms and four double bonds per molecule. Several isomers occur naturally, including the 3*E*,9*Z*,12*Z*,15*Z*-isomer, found in seed oil, and the *all-Z*-(6,9,12,15)-isomer, found in fish oil. The 9*Z*,11*E*,13*E*,15*Z*-isomer and the *all-E*-(9,11,13,15)-isomer are α - and β -parinaric acid respectively, from *Parinarium laurinum*; both isomers are used as fluorescent probes.

octadecatrienoic acid any straight-chain fatty acid having eighteen carbon atoms and three double bonds per molecule. A number of isomers occur naturally, the most prominent being the *all-Z*-(9,12,15)-isomer, (9,12,15)-linolenic acid; the *all-Z*-(6,9,12)-isomer is (6,9,12)-linolenic acid (*see linolenate*). *See also columbinic acid, eleostearate*.

octadecenoic acid any straight-chain fatty acid having eighteen carbon atoms and one double bond per molecule. (*Z*)-Octadec-9-enoic acid (i.e. oleic acid; *see oleate*) is the isomer found widely, often in high proportion, in plant and animal lipids, especially in olive oil. The 5*E*- and 6*Z*-(petroselenic acid; *see petroselenate*) isomers are found in seed oils; the 6*E*- isomer is named **petroselaidic acid**, and the 9*E*- isomer is elaidic acid, a common constituent of fats and oils (*see elaidate*). The 11*E*- and 11*Z*-isomers are, respectively, *trans*- and *cis*-vaccenic acid (*see vaccenate*).

octaethyleneglycol mono-*n*-decyl ether a nonionic $C_{10}E_8$ detergent (*see C₈E₉*).

octaethyleneglycol mono-*n*-dodecyl ether a nonionic $C_{12}E_8$ detergent (*see C₈E₉*).

octahedral coordination coordination in which six ligands are disposed about a central atom as if at the vertices of an octahedron enclosing that atom.

octahedron (*pl. octahedra*) any solid geometrical figure having eight plane triangular faces, 12 edges, and six tetrameric vertices. In a regular octahedron the faces are congruent equilateral triangles and may be considered to consist of two equal

pyramids opposing each other on the same square base. —**octahedral adj.**

octamer 1 an eight-base sequence element that is common in eukaryotic **promoters**. The consensus sequence is ATTTGCAT. It binds a number of transcription factors (*see OCT*). 2 any polymer of a protein having eight subunits, e.g. β -lactoglobulin at pH 4.7 below 4°C. 3 an assembly of eight **histones** containing two each of H2A, H2B, H3, and H4, that forms the histone core of the nucleosome.

octanoate or (formerly) **caprylate** 1 systematic and preferred name for the anion, $CH_3-[CH_2]_6-COO^-$, derived from octanoic acid (formerly known as caprylic acid). 2 any mixture of octanoic acid and its anion. 3 any salt or ester of octanoic acid.

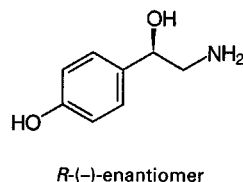
octanoyl or (formerly) **capryloyl** or **caprylyl** symbol: Oco; the univalent acyl group, $CH_3-[CH_2]_6-CO-$, derived from octanoic acid (formerly known as caprylic acid).

octet 1 any group of eight. 2 (in chemistry) (esp. valency) a stable group of eight electrons constituting an electron shell of an atom.

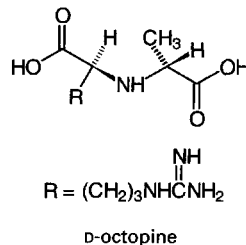
octo+ a variant form of **octa+**.

octonic acid any monocarboxylic **aldonic acid** formally derived from an octose by oxidation at C-1.

octopamine β -hydroxytyramine; 1-(*p*-hydroxyphenyl)-2-aminoethanol; the D(-)-enantiomer is a biogenic amine, about one-tenth as active as norepinephrine, formed by the β -hydroxylation of tyramine by dopamine β -hydroxylase. It is found in the salivary glands of *Octopus* spp. and of *Eledone moschata*.



octopine 1 *N*-(1-carboxy-4-guanidinobutyl)-L-alanine; N^2 -(1-carboxyethyl)-L-arginine; D-octopine (i.e. N^2 -(D-1-carboxyethyl)-L-arginine) is an **opine** found in the tumours of crown-gall disease in plants and in the muscles of certain invertebrates. The genes responsible for the synthesis of octopine are part of the **T-DNA** from a **Ti-plasmid**. 2 any opine that, like octopine itself, is an *N*-substituted derivative of alanine. Octopine was first isolated from the muscles of *Octopus*, but is found in other cephalopod species and lamellibranchs. The guanidine group can undergo phosphorylation to phosphooctopine, which acts as a **phosphagen**. Compare **octopinic acid**.



octopinic acid *N*-(1-carboxy-4-aminobutyl)-L-alanine; N^2 -(1-carboxyethyl)-L-ornithine; D-octopinic acid (i.e. N^2 -(D-1-carboxyethyl)-L-ornithine) is an **opine** found in the tumours of crown-gall disease of plants; it is a member of the **octopine** (def. 2) family.

octose any **aldose** having a chain of eight carbon atoms in the molecule.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



octulose

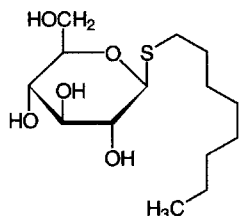
octulose any **ketose** having a chain of eight carbon atoms in the molecule.

octyl *symbol:* Oc; the alkyl group, $\text{CH}_3\text{--}[\text{CH}_2]_6\text{--CH}_2\text{--}$, derived from octane.

octyl glucoside *n*-octyl β -D-glucopyranoside; a mild nonionic detergent, aggregation number 84, CMC 20–25 mM. The α -glucoside isomer has also been used, e.g. for crystallization of membrane proteins. The lack of absorbance at 228 nm is an advantage in the use of these compounds. *Compare* **heptyl glucoside**, **hexyl glucoside**, **nonyl glucoside**, **octyl maltoside**.

octyl maltoside *n*-octyl β -D-maltopyranoside; a mild nonionic detergent, CMC 23 mM. *Compare* **octyl glucoside**.

octyl thioglucoside *n*-octyl 1-thio- β -D-glucopyranoside; a mild nonionic detergent, CMC 9 mM, with properties similar to other alkyl glucosides.



ocular **1** of or concerned with the eye or vision. **2** or **eyepiece** the lens or system of lenses in an optical instrument (especially a compound microscope) that is positioned nearest to the eye.

ocytocin (sometimes) a variant spelling of **oxytocin**.

od or **o.d.** *abbr.* for outside diameter.

OD or **O.D.** *abbr.* for optical density (now called **absorbance**).

Oe *symbol* for oersted.

oedema an alternative spelling (*esp. Brit.*) of **edema**.

oenanthate an alternative spelling (*esp. Brit.*) of **enanthate** (see **heptanoate**).

oenology an alternative spelling (*esp. Brit.*) of **enology**.

oersted *symbol:* Oe; the cgs unit of magnetizing force or magnetic field strength;

1 oersted = $10^3/4\pi$ amperes per metre (= 79.58 A m⁻¹).

[After Hans Christian Oersted (1777–1851), Danish physicist and chemist who discovered the phenomenon of electromagnetism.]

17 β -oestradiol an alternative spelling (*esp. Brit.*) of **17 β -estradiol**.

OEt *symbol* for the ethoxy group, $\text{CH}_3\text{--CH}_2\text{--O--}$ (alternative to EtO).

off-rate the constant, k_{-1} , for the dissociation step in any binding equilibrium, e.g. $\text{LB} \rightarrow \text{L} + \text{B}$ in the equilibrium $\text{L} + \text{B} = \text{LB}$, where L represents ligand and B the molecule it binds; the rate of dissociation of a ligand from the bound complex. The dimensions of k_{-1} are time⁻¹. *Compare* **dissociation constant**, **on-rate**.

OGCP *abbr.* for oxoglutarate/malate carrier protein; a mitochondrial inner membrane integral protein that plays an important role in several processes, including the malate-aspartate shuttle, gluconeogenesis from lactate, and nitrogen metabolism. Example from human: database code M2OM_HUMAN, 313 amino acids (33.91 kDa). *See also* **mitochondrial carrier proteins**.

+ogen a variant form of **+gen** (after a consonant).

Ogston concept or **three-point attachment hypothesis** a concept formulated to explain the inherent differing reactivities, often expressed in an enzymic reaction, of identical chemical groups in a **prochiral** molecule. It states that there must be at least three points of attachment of a substrate molecule to the active site of an enzyme. It is similar to the **Easson-Stedman model**. [After Alexander George Ogston (1911–96), British biophysicist who formulated it in 1948.]

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

25-OHD₃ *abbr.* for 25-hydroxycholecalciferol; **calcidiol**; use not recommended.

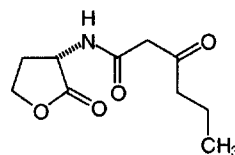
1,25-(OH)₂D₃ *abbr.* for 1,25-dihydroxycholecalciferol; **calcitriol**; use not recommended.

24,25-(OH)₂D₃ *abbr.* for 24(R),25-dihydroxycholecalciferol; (24R)-hydroxycalcidiol; use not recommended.

25,26-(OH)₂D₃ *abbr.* for 25,26-dihydroxycholecalciferol; 26-hydroxycalcidiol; use not recommended.

1,24,25-(OH)₃D₃ *abbr.* for 1,24(R),25-trihydroxycholecalciferol; **calcitriol**; use not recommended.

OHHL *abbr.* for *N*-(3-oxohexanoyl)-L-homoserine lactone.



ohm *symbol:* Ω ; the SI derived unit of electric resistance, defined as the resistance between two points on a conductor through which a current of one ampere flows as the result of a potential difference of one volt applied between the points, the conductor not being the source of any electromotive force; i.e. $1 \Omega = 1 \text{ V A}^{-1}$. [After Georg Simon Ohm (1787–1854), German physicist.]

Ohm's law a law stating that under constant conditions the current, I , flowing through a given conductor is proportional to the potential difference, U , applied across it. The law is often expressed in the form $U = IR$, where the proportionality constant, R , is the resistance of the conductor.

+oic *suffix (in chemical nomenclature)* denoting the carboxyl group, --COOH . It is added to the systematic name of an unbranched alkane (or substituted alkane) to denote its modification to the corresponding alkanic (i.e. carboxylic) acid by formal conversion of a methyl group to a carboxy group; e.g. hexanoic acid, $\text{CH}_3\text{--}[\text{CH}_2]_4\text{--COOH}$, formally derived from hexane, $\text{CH}_3\text{--}[\text{CH}_2]_4\text{--CH}_3$. It is used also in the trivial names of certain carbocyclic and heterocyclic carboxylic acids; e.g. benzoic acid, furoic acid.

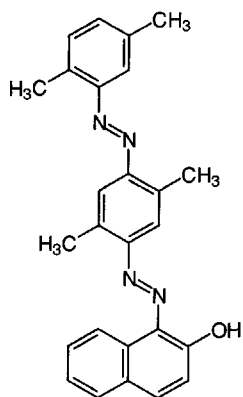
+oid *suffix forming adjectives and (associated) nouns* denoting likeness to, or having the form of (something specified); e.g. colloid (i.e. glue-like substance). **2** generated by rotation of (the geometrical figure specified); e.g. ellipsoid (i.e. solid figure derived from an ellipse). **3** belonging to the class or group represented by (the type member specified); e.g. steroid (i.e. compound belonging to the same chemical family as sterols). **—+oidal** *adj.*

oil any of a large, heterogeneous group of neutral, flammable substances that are liquid at room temperature and are characteristically soluble in relatively nonpolar solvents but only sparingly soluble in aqueous solvents. There are three main distinctive groups: (1) animal and vegetable oils, which usually consist predominantly of triacylglycerols but may contain varying amounts of fatty-acid esters of other alcohols; (2) mineral oils, derived from petroleum, coal, shale, etc., which consist predominantly of hydrocarbons; and (3) **essential oils**.

oil body an alternative name for **spherosome**.

oil-immersion objective an objective lens in a light microscope designed to be used with a layer of a special oil between the object (or its coverslip) and the lens. The oil is of the same refractive index as the glass of the lens. This arrangement maximizes the numerical aperture and hence the resolving power of the lens.

Oil Red O a stain for lipid it is widely used for staining lipoproteins on cellulose acetate electrophoretic strips.



Oil Red O

oilseed rape or **colza** the common name for *Brassica campestris*, a species of rape grown as a source of rapeseed oil.

okadaic acid a polyether fatty acid, $M_r = 804.9$, used as a potent inhibitor of **protein phosphatases**, especially the PP-1 and PP-2A types. It does not inhibit protein tyrosine kinases and does not activate protein kinase C. It is a tumour promoter, and is implicated as the causative agent of diarrhetic shellfish poisoning. Its name derives from *Hilichondria okadaei*, a marine sponge that feeds on *Prorocentrum lima*, a dinoflagellate from which it is isolated. See structure below.

Okazaki fragment or **Okazaki piece** or **Okazaki segment** any of the relatively short polydeoxyribonucleotides that are formed, concomitantly with the continuous replication of one of the two strands of a duplex DNA molecule, as intermediates during the discontinuous replication of the other strand. Such fragments are synthesized from mononucleotides in a direction opposite to that of the movement of the replication fork; then, as replication proceeds, they are covalently joined through the action of a polynucleotide ligase to form a long daughter polynucleotide chain, the lagging strand. These fragments appear to be 1000–2000 residues long in *Escherichia coli* and 100–200 residues long in mammalian cells; a proportion of them may have RNA primer attached. [After Reiji Okazaki (1930–1975), Japanese molecular biologist.]

OK cell *abbr.* for opossum kidney cell; any cell of a culture derived from this organ and used to study α_2C adrenoceptors.

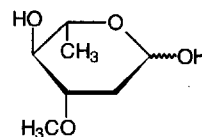
+ol suffix (in chemical nomenclature) indicating the presence of a hydroxyl group attached to a carbon atom.

+olate suffix used as an alternative to **+oxide** (def. 1).

old yellow enzyme an obsolete name for the enzyme NADPH dehydrogenase (EC 1.6.99.1), a yellow flavoprotein (FMN in yeast, FAD in plants). Compare **new yellow enzyme**.

Ole symbol for the oleoyl (i.e. (Z)-octadec-9-enoyl) group.

oleandrose 2,6-dideoxy-3-O-methyl-arabino-hexose; the L enantiomer is a component of some **cardiac glycosides**.



L-oleandrose

oleate 1 numerical symbol: 18:1(9); the trivial name for Z-octadec-9-enoate, $CH_3-[CH_2]_7-CH=CH-[CH_2]_7-COO^-$ (cis isomer), the anion derived from oleic acid (i.e. Z-octadec-9-enoic acid), a monounsaturated straight-chain higher fatty acid. See also **octadecenoic acid**, **oleic family**. 2 any mixture of free oleic acid and its anion. 3 any salt or ester of oleic acid. Compare **elaidate**, **vaccenate**.

oleate desaturase see **phosphatidylcholine desaturase**.

olefin an older name for an alkene. —olefinic *adj.*

oleic family (of polyunsaturated fatty acids) a series of polyenoic acids in which the hydrocarbon chain terminates with the alkenyl grouping $CH_3-[CH_2]_7-CH=CH-$ (as in oleic acid); there are three other series, i.e. the **linoleic family**, **linolenic family**, and **palmitoleic family**. The oleic family can be synthesized from acetyl-CoA: palmitoyl-CoA is first synthesized by the fatty-acid synthase (see **fatty-acid synthase complex**); next, by chain elongation and action of **desaturases**, stearoyl-CoA, oleoyl-CoA, and longer chain fatty acyl-CoA compounds are synthesized; then from their respective CoA derivatives, the cognate fatty acids are formed. The series includes (all-Z)-eicosa-5,8,11-trienoic acid. In **essential fatty acid deficiency** in mammals, longer chain polyunsaturated fatty acids of the oleic family are synthesized, and partially substitute for long-chain polyunsaturated fatty acids of the linoleic family. See also **eicosenoic acid**, **eicosatrienoic acid**, **Mead acid**.

olein an old name for any of the glyceryl esters of oleic acid (see **oleate**), especially the triester, **triolein** (i.e. trioleoylglycerol).

oleo (US) *abbr.* for oleomargarine.

oleomargarine an alternative name (esp. US) for **margarine**.

oleosome an alternative name for **spherosome**.

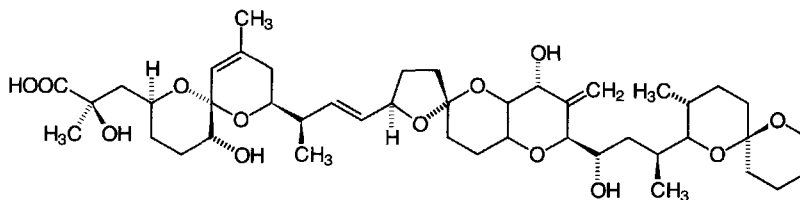
oleoyl 1 a common name for Z-octadec-9-enyl, $CH_3-[CH_2]_7-CH=CH-[CH_2]_7-CO-$ (cis isomer), the acyl group derived from oleic acid (i.e. (Z)-octadec-9-enoic acid). It occurs very widely in natural lipids. Compare **elaidoyl**, **vaccenoyl**. See also **oleic family**.

1-oleoyl-2-acetylglycerol *abbr.*: OAG; an activator of protein kinase C.

oleum fuming sulfuric acid; a solution of SO_3 in concentrated $H_2SO_4 \rightarrow H_2S_2O_7$.

oleyl 1 a common name for Z-octadec-9-enyl, $CH_3-[CH_2]_7-CH=CH-[CH_2]_8-$ (cis isomer), the monoalkenyl group derived from Z-octadec-9-ene; use deprecated. 2 an old and misleading trivial name for Z-octadec-9-enoyl; oleoyl now used.

olfactomedin a major glycoprotein component of the extracellular matrix of olfactory neuroepithelium. Example (precursor) from *Rana catesbeiana*: database code OLFM_RANCA, 464 amino acids (52.51 kDa).



okadaic acid



olfactory receptor

olfactory receptor any of a group of integral membrane proteins thought to be involved in effecting the sense of smell. They are similar to other G-protein-coupled receptors; several are known (or suspected). Example, presumed olfactory receptor-like protein HGMP07E from human: database code OLFE_HUMAN, 312 amino acids (35.17 kDa); five motifs.

+olide suffix (to the chemical name of an aliphatic hydrocarbon) denoting conversion to a lactone having the same number of carbon atoms as the parent hydrocarbon.

oligo abbr. (colloquial) for **oligonucleotide**.

oligo+ or (before a vowel) **olig+** comb form denoting 1 few, little, not much, or not many. 2 (in chemistry and biochemistry) a small number of (the component specified); e.g. **oligopeptide**. 3 (in pathology) deficiency or insufficiency of (some thing or some attribute); e.g. **oliguria**. Compare **multi+**, **poly+**.

oligo-2',5'-adenylate synthetase an enzyme of the sub-subclass EC 2.7.7 that binds double-stranded RNA and polymerizes ATP into pppA(2'p5'A)_n oligomers; the latter activate the latent RNase L causing it to cleave single-stranded RNAs. The enzyme is induced by **interferons**. It is a homotetramer, associated with many different subcellular fractions. In humans, two forms, E16 and E18, are produced by alternative splicing of the same gene. Examples from human, E16: database code 25A1_HUMAN, 364 amino acids (41.73 kDa); E18: database code 25A2_HUMAN, 400 amino acids (46.00 kDa).

oligodendrocyte a type of glial cell that forms and supports the myelin sheath around axons in the central nervous system of vertebrates. In contrast to the **Schwann cell**, which performs this role in peripheral neurons for a single axon, one oligodendrocyte is involved in the myelination of several axons.

oligo-1,6-glucosidase (EC 3.2.1.10), other names: sucrose-isomaltase; limit dextrinase; isomaltase; an enzyme that catalyses the hydrolysis of 1,6- α -D-glucoside linkages in isomaltose and dextrans produced from starch and glycogen by α -amylase. Example, isomaltase from *Bacillus cereus*: database code O16G_BACCE, 558 amino acids (65.94 kDa). A similar enzyme is intestinal human enzyme, which has sucrose/maltase activity; EC 3.2.1.48: database code SUI5_HUMAN, 1827 amino acids (209.17 kDa).

oligohistidine the homopolypeptide of histidine; the term is used in referring to a **histidine tail**.

oligomer 1 (in chemistry and biochemistry) any substance or type of substance that is composed of molecules containing a small number – typically two to about ten – of constitutional units in repetitive covalent linkage; the units may be of one or of more than one species. Depending on the branch of chemistry concerned, an oligomer may or may not be considered to be a variety of **polymer**. 2 (in molecular biology and enzymology) any multimeric protein that contains a finite, relatively small, number of identical subunits (i.e. protomers) that are not in covalent linkage but in a state of reversible association with each other; the protomers may themselves each consist either of single polypeptide chains (i.e. monomers) or of two or more reversibly associated (identical or nonidentical) monomers. This definition is based exclusively on the identity of the subunits and does not restrict the number of polypeptide chains each identical subunit may contain. —**oligomeric** adj.; **oligomerize** or **oligomerise** vb.; **oligomerization** or **oligomerisation** n.

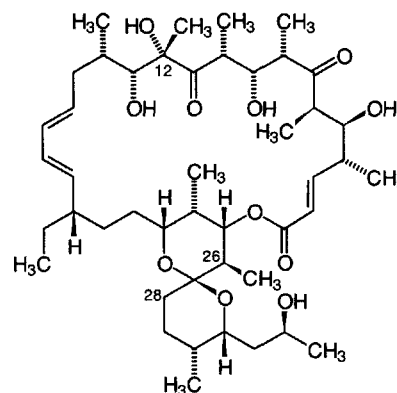
oligomycin any of a group of closely related **macrolide** antibiotics produced by an actinomycete similar to *Streptomyces diastatochromogenes*. Oligomycin B is an inhibitor of oxidative phosphorylation in mitochondria, interfering with the passage of hydrons through the F₀ component of **H⁺-transporting ATP synthase**, and of photophosphorylation in chloroplasts. Oligomycin D (**rutamycin**) is useful as an antifungal agent. (Illustrated opposite.)

oligomycin sensitivity-conferring protein abbr.: OSCP; a protein component of mitochondrial ATP synthase that is required for the binding of the F₁ component to the F₀ component of the enzyme system. See also **H⁺-transporting ATP synthase**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



oligomycin A
oligomycin B: 28-oxo
oligomycin C: 12-deoxy
oligomycin D: 26-demethyl

oligonucleotide abbr. (colloquial): oligo; any molecule that contains a small number (two to about ten) of nucleotide units connected by phosphodiester linkages between (usually) the 3' position on the glycoside moiety of one nucleotide unit and the 5' position on the glycoside moiety of the adjacent one.

oligopeptide any molecule that contains a small number (2 to about 20) of amino-acid residues connected by peptide linkages.

oligophosphate 1 any ion or salt of an **oligophosphoric acid** — an acid containing commonly two or three, but sometimes four or more, phosphoric residues connected by anhydride linkages. 2 any mixture of a free oligophosphoric acid and its various anions. 3 or **oligophosphoric ester** any ester of an oligophosphoric acid.

oligosaccharide any molecule that contains a small number (2 to about 20) of monosaccharide residues connected by glycosidic linkages.

oligosaccharin any of a group of regulatory oligosaccharides, including a fungal heptagluco-side and a plant oligogalacturonide, that are released from cell walls and control functions such as growth, development, reproduction, and defence against disease. They also include xyloglucans.

***N*-oligosaccharyltransferase** an enzyme that transfers an oligosaccharide with high mannose content from a lipid-linked oligosaccharide donor to an asparagine acceptor site on newly synthesized glycoproteins. It has three subunits: **ribophorins** I and II, and a 48 kDa subunit, database code (precursor) OST4_HUMAN, 456 amino acids (50.71 kDa).

oligotrophic describing a habitat, especially a lake or other mass of water, that is poor in nutrients capable of supporting the growth of aerobic plants and microorganisms, and (in the case of a lake, etc.) that hence contains abundant dissolved dioxygen at all depths.

olivomycin one of several related components of a complex of antibiotics produced by *Streptomyces olivoreticuli*. It has been investigated for antineoplastic activity. It is used as a dye specific for G–C base pairs, and stains the R bands first demonstrated in Giemsa staining.

+oma (pl. **+omas** or **+omata**) suffix denoting a tumour.

OMe symbol for the methoxy group, CH₃–O– (alternative to MeO).

omega symbol: ω (lower case) or Ω (upper case); the twenty-fourth and last letter of the Greek alphabet. For uses see **Appendix A**.

Omega an alternative name for **Mega-8**.

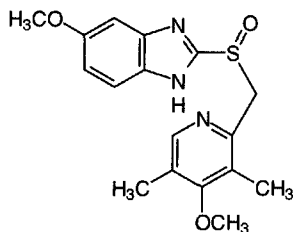
omega-oxidation or ω -oxidation a minor oxidative metabolic pathway of fatty acids, particularly those of medium chain-length, in which the methyl group that contains the carbon atom furthest from the carboxyl group of such an acid is oxidized.



dized first to a hydroxymethyl group and ultimately to another carboxyl group.

omega protein or ω -protein an alternative name for type I DNA topoisomerase.

omeprazole the nonproprietary pharmaceutical name for 5-methoxy-2-[[[4-methoxy-3,5-dimethyl-2-pyridinyl)methyl]sulfinyl]-1H-benzimidazole, a gastric proton-pump inhibitor. It blocks both basal and stimulated secretion of acid in the stomach by reversibly inhibiting H^+/K^+ -transporting ATPase (EC 3.6.1.36) in the cell membranes of parietal cells within the oxyntic glands of the gastric mucosa. It is useful clinically in the treatment of erosive reflux esophagitis, benign peptic ulcer, and Zollinger–Ellison syndrome. *Proprietary name*: Losec.



omicron symbol: o (lower case) or O (upper case); the fifteenth letter of the Greek alphabet.

ommatidium any of many identical structural units of the compound eye found in many invertebrates. In *Drosophila*, each consists of eight photoreceptor cells and 12 accessory cells. In sevenless and bride-of-sevenless mutants, one of the photoreceptor cells fails to develop.

ommatin a type of **ommochrome**.

ommatochrome an alternative name for **ommochrome**.

ommin a type of **ommochrome**.

ommochrome or **ommatochrome** any of a large group of yellow, brown, red, or violet natural polycyclic pigments especially commonly found in the Arthropoda, particularly in the ommatidia of the compound eye. They all contain a phenoxazine nucleus and are formed biologically from tryptophan via kynurenine. Ommochromes are divided into two groups: **ommatis**, which are alkali-labile, of low relative molecular mass, and rather weakly coloured; and **ommins**, which are alkali-stable, and of higher relative molecular mass and more strongly coloured than the ommatis.

OMP abbr. for orotidine monophosphate; see **orotidine phosphate**.

OMP decarboxylase abbr. for orotidine-5'-phosphate decarboxylase, a reaction carried out by the multifunctional enzyme **uridine 5'-monophosphate synthase**.

+onate noun suffix denoting the anion, a salt, or an ester of an aldonic acid. See also **uronate**.

onco+ comb. form denoting 1 tumour. 2 volume.

oncofetal parvalbumin see **oncomodulin**.

oncogen any agent that causes **oncogenesis**; the term embraces **carcinogen**.

oncogene any gene associated with the causation of cancer. The first oncogenes to be characterized were isolated from acutely transforming **retroviruses**; it was found subsequently that the retroviral oncogenes (v-onc) were related to, and probably derived from, normal cellular genes (c-onc), which came to be termed **protooncogenes**; these in most cases are involved with growth control. The derivation of oncogenes from protooncogenes is presumed to be the consequence of mutations in protooncogenes after capture by the virus, removing regulatory constraints from the protooncogene or its product. This may result from (1) deletions of portions of the oncogene; (2) mutation of the oncogene resulting in a protein product that is in an active state even in the absence of any normally required stimulus; (3) translocation of the gene from the normal site of expression, resulting in an altered gene product.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

with aberrant function; or (4) amplification of the gene resulting in inappropriate expression or overexpression of the product. Protooncogenes function in normal cells in growth and differentiation pathways. They and their oncogenic counterparts can be grouped into categories based on their roles in these processes. Some have products that act as growth factors (e.g. *sis*, *int-2*) or growth-factor receptors (e.g. *fms*, *neu*, *kit*). Others encode signal-transduction molecules normally activated by growth-factor receptors (e.g. *ras*, *raf*, *mos*, and *src* family members). Others encode proteins that function in the nucleus (e.g. *myc*, *myb*, *fos*, *jun*) where they influence the expression of other genes. For further details see under individual oncogenes. Strictly speaking, those genes carried by DNA tumour viruses, such as **SV40** and **adenoviruses**, should also be construed as oncogenes. Unlike the retroviral genes, however, they appear to have no cellular counterparts but instead operate by subverting the actions of cellular genes (such as protooncogenes) that normally function to control cell growth. See also **FMS**, **FOS**, **int-2**, **JUN**, **KIT**, **MOS**, **MYB**, **MYC**, **neu**, **RAF**, **RAS**, **sis**, **src**.

oncogenesis the production or formation of neoplastic tumours or a neoplastic tumour, whether benign or malignant; the term embraces **carcinogenesis**.

oncogenic (of an agent) causing the formation of a neoplastic tumour or tumours; of or pertaining to **oncogenesis**; the term embraces carcinogenic. —**oncogenicity** *n*.

oncology the branch of medicine concerned with the study and treatment of neoplastic tumours. —**oncological** *adj*; **oncologist** *n*.

oncolysis the destruction of neoplastic tumours or tumour cells, whether spontaneous or in response to treatment. —**oncolytic** *adj*.

oncomodulin other names: parvalbumin beta; oncofetal parvalbumin; an **EF-hand** Ca^{2+} -binding protein that may function in a similar manner to **calmodulin**. Expression is limited to early embryonic stages, the placental cytotrophoblasts, and neoplastic tissues. Example from human: database code ONCO_HUMAN, 108 amino acids (12.01 kDa).

oncornavirus an old name for a group of RNA tumour viruses now classified in the subfamily **Oncovirinae**. See **retrovirus**.

oncostatin M abbr.: OSM; a polypeptide cytokine, produced by activated T cells and PMA-treated monocytes, that inhibits the growth of a variety of cancer cells, but stimulates growth of AIDS-Kaposi's sarcoma-derived cells and normal fibroblasts. It is a member of the **interleukin 6** subfamily. It exists as a monomer. Example from human (precursor): database code ONCM_HUMAN, 252 amino acids (28.48 kDa); residues 1–25 are the signal, 26–234 oncostatin M, and 235–252 a propeptide.

oncotic pressure the **colloid osmotic pressure** of plasma proteins; it is important as an index of the flow of water between the blood and the fluid in the tissues.

+one noun suffix denoting ketone.

one-carbon compound an alternative name for **C₁ compound**.

one cistron-one polypeptide hypothesis an alternative name for the **one gene-one polypeptide hypothesis**.

one gene-one antigen hypothesis a hypothesis stating that each of the genes in a tissue transplant governing the host's reaction to it is responsible for the manufacture of a particular transplantation antigen and for that antigen alone. It was first formulated by W. Beadle and E.L. Tatum.

one gene-one enzyme hypothesis a hypothesis, developed from the **one gene-one reaction hypothesis**, stating that every enzyme is synthesized under the control of a different individual gene. It was subsequently refined into the **one gene-one polypeptide hypothesis**.

one gene-one polypeptide hypothesis or **one cistron-one polypeptide hypothesis** a hypothesis stating that each naturally occurring polypeptide is synthesized biologically under the control of a different corresponding gene, which is transcribed into a unique species of messenger RNA specifying the polypeptide in question. Subsumed within this hypothesis are various earlier ones, formulated before the molecular basis of



one gene—one reaction hypothesis

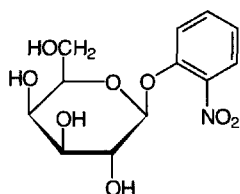
the genetic control of polypeptide synthesis had been elucidated.

one gene—one reaction hypothesis a hypothesis stating that every specific biochemical reaction is under the ultimate control of a different gene. It was later redefined as the **one gene—one enzyme hypothesis**.

+onic *suffix* denoting the formal conversion of an aldose to the corresponding aldonic acid; e.g. glucose gives rise to gluconic acid.

+onium *noun suffix* denoting a cationic molecular entity in which a heteroatom is bonded to one more univalent ligand than is normal for a neutral molecule containing that heteroatom and therefore bears a formal positive charge; e.g. ammonium, H_4N^+ ; dimethylammonium, $(\text{CH}_3)_2\text{H}_2\text{N}^+$; diphenyliodonium, $(\text{C}_6\text{H}_5)_2\text{I}^+$; trialkylsulfonium, R_3S^+ (where R = alkyl). *See also carbonium ion* (def. 2).

ONPG *abbr.* for *ortho*-nitrophenyl- β -D-galactopyranoside; a chromogenic substrate used for the detection and assay of β -galactosidase activity.



on-rate the constant, k_{+1} , for the association step $\text{L} + \text{B} \rightarrow \text{LB}$ in the equilibrium $\text{L} + \text{B} \rightleftharpoons \text{LB}$, where L represents ligand and B the molecule it binds; the rate of association of ligand to form a complex. The dimensions of k_{+1} are time^{-1} . *Compare association constant, off-rate*.

Onsager, Lars (1903–76), Norwegian-born US physical chemist noted for his work on the theory of dielectrics and for his research into nonequilibrium thermodynamics; Nobel Laureate in Chemistry (1968) for 'the discovery of the reciprocal relations bearing his name, which are fundamental for the thermodynamics of irreversible processes'.

Onsager theory an extension of the **Debye-Hückel theory** of dielectrics. *See also Kirkwood theory*.

ontogeny or **ontogenesis** the entire sequence of events in the development of an individual organism during its lifetime. *Compare phylogeny*. —**ontogenic** or **ontogenetic** *adj.*

oo+ *comb. form* denoting ovum or egg.

oocyte (*in zoology*) a cell of an animal ovary that undergoes meiosis to form an ovum. A **primary oocyte** is formed by differentiation of an oogonium and then undergoes the first division of meiosis to form a polar body and a secondary oocyte. Following fertilization of the egg, the **secondary oocyte** undergoes the second meiotic division to form the mature ovum and a second polar body.

oogenesis the complete process of formation and maturation of an ovum from a primordial female germ cell. —**oogenetic** *adj.*

oogonium (*pl. oogonia*) **1** (*in zoology*) a cell of an animal ovary that undergoes repeated mitosis, eventually forming **oocytes**. **2** (*in botany*) the female reproductive organ in certain algae and fungi. —**oogonial** *adj.*

OP *abbr.* for organophosphorus. *See organophosphate*.

opal codon or **opal triplet** *symbol*: UGA; one of the three terminator codons or **nonsense codons** in a molecule of messenger RNA.

opalescence a milky iridescence, similar to that of opal. The opalescence of some solutions is due to the reflection of light by very fine suspended particles. —**opalescent** *adj.*; **opalesce** *vb.*

opal mutant any cell or virus carrying an **opal mutation**.

opal mutation a mutation that gives rise to an **opal codon** at an abnormal position in a molecule of messenger RNA.

opal suppressor any of a number of mutations, occurring in *Escherichia coli*, that suppress an **opal codon** in messenger RNA thereby allowing insertion of one of several alternative amino acids into a polypeptide chain at that site. The mutation results in changed anticodons of the transfer RNA molecules, consequent on single-base substitution in their DNA coding sequences.

open chain (*in chemistry*) **1** a chain of atoms in a molecule that is not formed into a ring, i.e. one that has two (or more) free ends. **2** **open-chain** an alternative term for **acyclic**.

open circle any circular duplex DNA structure containing one or more single-stranded nicks. Such a structure will dissociate at high pH into two (or more) single-stranded structures.

open circuit (*in physics*) an incomplete electrical circuit, i.e. one with infinite impedance.

open-loop system a type of system, e.g. a metabolic pathway or an electrical circuit, in which control is not exerted by feedback, each operation or activity being affected only by those earlier in the sequence. *Compare closed-loop system*.

open reading frame *abbr.*: ORF; a series of codon triplets, deduced from a DNA sequence, that include a 5' **initiation codon** running through to a **termination codon**, and representing a putative or known gene. The problem often arises, e.g. in screening DNA libraries, of determining whether a stretch of DNA represents a gene. For this to be the case, it must, apart from including appropriately placed initiation and termination codons, be of a suitable length and should fulfil certain criteria indicating that the correct **reading frame** has been selected. Methods for checking this include the **RYN method**.

open system (*in thermodynamics*) any geometrically defined volume that can exchange both energy (e.g. heat or work) and matter with its surroundings. *Compare adiabatic system, closed system*.

open tetrapyrrole *see* **phycobilin**.

operand a quantity or function on which a mathematical operation is to be performed.

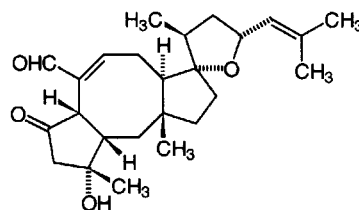
operating system (*in computing*) the software that handles input-and-output, file handling, and provides the **command language** and other user interfaces.

operator **1** (*in genetics*) a DNA sequence within an **operon** (under negative promoter control) to which a specific repressor protein can bind, thereby regulating the functioning of a **structural gene** or group of genes. **2** (*in mathematics*) a symbol (i.e. an alphabetic character, term, or special symbol) representing a particular operation to be performed on an operand or operands; e.g. Σ (= sum of), d/dx (= differentiate with respect to the variable x), $+$ (= add to).

operon a unit of coordinated and regulated gene activity found in prokaryotes, by means of which the control of the synthesis of a protein or a group of (usually functionally associated) proteins is determined. It consists of a segment of genomic DNA containing a **structural gene** or a linear sequence of structural genes (which are transcribed as a single unit), together with one or more regulatory regions such as an **operator** (def. 1). *See also Jacob-Monod model, lac operon*.

ophidin a trivial name for 2-methylcarnosine; N^α -(β -alanyl)-2-methylhistidine; a constituent of snake muscle.

ophiobolin a class of sesterterpene produced only by fungi; there are six types based on the ophiobolane structure shown.



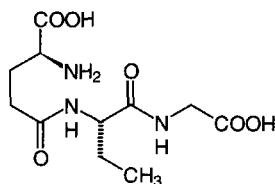
Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.



ophthalmic of, or pertaining to, the eye.

ophthalmic acid *N*-[*N*-(γ -glutamyl)- α -aminobutyl]glycine; an analogue of glutathione in which the thiol group is replaced by a methyl group. It was originally isolated from the lens of the eye.



opiate 1 any of a group of narcotic drugs structurally related to and including **morphine** all of which are derived from opium. Apart from morphine the group includes **codeine**, **papaverine**, and **thebaine**. The term is sometimes extended to include the **opioids** (def. 1) and any other naturally occurring, semisynthetic, or synthetic narcotic substance with similar properties and whose morphine-like effects may be reversed by recognized morphine antagonists. 2 consisting of or containing **opium**; having narcotic properties similar to those of opium.

opiate receptor a membrane receptor for opiates (see **opioid receptor**.)

opine any of a class of unusual derivatives of basic amino acids not found in normal plant tissue but found in plant tissue that has been transformed by oncogenic plasmids from agrobacteria. The class includes the **nopaline** and **octopine** families.

opiocortin an alternative name for **opiomelanocortin**.

opioid 1 any nonalkaloid having **opiate**-like pharmacological effects that can be reversed by **naloxone** or other recognized morphine antagonists. 2 describing any substance having such properties.

opioid peptide any naturally occurring peptide that is an **opioid** (def. 1). Five main groups are recognized: (1) [Leu]-enkephalin and [Met]enkephalin; (2) dynorphin, α and β neoendorphin, and several other peptides that arise (or are presumed to arise) from **enkephalin** precursors; (3) α , β , γ , and δ **endorphins**, all of which are formed from β lipotropin; (4) various pronase-resistant peptides present in body fluids, e.g. β **casomorphin** (in bovine milk); (5) sundry other peptides whose opiate-like action appears to be indirect. [Note: In the past, the terms 'endorphin' and 'opioid peptide' were often used synonymously; it is now recommended that the former should be restricted to those opioid peptides that are derived from β lipotropin.]

opioid receptor any site on a cell surface that interacts in a highly selective manner with an opiate or an **opioid** (def. 1) and that mediates its prime pharmacological action; types μ , δ , and κ , and possibly σ have been distinguished on the basis of their response to β endorphin (End), dynorphin A (Dyn), [Met]enkephalin (Met), and [Leu]enkephalin (Leu); the potency being, for μ : End > Dyn > Met > Leu; for δ : End = Leu = Met > Dyn; and for κ : Dyn >> End >> Leu = Met. Morphine and its analogues are primarily μ agonists, but also have activity on δ and κ receptors. Effector pathways for all of these types are G-protein-mediated, μ and δ depressing cyclic AMP levels and opening a G-protein-modulated K⁺ channel, κ inhibiting a G-protein-modulated Ca²⁺ channel. Some opioids also bind to other receptors known as σ , but morphine has a very low affinity for these receptors and naloxone does not block its action.

opiomelanocortin or **opiocortin** a polypeptide, present in the intermediate lobe of the pituitary gland, that includes within it the entire sequences of the peptide hormones **corticotropin**, α and β melanocyte stimulating hormone, β and γ **lipotropin**, and β **endorphin**. It is formed from **proopiomelanocortin**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

opium the milky, air-dried exudate from incised, unripe seed capsules of the opium poppy, *Papaver somniferum*, and some other species. Opium contains approximately 20 alkaloids, the most significant of which are **codeine**, **morphine**, **papaverine**, and **thebaine**.

OPLC *abbr.* for overpressure(d) layer chromatography.

OPRTase see **orotate phosphoribosyltransferase**.

opsin 1 any of a group of hydrophobic glycoproteins that occur in the visual pigments of vertebrates in equimolecular combination (as Schiff's bases) either with 11-*cis*-retinal – in **iodopsin** and **rhodopsin** – or with 3,4-didehydro-11-*cis*-retinal – in **cyanopsin** and **porphyropsin**. These opsins are of ≈ 35 –40 kDa and consist of single-chain polypeptides glycosylated with two branched oligosaccharides. See also **photopsin**, **scotopsin**. 2 any of a group of proteins of ≈ 25 kDa that occur in combination with retinal in the purple-membrane pigments present in some strains of *Halobacterium halobium*; e.g. bacteriorhodopsin (in bacteriorhodopsin), haloopsin (in halorhodopsin), and slow-opsin (in slow-cycling rhodopsin).

opsonify a less common word for **opsonize**.

opsonin any blood-serum protein that, when combined with microorganisms or other particulate material (e.g. foreign erythrocytes), increases the latter's susceptibility to phagocytosis. The two main classes are immunoglobulin G antibodies and the C3b and IC3b fragments formed from the C3 component of **complement**; other classes include **fibronectin** and the soluble mannose-binding protein of serum. An opsonin molecule acts by linking the surface of the particulate matter to a specific receptor on the surface of a phagocytic cell; commonly this receptor is either an Fc receptor (recognizing the Fc domain of an IgG immunoglobulin molecule) or a CR3 receptor (recognizing the IC3b fragment of complement). —**opsonic** *adj.*

opsonize or **opsonise** or (less commonly) **opsonify** to render (microorganisms or other particulate material) more susceptible to phagocytosis by coating with **opsonin**. —**opsonization** or **opsonisation** or **opsonification** *n.*

optic 1 or **optical** of or pertaining to the eye or the sense of sight; aiding vision. 2 a less common word for **optical** (def. 1, 2).

optical 1 or **optic** of, pertaining to, or involving light. 2 or **optic** of or pertaining to **optics**. 3 a less common word for **optic** (def. 1).

optical activity the ability (of chiral substances or their solutions, and chiral crystals) to rotate the plane of polarization of a transmitted beam of plane-polarized light. There is usually a correspondence between **chirality** of molecules and optical activity. See also **chiroptical**.

optical antipode an older term for **enantiomer**.

optical density a term (no longer recommended) for **absorbance**.

optical isomer an older term for **enantiomer**.

optical isomerism an older term for **enantiomerism**.

optically active having or displaying optical activity; the term is often used interchangeably with **chiral**.

optically inactive lacking optical activity (through being either **achiral** or **racemic**).

optical purity the ratio (usually expressed as a percentage) of the **specific optical rotation** of a sample of an enantiomeric substance to the specific optical rotation of a pure single enantiomer of the substance. Compare **enantiomeric purity**.

optical rotation or **rotation** 1 the phenomenon of rotation of the plane of polarization of plane-polarized light on its transmission by (solutions of) chiral substances, and by chiral crystals. By convention, optical rotation is designated as rightwards, or positive – and termed **dextrorotation** – when it is clockwise as seen by an observer looking towards the light source, and designated as leftwards, or negative – and termed **levorotation** – when it is anticlockwise. 2 or **optical rotatory power** the ability of (a solution of) a chiral substance to effect such rotation, an effect that is normally associated with **optical activity**. 3 *symbol:* α ; any measure, expressed in angular degrees, of such ability under specified conditions; see **molar optical rotation**, **specific optical rotation**.

optical rotatory dispersion

optical rotatory dispersion *abbr.*: ORD; the variation of **optical rotation** with the wavelength or frequency of the transmitted light. The shape of an optical rotatory dispersion spectrum of a solution of a macromolecule often gives useful information about the conformation of the solute. *See also* **circular dichroism**, **Cotton effect**.

optical rotatory power *an alternative term for optical rotation* (def. 2).

optical system any arrangement of diffraction gratings, lenses, mirrors, prisms, etc., frequently within a specific instrument and used for a specific purpose.

optics **1** (*functioning as sing.*) the branch of physics concerned with the nature, properties, and behaviour of (visible) light and with the characteristics of optical devices, instruments, and systems. The term is extended to other kinds of electromagnetic or particulate radiation in so far as they have similar properties to light. **2** *a collective term for* an arrangement of optical components (in an instrument); the optical properties of such an arrangement.

OPTLC *abbr. for* overpressured thin-layer chromatography. *See* **overpressure(d) layer chromatography**.

oral of or pertaining to the mouth; administered by mouth.

orbit **1** a closed circular or near-circular course taken by a moving body (often around something). **2** the depression in the skull that contains the eye. **3** to move in or as if in an **orbit** (def. 1). *See also* **orbital**.

orbital **1** any **wave function**, Ψ , of an electron in a molecular entity. It is characterized by a particular energy value and a unique set of three quantum numbers; the counterpart in wave mechanical theory of an electron orbit in Niels Bohr's concept of atomic structure. Any particular orbital may be occupied either by one electron or by two electrons of opposed spins, or it may be unoccupied. The square of the value of Ψ at any point can be interpreted as the probability of the electron being located at that point, or as the average fractional charge density at that point. *See also* **atomic orbital**, **molecular orbital**, **pi orbital**, **sigma orbital**. **2** of, relating to, occupying, or moving in or as if in an **orbit** (def. 1). **3** belonging to the **orbit** (def. 2).

orbital shaker a device designed to impart an orbital motion to flasks, tubes, etc. with the purpose of mixing their contents and facilitating the transfer of gases from the gaseous to the liquid phase. It is useful especially for growing cultures of cells.

orcinol test or **Bial's test** a colorimetric test specific for aldopentoses. The sample is heated with strong acid to convert any aldopentose present to furfural (i.e. 2-furaldehyde), which is then reacted with orcinol (i.e. 5-methyl-1,3-benzenediol) in the presence of ferric chloride to give a blue-green compound extractable into amyl alcohol. The test is used for the determination of RNA and the detection of **pentosuria**. [After Manfred Bial (1870–1908), German physician.]

ORD *abbr. for* optical rotatory dispersion.

Ord *symbol for* a residue of the ribonucleoside orotidine (alternative to O).

order **1** (*in chemistry*) sequence (of residues in a polymer). **2** **order of reaction**. **3** (*in taxonomy*) a category consisting of a number of similar families (sometimes only one family). One or more orders make up a class. **4** (*in mathematics*) the number of times a function must be differentiated to obtain a particular derivative. **5** (*in mathematics*) the number of columns or rows in a determinant or a square matrix. **6** **order of magnitude**.

ordered mechanism or **ordered pathway** (for an enzymic reaction involving two substrates, $A + B \rightleftharpoons C$) a mechanism in which the enzyme, E, must necessarily first bind to one substrate, e.g. $E + A \rightleftharpoons EA$, and then to the second substrate, thus $EA + B \rightleftharpoons EAB$, this ternary complex then reacting to form the product, $EAB \rightleftharpoons E + C$. *Compare* **random mechanism**.

order of magnitude **1** a tenfold range of magnitude extending from a particular value either upwards or downwards. **2** a magnitude expressed to the nearest power of ten.

order of reaction *symbol*: n ; the sum of the exponents of the concentration terms in the rate equation for an elementary

chemical reaction considered in one direction only. It may also be applied to certain composite reactions.

order of reflection *symbol*: n ; the integer, n , in the equation of **Bragg's law**.

ordinate the vertical, or y , coordinate in a plane rectangular (Cartesian) coordinate system. *Compare* **abscissa**.

ORF *abbr. for* open reading frame.

organ any part of the body of a multicellular organism that is adapted and/or specialized for the performance of one or more vital functions. The term is sometimes extended to include organelles of unicellular organisms.

organ culture a category of **tissue culture**, in which an organ, part of an organ, or an organ primordium, after removal from an animal or plant, is maintained *in vitro* in a nutrient medium with retention of its structure and/or function.

organelle any discrete structure in a unicellular organism, or in an individual cell of a multicellular organism, that is adapted and/or specialized for the performance of one or more vital functions.

organic **1** of, pertaining to, or derived from an organism or organisms. **2** of, pertaining to, or affecting an organ or organs. **3** of, relating to, or being a compound of carbon, whether or not of natural origin. **4** of any element contained in an organic compound. **5** any organic compound.

organic chemistry the branch of chemistry concerned with the compounds of carbon in covalent linkage, whether natural or artefactual, including most compounds of biological origin but excluding some simple carbon compounds, e.g. oxides, sulfides, and cyanides. *See also* **organic compound**.

organic compound any compound containing carbon in covalent linkage; various simple carbon compounds, especially carbon dioxide, carbonic acid, and salts and ions of carbonic acid, are usually excluded.

organiser *a variant spelling of organizer*.

organism any unicellular or multicellular prokaryote or eukaryote, usually extended to include various noncellular nucleic-acid-containing infective agents such as viruses.

organizer or **organiser** (*in embryology*) any part of an embryo that, through substances produced by it, stimulates the morphological development and differentiation of other parts.

organo+ *comb. form* denoting **organic** (def. 1, 2, 3).

organogenesis the formation and development of organs in an organism. —**organogenetic** *adj.*

organoleptic affecting, involving, or making use of the sense organs, especially those of smell and taste.

organomercurial **1** or **organomercury** of, pertaining to, or being an organic compound containing mercury, especially one in which a mercury atom is linked directly to one or more carbon atoms. **2** or **organomercury compound** any such compound.

organometallic **1** or **organometal** of, pertaining to, or being a metal-containing organic compound, especially one in which a metal atom is linked directly to one or more carbon atoms. **2** or **organometal compound** any such compound.

organophosphate **1** any organic compound containing a phosphoric (including fluorophosphoric, thiophosphoric, etc.) residue in amide, anhydride, ester, or thioester linkage. **2** any of a number of organophosphorus compounds originally developed as nerve gases, but formerly used as agricultural insecticides and as anthelmintics, the best known being diisopropylphosphorofluoridate. These compounds inhibit serine proteases (including acetylcholinesterase, crucial for nerve transmission) by blocking an essential serine residue. *See also* **aminoethylbenzenesulfonyl fluoride**, **diisopropyl fluorophosphonate**, **phenylmethylsulfonyl fluoride**.

organotroph *an alternative term for heterotroph(e)*.

organotrophic **1** heterotrophic; *see* **heterotroph(e)**. **2** of, pertaining to, or influencing the formation or nourishment of bodily organs.

organyl any organic substituent group, regardless of functional type, that has a single free valence at a carbon atom; e.g. ethyl, CH_3CH_2- ; glycol, $\text{H}_2\text{NCH}_2\text{C}(=\text{O})-$; ribosyl, $\text{C}_5\text{H}_9\text{O}_4-$.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

origin

origin (in mathematics) the point of intersection of a set of (usually two or three) coordinate axes; the point whose coordinates are all zero. —original *adj.*

O-ring a toroidal ring, usually of circular cross-section, made of neoprene or similar material and useful as a fluid- or gas-tight seal between particular parts of a device while permitting relative movement between them.

Orn symbol for a residue of the α -amino acid L-ornithine.

ornaline an alternative name for nopalinic acid.

ornithine symbol: Orn; α,δ -diaminopentanoic acid; 2,5-diaminopentanoic acid; an amino acid only rarely found in proteins. L-Ornithine is an intermediate in the biosynthesis of arginine (see **ornithine-urea cycle**) and, in plants, of pyrrolidine and tropane alkaloids; it was first found as its dibenzoyl conjugate, **ornithuric acid**. D-ornithine is a component of **bacitracin A**.

ornithine acetyltransferase see **glutamate N-acetyltransferase**.

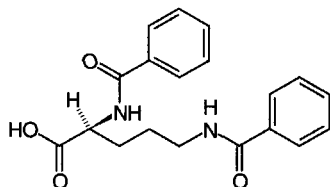
ornithine carbamoyltransferase *abbr.* (in clinical chemistry): OCT; EC 2.1.3.3; *systematic name*: carbamoyl-phosphate:L-ornithine carbamoyltransferase; *other names*: ornithine transcarbamylase; citrulline phosphorylase. An enzyme of the **ornithine-urea cycle** that catalyses a reaction between carbamoyl phosphate and L-ornithine to form L-citrulline and orthophosphate. It is deficient in congenital **hyperammonemia**. Example, product of *arg1* gene in *Escherichia coli*: database code ECARG1, 334 amino acids (36.93 kDa).

ornithine decarboxylase EC 4.1.1.17; *systematic name*: L-ornithine carboxyl-lyase; an enzyme that catalyses the decarboxylation of L-ornithine to putrescine and CO_2 ; pyridoxal-phosphate is a coenzyme. It is the first enzyme of polyamine biosynthesis; its activity increases in the cell in response to many agonists, especially those stimulating cell growth. Example from *Neurospora crassa*: database code DCOR_NEUCR, 484 amino acids (53.24 kDa).

ornithine transacetylase see **glutamate N-acetyltransferase**.

ornithine-urea cycle or **ornithine cycle** or **arginine-urea cycle** or **urea cycle** a cyclic metabolic pathway, present in mammals and other ureotelic animals, that converts waste nitrogen, in the form of highly toxic ammonium, to essentially nontoxic urea, which may then be excreted. Urea is formed by hydrolysis of L-arginine to L-ornithine, the cycle being completed by conversion of ornithine to L-citrulline by carbamoylation with carbamoyl phosphate (previously formed by enzymic combination of carbon dioxide and ammonium with phosphate from ATP), reaction of the citrulline with L-aspartate to form L-argininosuccinate, and cleavage of this to produce fumarate and regenerate arginine. The cycle is also sometimes called the **Krebs cycle** or the **Krebs-Henseleit cycle**, after its discoverers. The role of argininosuccinate was revealed by the US biochemist, Sarah Ratner (1903–), in 1954.

ornithuric acid *N,N'*-dibenzoylornithine; a detoxication product of benzoic acid found in the excreta of birds.



Oro symbol for a residue of the pyrimidine base orotate.

orosomucoid or α_1 -acid glycoprotein a ≈ 40 kDa, water-soluble plasma protein; it consists of a single polypeptide chain with two intrachain disulfide bonds and carrying five branched, highly sialylated oligosaccharide units attached to the β -carboxyl groups of asparagine residues. Certain alleles of the gene encoding this protein have been found to be significantly increased in patients with different types of carcinoma. Exam-

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

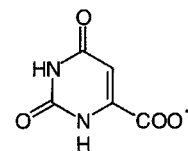
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

orotidine-5'-phosphate decarbox

ple from human: database code A1AG_HUMAN, 201 amino acids (23.51 kDa).

orotate 1 symbol: Oro; uracil-6-carboxylate; 1,2,3,6-tetrahydro-2,6-dioxo-4-pyrimidine carboxylate; the anion of orotic acid (i.e. uracil-6-carboxylic acid). It is an intermediate in the biosynthesis of pyrimidine nucleotides from aspartate. 2 any mixture of orotic acid and its anion. 3 any salt or ester of orotic acid.



orotate phosphoribosyltransferase *abbr.*: OPRTase; EC 2.4.2.10; *other names*: orotidylic acid phosphorylase; orotidine-5'-phosphate pyrophosphorylase; an enzyme of the pathway for *de novo* biosynthesis of pyrimidines. It catalyses the formation of orotidine 5'-phosphate from orotate and 5-phospho- α -D-ribose 1-diphosphate, releasing pyrophosphate. Orotidine 5'-phosphate is decarboxylated to uridine 5'-phosphate by **orotidine-5'-phosphate decarboxylase**. Example, human protein: database code PYR5_HUMAN, 480 amino acids (52.16 kDa) – this contains domains for OPRTase (residues 1–214) and orotidine 5'-phosphate decarboxylase (residues 221–480). Deficiency causes the recessive disease **orotic aciduria** type 1.

orotic aciduria a rare hereditary disorder of pyrimidine synthesis in humans that is characterized by retarded growth, hyperchromic anemia unresponsive to the usual hematinics, and an excessive excretion of orotate in the urine. It is due to deficiency in the activities of either **orotate phosphoribosyltransferase** (EC 2.4.2.10) or **orotidine-5'-phosphate decarboxylase** (EC 4.1.1.23), or both. The condition results in the subject being a 'pyrimidine auxotroph' and can be effectively treated by administration of uracil, which both rectifies the pyrimidine deficiency and indirectly leads, through the effect of cytidine triphosphate on aspartate transcarbamoylase, to inhibition of orotate synthesis.

orotidine symbol: O or Ord; 6-carboxylatouridine; orotate riboside; 1- β -D-ribofuranosylorotate; a ribonucleoside occurring primarily as its 5'-phosphate, orotidylic acid but produced by certain mutants of *Neurospora crassa* and excreted in relatively large amounts in the urine of cancer patients treated with 6-azauridine. It is not known to be a component of ribonucleic acid.

orotidine monophosphate *abbr.*: OMP; an alternative name for any **orotidine phosphate**, but in particular for orotidine 5'-phosphate.

orotidine phosphate orotidine monophosphate (*abbr.*: OMP); any phosphoric monoester or diester of orotidine. Of the three possible monoesters – the 2'-phosphate, the 3'-phosphate, and the 5'-phosphate – and the two possible diesters – the 2',3'-phosphate and the 3',5'-phosphate – only **orotidine 5'-phosphate** is known to occur naturally (the locant therefore being omitted if no ambiguity may arise).

orotidine 5'-phosphate or 5'-orotidylic acid or 5'-phosphoorotidine or 5'-O-phosphoorotidine symbol: Ord5'P; *alternative recommended names* for orotidine monophosphate (*abbr.*: OMP), orotidine 5'-(dihydrogen phosphate), orotate ribonucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation.) It is the immediate biosynthetic precursor of uridine monophosphate, uridylic acid, UMP, to which it is converted by decarboxylation. Thus it is important also as an intermediate in the biosynthesis of other pyrimidine nucleotides from UMP.

orotidine-5'-phosphate decarboxylase EC 4.1.1.23; *other name*: orotidylic decarboxylase; an enzyme that catalyses the

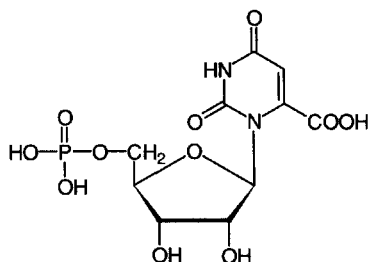
orotidine-5'-phosphate pyrophosphorylase

decarboxylation of orotidine 5'-phosphate to UMP a reaction carried out by the multifunctional enzyme **5'-monophosphate synthase**. Example from *Neurospora crassa*: database code DCOP_NEUCR, 397 amino acids (43.85 kDa).

orotidine-5'-phosphate pyrophosphorylase *see* **orotate phosphoribosyltransferase**.

orotidylate **1** either the monoanion or the dianion of orotidylic acid. **2** any mixture of orotidylic acid and its anions. **3** any salt or ester of orotidylic acid.

orotidylic acid *the trivial name for any phosphoric monoester of orotidine*. The position of the phosphoric residue on the ribose moiety of a given ester may be specified by a prefixed locant – *see* **orotidine phosphate**. However, 5'-orotidylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-Orotidylic acid is also an alternative recommended name for **orotidine 5'-phosphate**.



orotidylic acid phosphorylase *see* **orotate phosphoribosyltransferase**.

orotidylic decarboxylase *see* **orotidine-5'-phosphate decarboxylase**.

ortho (*followed by to*) at a position in a benzene ring adjacent to a particular (specified) substituent.

ortho- *abbr.: o-; prefix (in chemical nomenclature) denoting an isomer of a disubstituted monocyclic aromatic compound in which the substituents are attached at positions 1 and 2 of the ring. Compare meta-, para-*.

ortho+ or (*before a vowel*) **orth+** *comb. form 1* straight or upright; perpendicular (to) or at a right angle (to); correct. **2 a** (*in inorganic chemistry*) denoting the most hydrated acid of a series of oxoacids formed from any one of certain anhydrides, or the anion or a salt formed from such an acid. The term is now approved only for orthoboric acid, H_3BO_3 ; orthosilicic acid, H_4SiO_4 ; orthophosphoric acid, H_3PO_4 ; orthoperiodic acid, H_5IO_6 ; and orthotelluric acid, H_6TeO_6 . Compare **meta+** (def. 2), **pyro+**. **b** (*in organic chemistry*) similarly denoting hypothetical acids of general formula $RC(OH)_3$ or $C(OH)_4$ or real esters of such acids, e.g. trimethyl orthoacetate, $CH_3-C(OCH_3)_3$; tetramethyl orthocarbonate, $C(OCH_3)_4$.

ortho- and peri-fused describing a polycyclic compound in which one ring contains two, and only two, atoms in common with each of two or more rings of a contiguous series of **ortho-fused** rings. Such compounds have n common faces and $<2n$ common atoms.

orthochromatic 1 (of a dye or stain) imparting, or capable of imparting, its own colour to a structure. **2** (of a structure) assuming, or capable of assuming, the same colour as that of a dye or stain with which it is treated. Compare **metachromatic**. — **orthochromatically adv.**

orthodox conformation the conformation adopted by a mitochondrion in the resting state when the [ATP]/[ADP] ratio is high. This is the usual conformation of mitochondria seen in electron micrographs of thin sections of intact tissue.

ortho-fused describing a polycyclic compound in which any two adjacent rings contain two, and only two, atoms in common. Such compounds have n common faces and $2n$ common atoms.

orthogonal 1 (of axes, lines, planes, surfaces, etc.) mutually at right angles; perpendicular to one another. **2** (*in statistics*) (of

a set of variates) being, or regarded as being, statistically independent.

orthograde *see* **anterograde**.

orthologous describing genes of similar function but occurring in different species.

orthomyxovirus any of a group of RNA animal viruses, in class V of the **Baltimore classification**, consisting of enveloped, pleomorphic particles 80–120 nm in diameter. The nucleocapsids are helical, 6–9 nm in diameter, and of variable length, and contain segmented RNA (minus strand); this is transcribed by a virion polymerase into complementary RNA molecules, which act as messengers. The group includes the influenza A, B, and C viruses.

orthophosphate *an alternative name for phosphate* (used especially to distinguish from **metaphosphate** or **pyrophosphate**). *See also* **ortho+** (def. 2a).

orthotopic transplant any tissue graft transplanted into the normal position of that tissue in the body of the recipient. Compare **heterotopic transplant**.

Os *symbol for osmium*.

osamine *an alternative name for aminodeoxysugar*.

osazone any 1,2-bis(arylhydrazone); i.e. any condensation product of the type $RNH-N=CR-CR'=N-NHR$ formed between an α -dicarbonyl compound (e.g. a 2-ketoaldose) and hydrazine or a substituted hydrazine. α -Hydroxy- and α -amino-carbonyl compounds will also form osazones through oxidation of the corresponding [mono]hydrazones initially formed. *See also* **phenylosazone**.

oscillator an instrument or device for producing oscillations, especially an electronic device that produces an alternating output of known frequency.

oscillin a 33 kDa protein identified in sperm that induces oscillations in intracellular Ca^{2+} concentrations in mammalian eggs. It is considered to be involved in Ca^{2+} oscillations which serve as the essential trigger for egg activation and early development of the embryo.

oscillograph an instrument for producing a graphical record of an oscillating value.

oscilloscope an instrument for producing a temporary visual record of an oscillating value. A **cathode-ray oscilloscope**, based on the **cathode-ray tube**, provides a visual image of electrical signals, in the form of a graph of the signal over time.

OSCP *abbr. for oligomycin sensitivity-conferring protein*.

+ose *noun suffix 1* originally indicating related to glucose, as either an isomer (e.g. galactose) or a polymer (e.g. cellulose, maltose), now also extended to other sugars (e.g. arabinose, fructose, sucrose) or classes of sugars (e.g. aldose, furanose, ketose, hexose, pentose, pyranose). **2** indicating the primary products of hydrolysis of a biopolymer; e.g. proteose, peptose.

Ose (*in glycolipid nomenclature*) *symbol for a monosaccharide residue*. A numerical subscript denotes the number of connected monosaccharide units in the oligosaccharide component, and a prefix of two or three characters represents the trivial name of the oligosaccharide; e.g. $GbOse_3$ (i.e. globotriaose), $nLcOse_4$ (i.e. neolactotetraose), $McOse_3Cer$ (i.e. mucotriosylceramide).

oside *an alternative term for glycoside*.

+oside *noun suffix* denoting **glycoside**.

+osis *noun suffix* denoting **1** a process or state (e.g. metamorphosis, narcosis, necrosis, osmosis). **2** a diseased condition (e.g. nephrosis, tuberculosis). **3** an increase or excess (of something) (e.g. glycogenosis, leukocytosis). *See also* **+otic**.

oskar one of the **maternal effect genes** in *Drosophila melanogaster*. It organizes the germplasm and directs localization of the germ-cell determinants. Oskar protein is required to keep *oskar* RNA and *staufen* (another maternal effect gene) protein at the posterior pole. Three other genes (*capu*, *spir*, and *stau*) are required for the initial localization of the *oskar* product to the posterior pole of the oocyte. Database code of expressed protein OSKA_DROME, 606 amino acids (69.21 kDa).

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynghuyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynghuyen@huynghuyentld.com.



osM or **osm** symbol for osmolar (osmol L⁻¹ now recommended). **OSM** abbr. for oncostatin M.

osmophilic (of a structure or substance) having an affinity for, or staining readily with, osmium tetroxide; e.g. osmophilic globules (plastoglobuli) in plastids.

osmium tetroxide osmium(VIII) oxide; OsO₄; a volatile, evil-smelling, moderately water-soluble, electron-dense substance used for fixing specimens for microscopy, especially electron microscopy, and used chemically in the hydroxylation of alkenes. The vapour is highly poisonous.

osmo+ comb. form denoting **1** osmosis. **2** smell, odour.

osmoceptor see **osmoreceptor**.

osmol symbol for osmole.

osmolal **1** describing a solution that contains one **osmole** (of a specified solute), or an indicated number of osmoles, per kilogram of solvent; used also of a specified solute in such a solution. **2** describing an amount-of-substance concentration expressed in osmoles (of a specified solute) per kilogram of solvent. Compare **osmolar**. See also **osmolality**.

osmolality the concentration of osmotically active particles in an aqueous solution expressed in **osmoles** per kilogram of solvent. For a nondissociating substance the osmolality is equal to the **molarity**, whereas for a substance that dissociates in aqueous solution the osmolality is greater than the molarity to an extent dependent on the number of particles into which the substance dissociates and on the position of the equilibrium.

osmolar **1** describing a solution that contains one **osmole** (of a specified solute), or an indicated number of osmoles, per litre of solution; used also of a specified solute in such a solution. **2** abbr.: osM or osm; describing an amount-of-substance concentration expressed in osmoles (of a specified solute) per litre of solution. Compare **osmolal**. See also **osmolality**.

osmolarity the concentration of osmotically active particles in an aqueous solution expressed in **osmoles** per litre of solution. For a nondissociating substance the osmolarity is equal to the molarity, whereas for a substance that dissociates in aqueous solution the osmolarity is greater than the molarity to an extent dependent both on the number of particles into which the substance dissociates and on the position of the equilibrium. Compare **osmolality**.

osmole symbol: osmol; the amount of substance that contains, or gives rise to, one mole of osmotically active particles when in aqueous solution.

osmometer **1** or **membrane osmometer** any device for demonstrating the phenomena of osmosis and osmotic pressure; any instrument for measuring osmotic pressure. **2** an apparatus for measuring the acuteness of the sense of smell. —**osmometric** adj.; **osmometry** n.

osmophile any organism that grows preferentially (or only) in media of relatively high osmotic pressure, such as ones with a high salt or sugar content. —**osmophilic** adj.

osmoreceptor or **osmoceptor** a sensory cell, or group of such cells, that is specialized to react to changes in environmental osmotic pressure, e.g. of blood plasma or of a tissue fluid. Osmoreceptors are located in the hypothalamus, in the main arteries, and at other sites.

osmoregulation the sum of the processes by which organisms regulate their internal contents of water and solutes. —**osmoregulatory** adj.

osmosis the spontaneous net flow of solvent by diffusion through a semipermeable membrane from a phase where the solvent has a higher chemical potential to one where the solvent has a lower chemical potential. The flow continues until the chemical potential of the solvent becomes the same on both sides of the membrane or until it is countered by a difference in hydrostatic pressure either generated or applied. —**osmotic** adj.; **osmotically** adv.; **osmose** vb.

osmotic coefficient symbol: ϕ_m (on a molality basis) or ϕ_x (on a mole fraction basis); the ratio of the actual to the ideal osmotic pressure of a given solution at a specified temperature;

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

i.e. a representation of the departure of the solution from ideality.

osmotic pressure symbol: Π ; the excess hydrostatic pressure that must be applied to a solution in order to reduce to zero the net flow of pure solvent into it by osmosis. The osmotic pressure of a dilute solution is equal to the pressure the solute would exert if it could exist as a gas at the same temperature, T , and the same amount-of-substance concentration, c ; thus it may be expressed by a relation analogous to the gas equation, i.e. $\Pi = \nu cRT$ where ν is the (average) number of particles into which one solute molecule is dissociated under the given conditions, and R is the gas constant. Osmotic pressure is a **colligative property** of a solution. See also **colloid osmotic pressure**, **oncotic pressure**.

osmotic shock any disruption or other disturbance caused to suspended cells or cell organelles by a sudden change in the osmotic pressure of their suspending medium.

osmotolerant describing a cell or organism that is tolerant to a wide range of osmotic pressures in its environment.

osone any 2-ketoaldose; use not recommended.

osseine the organic material, consisting mainly of collagen, that remains after treatment of bone with dilute acid to dissolve out the inorganic components.

ossify to make into bone or to become bone. —**ossification** n.

osteo+ or (before a vowel) **oste+** or (sometimes) **ost+** comb. form denoting bone.

osteoblast a mesenchymal cell that secretes the organic matrix of bone (i.e. osteoid, consisting chiefly of collagen); as the matrix becomes calcified the osteoblast becomes trapped and is known as an osteocyte.

osteocalcin other names: γ -carboxyglutamic acid-containing protein; bone GLA-protein (abbr.: BGP); a protein that binds Ca²⁺ and apatite; the binding depends on γ -carboxyglutamic acid residues (three per molecule) formed by vitamin K-dependent carboxylation. Example (precursor) from human: database code OSTC_HUMAN, 100 amino acids (10.95 kDa).

osteoclast a large, multinucleate **macrophage** that erodes bone matrix and is able to tunnel deep into the substance of compact bone. Osteoclasts are formed by the fusion of blood monocytes and play an important role in the remodelling of bone and the removal of dead bone.

osteocyte an osteoblast when it has become trapped in a lacuna of the bone matrix and is no longer able to divide or secrete appreciable amounts of new matrix. Osteocytes are connected by fine cellular processes that pass through the bone canaliculi.

osteogenesis the processes of bone formation; ossification.

osteogenesis imperfecta a genetically heterogeneous group of human diseases in which there is disorder of connective tissue affecting not only bone but also tendons, ligaments, fascia, etc. The bones are unusually brittle and liable to fracture and in **osteogenesis imperfecta congenita** in the neonate there is evidence that the collagen of bone matrix may be abnormal.

osteoid **1** the unmineralized bone matrix secreted by osteoblasts; it consists chiefly of collagen. **2** resembling bone.

osteolysis the breakdown and dissolution of bone, especially through disease. —**osteolytic** adj.

osteomalacia a clinical condition of adults characterized by softening and deformation of the bones due to a failure to maintain their mineralization. It has a wide variety of possible causes, of which dietary deficiencies of vitamin D, calcium, or phosphate are the commonest.

osteonectin see **SPARC**.

osteopontin other names: bone sialoprotein 1; urinary stone protein; secreted phosphoprotein 1 (abbr.: SPP-1); nephropontin; uropontin; a bone-specific sialoprotein produced by cells in osteoid matrix but independently discovered from several different functions. It can form a bridge between cells and the mineral in the matrix, and is found at high concentrations on the cement lines where bone formation follows resorption. It is also found in the kidney, and plays a principal role in urinary

osteoporosis

stone formation. In arterial smooth muscle cells, its level increases in atherosclerotic plaques. It is secreted, binds calcium, and binds covalently to fibronectin. It has an RGD integrin-binding sequence and is one of the major ligands for the vitronectin receptor. It is phosphorylated on serine, and is an *O*- and *N*-glycosylated protein. There are two forms from alternative splicing. Example (precursor) from human: database code OSTP_HUMAN, 314 amino acids (35.38 kDa).

osteoporosis a clinical condition characterized by a general or local reduction in bone mass with little alteration in gross composition. This causes the affected bones to become porous, brittle, and liable to fracture. Generalized osteoporosis is often hormonal in origin.

Ostwald, Friedrich Wilhelm (1853–1932), Russian-born German chemist; noted also for his work on electrochemistry and for the invention of an industrially important process for the catalytic oxidation of ammonia to nitric acid; Nobel Laureate in Chemistry (1909) 'in recognition of his work on catalysis and for his investigations into the fundamental principles governing chemical equilibria and rates of reaction'.

Ostwald dilution law a relationship, deduced by Ostwald (1888), between the amount-of-substance concentration, *c*, and the degree of dissociation, α , of a weak electrolyte. It is given by:

$$\alpha^2 c / (1 - \alpha) = K,$$

where *K* is the apparent dissociation constant of the electrolyte.

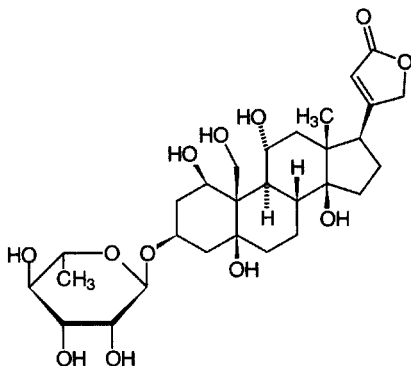
Ostwald pipette a pipette for measuring small volumes of liquids, especially viscous liquids. It features a capillary stem and delivery jet and a shoulderless bulb near the jet. Various modifications have been described.

Ostwald viscometer a simple capillary viscometer.

+osyl suffix indicating a glycosyl group.

+otic suffix forming adjectives and nouns 1 affected by (e.g. necrotic) or relating to (e.g. osmotic). 2 causing (e.g. mitotic, narcotic). —**otically** *adv.*

ouabain or **G-strophanthin** a cardiac glycoside whose carbohydrate moiety is L-rhamnose, obtained from the seeds of *Strophanthus gratus*. It is a specific inhibitor of the membrane-bound sodium pump Na⁺,K⁺-ATPase (see sodium/potassium ATPase) of animal cells.



Ouchterlony technique a development of double immunodiffusion, effected in two dimensions so that comparisons can be made between two or more antigen-containing samples in their reaction with one antiserum. The particular pattern of precipitation lines or arcs formed enables conclusions to be drawn regarding the identity, partial identity, or nonidentity of antigens in the various samples examined. [After O. Ouchterlony, French immunologist.]

Oudin technique a simple, single immunodiffusion technique, effected in one dimension, in which a solution of an antigen-containing sample is placed above a column of antiserum in

overpressure(d) layer chromatography

corporated in a gel in a tube; antigen diffuses into the gel, where it reacts with antibody to form one or more precipitation bands. [After Jacques Oudin (1923–).]

ounce *symbol:* oz; a unit of mass, equal in the avoirdupois system to one-sixteenth of a pound (= 28.3495 grams). This must be distinguished from the **troy ounce** (equal to 480 grains, 31.1035 grams).

outer membrane 1 (*in eukaryotes*) the more external of the two membranes that surround a chloroplast, a mitochondrion, or a nucleus. 2 (*in prokaryotes*) the layer of lipopolysaccharide, lipoprotein, and protein that lies external to the peptidoglycan in the cell wall of a Gram-negative bacterium.

outer membrane protein P68 see **peractin**.

outlier (*in statistics*) any member of a set of observations that is so far removed in value from the remaining members that its validity is questionable.

ov+ a variant form of **ovi+** (before a vowel).

ova the plural of **ovum**.

ovalbumin *abbr.:* OA; a ≈44.5 kDa glycoprotein, also called egg albumin, that is the major protein of egg white from hens' eggs. Each molecule consists of a single polypeptide chain and carries a single oligosaccharide chain containing only mannose and *N*-acetylglucosamine residues; one or two phospho groups also may be present. It is included in the **serpin** family. It is synthesized by the oviduct under hormonal control. Although it is a secreted protein, it has no N-terminal signal peptide. Example from chicken: database code OVAL_CHICK, 385 amino acids (42.70 kDa); 3-D structure known.

ovarian hormone any of the numerous steroid hormones secreted by the ovary.

ovary 1 (*in zoology*) the gonad of a female animal; vertebrates have a pair of ovaries. The ovary contains oocytes from which ova are produced, and it also secretes various steroid hormones. Its activity is cyclical and is controlled by gonadotropin(s). 2 (*in botany*) the enlarged basal part of a carpel or a syncarpous gynecium of a flower that contains one or more ovules. —**ovarian** *adj.*

overgrowth 1 excessive growth; hyperplasia. 2 a later but more vigorous growth by another strain or species of organism that supplants one already existing.

Overhauser effect or **Overhauser enhancement** see **nuclear Overhauser effect**.

overlap peptide a fragment, produced by partial hydrolysis of a polypeptide or protein, that contains both the N-terminal part of the sequence of one further peptide fragment and the C-terminal part of the sequence of a second further peptide fragment, and enables those two to be placed in the correct position in the overall sequence of the parent compound during protein sequencing procedures.

overlapping code a code in which one element of a code-word forms part of an adjacent code-word in an encoded piece of information. A triplet code (such as the **genetic code**, where the code-words are codons) may be either: a **fully overlapping code**, in which one code-word consists of the elements at positions *n*, (*n*+1), (*n*+2), and the next code-word consists of the elements at positions (*n*+1), (*n*+2), (*n*+3); or a **partially overlapping code**, in which adjacent code-words comprise elements at *n*, (*n*+1), (*n*+2), and those at (*n*+2), (*n*+3), (*n*+4). In a **nonoverlapping code**, such as the genetic code proved to be, no element of one code-word forms part of an adjacent code-word.

overlapping gene any gene whose polynucleotide sequence overlaps, wholly or in part, that of another gene in the same stretch of a genome. The codons may or may not be in the same **reading frame**. In mitochondria and some viruses, some mRNAs are synthesized by the transcription of the same DNA using different reading frames.

overpressure(d) layer chromatography *abbr.:* OPLC; a chromatographic technique that gives some of the advantages of **high-pressure liquid chromatography** to **thin-layer chromatography**, with resultant increases in both speed and efficiency of separation.



overshoot

tion. The sorbent layer is completely covered by a flexible membrane under external pressure and the solvent is pumped continuously through the sorbent layer.

overshoot the extent to which a control system causes the variable being controlled to go initially beyond its final equilibrium position.

ovi+ or **ovo+** or (before a vowel) **ov+** comb. form denoting ovum or egg.

oviduct (in zoology) the tube through which ova are conveyed from the ovary to the exterior or to some intermediate organ such as the uterus. In mammals it is known more commonly as the **Fallopian tube** or **uterine tube**.

oviductin other name: gp43 processing protease; a high molecular weight glycoprotein that is secreted by the nonciliated secretory cells of the oviduct and later transferred to the zona pellucida of the oocyte during oviductal transit. It processes a 43 kDa glycoprotein of the oocyte envelope during its transit through the duct. Example from golden hamster: database code MAU15048, 635 amino acids (69.36 kDa).

ovine abbr.: o or (formerly) O; of, belonging to, characteristic of, obtained from, or being sheep or a sheep; resembling a sheep, sheeplike.

ovo+ a variant form of **ovi+**.

ovoflavin or **oviflavine** a former name for **riboflavin**.

ovoinhibitor a contaminant of crude **ovomucoid** that, unlike pure ovomucoid, inhibits chymotrypsin as well as trypsin.

ovomucoid a glycoprotein found in egg white from hens' eggs; it is an inhibitor of trypsin. Example from *Gallus gallus*: database code IOVO_CHICK, 210 amino acids (22.59 kDa).

ovothiol a novel thiohistidine compound from sea-urchin eggs that confers NAD(P)H-O₂ oxidoreductase activity on ovoperoxidase.

ovotransferrin an alternative name for **conalbumin**.

ovulate to discharge an egg or eggs from an ovary. —**ovulation** n.

ovule 1 (in botany) a structure in the ovary of a seed plant that develops into a seed after fertilization of the egg cell within it. 2 (in zoology) a small, especially immature, ovum or egg.

ovum (pl. **ova**) (in zoology) strictly, an unfertilized egg cell, i.e. female gamete, which contains a haploid nucleus and which on fertilization has the unique ability to develop into a new individual. In many animals, mammals included, **oogenesis** is arrested and meiosis is not completed until after fertilization of the secondary **oocyte**, hence in these animals an ovum in the strict sense cannot exist. However, the term is often applied instead either to the secondary oocyte or to the product resulting from the second meiotic division after fertilization.

OWL a composite protein database (University of Leeds, UK). See **Appendix E**.

oxa- or (sometimes before a vowel) **ox+** prefix (in chemical nomenclature) denoting replacement of a methylene group in a specified acyclic or monocyclic hydrocarbon by an oxygen atom. For certain natural products, (e.g. tetrapyrroles), the meaning is extended to denote replacement of a noncarbon atom (with any attached hydrogen atoms) by an oxygen atom.

oxalacetate an alternative name for **oxaloacetate**.

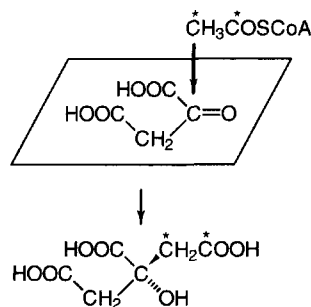
oxalate 1 the trivial name for ethanedioate, $^{-}\text{OOC-COO}^{-}$, the dianion of oxalic acid, i.e. ethanedioic acid. It occurs in many plants, especially sorrels of the genera *Oxalis* and *Rumex*, and is highly toxic to animals. Normally, small amounts are formed in humans and excreted in the urine; larger amounts occur in patients with **hyperoxaluria**. The calcium salt is virtually insoluble in water, a property that is valuable in the determination of calcium and in the use of oxalate *in vitro* as an anticoagulant. 2 any mixture of free oxalic acid and its mono- and dianions. 3 any salt or ester of oxalic acid.

oxalo the univalent acyl group, HOOC-CO- , derived from oxalic acid by loss of one hydroxyl group. See also **oxalyl** (def. 1).

oxalo+ prefix indicating the presence of an **oxalo** group.

oxaloacetate or **oxalacetate** abbr.: OAA; 1 the trivial name for oxosuccinate (formerly ketosuccinate); oxobutanedioate;

$^{-}\text{OOC-CO-CH}_2\text{-COO}^{-}$ (keto form) or $^{-}\text{OOC-C(OH)=CH-COO}^{-}$ (enol form); the dianion of oxaloacetic acid. It is an important intermediate in metabolism, especially as a component of the **tricarboxylic-acid cycle**. The stereochemistry of the **citrate (S)-synthase** reaction is shown below.



2 any mixture of free oxaloacetic acid and its mono- and dianions. 3 any salt or ester of oxaloacetic acid.

oxalosis the renal and extrarenal deposition of calcium oxalate (see **oxalate**) seen in patients with **hyperoxaluria**.

oxalosuccinate 1 the trivial name for 2-oxotricarballylate (formerly α -ketotricarballylate); 1-oxopropane-1,2,3-tricarboxylate; $^{-}\text{OOC-CO-CH(COO-)-CH}_2\text{-COO}^{-}$; the trianion of oxalosuccinic acid. It is an enzyme-bound intermediate in the conversion of isocitrate to 2-oxoglutarate by **isocitrate dehydrogenase (NAD⁺)** in the **tricarboxylic-acid cycle**. 2 any mixture of free oxalosuccinic acid and its mono- and dianions. 3 any salt or ester of oxalosuccinic acid.

oxaluria an alternative name for **hyperoxaluria**.

oxalyl 1 (in biochemistry) (sometimes) an alternative name for **oxalo**. 2 (in chemistry) the divalent diacyl group, $^{-}\text{CO-CO-}$, derived from oxalic acid by loss of both hydroxyl groups.

OXBOX a 13 bp sequence (present in the region -123 to -674 bp) of the heart and skeletal muscle gene for the adenine nucleotide translocator and also in the promoter region of the gene for human ATP synthase β subunit. It appears to be a positive translational element.

oxene a reactive species of oxygen involved in the hydroxylation reaction catalysed by cytochrome P450; it is iron-bound oxygen which is electron deficient but electrically neutral $^{-}\text{P450-Fe}^{3+}\text{-O}$. It reacts with the bound substrate to release the hydroxylated substrate with the regeneration of P450-Fe^{3+} .

oxidant the chemical species that accepts one or more electrons in a redox reaction; i.e. the species that undergoes reduction in such a reaction. Compare **reductant**.

oxidase any **oxidoreductase** that catalyses a reaction in which dioxygen is the electron acceptor (i.e. **oxidant**).

oxidase test a test for the presence of cytochrome c oxidase in bacteria that depends on the oxidation of tetramethyl-*p*-phenylenediamine to a violet-coloured product. It is useful for primary characterization, especially of Gram-negative organisms.

oxidation 1 the action or process of reacting with oxygen, especially the addition of oxygen to a substance. 2 the loss or removal of hydrogen from a substance. 3 the loss or removal of one or more electrons from a molecular entity, with or without concomitant loss or removal of a proton or protons. In this sense, oxidation is the opposite of, and is always coupled to, **reduction**. 4 the increase to a more positive value of the **oxidation number** of an atom, whether the atom is uncharged or charged, and whether free or covalently bound. —**oxidative** adj.

α -oxidation see **alpha-oxidation**.

β -oxidation see **beta-oxidation**.

ω -oxidation see **omega-oxidation**.

oxidation level or **oxidation state** the status of an organic compound in terms of whether it can be considered to undergo oxidation.



oxidation number

idation or reduction (or neither) upon conversion to a related compound. If conversion involves oxidation, the parent compound is said to have a higher oxidation level than the product; if conversion involves reduction, the parent compound is said to have a lower oxidation level. The term is thus used in a relative rather than an absolute sense. *See also* **oxidation number**.

oxidation number or oxidation state a number assigned to a particular atom in a molecular entity that represents, actually or notionally, the charge on that atom. Thus, atoms of pure elements have an oxidation number of zero; in molecules of simple electrovalent compounds the oxidation number of each component atom is equal to the charge on the respective ion; and in molecules of covalent compounds the notional charge on a particular atom is derived by assigning electrons to the more electronegative partner in each bond involving the atom in question. By convention, hydrogen is considered to have an oxidation number of +1 when in combination with nonmetals. The oxidation number of a carbon atom can range from -4 (e.g. in CH_4) through zero (e.g. in CH_2O or solid C) to +4 (e.g. in CO_2). The oxidation number of an atom may be indicated by a Roman numeral; e.g. iron(III) oxide, ferric oxide, Fe_2O_3 . The algebraic sum of the oxidation numbers of all the atoms in any molecular entity must equal the charge on that entity.

oxidation potential a measure of the tendency of a substance to oxidize spontaneously through loss of electrons. It is the standard potential of the half-reaction: $\text{A} \rightarrow \text{A}^{n+} + ne$ and is equal to the **reduction potential** but of opposite sign.

oxidation–reduction an alternative term for **redox**.

oxidation–reduction indicator an alternative term for **redox indicator**.

oxidation–reduction potential an alternative term for **redox potential**.

oxidation state 1 an alternative term for **oxidation number**. 2 an alternative term for **oxidation level** (less preferred).

oxidative deamination deamination with concomitant oxidation, as in the enzymic conversion, regardless of the nature of the electron acceptor, of an α -amino acid to an α -oxo acid. *Compare* **reductive amination**.

oxidative decarboxylation decarboxylation with concomitant oxidation, as in the enzymic conversion, regardless of the nature of the electron acceptor, of an α -oxo acid (actually or formally) to a carboxylic acid containing one less carbon atom.

oxidative metabolism an alternative term for **respiration** (def. 1).

oxidative pentose phosphate pathway or hexose monophosphate shunt an alternative route additional to glycolysis in plants that generates: (1) NADPH when it is not being formed by photosynthesis; (2) ribose 5-phosphate required in the biosynthesis of nucleosides; and (3) erythrose 4-phosphate required for biosynthesis of shikimic acid, a key intermediate in the formation of aromatic rings. *See also* **pentose phosphate pathway**.

oxidative phosphorylation or respiratory-chain phosphorylation the phosphorylation of ADP to ATP that accompanies the oxidation of a metabolite through the operation of the **respiratory chain**. The overall reaction may be summarized by the equation:



where AH_2 is an oxidizable metabolite, A is the product of the oxidation, P_i is inorganic orthophosphate, and x has a value between 1 and 3 depending on the nature of AH_2 and the functional state of the respiratory chain. The proton gradient built up across the inner mitochondrial membrane by the proton pumps of the respiratory chain drives ATP synthesis by the **H^+ -transporting ATP synthase**. *See also* **chemiosmotic coupling hypothesis**.

oxide 1 (in inorganic chemistry) any binary compound of oxygen and some other element. 2 an alternative generic name for any cyclic ether that is an **oxirane**; such a compound is named

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

as the oxide of the alkene from which it may be formed through addition of an atom of oxygen across the double bond; e.g. ethylene oxide, $\text{C}_2\text{H}_4\text{O}$. 3 a term used in some languages for **ether**. 4 any compound in which an oxygen atom has been attached to a heteroatom of a precursor compound. This definition includes compounds of general formulae: $\text{RC}\equiv\text{NO}$ (nitrile oxides; e.g. benzonitrile oxide), $>\text{C}=\text{N}(\text{O})\text{R}$ or $\text{R},\text{R}',\text{R}''\text{NO}$ (N-oxides, amine oxides; e.g. trimethylamine oxide); and $\text{R},\text{R}'\text{SO}$ (S-oxides, sulfoxides; e.g. methionine S-oxide; dimethyl sulfoxide).

+oxide or +olate suffix denoting 1 an anion formed by loss of a proton from the hydroxyl group of an alcohol or phenol; e.g. ethoxide ion, $\text{C}_2\text{H}_5\text{O}^-$. 2 a compound of such an anion with a cation; e.g. sodium phenoxide (i.e. sodium phenolate), $\text{C}_6\text{H}_5\text{ONa}$. *See also* **epoxide**, **hydroperoxide**, **hydroxide**, **nitroxide**, **peroxide**, **sulfoxide**, **superoxide**.

oxidize or oxidise 1 to undergo or effect **oxidation**. 2 to form or cause to form a superficial layer of oxide on a metal. *Compare* **reduce**. —**oxidization or oxidisation** *n*.

oxidizer or oxidiser 1 a less common name for **oxidizing agent**. 2 a device for effecting a particular oxidation.

oxidizing agent or oxidising agent any substance or mixture used for effecting **oxidation**.

oxidoreductase the systematic name for any enzyme of class EC 1. These enzymes catalyse oxidation–reduction (i.e. **redox**) reactions and the class comprises **dehydrogenases**, **oxidases**, and **reductases**.

oxidoreduction an alternative term for **redox**.

oximeter any photometric device for (continuously) measuring the oxyhemoglobin/hemoglobin ratio in blood, either *in vivo* or *in vitro*.

oxine the trivial name for 8-hydroxyquinoline; 8-quinolinol. With many species of metal ion this compound forms water-insoluble chelate complexes and thereby is useful in analysis. It is also a disinfectant, and, as the copper chelate, is an effective fungicide.

oxirane 1 the class name for any compound containing a three-membered saturated ring comprising two carbon atoms and one oxygen atom. 2 the systematic name for the type member of this class, known commonly as ethylene oxide, $\text{C}_2\text{H}_4\text{O}$.

oxo 1 the chemical group consisting of an oxygen atom, $=\text{O}$, doubly bonded to another atom (often of carbon) in a molecular structure. When bonded to a carbon atom it forms the carbonyl group, $>\text{C}=\text{O}$. *Compare* **oxy**. 2 or (formerly) **oxo+** modifying the class-name of a chemical compound to indicate the presence of an oxo group as a substituent in place of two hydrogen atoms; e.g. **oxocarboxylic acid**. The position of substitution may be specified by a prefixed locant; e.g. 17-oxosteroid. *See also* **keto** (def. 3). *Compare* **oxo** (def. 1).

oxo+ or (sometimes before a vowel) **ox+** 1 prefix (in chemical nomenclature) indicating substitution of a doubly bonded oxygen atom into a specified compound or group by the (formal or actual) oxidation of a methylene group, $>\text{CH}_2$, to a keto group, $>\text{C}=\text{O}$, or of a methyl group, $-\text{CH}_3$, to a formyl group, $-\text{CH}=\text{O}$, with the production of a ketone or an aldehyde respectively; e.g. oxoglutaric acid (*see* **oxoglutarate**), 3,6-dioxohexanoic acid (or 5-formyl-3-oxopentanoic acid). *Compare* **oxo** (def. 2). 2 *comb. form* denoting presence of one or more (singly or doubly bonded) oxygen atoms in a molecular structure; e.g. **oxoacid**, **oxonium**.

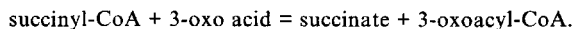
oxoacid 1 the recommended term for any compound whose molecular structure contains oxygen, at least one other element, and at least one atom of hydrogen bound to oxygen, and also that yields a conjugate base (*see* **oxoanion**) by loss of a hydrogen. Examples include: hypochlorous acid, ClOH ; sulfurous acid, $\text{O}=\text{S}(\text{OH})_2$; nitric acid, $(\text{O}=\text{N})\text{OH}$; [orthol]phosphoric acid, $\text{O}=\text{P}(\text{OH})_3$; and also any phosphonic acid, $\text{RP}(\text{O})(\text{OH})_2$, and any carboxylic acid, $\text{RC}(\text{O})\text{OH}$ (where $\text{R} = \text{H}$ or hydrocarbyl). *See also* **oxocarboxylic acid**. 2 or **oxo acid** or **oxyacid** or **oxy-acid** the traditional name, now obsolete, for any acid having oxygen in its acidic function, as opposed



oxo acid

to a 'hydracid' where oxygen is lacking (e.g. hydrochloric acid, HCl).

oxo acid 1 *see* **oxocarboxylic acid**. **2** *a variant of oxoacid* (def. 2). **3-oxoacid CoA-transferase** EC 2.8.3.5; *other name*: succinyl-CoA:3-ketoacid-CoA transferase; an enzyme that catalyses the reaction:



In mammals it converts plasma acetoacetate that has been taken up by tissues into acetoacetyl-CoA for further metabolism. Example from pig: database code A41771, 520 amino acids (56.34 kDa).

3-oxoacyl-[acyl-carrier protein] reductase (NADH) EC 1.1.1.212; an enzyme that catalyses the reduction by NADH of 3-oxoacyl-[acyl-carrier protein] to (3*R*)-3-hydroxyacyl-[acyl-carrier protein] with formation of NAD⁺. This is one of the steps of fatty-acid synthesis in plants. *Compare* **fatty acid synthase complex**.

oxoanion or (formerly) **oxo anion** or **oxyanion** or **oxy-anion** any of the possible anions formable from a molecule of an **oxoacid** (def. 1) by loss of one or more hydrons.

oxobutanedioate *see* **oxaloacetate**.

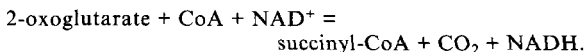
oxocarboxylic acid *the recommended name for any carboxylic acid* that contains an aldehydic group attached to, or/and one or more ketonic groups contained in, its principal chain or parent ring system, i.e. any into which one or more oxo groups have been substituted. The term is commonly shortened to **oxo acid**, but the full name should be used wherever confusion with **oxoacid** is possible. The older term keto acid is often still used (*see* **keto** (def. 3)).

17-oxogenic steroid or (formerly) **17-ketogenic steroid** any urinary corticosteroid or its metabolites that can be estimated by the **Zimmermann reaction** after oxidation to the corresponding 17-oxo steroid.

oxoglutarate or (formerly) **ketoglutarate**; **1** 2-oxoglutarate or α -oxoglutarate 2-oxopentanedioate; $^-\text{OOC}-[\text{CH}_2]_2-\text{CO}-\text{COO}^-$; the dianion of 2-oxoglutaric acid (i.e. 2-oxopentanedioic acid), an α -oxo (carboxylic) acid. It is a key constituent of the tricarboxylic-acid cycle and a key intermediate in amino-acid metabolism. In the cycle it is formed by oxidative decarboxylation of isocitrate and converted into a further oxidative decarboxylation reaction to succinyl-coenzyme A. It may also be formed from or converted into glutamate (and thus several other amino acids indirectly) by transamination, or formed by degradation of lysine via glutarate and 2-hydroxyglutarate. The older name **α -ketoglutarate** is often still used. **2** 3-oxoglutarate or β -oxoglutarate acetonedicarboxylate; 3-oxopentanedioate; $^-\text{OOC}-\text{CH}_2-\text{CO}-\text{CH}_2-\text{COO}^-$; the dianion of 3-oxoglutaric acid (i.e. 3-oxopentanedioic acid), a β -oxo (carboxylic) acid. It is a product of the action of the mould *Aspergillus niger* on citrate.

oxoglutarate decarboxylase *see* **oxoglutarate dehydrogenase (lipoamide)**.

oxoglutarate dehydrogenase complex an enzyme complex, closely analogous in mechanism to the **pyruvate dehydrogenase complex**. It accomplishes the overall reaction:



It consists of three activities: **oxoglutarate dehydrogenase (lipoamide)**, the E₁ component; **dihydrolipoamide S-succinyltransferase**, the E₂ component; and **dihydrolipoamide dehydrogenase**, the E₃ component (this being identical to the E₃ component of the pyruvate dehydrogenase complex).

oxoglutarate dehydrogenase (lipoamide) EC 1.2.4.2; *other names*: oxoglutarate decarboxylase; α -ketoglutaric dehydrogenase; an enzyme that catalyses the reaction between 2-oxoglutarate and lipoamide to form S-succinyl-dihydrolipoamide with release of CO₂. This is the E₁ component of the **oxoglutarate dehydrogenase complex**. Thiamine diphosphate is a coenzyme. Examples: 2-oxoglutarate dehydrogenase E₁ com-

ponent (α -ketoglutarate dehydrogenase) from *Escherichia coli*: database code ODO1_ECOLI, 933 amino acids (104.93 kDa); E₁ component (precursor) from human: database code ODO1_HUMAN, 1003 amino acids (113.11 kDa).

oxoglutarate/malate carrier protein *see* **OGCP**.

2-oxoisovalerate dehydrogenase *see* **3-methyl-2-oxobutanoate dehydrogenase (lipoamide)**.

oxolinic acid 5-ethyl-5,8-dihydro-8-oxo-1,3-dioxolo[4,5-g]quinoline-7-carboxylic acid; a **4-quinolone antibiotic** with antibacterial activity through inhibition of bacterial DNA gyrase.

oxonium 1 or (formerly) **hydronium** or **hydroxonium** the ion, H₃O⁺, formed by covalent linkage of a hydron to a single water molecule. *Compare* **H⁺(aq)**. **2** any organic entity in which an oxygen atom is linked to three groups (one or two of which may be -H) and bears a formal single positive charge.

(5)-loxoproline *symbol*: Glp or <Glu; *the recommended trivial name* for pyrrolutamic acid; pyrrolidone carboxylic acid; 5-oxopyrrolidine-2-carboxylic acid; a **noncoded amino acid**. It occurs at the N terminus of certain peptide hormones and is an intermediate in the γ -glutamyl cycle. *Compare* **oxyproline**.

oxo steroid or **keto steroid** any steroid having an oxo substituent in its molecular structure, whether in the ring system (as part of a ketonic group) or in a side chain (as part of either a ketonic or an aldehydic group). The position of substitution may be specified by a prefixed locant. The term **17-oxo steroid** (or 17-keto steroid) usually refers specifically to any of a group of androgens and their metabolites that are present in urine and that can be extracted and estimated by the **Zimmermann reaction**. The amount of steroids excreted in the urine has been used as an index of androgen production in the body.

oxosuccinate *see* **oxaloacetate**.

oxphos *a colloquial term for oxidative phosphorylation*.

OXT *abbr. for oxytocin*.

oxy the chemical group consisting of an oxygen atom, -O-, singly bonded to two carbon atoms in a molecular structure, e.g. of an epoxide, an ester, or an ether. *Compare* **oxo**.

oxy+ *prefix denoting 1 (in organic chemical nomenclature) the presence of an oxygen atom directly attached (by single covalent bonds) to two moieties of the same indicated parent compound, with formation of a symmetrical ether, where there is a functional group or substituent in the parent compound that has priority over 'ether' for citation as the principal characteristic group; e.g. 3,3'-oxydipropionic acid (not bis(3-carboxypropyl) ether). See also epoxy+. 2 (obsolete in organic chemical nomenclature but retained in pharmaceutical nomenclature) substitution of a hydroxyl group for a hydrogen atom; e.g. oxyproline (now hydroxyproline), oxytetracycline. 3 (in inorganic chemical nomenclature) formation of a 'basic' (i.e. double) salt containing the oxide anion, O²⁻, and a specified second anion; e.g. bismuth oxychloride (i.e. bismuth chloride oxide). 4 or oxy- presence of oxygen in the acidic function of an acid or in the corresponding moiety of a derived anion (obsolete; see **oxoacid** (def. 2), **oxoanion**). comb. form indicating. 5 dioxygen; e.g. oxybiotic, oxyhemoglobin, oxyhydrogen. 6 oxidation; e.g. oxycellulose. 7 acid; e.g. oxyntic.*

oxyacid or **oxy-acid** *a variant of oxoacid* (def. 2).

oxyanion or **oxy-anion** (formerly) *a variant of oxoanion*.

oxybiotic or **oxybiotic** living in the presence of oxygen; aerobic.

oxycellulose any of the various materials obtained by the oxidation of cellulose. Such materials are useful as cation-exchangers or for fabrication into lint and gauzes.

oxygen 1 *symbol*: O; a nonmetallic element of group 16 of the (IUPAC) **periodic table**; atomic number 8; relative atomic mass 15.9994. Naturally occurring oxygen consists of a mixture of three stable nuclides of relative masses 16 (99.759 atom percent), 17 (0.037 atom percent), and 18 (0.204 atom percent). The latter, commonly known as **heavy oxygen**, is useful as a tracer in studies of chemical and biochemical reaction mechanisms. The commonest valence is two. Oxygen is the most



oxygen-15

abundant element in the biosphere; in the atmosphere it occurs elementally in the form of molecules of the diatomic gas dioxygen, which comprises 20.95% by volume of dry air, and as traces of the gas **ozone** (i.e. trioxygen). The atmosphere also contains small amounts of the oxygen compounds carbon dioxide and water (vapour); almost half by weight of the surface of the Earth consists of oxygen in combined form, principally as water and silicates. **2 or molecular oxygen** *symbol*: O₂; *the common name* for the gas correctly known as **dioxygen**.

oxygen-15 the radioactive nuclide of oxygen, ¹⁵O. It is prepared in a cyclotron by bombarding ¹⁷N with deuterons, the products being ¹⁵O and neutrons. It emits a positron, β⁺ (0.511 MeV), and has a half-life of 2 min; it is used in positron emission **tomography**.

oxygenase any oxidoreductase enzyme catalysing a reaction in which dioxygen is the electron acceptor. Such enzymes may be either **monooxygenases** or **dioxygenases** (with the exception of certain uncharacterized enzymes forming EC sub-subclass 1.13.99).

oxygenate **1** to mix, supply, or treat with dioxygen; to saturate a fluid, especially blood, culture medium, etc., with dioxygen. **2** to cause the combination of an oxygen-carrying or oxygen-storing protein with dioxygen; to undergo such a reaction. —**oxygenation** *n*.

oxygenator an apparatus for oxygenation, especially of blood.

oxygen cycle the sum total of the processes via which atmospheric dioxygen is exchanged with the oxygen combined in organic compounds and carbon dioxide. Photosynthetic organisms, using solar energy, generate dioxygen from water and liberate it into the atmosphere, while heterotrophic cells take up atmospheric dioxygen to oxidize organic compounds and thereby provide energy. *Compare* **carbon cycle**.

oxygen debt the oxygen deficit that develops when a normally aerobic organism or tissue cannot increase its oxygen uptake sufficiently to match a temporarily increased energy requirement, as in muscular exercise. After the increased energy requirement ceases, this deficit has to be made good by a continuing increased rate of oxygen uptake until normal aerobic metabolism is restored.

oxygen demand *see* **biochemical oxygen demand**, **chemical oxygen demand**.

oxygen electrode a device for measuring dioxygen activity in a liquid or gas. It consists essentially of a silver-silver chloride anode in saturated potassium chloride solution and a platinum or gold cathode. When an electric potential is applied between these electrodes dioxygen is electrolytically reduced at the cathode, the current flowing being proportional to the dioxygen activity when the applied potential is 0.5–0.8 V. The electrodes are either directly exposed to the sample or, in the **Clark electrode**, separated from it by a thin dioxygen-permeable membrane in order to prevent deposition of interfering substances on the cathode.

oxygen quotient (*sometimes*) the **metabolic quotient** with respect to dioxygen, *symbol*: Q_{O₂}.

oxyhemoglobin *symbol*: HbO₂; the fully oxygenated form of **hemoglobin**. It is bright red in colour and has a characteristic absorption spectrum.

oxyhydrogen reaction *an alternative name for* **knallgas reaction**.

oxymyoglobin *symbol*: MbO₂; the fully oxygenated form of **myoglobin**. It is bright red in colour and has a characteristic absorption spectrum.

oxyntic acid-secreting; applied especially to the hydrochloric acid-secreting parietal cells of the gastric mucosa.

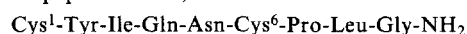
oxyntomodulin *an alternative name for* glucagon-37; *see* **enteroglucagon**.

oxyproline *an older name for* **hydroxyproline**. *Compare* **5-oxoproline**.

oxysome *an alternative name for* **elementary particle** (def. 2).

oxytetracycline 5-hydroxytetracycline; an antibiotic that is similar in action to **tetracycline**. Proprietary names include Terramycin.

oxytocin *or* (*sometimes*) **ocytocin** *abbr.*: OXT; a heterodetic cyclic nonapeptide amide,



(Cys¹ and Cys⁶ form a disulfide bond). It is one of the two (or possibly more) hormones secreted by the neurohypophysis of the pituitary gland. It facilitates the ejection of milk by stimulating the myoepithelial cells in the mammary gland, and it may also aid parturition by stimulating contraction of the uterus. It is synthesized as a polypeptide precursor that is cleaved to oxytocin and neurophysin 1 (for database code *see* **neurophysin**). Oxytocin was the first peptide hormone to be chemically synthesized. [Note: the spelling ocytocin is preferred on etymological grounds, the name being derived from the Greek *okytokos*, fast birth, rather than from the Greek *oxys*, acid, sharp; however, the spelling oxytocin is in wide use, esp. in the English language.] *See also* **vasopressin**.

oxytocin receptor any of a number of membrane proteins that bind oxytocin and mediate its intracellular effects. Structurally they are of the seven-transmembrane-domain type, characteristic of G-protein-coupled receptors, and activate the **phosphatidylinositol cycle**. Example from *Sus scrofa*: database code OXYR_PIG, 386 amino acids (42.52 kDa).

+oyl *suffix (in chemical nomenclature)* denoting conversion (of a mono- or dicarboxylic acid) to the corresponding mono- or divalent acyl group by removal of hydroxyl from the carboxyl group; e.g. the univalent acyl group, hexanoyl, CH₃–[CH₂]₄–CO–, is derived from hexanoic acid, CH₃–[CH₂]₄–COOH. It is used also when the carbonyl oxygen atom of the acyl group has been replaced by =NH, =N–NH₂, =N–OH, or =S. *See also* **+yl**.

oz *symbol for* ounce.

ozone *symbol*: O₃; trioxygen; a colourless, highly reactive gas present in the stratosphere, where it is formed under the influence of short-wavelength ultraviolet radiation. It is also normally present in trace amounts in air, and is formed by the action of sunlight on hydrocarbons and nitrogen oxides – present in vehicle emissions. The latter process can lead to harmful accumulations of ozone at or near ground level. In contrast, stratospheric ozone protects the biosphere from highly damaging ultraviolet radiation, especially UV-C.

ozonolysis a method used for locating a double bond in an alkene. The alkene is treated with ozone, and the resulting ozonide is then cleaved to form two carbonyl compounds, one from each moiety of the parent compound.

Pp

p symbol for 1 pico+ (SI prefix denoting 10^{-12} times). **2** the negative logarithm to the base ten, as in pH, pK. **3** proton; it may be written as p^+ to indicate its charge. **4** a phosphoric residue (alternative to *P*). **5** the protein product of a gene; it is prefixed to the product's molecular mass to designate a particular entity, e.g. **p53**. **6** (in atomic spectroscopy) electron state $l = 1$.

p symbol for 1 pressure (alternative to *P*). **2** partial pressure of a gas. The gas may be specified by a subscript or in parentheses; e.g. p_{O_2} or $p(O_2)$. **3** pyranose form (in monosaccharide symbolism); e.g. Glcp, glucopyranose; GalpA, galactopyranuronic acid. **4** pitch height (of a helix). **5** (bold italic) (electric) dipole moment (of a molecule), alternative to μ (bold italic). **6** (bold italic) momentum. **7** probability (alternative to *P*).

p- prefix (in chemical nomenclature) denoting *para-*. Compare *m-*, *o-*.

P₅₀ symbol for the partial pressure of a gas at which a binding protein is 50% saturated with it.

P_{LR} symbol for the proportion of receptors or binding sites occupied by ligand L.

P_{LR}* symbol for the proportion of receptors in which ligand L occupies its binding site(s), and which are in an active state.

P_R symbol for the proportion of receptors or binding sites free of ligand.

P symbol for 1 peta+ (SI prefix denoting 10^{15} times). **2** poise. **3** phosphorus. **4** a residue of the α -amino (strictly, cyclic α -alkylamino) acid proline (alternative to Pro). *abbr. for 5* phosphate or a phosphoric group, as in ATP. **6** (in genetics) parental generation. **7** a site near a peptidase cleavage site (see **peptidase P-site**).

P_i symbol for inorganic phosphate.

P symbol for 1 pressure (alternative to *p*). **2** radiant power (alternative to Φ). **3** probability (alternative to *p*). **4** parity (of a wave function).

p53 a 53 kDa nuclear phosphoprotein originally discovered attached to large T-antigen in cells transformed with **simian virus 40**. It is a probable human cell cycle regulator and *trans*-activator (see **trans-acting**) that acts negatively to regulate cellular division by controlling a set of genes required for this process; one of the genes activated is an inhibitor of **cyclin-dependent protein kinases**. p53 acts as a tumour suppressor in some (not all) tissues; mutations in the p53 gene are the most common genetic alterations in human cancers. Wild-type p53 can suppress or inhibit the transformation of cells in culture either by viral or cellular oncogenes, and introduction of wild-type cDNA into a transformed cell in culture stops growth. Hence the p53 gene is classed as a **tumour-suppressor gene**. Three functional domains of p53 have been defined: an N-terminal transcriptional activation domain (residues 1–42), a central DNA-binding domain (residues 120–290), and a C-terminal regulatory domain (residues 311–393). The DNA-binding domain recognizes a DNA motif in genes that are activated by p53, and mutations in p53 that are associated with human cancers generally cluster in its DNA-binding domain. Database code WT1_HUMAN, 449 amino acids (49.13 kDa).

PO a myelin protein.

P1 a temperate bacteriophage whose natural hosts are *Shigella* spp. but which was used formerly for generalized transduction in *Escherichia coli* and is currently used as the basis for the **P1 cloning vector**, which can carry up to 100 kbp insert DNA. P1 prophage is a plasmid.

P2 a myelin protein.

P65 see **synaptotagmin I**.

P81 see **ezrin**.

P450 or **P-450** any of a group of widely occurring heme-containing proteins that embraces various separately classified enzymes. They are named from the location of the Soret band

(see **cytochrome absorption bands**) of the reduced CO complex at 450 nm. The group includes unspecific monooxygenase (EC 1.14.14.1), camphor 5-monooxygenase (EC 1.14.15.1), alkane 1-monooxygenase (EC 1.14.15.3), steroid 11 β -monooxygenase (EC 1.14.15.4), **cholesterol monooxygenase (side-chain-cleaving)** (EC 1.14.15.6), prostacyclin synthase (EC 5.3.99.4), **thromboxane-A synthase** (EC 5.3.99.5), and probably chloride peroxidase (EC 1.11.1.10). This type of enzyme is also known as **cytochrome P450** (see this for more detail) and considered as a *b*-type **cytochrome**. The name **heme-thiolate protein** has been proposed on the grounds of a thiolate ligand at the heme being responsible for the group's unusual spectral and catalytic properties.

P680 or **pigment 680** a type of chlorophyll *a*, having an absorption maximum at 680 nm, that constitutes the reaction centre of **photosystem II** in most species. It has a redox potential of $>+0.81$ V; the oxidized form, P680⁺, is a powerful oxidant that can abstract electrons from water with the evolution of dioxygen. See also **chlorophyll**.

P700 or **pigment 700** a type of chlorophyll *a*, having an absorption maximum at 700 nm, that constitutes the reaction centre of **photosystem I**. It has a redox potential of approximately +0.5 V, its electron being used ultimately to reduce NADP via ferredoxin. See also **chlorophyll**.

P870 the special pair of bacteriochlorophyll *a* in the **photosynthetic reaction centre** of bacteria, so named because they have an absorption maximum of 870 nm.

P960 the special pair of bacteriochlorophyll *b* in the **photosynthetic reaction centre** of bacteria, so named because they have an absorption maximum of 960 nm.

Pa symbol for pascal.

pA₂ the negative logarithm to the base 10 of the molar concentration of antagonist that makes it necessary to double the concentration of agonist needed to elicit the original submaximal response.

p[A]₅₀ the negative logarithm to the base 10 of the [A]₅₀ of an agonist. pEC₅₀ is the corresponding term for the EC₅₀ of an agonist (often used interchangeably with [A]₅₀).

PAB or **PABA** *abbr. for para-aminobenzoate; para-aminobenzoic acid*. See **p-aminobenzoic acid**.

PABA test an indirect test of pancreatic function in which the synthetic peptide *N*-benzoyl-L-tyrosyl-*p*-aminobenzoic acid (BT-PABA) is administered orally, usually with a small amount of [¹⁴C]-*p*-aminobenzoic acid (PABA). Normally the peptide is hydrolysed by chymotrypsin to PABA, which is absorbed and excreted in the urine, but in pancreatic insufficiency hydrolysis is much reduced.

pacemaker reaction any metabolic reaction whose rate depends on factors other than the amounts of enzyme or substrate. An enzyme catalysing such a reaction may be termed a **pacemaker enzyme**.

pachyman see **callose**.

pachytene the third phase of prophase I in **meiosis**.

package 1 (in cell biology) to surround (one or a number of macromolecules) with a membrane in the formation of a mature secretory granule. **2** (in virology) to enclose the nucleic acid core of a virus particle with a protein shell (the capsid) to form a mature virion; to encapsidate.

packing (in chromatography) the material (whether active solid, liquid held on solid support, or swollen gel) that is introduced into a column and that consists of or contains what is to become the stationary phase during a chromatographic separation.

packing density the ratio of the minimum (or actual) volume of an object, such as a molecule in a crystal or in solution, to the total volume it occupies. For a macromolecule or region of



packing fraction

a macromolecule, it is the ratio of the volume enclosed by the van der Waals envelope of all atoms in the molecule or region to the total volume of the molecule or region. Theoretical values are 0.74 for close-packed spheres, 0.91 for infinite cylinders, and 1.0 for a continuous solid.

packing fraction the ratio of the relative atomic mass of a particular nuclide minus its nucleon number divided by its nucleon number. It is positive for the lightest and heaviest nuclides and negative for nuclides of intermediate mass.

Paclitaxel a proprietary name for **taxol**.

PAF abbr. for platelet-activating factor.

PAGE abbr. for polyacrylamide gel electrophoresis.

PAGIF abbr. for polyacrylamide gel isoelectric focusing. See **polyacrylamide gel electrophoresis**.

PAH abbr. for para-aminohippurate; para-aminohippuric acid.

pair see **base pair**.

paired the gene for a sequence-specific DNA-binding homeobox segmentation protein in *Drosophila melanogaster*. It is one of a number of **pair-rule genes**, each of which specifies a simple alternation with a repeat distance of two segments. Their protein products are characterized by four motifs, referred to as the paired box. Example from *D. melanogaster*: database code HMPR_DROME, 613 amino acids (65.43 kDa).

paired box see **paired**, **Pax genes**.

pair-rule gene see **paired**.

PaJaMa experiment the nickname for a bacterial mating experiment carried out at l'Institut Pasteur, Paris, that showed that induction and repression of the enzyme β -galactosidase in *Escherichia coli* are regulated by two closely linked genes, one of which produces a cytoplasmic repressing substance that blocks the expression of the other. [From the names of the experimenters: Pardee, Jacob, and Monod.]

Palade, George Emil (1912–), Romanian-born US cell biologist; Nobel Laureate in Physiology or Medicine (1974) jointly with A. Claude and C. R. M. J. de Duve 'for their discoveries concerning the structural and functional organization of the cell.'

palindrome **1** any linear arrangement of symbols, such as letters or digits, that has the same sequence from either end; e.g. noon, radar; 75311357; 'Madam, in Eden I'm Adam'. **2** (in molecular biology) or **palindromic sequence** a popular name for a DNA sequence with a twofold rotational axis of symmetry (**dyad symmetry**). Confusingly the term is used in at least two distinguishable senses: (1) to describe a region of local twofold rotational symmetry in duplex DNA, e.g.



and (2) to describe a sequence in a single polynucleotide strand that contains an inverted repeat, e.g. 5'-TGA-AGT-3'. It is also sometimes applied to a region of either a single- or a double-stranded polynucleotide that is capable of forming one or more **hairpins** containing palindromes of the first type; an example is shown at **cruciform**. Examples of palindromes are **operators** and **restriction sites** but, in general, any stem-loop in RNA must have a corresponding palindrome in the DNA from which the RNA was transcribed. —**palindromic adj.**

pallidin any of a group of endogenous carbohydrate-binding proteins (**lectins**) produced by cells of the slime mould *Polysphondylium pallidum* during differentiation. Similar to **discoïdin**, they may be concerned in cell adhesion.

palmitate **1** numerical symbol: 16:0; the trivial name for hexadecanoate, $\text{CH}_3\text{-(CH}_2\text{)}_{14}\text{-COO}^-$, the anion derived from palmitic acid (i.e. hexadecanoic acid), a saturated straight-chain higher fatty acid. **2** any mixture of free palmitic acid and its anion. **3** any salt or ester of hexadecanoic acid. See also **palmitoyl**.

palmitate synthase see **fatty-acid synthase complex**.

palmitic acid hexadecanoic acid; a saturated straight-chain fatty acid having sixteen carbon atoms per molecule, and a major component of plant and animal fats. See **palmitate**.

palmitoleate **1** numerical symbol: 16:1(9); the trivial name for (Z)-hexadec-9-enoate, $\text{CH}_3\text{-(CH}_2\text{)}_5\text{-CH=CH-(CH}_2\text{)}_7\text{-COO}^-$, the anion derived from palmitoleic acid, (Z)-hexadec-9-enoic acid, a monounsaturated straight-chain higher fatty acid. **2** any mixture of free palmitoleic acid and its anion. **3** any salt or ester of 9-hexadecenoic acid. See also **palmitoleoyl**.

palmitoleic family (of polyunsaturated fatty acids) a series of polyenoic acids in which the hydrocarbon chain terminates with the alkenyl grouping $\text{CH}_3\text{-(CH}_2\text{)}_5\text{-CH=CH-}$ (as in palmitoleic acid (see **palmitoleate**)); there are three other series of such acids, i.e. the **linoleic family**, **linolenic family**, and **oleic family**. Members of the palmitoleic family can be synthesized from palmitic acid via palmitoleic acid by chain elongation and/or desaturation, but in mammals not from linoleic, (9,12,15)-linolenic, or oleic acids; the series includes *cis*-vaccenic acid.

palmitoleoyl symbol: ΔPam ; the trivial name for (Z)-hexadec-9-enoyl, $\text{CH}_3\text{-(CH}_2\text{)}_5\text{-CH=CH-(CH}_2\text{)}_7\text{-CO-}$ (*cis* isomer), the acyl group derived from palmitoleic acid (i.e. (Z)-hexadec-9-enoic acid). It is a relatively minor component of plant and animal lipids. The chain is synthesized metabolically by the action of a Δ^9 -desaturase on palmitoyl-CoA.

palmitoyl symbol: Pam; the trivial name for hexadecanoyl, $\text{CH}_3\text{-(CH}_2\text{)}_{14}\text{-CO-}$, the acyl group derived from palmitic acid (i.e. hexadecanoic acid). This is one of the major fatty-acid components of plant and animal lipids; together with stearic acid it represents a high proportion (often around 30%) of the fatty-acid content of dietary and other lipids. It acts as a precursor in mammals for a family of unsaturated fatty acids in which all double bonds are nine or more carbons from the methyl terminus. The chain is synthesized metabolically from acetyl-CoA as palmitate (in mammals) or palmitoyl-CoA (in yeast), the product of fatty-acid synthase. See **fatty-acid synthase complex**.

palmityl **1** or **cetyl** a common name for hexadecyl, $\text{CH}_3\text{-(CH}_2\text{)}_{14}\text{-CH}_2\text{-}$, the alkyl group derived from hexadecane. **2** a former name (now incorrect) for **palmitoyl**.

palytoxin any of a group of structurally very similar potent neurotoxins isolated from the marine zoanthid *Palythoa toxica* or other *Palythoa* spp. Each consists of a single carbon chain of 115 carbon atoms containing 64 chiral centres and seven double bonds, and terminating in nitrogen-containing groups; there are 128 carbons in all and a further double bond in a methylene group. The chain bears numerous hydroxyl groups and a number of methyl groups, and in several places it is folded and formed into pyran rings. Palytoxin is the most deadly nonproteinaceous material isolated, having 50 times the toxicity of **tetrodotoxin** in mice. The number of stereoisomers is $2^{64} = 18\,446\,744\,073\,709\,551\,620$ (not including possible *E/Z* variations at the double bonds!).

Pam symbol for **palmitoyl** (i.e. hexadecanoyl).

pancreas a compound gland, occurring in the abdominal cavity of most vertebrates, that has both exocrine and endocrine functions. The major (exocrine) part consists of acinar tissue, which secretes **pancreatic juice** into the upper part of the gut (duodenum in mammals). Within this acinar tissue are scattered numerous (endocrine) **islets of Langerhans**, containing notably A cells, secreting **glucagon**, B cells, secreting **insulin**, and D cells, secreting **somatostatin**. —**pancreatic adj.**

pancreastatin a 49-residue peptide hormone that acts as an inhibitor of glucose-induced insulin release from the pancreas. It also inhibits exocrine pancreatic excretion. It was originally isolated from porcine pancreas, where it is colocalized with insulin, glucagon and somatostatin, but is synthesized as part of the sequence of **chromogranin A** and is thus found in tissues where the chromogranin gene is expressed, including most endocrine cells.

pancreatic cholera an alternative name for **Verner-Morrison syndrome**.

pancreatic diabetes an older term for insulin-dependent **diabetes mellitus**.

pancreatic hormone or **pancreatic polypeptide** abbr.: PP; a

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

pancreatic islet

crystallizable polypeptide present in the **PP cells** of the pancreas, first detected as an impurity in insulin preparations. Avian pancreatic polypeptide, known as aPP, has been isolated from chicken and turkey pancreas; bovine, human, ovine, and porcine homologues, known as bPP, hPP, oPP, and pPP, respectively, have also been identified. All have 36 amino-acid residues with C-terminal tyrosinamide and show considerable sequence homologies. The PP level in the plasma rises rapidly after feeding, especially with protein foods; the hormone appears to act as a regulator of pancreatic and gastrointestinal functions; sequence from human: APLEPVYPGDNATPEQMAQYAADLRRYINMLTRPRY(NH₂); database code (precursor) PAHO_HUMAN, 95 amino acids (10.43 kDa) also contains an icosapeptide of uncertain physiological role (residues 69–88).

pancreatic islet an alternative name for **islet of Langerhans**.

pancreatic juice a slightly alkaline digestive juice secreted by the exocrine pancreas into the upper part of the small intestine. It contains numerous enzymes and inactive enzyme precursors including **a-amylase**, chymotrypsinogen, **lipase**, procarboxypeptidase, proelastase, phospholipase A₂, ribonuclease, and trypsinogen. Its high concentration of hydrogen-carbonate ions helps to neutralize the acid digesta from the stomach.

pancreatic lipase a Ca²⁺-requiring triacylglycerol lipase (EC 3.1.1.3) that is secreted into the intestine by the exocrine pancreas when this is stimulated by cholecystokinin in response to ingestion of food. Its role is to degrade triacylglycerols, partially or completely, to fatty acids and glycerol in the intestine; it acts at an ester–water interface. It is a glycoprotein with sequence similarity with other lipases. Example (precursor) from human: database code LIPP_HUMAN, 465 amino acids (51.10 kDa).

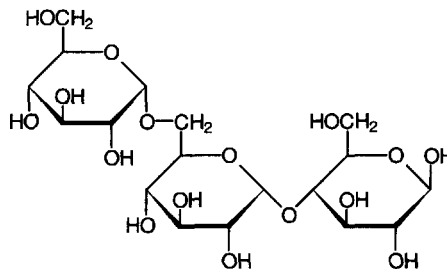
pancreatic polypeptide *abbr.*: PP; another name for **pancreatic hormone**.

pancreatic thread protein or **pancreatic stone protein** (*abbr.*: PSP); a C-type lectin that may inhibit spontaneous calcium carbonate precipitation, and one of the major secretory proteins of the human exocrine pancreas. It is found in acinar cells of pancreas and (in smaller amounts) in the brain. It is a major soluble protein of human pancreatic calculi and a Ca²⁺-binding phosphoprotein present in zymogen granules of pancreatic acinar cells and secreted in pancreatic juice of normal subjects and calculus formers. It inhibits calcium carbonate precipitation from the juice. The protein is rich in aromatic amino acids. Between pH 5.4 and 9.2 it undergoes reversible fibril formation. It is also typically characteristic of infant brains. There is increased expression of PSP-like proteins in Down's syndrome and Alzheimer's disease. Example (precursor) from human: database code LITH_HUMAN, 166 amino acids (18.71 kDa).

pancreatic trypsin inhibitor *abbr.*: PTI; a serine proteinase inhibitor, correctly known as **pancreatic secretory trypsin inhibitor**. A crystallizable protein trypsin inhibitor isolated from bovine pancreas, it forms a crystalline complex with trypsin and is relatively heat-stable in trichloroacetic acid solution. Its physiological role is to control trypsin activation of zymogens. The bovine protein is of known sequence: NILGRE-AKCTNEVNGCPRIYNPVC GTDGV TYSNECLLCMENK-ERQTPVLIQKSGPC. The 3-D structure is also known, and it is a favoured model for testing protein-folding algorithms. Example (bovine): database code IPST_BOVIN, 56 amino acids (6.16 kDa). It was also formerly known as **Kunitz** and **Northrop inhibitor**.

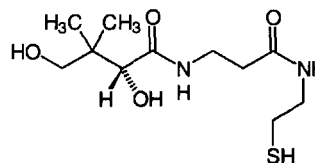
pancreozymin *abbr.*: PZ; a putative hormone having secretagogic activity on the exocrine pancreas and extractable from duodenal mucosa. It is now known to be identical with **cholecystokinin**.

panose an oligosaccharide, Glc(α1-6)Glc(α1-4)Glc; it is found in the fungus *Aspergillus niger*.

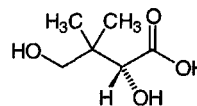


panose

pantetheine *N*-pantothenylcysteamine; the D-enantiomer is a growth factor for *Lactobacillus bulgaricus*, and is an intermediate in the pathway for the biosynthesis of coenzyme A in mammalian liver and some microorganisms.

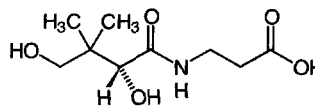


pantoate 1 the anion, HO–CH₂–C(CH₃)₂–CH(OH)–COO[–], derived from pantoic acid (i.e. (R)-2,4-dihydroxy-3,3-dimethylbutyric acid). **2** any salt or ester of pantoic acid.



pantoic acid

pantothenate 1 the anion, HO–CH₂–C(CH₃)₂–CH(OH)–CO–NH–CH₂–CH₂–COO[–] (i.e. *N*-pantoyl-β-alanine anion), derived from pantothenic acid, a B vitamin complex; only the D-(R)-enantiomer is biologically active. **2** any salt of pantothenic acid.



pantothenic acid

pantothenyl the acyl group, HO–CH₂–C(CH₃)₂–CH(OH)–CO–NH–CH₂–CH₂–CO–, derived from pantothenic acid; coenzyme A is a D-(R)-pantothenyl derivative.

pantoyl 1 the acyl group, HO–CH₂–C(CH₃)₂–CH(OH)–CO–, derived from pantoic acid, 2,4-dihydroxy-3,3-dimethylbutyric acid.

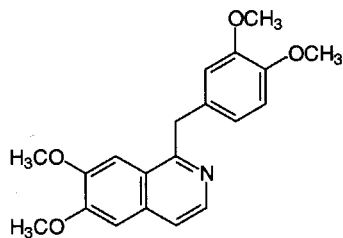
PAP *abbr. for peroxidase-antiperoxidase*.



papain

papain or **papaya peptidase I** EC 3.4.22.2; a cysteine endopeptidase obtained from the latex, leaves, and unripe fruit of the papaya (or pawpaw) tree, *Carica papaya*. It will preferentially hydrolyse peptide bonds at the carboxyl end of Arg, Lys, and Phe residues (but never Val), with preference for large hydrophobic residues at the P2 position. It also has esterase, thiolesterase, transamidase, and transesterase activity. Papain is unusually stable to elevated temperatures and to denaturing agents and consists of a single polypeptide chain of 212 amino-acid residues. Example from *C. papaya*: database code NRL_1PPD, 212 amino acids (23.40 kDa); 3-D structure known; α and β regions.

papaverine 1-[(3,4-dimethoxyphenyl)methyl]-6,7-dimethoxyisoquinoline; a constituent of **opium** that acts as a smooth muscle relaxant; this action is thought to be due to phosphodiesterase inhibitory activity and blockade of membrane calcium channels.



paper chromatogram the visible result of a chromatographic separation effected on paper.

paper chromatography *abbr.*: PC; a technique of **chromatography**, applicable to microgram quantities of soluble substances, in which specially prepared filter paper, **chromatography paper**, forms the support for the stationary phase. The latter is commonly a film of water held by adsorption on the cellulose fibres of the paper and in equilibrium with a water-immiscible liquid or liquid mixture, which forms the mobile phase. A solution of a sample is applied near one end of a strip of paper, allowed to dry, and the mobile phase allowed to flow over the strip from that end. If the sample consists of only one component, development using an appropriate method of detection will reveal a single spot at a characteristic distance from the origin relative to that travelled by the mobile phase; in general, a mixture will give rise to a number of spots each occupying a different and characteristic position. Separations occur essentially by partition between the mobile and stationary phases, with some contribution, under certain conditions, attributable to adsorption or ion-exchange onto the paper support. An elaboration of the technique uses a square of paper on which two successive separations are effected at right-angles using mobile phases of differing separatory powers to produce a two-dimensional pattern of spots on the paper. With suitable treatment of the paper and an appropriate choice of the composition of the two phases, reversed-phase partition chromatography on paper may be effected. Since cellulose is chiral, some enantiomer separations are possible.

paper electrophoresis a type of **zone electrophoresis** on (filter) paper.

Papovaviridae a family of DNA viruses, most or all of which are, under suitable conditions, oncogenic in vertebrate hosts. The virion, 45–55 nm in diameter, consists of a nonenveloped icosahedral capsid with 72 capsomeres, and contains a circular DNA genome.

papovavirus any virus belonging to the family **Papovaviridae**. The name derives from papilloma, polyoma, and vacuolating agent, an early name for **simian virus 40**.

PAPS *abbr.* for 3'-phosphoadenosine-5'-phosphosulfate; i.e. adenosine 3'-phosphate 5'-phosphosulfate.

par+ a variant form of **para+** (before a vowel).

para 1 characterized by or relating to (substitution at) two ring-carbon atoms separated by two others in a monocyclic aromatic compound. 2 (followed by to) at a position in a benzene ring next but two to a particular (specified) substituent. *See also para-*.

para+ or (before a vowel) **par+** prefix indicating beside or near; beyond; resembling; defective or abnormal.

para- *abbr.*: p-; prefix (in chemical nomenclature) denoting an isomer of a disubstituted monocyclic aromatic compound in which the substituents are attached at positions 1 and 4 of the ring. *Compare meta-, ortho-*.

parabiont either individual of a pair united in **parabiosis**.

parabiosis the state or condition of two individual animals being functionally united, either naturally, as in Siamese twins, or as a result of an experimental procedure. —**parabiotic** *adj.*

paracasein an alternative name (*esp. US*) for **casein** (def. 2).

paracetamol *see acetaminophen*.

paracrine 1 describing or relating to a regulatory cell that secretes an agonist into intercellular spaces in which it diffuses to a target cell other than that which produces it; describing or relating to such an agonist. 2 such an agonist; a **paramone**. *Compare endocrine, neurocrine, neuroendocrine*.

paracrystal any assemblage of molecules or particles having a certain degree of order but one that is insufficient for it to be considered a true crystal. —**paracrystalline** *adj.*; **paracrystallinity** *n*.

paradaxin or **pardaxin** any of several ichthyotoxic, neurotoxic peptides, isolated from the defence secretion of the Red Sea Moses sole fish, *Paradachirus marmoratus*, or the Pacific sole, *Paradachirus pavoninus*, that act as a shark repellent. Each consists of 33 amino acids, with hydrophilic C-terminal and hydrophobic N-terminal regions and with physical and pharmacological similarities with **melittin** but no sequence homology. Example, paradaxin P-2 from *P. pavoninus*: database code PAP2_PARPV, 33 amino acids (3.32 kDa).

paraffin 1 an obsolete term for **alkane**. 2 or **paraffin oil** or **kerosene** or **kerosine** any liquid mixture, consisting mainly of alkanes, with a boiling point range of 150–300 °C and relative density range of 0.78–0.82. 3 an alternative name for **paraffin wax**. *Compare liquid paraffin*.

paraffin wax any waxy mixture of higher alkanes, obtained as a residue from the distillation of petroleum, with a relative density of ≈ 0.9 and a melting point range of 45–65 °C.

Parafilm the proprietary name for a flexible, moisture-proof, thermoplastic tissue useful for temporarily sealing small containers.

parafollicular cell an alternative name for **C cell** (when found in the thyroid gland).

parafusin an evolutionarily conserved phosphoglycoprotein involved in exocytosis. It is rapidly dephosphorylated via a Ca^{2+} -dependent process when secretagogues induce exocytosis in competent cells. Example from *Paramecium tetraurelia*: database code PARPARAFUS, 475 amino acids (52.94 kDa).

parahematin or (*esp. Brit.*) **parahaematin** a former name for **ferrihemochrome**.

parallel 1 (in **biochemistry**) describing a pair of linear structures, such as two polynucleotide or polypeptide chains, that are polarized or asymmetric in the same direction. 2 (in **physical chemistry**) denoting the spins of a pair of electrons, occupying the same atomic or molecular orbital, that are described by the same spin quantum number. *Compare antiparallel*. 3 (in **electricity**) describing an electrical component that is connected to the same two points in a circuit as another component; describing two or more components that are so connected.

parallel dichroism *see dichroic ratio*.

parallelism the quality or state of being parallel; similarity in corresponding details or in evolutionary pattern.

paramagnetism the property displayed by substances that have a positive but small magnetic susceptibility, due to the presence in them of atoms with permanent magnetic dipoles



paramecium

caused by unpaired electron spins (with a contribution from the orbital motion of the electrons). These dipoles tend to align themselves in the direction of an applied magnetic field but no permanent magnetism is conferred on such substances. *Compare diamagnetism, ferromagnetism. —paramagnetic adj.*

paramecium (*pl. paramecia*) any freshwater protozoan belonging to the genus *Paramecium*. Typically 50–300 µm in length, they are ovoid and uniformly ciliated. *See also kappa particle.*

parameter **1** (*in mathematics*) an unknown quantity that is a constant in a particular context but that may have different values in other (similar) contexts; e.g. the coefficient *b* in the general equation, $y = bx + c$, representing a family of straight lines. **2** (*in statistics*) a numerical characteristic of a population (as opposed to that of a sample of such a population). **3** any distinguishing characteristic of something, especially one to which a measured value is or can be ascribed. —**parametric adj.**

paramomycin an aminoglycoside antibiotic produced by *Streptomyces paromomycinus*. Its actions are similar to those of **neomycin**.

paramone a proposed name for **paracrine** agonist, derived as a contraction of 'paracrine hormone'.

paramylon a storage polysaccharide occurring in *Euglena gracilis*. It comprises a backbone of $\beta 1 \rightarrow 3$ -glucopyranose units with side chains joined by $\alpha 1 \rightarrow 6$ linkages.

paramyosin a two-chain, coiled-coil, α -helical ≈ 200 kDa protein that forms the core protein of the thick filaments of a number of invertebrate muscles.

paramyxovirus any of a group of RNA animal viruses of the family Paramyxoviridae (distinguish from the genus Paramyxovirus), in class V of the **Baltimore classification**, consisting of enveloped, pleomorphic particles ≈ 150 nm in diameter with helical nucleocapsids, 1000 nm long and 18 nm in diameter. They contain single-stranded RNA (minus strand) of 7 MDa plus virion polymerase. The group includes viruses causing mumps, measles, Newcastle disease of chickens, and distemper of dogs, and the morbilliviruses, pneumoviruses, and parainfluenza viruses.

paranemic describing the nature of the coiling of a double-stranded helix in which the two strands are not intertwined, i.e. they may be separated without uncoiling. *Compare plectonemic.*

parapepsin I an alternative name for **pepsin B**.

parapepsin II an alternative name for porcine **gastric** (i.e. pepsin C).

paraprotein name originally given to any plasma protein that gave an abnormal band on electrophoresis, derived from **para** + protein. Usually such a protein is a monoclonal immunoglobulin derived from neoplastic plasma cells and present at abnormally high concentration in the blood plasma. Such proteins are seen as a discrete band in the gamma-globulin region, but they may appear elsewhere if the paraprotein is IgA or IgM. Examples are proteins characteristic of a myeloma, e.g. **Bence-Jones protein**, amyloid proteins, **Waldenström's macroglobulinemia**, or cryoglobulins. Some paraproteins are not abnormal, e.g. the immunoglobulin that arises as a result of a severe bacterial infection.

paraproteinaemia or (*esp. Brit.*) **paraproteinaemia** the presence in blood plasma of **paraprotein**.

Paraquat methyl viologen; 1,1'-dimethyl-4,4'-bipyridinium dichloride; a nonselective herbicide whose action requires direct contact with the plant. It is highly toxic to humans and other animals if ingested in undiluted form.

parasite any organism that spends all or part of its life cycle in (**endoparasite**) or on (**ectoparasite**) another living organism of a different species (its **host**), from which it obtains nourishment and/or protection, and to which it is usually detrimental. *Compare symbiont.*

parastatin a 73-amino-acid peptide hormone, M_r 11 000, that inhibits **parathyrin** secretion. Porcine parastatin has the amino-acid sequence: LSFAPAYGFRGPGLQLRRGWPSRED-

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

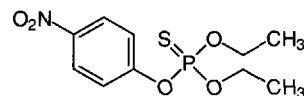
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

SVEAGLPQVRXYLEEKKEEEGSANRRPEDQELESLSAI-EAELEK. Activity resides in a 19-amino-acid N-terminal fragment; that from rat has the sequence: LSFARAYGF-RDPGPQLRR. It is synthesized as part of the sequence of **chromogranin A**.

paratartaric acid an old name for racemic (DL)-tartaric acid. The resolution of this acid into the D- and L-enantiomers by Pasteur was a landmark experiment in stereochemistry. *See tartaric acid.*

parathion O,O-diethyl O-p-nitrophenyl phosphorothioate; a compound that acts as an inhibitor of cholinesterase through covalent bonding to the esteratic site of the enzyme. It is used as a chemical warfare agent and insecticide.



Parathormone a proprietary name for **parathyrin**.

parathyrin or **parathyroid hormone** (*abbr.*: PTH); a 9.5 kDa polypeptide hormone, consisting of a single chain of 83 amino-acid residues (bovine). It is synthesized in parathyroid tissue by proteolysis of a 12.5 kDa prohormone. Parathyrin acts in conjunction with **calcitonin** and other hormones to control calcium and phosphate metabolism; it elevates the blood calcium level by dissolving the salts in bone and preventing their renal excretion. Example (precursor) from human: database code PTH_HUMAN, 115 amino acids (12.85 kDa). *Proprietary name:* Parathormone.

parathyroid **1** situated near the thyroid gland. **2** of, pertaining to, or produced by the parathyroid gland. **3** the parathyroid gland.

parathyroid gland an endocrine gland of higher vertebrates; there are typically two pairs, located near or within the thyroid. They develop from the gill pouches, and secrete **parathyrin**.

parathyroid hormone *abbr.*: PTH; an alternative name for **parathyrin**.

paratope the part of an antibody, formed by the hypervariable loops of the **variable regions**, that binds to an **epitope**.

paratose 3,6-dideoxy-D-glucose; 3,6-dideoxy-D-ribo-hexose; a monosaccharide that occurs in type A O-antigen chains of the lipopolysaccharide on the outer membrane of certain species of *Salmonella*.

Parazoa a subkingdom of multicellular invertebrate animals comprising the sponges (phylum Porifera).

parenchyma **1** (*in botany*) in higher plants, any soft tissue consisting of thin-walled, relatively undifferentiated living cells. **2** (*in zoology*) the tissue constituting the essential or specialized part of an organ, as distinct from supportive tissue, blood vessels, etc. —**parenchymal adj.**; **parenchymatous adj.**

parenchymal liver cell *see* **hepatocyte**.

parent **1** any organism that has given rise to another such organism, whether sexually or asexually. **2** a precursor of some derived entity; applied, e.g., to a cell, a virus, a chromosome or molecule of DNA, or a radionuclide. **3** **parent compound** (*in chemical nomenclature*) the member of a group of related compounds that has the simplest chemical structure and that forms the basis for naming the others. —**parental adj.**

parenteral **1** located or occurring outside the intestine. **2** any route other than via the gastrointestinal tract, especially by injection. —**parenterally adv.**

parietal **1** of, pertaining to, or forming part of the wall of a body cavity or similar structure; e.g. the hydrochloric-acid-secreting **parietal cells** of the gastric mucosa, also known as oxyntic cells. **2** of or relating to the parietal bones of the skull.

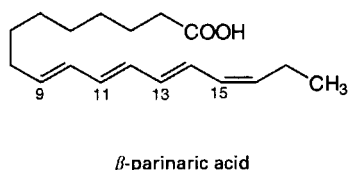
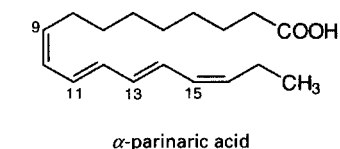
parietal layer *see* **serous membrane**.

parinaric acid any octadeca-9,11,13,15-tetraenoic acid; two isomers are found in *Parinarium laurinum*, namely the (9Z,11E,13E,15Z)-isomer (*g-* or *cis*-parinaric acid) and the



Parkinson's disease

(9E,11E,13E,15Z)-isomer (β - or *trans*-parinaric acid). Both are useful as probes in membrane studies, *trans*-parinaric acid especially, as it partitions preferentially into the less fluid regions of membrane lipids with increased quantum yield. They inhibit neutrophil elastase.



Parkinson's disease or **parkinsonism** a progressive disorder of the central nervous system characterized by tremor and impaired muscular coordination, thought to be due to defective dopaminergic transmission in some parts of the brain. There is a loss of dopaminergic neurons connecting the substantia nigra with the striatum, resulting in loss of dopaminergic inhibition in the striatum, with resulting cholinergic hyperactivity. Therapy is aimed at replacement of dopamine, achieved by administration of **L-dopa**, which is converted to dopamine within the brain (dopamine itself does not cross the blood-brain barrier). *See also* **MPTP**. [After James Parkinson (1755–1824), British palaeontologist and surgeon, who first described the disease in 1817.]

Park nucleotide UDP-*N*-acetylmuramyl pentapeptide; an intermediate in the biosynthesis of peptidoglycan in the cell walls of most bacteria. [After James Theodore Park (1922–).]

parotid gland either of a pair of **salivary glands** situated near each ear in mammals. The duct runs forwards and empties into the oral cavity.

PARP *abbr.* for 1 procyclin. 2 proline/arginine-rich protein.

parsimony 1 (*in computing*) an algorithm used for determining sequence relationships in macromolecules. 2 (*in genetics*) the construction of evolutionary trees from protein and nucleic acid sequences. It maximizes the genetic likeness associated with common ancestry, whilst minimizing the incidence of convergent mutation, and provides a way to find the best genealogical hypothesis from sequence data.

parthenocarp the formation of fruit without the setting of seeds. It occurs naturally, but can be induced by spraying flowers with gibberellic acid (*see* **gibberellin**) to produce crops of seedless fruit.

parthenogenesis the development of a new individual from an unfertilized female gamete. —**parthenogenetic** *adj.*; **parthenogenetically** *adv.*

partial agonist an agonist that is unable to evoke the maximal response of a biological system, even at a concentrations sufficient to saturate the specific receptors.

partial derivative (*in mathematics*) the derivative of an expression containing two or more independent variables that is obtained by differentiating the expression with respect to only one of them, the remaining variable or variables being considered constant.

partial hydrolysate any mixture of monomers and oligomers resulting from a limited degree of chemical hydrolysis of a biopolymer or from its exposure to a specific hydrolytic enzyme.

partial inhibition (*sometimes*) a type of nonlinear enzyme inhibition in which saturation of the enzyme with inhibitor does not decrease the rate of the enzyme-catalysed reaction to zero.

partial inverse agonist an **inverse agonist** that evokes a submaximal response in a biological system, even at a concentration sufficient to saturate the specific receptors. At high concentrations it will diminish the actions of a full inverse agonist. *Compare* **antagonist**, **partial agonist**.

partially overlapping code *see* **overlapping code**.

partially overlapping gene *see* **overlapping gene**.

partial molar quantity a function of any particular extensive property (e.g. enthalpy, entropy, Gibbs energy, heat capacity, internal energy, volume) of a specified component of a single-phase multicomponent system that describes how that property varies, at constant pressure and temperature, with variation of the concentration of the component when the change in the property is expressed per mole of the component. Partial molar Gibbs energy is more usually known as **chemical potential**. *Compare* **partial specific quantity**.

partial pressure *symbol:* p_B or $p(B)$ (for a gas B); the pressure exerted by one particular gas in a mixture of nonreacting gases. It is defined as the pressure the gas would exert if it alone were present and occupied the whole volume of the mixture at the same temperature, and is equal to the product of the amount-of-substance fraction of the gaseous component in question and the total pressure of the gaseous system. *Compare* **tension**. *See also* p_{50} .

partial pressures, law of *see* **Dalton's law of partial pressures**.

partial specific quantity a function of any particular extensive property (e.g. enthalpy, entropy, Gibbs energy, heat capacity, internal energy, volume) of a specified component of a single-phase multicomponent system that describes how that property varies, at constant pressure and temperature, with variation of the concentration of the component when the change in the property is expressed per unit mass of the component. *Compare* **partial molar quantity**.

partial specific volume (of a solute) *symbol:* $\bar{v}_B$; an instance of a **partial specific quantity**, defined as the change in volume of a solution on addition of a small amount of a solute B, expressed per unit mass of added solute and extrapolated to infinite dilution. The cgs unit (in which values are often listed) is $\text{cm}^3 \text{g}^{-1}$ and the SI unit is $\text{m}^3 \text{kg}^{-1}$.

particle 1 an extremely small piece or portion of (esp. solid or colloidal) matter. 2 **elementary particle**. —**particulate** *adj.*

particulate enzyme any enzyme that is bound to an insoluble cellular or subcellular structure or that has been artefactually or artificially attached to particulate material.

particulate radiation any form of radiation consisting of a stream of particles, especially alpha radiation, beta radiation, or neutron radiation emanating from a radioactive source.

partition 1 (*in chemistry*) or **distribution** the equilibration of a solute between two phases (usually of differing composition); the result of such a process. 2 (*in genetics*) the events that lead to the separation of chromosomes or to the segregation of plasmids into daughter cells in prokaryotic organisms.

partition chromatography any form of chromatography in which separation of the components of a mixture is based mainly on differences between the solubilities of the components in the stationary phase (in gas chromatography) or on differences between the solubilities of the components in the mobile and stationary phases (in liquid chromatography).

partition coefficient or **distribution coefficient** *symbol:* α or K_D ; the ratio of the equilibrium concentrations of a pure substance dissolved in two phases that are in contact, at a specified temperature. The partition coefficient is independent of the concentrations provided that no chemical interaction occurs with either (or both) of the phases and that the presence of the solute does not affect their immiscibility.

partition function *symbol:* Z ; a dimensionless mathematical function, derived by the application of statistical mechanical theory, that can be used to derive the average distribution of

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

partition law

the internal energy of a system in terms of its partition between all its molecular inhabitants.

partition law or **distribution law** the principle that a solute, when added to a system consisting of two phases, will, at a given temperature, distribute itself between the two phases in proportion to its solubility in each of the two phases separately. **Henry's law** is a particular case of the partition law. See also **Nernst distribution law**.

parturition the act or process of giving birth to offspring.

parvalbumin any of a group of closely related ≈ 12 kDa muscle proteins that bind calcium ions. Parvalbumin α is a muscle EF-hand Ca^{2+} -binding protein, that may be involved in muscle relaxation. Example from human: database code PRVA-HUMAN, 109 amino acids (11.93 kDa). Parvalbumin β is known as **oncomodulin**.

parvovirus any of a group of small, animal viruses, in class II of the **Baltimore classification**, consisting of a naked capsid, 20–25 nm in diameter, and containing linear, single-stranded DNA of 1.2–1.8 MDa. Parvoviruses are highly resistant to chemical and physical agents.

parvulin a **peptidylprolyl isomerase** isolated from *Escherichia coli*. It does not have any function as an **immunophilin**. Database code CYPC_ECOLI, 92 amino acids (10.10 kDa).

PAS *abbr.* for 1 *p*-aminosalicylic acid. 2 periodic acid–Schiff; see **periodic acid–Schiff reaction**.

pascal *symbol:* Pa; the SI derived unit of pressure or stress. It is equal to a force of one newton per square metre; i.e.

$$1 \text{ Pa} = 1 \text{ N m}^{-2} = 1 \text{ J m}^{-3}.$$

[After Blaise Pascal (1623–62), French mathematician and philosopher, who discovered the relationship between barometric pressure and altitude.]

passage (*in biology*) the action or process of (serially) subculturing cells or microorganisms, especially with a view to adapting them to a changed environment.

passage number (*in cell culture*) the number of subcultures performed after the original isolation of the cells from a primary source.

passenger or **passenger DNA** any specific DNA fragment introduced into a **cloning vector** or other cloning vehicle.

passive diffusion an *alternative name* for **simple diffusion** (used especially to emphasize its distinction from **active transport**). Compare **facilitated diffusion**.

passive immunity immunity in an individual due to the presence of antibody or primed lymphocytes derived from another, already immune, individual.

Pasteur, Louis (1822–95), French chemist, microbiologist, and immunologist; distinguished for his pioneer work in stereochemistry (in terms of symmetry rather than molecular constitution), for his recognition of the role of microorganisms in fermentation, for his invention of the technique of heat treatment of certain fluids to prevent microbial spoilage, now known as **pasteurization**, and for devising methods of immunization against certain diseases.

Pasteur effect or (*formerly*) **Pasteur–Meyerhof effect** either the phenomenon that occurs in facultatively anaerobic cells whereby oxygen inhibits glycolysis or fermentation, or its converse whereby the rate of glycolysis or fermentation increases when oxygen is excluded. Compare **Crabtree effect**.

pasteurization or **pasteurisation** a method of heat-treating milk or similar fluids to improve storage qualities and destroy pathogenic bacteria, without markedly altering taste and nutritional characteristics. Pasteurization of milk involves heating to 65 °C for 30 minutes or to 72 °C for 15 minutes followed by rapid cooling to below 10 °C. —**pasteurize** or **pasteurise** *vb*.

Pasteur pipette or **pasteur pipette** a simple, disposable, ungraduated, dropping or transfer pipette constructed by drawing out a short length of narrow glass tubing to form a long fine tip and operated commonly by means of an attached rubber or PVC bulb or teat. Analogous devices are available.

moulded in low-density polyethylene, with integral bulb and sometimes calibrated. *Also (esp. US):* **Pasteur pipet**, **pasteur pipet**.

Pasteur reaction the putative chemical reaction once considered to link fermentation and respiration and to account for the **Pasteur effect**.

patatin any of a family of glycoproteins that form 40% of the soluble protein of potato tubers. They serve as potato somatic storage proteins, and also have enzyme activity involved in host resistance. They have lipid acylhydrolase activity. Example (precursor): database code PAT0_SOLTU, 386 amino acids (42.44 kDa).

patch an intermediate stage in the formation of a **cap** (def. 2) on the surface of a cell, e.g. of a lymphocyte.

patch-clamp technique a technique whereby the conductance change of a single ion channel of an excitable membrane is measured directly. A very small electrode tip is sealed onto a patch of cell membrane, thereby making it possible to record the flow of current through individual ion channels in the patch. See also **Neher**, **Sakmann**.

path see **light path**.

+pathic see **+pathy**.

patho+ or (*before a vowel*) **path+** *comb. form* denoting disease or clinical disorder.

pathobiochemistry 1 the study of the biochemical processes associated with disease. 2 the processes themselves.

pathogen any agent, especially any living organism, that can cause disease.

pathogenesis the generation and production, or the origin and development, of a disease. —**pathogenetic** *adj*.

pathogenesis-related proteins proteins synthesized in plants in response to attack by fungal or bacterial pathogens. They are small proteins that belong to a limited number of families, characterized by six motifs. Example, pathogenesis-related leaf protein P14 from tomato: database code PR14_LYCE, 130 amino acids (14.24 kDa).

pathogenic (capable of) causing or producing disease. —**pathogenicity** *n*.

pathological or **pathologic** 1 of, or pertaining to, pathology. 2 causing, relating to, involving, or caused by disease.

pathology 1 the branch of medical science concerned with the causes and nature of disease and with the effects of disease on the structure and functioning of the organism. 2 the sum of the changes that occur in an organism as the result of a specific disease. —**pathologist** *n*.

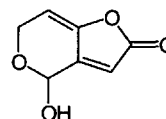
pathophysiology 1 the study of the physiological processes associated with disease. 2 the processes themselves.

pathway 1 the chain of reactions undergone by a given molecular entity or class of such entities in a particular ecosystem. 2 **metabolic pathway**.

pathway engineering the use of **genetic engineering** to change either the kinetics or the products of a metabolic pathway.

+pathy *comb. form* denoting 1 feeling, sensitivity. 2 disease of a (specified) part or kind. 3 a method of treating disease. —**pathic** *adj*.

patulin 4-hydroxy-4*H*-furo[3,2-*c*]pyran-2(6*H*)-one; a carcinogenic metabolite with antibiotic activity isolated from several fungi, particularly *Aspergillus* and *Penicillium* spp.



pauci+ *comb. form* indicating few.

paucidisperse describing a colloidal system whose dispersed phase consists of particles of only a few sizes. —**paucidispersity** *n*.

Pauli–Wolfgang (1900–58), Austrian-born US physicist; Nobel



Pauli exclusion principle

Laureate in Physics (1945) 'for the discovery of the Exclusion Principle, also called the Pauli Principle'.

Pauli exclusion principle or **exclusion principle** or **Pauli principle** the principle that no two electrons in an atom can be described by the same set of four quantum numbers.

Pauling, Linus Carl (1901–94), US chemist and peace campaigner; Nobel Laureate in Chemistry (1954) 'for his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances'; Nobel Laureate for Peace (1962).

Paulus filtration method a method for studying the binding of a ligand microsoluble to a macrosolute. It is applicable to small amounts of material and is particularly useful for a system having a relatively low binding constant. It involves equilibration of a series of mixtures of macrosolute and ligand in a suitable solvent, then separation of unbound ligand from macrosolute-plus-bound-ligand by ultrafiltration through anisotropic membrane filters of an appropriate retentivity. The amount of microsoluble retained with the macrosolute on each filter is then estimated by, e.g., radioactivity measurement or another sensitive technique. [After H. Paulus.]

Pauly's reagent a chromogenic reagent for imidazoles and phenols, consisting of a freshly prepared solution of diazotized sulfanilic acid (or other aromatic amine). In the presence of alkali, this reacts with histidine to give a red colour and with tyrosine to give an orange colour. It is useful for the identification of these amino acids on paper chromatograms.

Pavlov, Ivan Petrovich (1849–1936), Russian physiologist and experimental psychologist; Nobel Laureate in Physiology or Medicine (1904) 'for his work on the physiology of the saliva glands'. See also **salivary juice**.

Pax genes *abbr.* for paired box genes. The **paired box** is a conserved DNA sequence that plays a role in development. *Pax* genes encode a family of transcriptional regulators specifically expressed during the development of a wide range of structures and organs, including: *Pax-1*, paraxial mesoderm derivatives; *Pax-2*, urogenital system; *Pax-3*, neural crest and pre-myoblast migration; and *Pax-6*, eye and nose.

paxillin an adhesion protein, associated with **vinculin**, that becomes phosphorylated in response to a number of stimuli, phosphorylation being associated with focal adhesion. It is also associated with dysfunction of **cadherin**-dependent cell-cell contacts and with cell spreading. Example from human; database code HSU14588, 557 amino acids (60.87 kDa).

Pb *symbol* for lead.

PBD *see* **biphenylphenyloxadiazole**.

PBG *abbr.* for porphobilinogen.

PBI *abbr.* for protein-bound iodine.

PBP *abbr.* for penicillin-binding protein.

pBR322 a cloning vector.

PBS *abbr.* for phosphate-buffered saline.

PC *abbr.* for 1 paper chromatography. 2 phosphocreatine. 3 phosphatidylcholine. 4 personal computer.

pCa *symbol* for the negative logarithm of the concentration (or strictly the activity) of calcium ions in a specified solution.

PCA *abbr.* for perchloric acid.

P-cadherin placental **cadherin**.

PCB *abbr.* for pyruvate carboxylase.

PCCase *abbr.* for propionyl-CoA carboxylase.

PCMB *abbr.* for *p*-chloromercuribenzoate, or *p*-chloromercuribenzoic acid.

PCMBs *abbr.* for *p*-chloromercuribenzenesulfonate, or *p*-chloromercuribenzenesulfonic acid.

PCP *abbr.* for 1-(1-phenylcyclohexyl)piperidine (*see* **phencyclidine**).

PCR *abbr.* for polymerase chain reaction.

p.d. *abbr.* for (electric) potential difference.

pd *symbol* for palladium.

PDB *abbr.* for protein database; a database containing 3-D protein and nucleic-acid structures. A database of the files that

include the atomic coordinates is referred to as 'PDB format' or 'Brookhaven format' (after the Brookhaven National Laboratory at Upton, Long Island, New York, USA).

PDBu *abbr.* for phorbol 12,13-dibutyrate, a **phorbol ester**.

PDD *abbr.* for phorbol 12,13-didecanoate, a **phorbol ester**.

PD-ECGF *abbr.* for platelet-derived endothelial cell growth factor.

PDGF *abbr.* for platelet-derived growth factor.

pDNA *abbr.* for plasmid DNA.

P domain a domain of highly conserved cysteine residues, disulfide-bonded in the pattern 1–5, 2–4, 3–6, associated with highly conserved alanine, glycine, and tryptophan residues. Such domains are found in secretory polypeptides from animals.

Pe *symbol* for the pentyl group.

PE *abbr.* for 1 phosphatidylethanolamine. 2 potential energy.

peak 1 a sharp increase followed by a sharp decrease in a physical quantity that varies with distance, frequency, time, etc. 2 (*in chromatography*) the portion of a differential chromatogram recording the changes in detector response or eluate concentration while a given component of the applied mixture emerges from a chromatographic column. 3 the maximum value of a **peak** (def. 1, 2).

pEC₅₀ *see* **p[A]₅₀**.

PEC-60 a 60-amino-acid endogenous regulatory polypeptide abundant in intestinal tissue and also found in the central nervous system. It inhibits the formation of cyclic AMP and activates Na⁺, K⁺-ATPase. It is structurally related to pancreatic secretory trypsin inhibitor and inhibits insulin secretion. Example from *Sus scrofa* (precursor); database code PE60_PIG, 86 amino acids (9.63 kDa).

PECAM *abbr.* for platelet endothelial cell adhesion molecule; a class of **adhesion molecules** represented by PECAM-1 (or CD31), a member of the immunoglobulin superfamily. It is constitutively expressed on endothelial cells, platelets, granulocytes, monocytes, and T cells, and participates in homophilic binding (i.e. binding to itself). It is required for transmigration of leukocytes across the endothelium during extravasation. Example from human (precursor); database code PEC1_HUMAN, 738 amino acids (82.54 kDa), residues 1–27, signal; 28–738, PECAM.

pectate any salt or ester of **pectic acid**.

pectic acid the form of **pectin** (def. 1) that is essentially free of methyl ester groups.

pectic substance an alternative name for **pectin** (def. 1).

pectin 1 or **pectic substance** any mixture of complex, colloidal, macromolecular plant **galacturonans** containing a large proportion of D-galactopyranosyluronic acid residues in (α1→4) linkage, the carboxyl groups of which may be esterified to varying degrees by methyl groups or be partially or completely converted to salts; some D-arabinose and D-galactose residues nearly always occur in addition, and residues of certain other sugars are sometimes present. *M_r* is usually in the range 1 × 10⁴–4 × 10⁵. Pectins are important matrix polysaccharides of plant cell walls, especially those of higher plants, sometimes constituting as much as one-third of the dry matter; they also occur in some plant juices. Certain fruits are particularly rich in pectin. 2 (*in commerce*) a whitish, water-soluble powder prepared from the rind of citrus fruits or apple **pomace** by extraction with dilute acid and consisting essentially of **pectin** (def. 1). It is used *inter alia* in the preparation of jams because of its ability to form gels in the presence of sucrose at high concentrations.

pectinase *see* **polygalacturonase**.

pectinesterase EC 3.1.1.11; *other names*: pectin methylsterase, pectin methoxylase; an enzyme that hydrolyses pectin to methanol and **pectic acid**. It is found in plant cell walls. One role may be concerned with cell-wall metabolism during fruit ripening. Example from *Lycopersicon esculentum* (tomato); PME1_LYCES, 317 amino acids (34.55 kDa).

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



pectinic acid

pectinic acid the form of **pectin** (def. 1) in which a high proportion of the carboxyl groups are esterified with methanol.

pectin lyase EC 4.2.2.10; an enzyme that catalyses a *trans* elimination to convert pectin to oligosaccharides with unsaturated (C-4,C-5) terminal residues. Although often described as 4-deoxy-6-methyl- α -D-galact-4-enuronosyl groups, they are more accurately termed 4-deoxy-6-methyl- β -L-threo-hex-4-enopyranuronosyl groups. Example (precursor) from *Aspergillus niger*: database code PLYA_ASPNG, 379 amino acids (39.85 kDa).

pectin sugar an old name for L-arabinose.

PEG abbr. for polyethylene glycol.

PEI abbr. for the poly(ethyleneimine) group,

pelargonate another name for **nonanoate**.

pelargonoyl another name for **nonanoyl**.

pellagra a disease of animals, including humans, arising from a dietary deficiency of **nicotinic acid**, especially in the absence of sufficient dietary tryptophan to generate adequate amounts of nicotinamide coenzymes metabolically. This situation can occur when the diet consists largely of maize (corn). Pellagra is characterized by scaly dermatitis on exposed skin surfaces, diarrhoea, and depression.

pellagra-preventative factor abbr.: PP factor or PPF; a former name for **nicotinic acid** or nicotinamide.

pellet a small compressed mass of material, especially such a mass deposited at the bottom of a tube by centrifugation of a suspension.

pellicle a thin skin, membrane, or film.

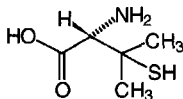
pellicular 1 of, relating to, or being in the form of a pellicle. 2 describing a type of column packing, used in high-pressure liquid chromatography, in which the stationary phase is present as a thin layer on the surface of tiny glass beads.

Peltier effect the phenomenon whereby heat is given out, or absorbed, when an electric current passes across a junction between two dissimilar materials. See **thermocouple**. Compare **Seebeck effect**. [After Jean Peltier (1785–1845), French physicist.]

penatin an obsolete name for **glucose oxidase**.

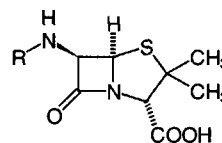
penetrance (in genetics) the degree to which a particular genotype is expressed in a population phenotype. If every individual carrying a dominant mutant gene shows the mutant phenotype, the gene is said to show complete penetrance.

penicillamine 3-mercapto-D-valine; β,β -dimethylcysteine; 2-amino-3-methyl-3-mercaptopropanoic acid; the D-enantiomer is a characteristic degradation product of penicillin-type antibiotics. It is a potent chelator of heavy metal ions, hence it is used clinically in the treatment of **Wilson's disease** and of poisoning by metals such as copper, lead, and mercury.



penicillic acid 3-methoxy-5-methyl-4-oxo-2,5-hexadienoic acid (exists in equilibrium with lactone form); a toxic antibiotic metabolite obtained from several fungi. It bears no relationship to penicillin or its degradation products. (Illustrated opposite.)

penicillin 1 the generic name for a range of related antibiotics, whether produced naturally during the growth of various microfungi of the genera *Penicillium* and *Aspergillus* or of semisynthetic origin, that contain the condensed β -lactam-thiazolidine ring system:

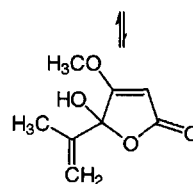
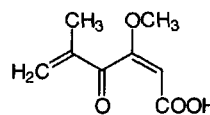


where R is any one of a variety of acyl groups such as: phenylacetyl, $C_6H_5-CH_2-CO-$ (in **penicillin G**, **penicillin II**, **benzylpenicillin**); phenoxyacetyl, $C_6H_5-O-CH_2-CO-$ (in **penicillin V**); and D-adipyl, $HOOC-CH(NH_2)-[CH_2]_3-CO-$ (in **penicillin N**, also called **cephalosporin N**). Only those enantiomers having the D-configuration at C-2 of the thiazolidine ring show antibiotic activity. The parent substance, **6-aminopenicillanic acid**, in which R is H, has little intrinsic antibiotic activity but is important as the naturally occurring starting point for the semisynthesis of artificial penicillins, e.g. **ampicillin**, in which R is aminophenylacetyl, $C_6H_5-CH(NH_2)-CO-$. All penicillins are readily inactivated by chemical or enzymic hydrolysis of the strained four-membered β -lactam ring, cold dilute acid yielding **penicillamine**, and alkali or penicillinases and other β -lactamases forming the corresponding substituted **penicilloic acids**. The penicillins generally have extremely low toxicity to animals and a high antibacterial activity, especially against Gram-positive organisms, the spectrum of activity varying somewhat with the nature of the group R. They act by inhibiting cross-linking of peptidoglycan in the cell wall of growing bacterial cells, the consequent structural weakness of the wall rendering the cells sensitive to osmotic lysis. 2 the (hypothetical) substance 6-formamidopenicillanic acid, used only in combination in trivial names of certain **penicillins** (def. 1), e.g. **benzylpenicillin**.

penicillin acylase EC 3.5.1.11; *recommended name*: penicillin amidase; an enzyme that catalyses the hydrolysis of penicillin to a fatty-acid anion and 6-aminopenicillinate. There are several enzymes that differ in specificity for the fatty acid. Examples: **penicillin V acylase** (or penicillin V amidase), a homotetramer from *Bacillus sphaericus*, database code PAC_BACSH, 338 amino acids (37.42 kDa); **penicillin G acylase** (or penicillin G amidohydrolase) from *Escherichia coli*, database code PAC_ECOLI, 846 amino acids (94.54 kDa) – the latter sequence represents a zymogen for a periplasmic enzyme comprising a signal (amino acids 1–26), an α subunit (27–235), a spacer (236–289) and a β subunit (290–846); the active enzyme is a heterodimer (α,β). See also **6-amino-penicillanic acid**.

penicillin amidase the recommended name for **penicillin acylase**.

penicillinase a trivial name for any β -lactamase that acts preferentially on penicillins.



penicillic acid

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



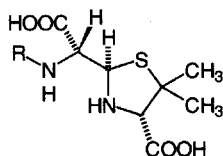
penicillin-binding protein

penicillin-binding protein *abbr.*: PBP; any of various bacterial proteins that are capable of binding penicillin(s) covalently. Such proteins appear to be located predominantly, if not solely, in the cytoplasmic membrane of all species of bacteria in which peptidoglycan occurs in the cell wall. These proteins are believed to catalyse reactions in the later stages of cell-wall synthesis and there is a strong presumption that at least one of them catalyses the insertion of nascent peptidoglycan into the cell wall. In Gram-negative bacteria some penicillin-binding proteins are known to be involved in the maintenance of cell shape or in cell elongation or septation. Another is a carboxypeptidase, example from *Haemophilus influenzae* (precursor): database code DACA_HAEIN, 393 amino acids (43.41 kDa).

penicillin G acylase *see* **penicillin acylase**.

penicillin V acylase *see* **penicillin acylase**.

penicilloic acid any of a range of compounds of structure:



and produced by cleavage of the β -lactam ring of **penicillins** (def. 1) by enzymic or alkaline hydrolysis.

penta+ or (before a vowel) **pent+** *comb. form* denoting five, fivefold, five times.

pentacosa+ *comb. form* denoting 25, 25-fold, 25 times.

pentadeca+ *comb. form* denoting 15, 15-fold, 15 times.

pentafunctional arom polypeptide a polypeptide that catalyses five consecutive enzymatic reactions in prechorismate polyaromatic amino-acid biosynthesis, i.e. the second to sixth steps in the biosynthesis from chorismate of the aromatic amino acids (the **shikimate pathway**). It occurs in some yeasts and other fungi and in *Euglena gracilis*. It contains: **3-dehydroquinase**, **3-dehydroquinase dehydratase**, **shikimate 5-dehydrogenase**, **shikimate kinase**, and **EPSP synthase**. Example from yeast: database code ARO1_YEAST, 1588 amino acids (174.55 kDa). Residues 1–392 = 3-dehydroquinase synthase; residues 404–866 = EPSP synthase; residues 887–1060 = shikimate kinase; residues 1061–1293 = 3-dehydroquinase dehydrogenase; and residues 1306–1588 = shikimate dehydrogenase.

pentagastrin the trivial name for (*N*-tert-butyloxycarbonyl- β -alanyl)-L-tryptophanyl-L-methionyl-L-aspartyl-L-phenylalaninamide; a synthetic analogue of **gastrin** that consists of the latter's C-terminal tetrapeptide amide sequence with an added protected N-terminal β -alanyl residue. It is used in the **pentagastrin test** as a gastrin analogue to stimulate gastric-acid secretion, which is then measured. Secretion is sometimes high in patients with duodenal ulceration and low in patients with gastric ulcer, but this is too variable to be of great use in diagnosis. *See also* **calcitonin**.

pentamer 1 any oligomer consisting of or derived from five monomers. 2 the group of five capsomeres in an icosahedral virus capsid.

pentanoate another name for **valerate**.

pentanoyl another name for **valeryl**.

pentaoase a pentasaccharide in which all the residues are of one type and uniform linkage. *Compare* **pentose**.

pentaric acid any **aldaric acid** obtained by oxidation of a **pentose** at C-1 and C-5.

pentasaccharide any oligosaccharide composed of five monosaccharide units. Pentasaccharides occur in small amounts in plants, e.g. **verbascose** in soybeans.

pentaxin or **pentaxin** any of a family of proteins with a discoid appearance under electron microscopy, having five non-covalently bound subunits. The family includes **C-reactive protein**, serum amyloid P-component, and neuronal pentaxins. Exam-

ple from human, neuronal pentaxin II (precursor): database code NPX2_HUMAN, 430 amino acids (46.81 kDa); residues 1–15 are the signal, 16–430 neuronal pentaxin II; residues 226–430 contain the pentaxin domain.

pentitol any **alditol** having a chain of five carbon atoms in the molecule.

pentobarbital 5-ethyl-5-(1-methylbutyl)barbiturate; a major barbiturate drug. *One proprietary name* is Nembutal.

pentolinium test a test in which **pentolinium**, a sympathetic ganglion blocking drug, is used to reduce catecholamine secretion; failure to achieve reduction may indicate a **pheochromocytoma**.

penton a capsomere that is surrounded by five other capsomeres in an icosahedral virus capsid.

pentonic acid any monocarboxylic acid (aldonic acid) formally derived from a pentose by oxidation at C-1.

pentosan a **glycan** that yields only pentoses on hydrolysis.

pentose any **aldose** having a chain of five carbon atoms in the molecule. *Compare* **pentaoase**.

pentose cycle an alternative name for **pentose phosphate pathway**.

pentose nucleic acid *abbr.*: PNA; a former term for **ribonucleic acid**.

pentose phosphate pathway or **hexose monophosphate pathway** a complex series of extramitochondrial metabolic reactions by which glucose 6-phosphate is oxidized with formation of carbon dioxide, ribulose 5-phosphate, and reduced NADP, the ribulose 5-phosphate then entering a series of reactions in which a number of sugar phosphates containing between three and seven (or eight) carbon atoms are successively interconverted. One consequence of these interconversions is the nonoxidative regeneration of glucose 6-phosphate, which then can enter the pathway for another cycle of oxidative decarboxylation and skeletal rearrangement, the eventual result being the complete oxidation of one molecule of glucose with the formation of six molecules of carbon dioxide and 12 molecules of NADPH. The pathway thus provides (in some organisms or tissues) an alternative route to (or shunt of) part of the glycolytic pathway; it also provides ribose 5-phosphate for nucleotide synthesis and (in some organisms) other intermediates for biosynthesis. Of special importance is D-erythrose 4-phosphate, required for the biosynthesis of aromatic amino acids and many other materials by the **shikimate pathway**. In higher plants many of the enzymes of the pathway function in the reverse direction as part of the **reductive pentose phosphate cycle**. *Other names*: hexose monophosphate shunt; pentose cycle; phosphogluconate (oxidative) pathway; Warburg–Dickens pathway; Warburg–Dickens–Horecker pathway. *See also* **glucose-6-phosphate dehydrogenase**, **phosphogluconate**, **phosphogluconate dehydrogenase** (decarboxylating), **transaldolase**, **transketolase**.

pentosuria the presence of excessive amounts of pentose in the urine. Two types may be distinguished: (1) **essential pentosuria**, in which daily excretion of 1–4 g L-xylulose (normal <60 mg) occurs due to an inherited diminished activity of the NADP-linked L-xylulose reductase (EC 1.1.1.10), an enzyme of the glucuronate pathway that converts L-xylulose to xylitol; (2) **alimentary pentosuria**, in which excretion of up to 100 mg of L-arabinose or L-xylulose occurs following ingestion of unusually large amounts of such fruits as cherries, grapes, and plums.

pentraxin *see* **pentaxin**.

pentulose any **ketose** having a chain of five carbon atoms in the molecule. There are two enantiomeric pairs of pentuloses having the carbonyl group at C-2, these being the D- and L-isomers of ribulose, *erythro*-2-pentulose, and of xylulose, *threo*-2-pentulose.

pentyl *symbol*: Pe; the alkyl group, $\text{CH}_3\text{—}[\text{CH}_2]_3\text{—CH}_2\text{—}$, derived from pentane.

PEP *abbr.* for phosphoenolpyruvate.

PEPCK *abbr.* for phosphoenolpyruvate carboxykinase.

peplomer a virus-coded proteinaceous knob or spike, numbers

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyenltd.com.



peplos

of which project from the peplos (i.e. **envelope** (def. 3)) of an enveloped virus particle. Peplomers may have cell-receptor, hemagglutinating, or neuramidinase activity.

peplos an alternative name for **envelope** (def. 3).

pepsin 1 any of a small family of closely related aspartic proteinase enzymes of 31–36 kDa that together give rise to the principal proteinase activity of gastric juice of vertebrates. The family includes **pepsin A**, **pepsin B**, **pepsin C** (i.e. **gastricsin**), and **pepsin D**. Pepsins have maximal activity at pH values in the range 2–3; they preferentially catalyse hydrolysis of peptide bonds formed from hydrophobic amino-acid residues, the detailed specificity varying somewhat between the different pepsins. 2 an alternative name for **pepsin A**.

pepsin A or **pepsin** abbr. (in clinical chemistry): PPS; the recommended name for the predominant enzyme (EC 3.4.23.1) of the **pepsin** family. It is a single-chain phosphoprotein of (in the pig) 327 amino-acid residues, formed by limited proteolysis of pepsinogen A. It preferentially cleaves peptide bonds where the residues on either side are formed from hydrophobic amino acids; for example, in the B chain of insulin it cleaves Phe¹-|-Val², Gln⁴-|-His⁵, Glu¹³-|-Ala¹⁴, Ala¹⁴-|-Leu¹⁵, Leu¹⁵-|-Tyr¹⁶, Tyr¹⁶-|-Leu¹⁷, Gly²³-|-Phe²⁴, Phe²⁴-|-Phe²⁵, and Phe²⁵-|-Tyr²⁶ bonds (compare **pepsin B**). The unphosphorylated form of porcine pepsin A is called **pepsin D**.

pepsin B or **parapepsin I** a pig enzyme (EC 3.4.23.2) of the **pepsin** family. It is a single-chain phosphoprotein of 332 amino-acid residues, formed by limited proteolysis of pig pepsinogen B. It has little activity towards hemoglobin as substrate (compare **gastricsin**), but degrades gelatin. Its specificity on the B chain of insulin is more restricted than **pepsin A**; it does not cleave Phe¹-|-Val², Gln⁴-|-His⁵, or Gly²³-|-Phe²⁴.

pepsin C an alternative name for **gastricsin**.

pepsin D unphosphorylated pig **pepsin A**.

pepsinogen a precursor (i.e. zymogen) of a **pepsin** enzyme; pepsinogens A, B, and C give rise to **pepsin A**, **pepsin B**, and **pepsin C** (i.e. **gastricsin**) respectively. Pepsinogens are secreted by the chief cells of the gastric mucosa of vertebrates. In pepsinogen, the active site is conformationally masked at neutral pH, and exposed as the pH falls after secretion into the stomach. The first bond breakage causes the release of residues 1–16, leaving a peptide known as **pseudopepsin**. Thereafter, a further 28 residues are removed in stages to form the active enzyme. Pepsinogen C is also secreted by the mucosa of the proximal duodenum. Example from human: database code PEPA_HUMAN, 388 amino acids (41.93 kDa); this is the pepsinogen precursor with a signal peptide; pepsin is from residue 63 to the end.

pepstatin A *N*-isovaleryl-L-valyl-L-valyl-3-hydroxy-6-methyl-4-aminoheptanoyl-L-alanyl-3-hydroxy-6-methyl-4-aminoheptanoic acid; a peptide, isolated from broth cultures of *Streptomyces* spp., that inhibits pepsin and other aspartic proteinases, notably cathepsin D, renin, and fungal acid proteinases.

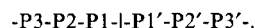
peptaibol or **peptaibol** any of a group of peptide amide antibiotics, 15–24 residues long and containing up to 40% α -aminoisobutyric acid residues. The members of this group, emerimicin, zervamicin IIA, antiameobin I, sizukacillin A, and **alamethicin**, are all capable of altering the permeability of phospholipid bilayers. The N termini are acylated and the C-terminal carboxyl groups are in amide linkage with phenylalaninol.

peptic 1 of, pertaining to, caused by, or containing **pepsin**. 2 of, pertaining to, or promoting digestion.

peptidase or **peptide hydrolase** or **protease** any enzyme that hydrolyses peptide bonds. The group includes the **exopeptidases**, such as the aminopeptidases and carboxypeptidases, (sub-classes EC 3.4.11–19) and the **endopeptidases** (sub-classes EC 3.4.21–99). The 1992 revision of *Enzyme Nomenclature* (published for IUBMB by Academic Press) should be consulted for more detail.

peptidase P-sites sites adjacent to or near the cleaved site in

peptidase substrates; in describing the specificity of peptidases, it is useful to have a convention for identifying the residues on either side of the cleaved bond; the following model is often followed (the cleaved, or scissile bond is indicated as -|-, and the N terminus is as usual to the left):



Compare **peptidyl site**.

peptide any compound containing two or more amino-acid residues joined by amide bond(s) (see **peptide bond**) formed from the carboxyl group of one amino acid (residue) and the amino group of the next one. The term peptide usually applies to compounds in which the amide bond(s) are formed between C-1 of one α -amino acid (residue) and N-2 of another, but it includes compounds in which the residues are linked by other amide bonds. See also **oligopeptide**, **polypeptide**. —**peptidic** adj.

peptide antibiotic any member of a diverse group of antibiotics containing amino-acid residues, some of which are often of the D-form, linked to one another by peptide bonds, to hydroxy acids by desipeptide bonds, and frequently also to additional components; the structure is often cyclic. The group includes the **bacitracins**, **gramicidins**, **polymyxins**, and **tyrocidins**.

peptide bond or **peptide linkage** any amide bond formed between two amino acids (or amino-acid residues). The term usually denotes the amide bond (sometimes called the **eupeptide bond**) formed between an α -amino group of one amino acid and an α -carboxyl group of another amino acid, but it also includes any amide bond (sometimes called an **isopeptide bond**) formed from an amino group and a carboxyl group either or both of which are in other positions in the contributing amino acids.

peptide chain or **polypeptide chain** any discrete linear sequence of amino-acid residues linked by eupeptide bonds, especially in a larger molecular structure. See **peptide bond**.

peptide HI abbr.: PHI; a 27-residue peptide amide, also known as PHI-27, found in intestinal tissue, brain, respiratory tract, and pancreas. It stimulates insulin secretion and pancreatic exocrine secretion, causes vasodilation, increases intestinal fluid transport, relaxes smooth muscle, and is a potent prolactin-releasing factor. It shows sequence homology with gastric inhibitory polypeptide, glucagon, secretin, and vasoactive intestinal polypeptide; human PHI has the sequence



(porcine PHI has Arg¹² and Ile²⁷; bovine has Tyr¹⁰, Arg¹², and Ile²⁷). It is named after the single-letter codes, H and I, for the histidine and isoleucine residues at the N and C termini respectively of the porcine and bovine peptides.

peptide histidine valine 42 abbr.: PHV 42; a peptide with N-terminal histidine and C-terminal valine, having 42 residues derived from residues 81–122 of prepro-vasoactive intestinal peptide.

peptide hormone or **polypeptide hormone** any peptide with hormonal activity in animals, whether endocrine, neuroendocrine, or paracrine. Such substances form a very diverse group physiologically, and the boundary between peptide hormones and protein hormones is somewhat indistinct.

peptide hydrolase another name for **peptidase**.

peptide linkage 1 an alternative name for **peptide bond**. 2 chemical linkage between structural components of a molecule by means of peptide bonds.

peptide map a pattern, characteristic of a particular polypeptide or protein, produced by partial hydrolysis followed by two-dimensional fractionation of the resultant peptides (and amino acids) by chromatography and/or electrophoresis.

peptide nucleic acid a synthetic DNA analogue with an *N*-(2-aminoethyl)glycine backbone that mimics DNA in forming a **heteroduplex**.

peptidergic 1 describing any nerve that is activated by a peptide agonist. 2 describing any nerve that acts by releasing a

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

peptide synthesis

peptide agonist from its nerve endings, whether into a synapse with another neuron, into intercellular fluid, or directly into the bloodstream.

peptide synthesis **1** (*in biology*) *see* **protein and peptide biosynthesis**. **2** (*in chemistry*) the synthesis, by chemical means, of peptides. A number of strategies are possible, including automated methods. The first step is the protection, in the amino-acid reactants, of reactive groups not involved in peptide bond formation (*see* **protecting groups**); peptide-bond formation is then achieved by activation of the carboxyl group, which is reacted with the free α -amino group of another amino acid. A variety of methods have been used for activation of the carboxyl group but a frequently used method is to react it with *N*-hydroxysuccinimide or a derivative to form an acyl succinimido ester, which reacts under mild conditions with free amino groups. Normally, and always in automated methods, the carboxyl group of the first amino acid is attached to a resin, by a bond that is easily detached later, so that the desired product, the growing peptide chain, remains attached to the resin which facilitates its purification; this technique, called **solid-phase peptide synthesis**, was pioneered by **Merri-field**.

peptide unit (*in crystallography*) any sequence of atoms in a peptide chain that may be represented by the general formula $-\text{CHR}-\text{CO}-\text{NH}-$, where R denotes the side chain of an amino-acid residue. *Compare* **amino-acid residue**.

peptide YY *abbr.*: PYY; a 36-residue peptide amide originally found in porcine upper small intestinal tissue. It is a gut hormone that inhibits exocrine pancreatic secretion. Members of this family regulate numerous physiological processes, including appetite, gastrointestinal transit, anxiety, and blood pressure. It shows sequence homology with **neurotensin** and **pancreatic hormone**, having the amino-acid sequence:

YPIKPEAPGEDASPEELNRYASLRHYLNLVTR-QRYNH₂.

It is named after the single-letter code, Y, for the tyrosine residues at its N terminus and the tyrosine amide at its C terminus.

peptido+ *comb. form* indicating the presence of peptide linked to some other (specified) component(s).

peptidoglycan *or* **murein** any of a class of glycoconjugates found only in bacterial cell walls and consisting of strands of glycosaminoglycan cross-linked by oligopeptides to form a huge and rigid network. There are characteristic differences between peptidoglycans of eubacteria and those of archaebacteria. The glycosaminoglycan strands are formed of alternating residues, usually *N*-acetylated or *N*-glycoloylated, of D-glucosamine and either muramic acid (in Bacteria) or L-talosaminouronic acid (in Archaea); the carboxyl groups of the muramic-acid residues are commonly joined in amide linkage to oligopeptides containing residues of both L- and D-amino acids whereas those of the L-talosaminouronic-acid residues are linked to peptides consisting of L-amino acids only. The peptide side-chains of adjacent polysaccharide strands are sometimes crosslinked between a carboxyl group of one peptide moiety and an amino group of another, either directly by a peptide bond or indirectly by an interpeptide bridge consisting of one to several amino-acid residues.

peptidyl the acyl group formed from a peptide by removal of the hydroxyl group from its C-terminal carboxyl group.

peptidylprolyl isomerase *abbr.*: PPIase; EC 5.2.1.8; *systematic name*: peptidylproline *cis-trans*-isomerase; an enzyme that irreversibly converts prolyl residues within a polypeptide chain from a *trans* to a *cis* configuration; the reaction is:

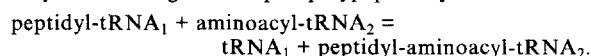
peptidylproline ($\omega = 180$) = peptidylproline ($\omega = 0$).

It accelerates protein folding. There are at least two families of enzymes with this activity. One is identical to **cyclophilin**, an **immunophilin** that is the major high-affinity binding protein of the immunosuppressive drug **cyclosporin A**. Another class of immunophilins that exhibit isomerase activity are the FK506-

binding proteins (*see* **FKBP**). The physiological role of these proteins is unresolved. The peptide Ala-Pro inhibits the reaction but is not a substrate. Example from human: database code FKB3_HUMAN, 141 amino acids (15.64 kDa).

peptidyl site *or* **P-site** *or* (*more correctly*) **peptidyl-tRNA site** the ribosomal site to which a newly elongated peptidyl-tRNA molecule is transferred from the aminoacyl site during the sequence of polypeptide biosynthesis reactions, and at which it displaces the tRNA molecule remaining from the previous elongation step. *Compare* **peptidase P-site**. *See also* **ribosome**.

peptidyltransferase EC 2.3.2.12; the ribosomal enzyme that catalyses the elongation step in polypeptide synthesis:



It is an activity of the large ribosomal subunit involving, in bacteria, **ribosomal proteins** L2, L3, L4, L15, and L16, and 23S rRNA. *See also* **peptidyl site**.

peptidyl-tRNA the form in which a growing polypeptide chain is held to a ribosome during polypeptide biosynthesis. The polypeptide is esterified through its C-terminal carboxyl group to the 3'-hydroxyl of the adenosine residue at the 3' end of a molecule of tRNA. *See* **protein synthesis**.

peptize *or* **peptise** to disperse (an aggregate) to form a colloidally stable suspension or emulsion; especially to convert (a gel) into a sol. —**peptization** *or* **peptisation** *n*.

peptolide (*sometimes*) any naturally occurring cyclodepsipeptide; *see* **depsipeptide**.

peptone any soluble mixture of products, 0.6–3 kDa, of the partial enzymic (or acid) hydrolysis of proteinaceous material. It is used as a constituent of many microbiological culture media.

peptonize *or* **peptonise** **1** to convert (proteinaceous material) into **peptone**. **2** to combine with peptone; e.g. peptonized iron.—**peptonization** *or* **peptonisation** *n*.

per+ *comb. form* denoting **1** through, throughout, beyond; e.g. perennial. **2** (*in chemistry*) **a** (of an indicated element) being in a higher than usual state of oxidation; e.g. periodate. *See also* **peracid** (def. 1). **b** containing a higher than normal proportion of (an indicated element); e.g. **peroxide**. *See also* **peroxoacid**, **peroxy acid**. **c** (in the trivial names performic, peracetic, and perbenzoic acids) **peroxy+**. **d** exhaustive replacement of hydrogen (except any in a characteristic group) in an organic compound by (an indicated halogen); e.g. perfluoropentane, C₅F₁₂. *See also* **perhydro**.

peracid *or* **per-acid** **1** a common name for any inorganic acid in which the central element of the acid is in its highest state of oxidation. **2** a common name for **peroxoacid**. **3** a common name for **peroxy acid**.

per cent *or* **percent** *symbol*: %; containing or consisting of a specified numerical proportion (of some component in relation to a whole) expressed as a number fraction multiplied by 100; bearing a specified numerical relationship (to some quantity) similarly expressed. When the term per cent or the symbol % is applied to a solution, one of the designations w/w, w/v, or v/v may be appended according as the units used are mass per mass, mass per volume, or volume per volume, respectively.

percentage proportionality expressed per cent; it is used especially to qualify a substantive, e.g. percentage error.

perchloric acid *abbr.*: PCA; chloric(VII) acid, HClO₄; a compound used in chemical analysis as an oxidizing agent.

Percoll the proprietary name for a polydisperse preparation of colloidal silica particles rendered nontoxic to cells by a coating of polyvinylpyrrolidone. Aqueous dispersions of the material can be used to generate density gradients for the separation of cells, viruses, or cell organelles by sedimentation.

percutaneous through the skin; used especially of the route of administration of drugs applied to the skin.

perforin a lymphocyte pore-forming protein; it is produced by cytotoxic T cells when these are in contact with target cells,

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



performic acid

being stored in secretory vesicles and released by local exocytosis. With Ca^{2+} , it polymerizes into transmembrane tubules that form pores in the target-cell membrane and is thus capable of lysing nonspecifically a variety of target cells. Example (precursor) from human: database code PERF_HUMAN, 555 amino acids (61.31 kDa).

peroxyformic acid the trivial name for peroxyformic acid, $\text{H}-\text{C}(=\text{O})-\text{O}-\text{OH}$; a strong oxidizing agent, useful in protein chemistry for cleavage of any disulfide bond, $\text{R}-\text{S}-\text{S}-\text{R}'$, to the corresponding pair of sulfonic acid groups, $\text{R}-\text{SO}_3\text{H} + \text{R}'-\text{SO}_3\text{H}$, e.g. of a cystine residue to two cysteic-acid residues. Tryptophan residues are also oxidized, and methionine residues are converted to methionine *S,S*-dioxide residues.

perfusate any fluid used for a perfusion.

perfuse to pass fluid through an organ or tissue, especially artificially. — **perfusion** *n*.

perhydro- prefix (in chemical nomenclature) signifying conversion of the indicated unsaturated organic compound to the corresponding fully hydrogenated compound; used especially where the parent compound is markedly unsaturated; e.g. perhydroanthracene, $\text{C}_{14}\text{H}_{24}$ (formed from anthracene, $\text{C}_{14}\text{H}_{10}$).

peri+ comb. form enclosing, surrounding; near.

pericellular surrounding a cell or cells.

perfusate any fluid used for a perfusion (see **perfuse**).

perfuse to pass a fluid around isolated cells or cell clusters in suspension. — **perfusion** *n*.

peri-fused see **ortho-** and **peri-fused**.

perikaryon or **cell body** the cytoplasm immediately surrounding the nucleus of a neuron.

perilipin a protein found in adipocytes exclusively at the surface of the lipid storage droplets that serves as the major A-kinase substrate. It is a major hormonally regulated adipocyte-specific phosphoprotein. Example, perilipin A from *Rattus norvegicus*: database code A49413, 517 amino acids (55.55 kDa).

perinuclear space the zone, 20–40 nm wide, between the two lipid bilayers of the **nuclear envelope** of a eukaryotic cell.

period 1 a specified or defined length of time, especially the time occupied by one complete cycle of some regularly recurring phenomenon or event. 2 (in chemistry) any of the horizontal rows of chemical elements in the **periodic table**.

periodate 1 any anion derived from a **periodic acid**, especially the anion, IO_4^- , derived from metaperiodic acid. 2 any salt of a periodic acid, especially of metaperiodic acid or orthoperiodic acid.

periodate oxidation a method for the investigation of the structure of organic molecules, based upon the ability of aqueous periodate (or periodic acid) to cleave specifically and quantitatively any carbon–carbon single bond where one carbon atom carries a hydroxyl group and the other carries an amino, carbonyl, carboxyl, or hydroxyl group, these groups being oxidized and the periodate reduced to iodate. The method is particularly useful in the structural analysis of carbohydrates. See also **periodic acid–Schiff reaction**.

periodic 1 of, relating to, or resembling a **period** (def. 1 or 2); happening or occurring in periods; intermittent; cyclical. — **periodically** *adv*. 2 (in chemistry) denoting any of several oxoacids of iodine containing a greater proportion of oxygen than iodic acid. See **peri-** (def. 2a), **periodic acid**.

periodic acid any of a number of iodic(vii) acids formed by the combination of iodine heptoxide, I_2O_7 , with water in different molecular proportions, especially **orthoperiodic acid** (formerly known as paraperiodic acid), H_5IO_6 ; **metaperiodic acid**, HIO_4 , exists mainly as its salts. These acids and their salts are strong oxidizing agents.

periodic acid–Schiff reaction *abbr.*: PAS reaction; a chromogenic reaction used as a sensitive test for carbohydrate, either histochemically or in chromatography. It is based on the periodate oxidation of vicinal $-\text{CHOH}-$ groups to $-\text{CHO}$ groups and the subsequent reaction of these with **Schiff reagent** to give a characteristic magenta colour.

periodic classification see **periodic system**.

periodicity the recurrence (of some phenomenon or event) at constant intervals in space or time; a measure of the frequency of such recurrence.

periodic law or **Mendeleev's law** the principle that the properties of the chemical elements are a periodic function of their atomic numbers (originally, of their atomic weights).

periodic system or **periodic classification** any classification of the chemical elements on the basis of the **periodic law**.

periodic table a mode of representing the **periodic system**; it is a tabular arrangement of the chemical elements in horizontal rows and vertical columns in order of increasing proton number row by row, each row, or **period**, consisting of those elements from an alkali metal to the rare gas of next highest proton number (except for the last, incomplete, period) and each column, or **group**, consisting of a family or set of those elements with similar chemical properties. Over the years, numerous versions of such a tabular arrangement have been proposed and used, the three commonest formats having 8, 18, or 32 columns of elements. Of these, the ones most widely adopted latterly have been variations of the so-called long periodic table, in which there are 18 columns. The particular version of the long table illustrated on the endpapers is the one currently recommended by IUPAC and adopted for use elsewhere in this dictionary. In this IUPAC periodic table, as similarly in other versions of the long table, the 15 elements of proton numbers 57 to 71, the **lanthanoids**, occupy a single position in period 6 of group 3, and those of numbers 89 to 103, the **actinoids**, a single position in period 7 of group 3, practical considerations requiring these elements to be listed separately at the bottom of the table.

In previous versions of the long table it was usual for 14 of the columns to be formed into seven groups, numbered I to VII and each divided into A and B subgroups. However, these subgroup designations were not always applied the same way round, with the consequence that two alternative conventions for the long table have concurrently had widespread support – one commonly in Europe (and hitherto followed in principle by IUPAC), the other in North America and elsewhere. Furthermore, the three columns containing the elements iron, cobalt, and nickel, originally designated together as group VIII, and the eighteenth column, containing the noble gases and originally designated as group O, have sometimes been labelled group VIIIA and group VIIIB respectively in the European version of the table, whereas in the North-American version the noble gases have in recent years been assigned usually to group VIIIA. The resultant confusion led IUPAC to recommend in 1990 that the columns of the long periodic table now be serially numbered as groups 1 to 18. The correspondence of the group designations in the new IUPAC periodic table with those of the European and North-American conventions in recent use is also shown in the endpapers.

peripheral (cell-membrane) protein or **extrinsic protein** an *operational term* for any protein that can be dissociated unchanged from a cell membrane by a mild treatment such as increase in ionic strength of the medium or addition of a chelating agent to it. Such proteins are presumed to be loosely attached to one or other surface of the cell membrane. Compare **integral (cell-membrane) protein**.

peripheral nervous system the part of an animal's nervous system other than the **central nervous system**, and generally located in the peripheral, or outer, parts of the body. It comprises the cranial nerves and spinal nerves and their branches, and includes the **autonomic nervous system**. It serves to link effector organs and sensory receptors with the central nervous system.

peripherin a glycoprotein, named from 'peripheral vision', that is essential for eye-disk morphogenesis; it is probably an adhesion molecule. Example from human: database code RDS_HUMAN, 346 amino acids (39.14 kDa).

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



periplasm

periplasm or **periplasmic space** the region in a bacterial cell between the cell membrane and the cell wall.

peristalsis a wavelike sequence of involuntary muscular contraction and relaxation that passes along a tubelike structure, such as the intestine, impelling the contents onwards. — **peristaltic adj.**

peristaltic pump a pump that functions, by analogy with the action of **peristalsis**, by the repeated compression of a length of elastic tubing in such a way as to cause the contents of the tubing to move in one direction in a pulsating fashion but with an overall flow rate that is constant and regulated by the rate of pulsation and the tubing diameter.

peritoneum the thin translucent membrane, consisting of a single layer of flattened mesothelial cells and fibrous connective tissue, that lines the abdominal cavity and covers most of the viscera. — **peritoneal adj.**; **peritoneally adv.**

perlecan another name for **heparan sulfate proteoglycan core protein**. See also **syndecan**.

permeability 1 the quality or degree of being permeable or porous. 2 or **magnetic permeability** (of a substance) *symbol*: μ ; the ratio of the magnetic induction, B , to the magnetic field strength, H , at any point within the substance, i.e. $\mu = B/H$. The SI unit is the henry per metre or newton per square metre.

permeability coefficient a measure of the ability of a specified permeant to diffuse across a permeability barrier. It is expressed as the amount of substance of the permeant passing through a unit area of the barrier in a unit time when a unit difference in amount-of-substance concentration is applied across it.

permeability of vacuum *symbol*: μ_0 ; a fundamental physical constant, of value $4\pi \times 10^{-7} \text{ H m}^{-1}$ exactly.

permeabilize or **permeabilise** to render permeable. In cell biology, agents such as **saponin**, streptolysin O (see **streptolysin**), or an applied electric field are used to open holes in the plasma membrane of cells. — **permeabilization** or **permeabilisation n.**

permeable capable of being permeated; pervious.

permeant 1 able to permeate or to diffuse into or through. 2 a substance having such properties.

permease (sometimes) any protein or protein system that effects a catalytic, stereospecific transfer of a substrate across a biological membrane.

permeate to pass through, or cause to pass through, by diffusion; to pervade. — **permeation n.**

permeation chromatography any form of chromatography in which a permeable stationary phase is used and in which separation of the components of a mixture is based mainly upon selective exclusion effects, such as differences in molecular size and/or shape (in **molecular-sieve chromatography**) or differences in charge (in **ion-exclusion chromatography**). See also **gel-permeation chromatography**.

permissible allowable. — **permissibility n.**; **permissibly adv.**

permissive 1 allowing; tolerant. 2 (in **virology**) describing a line or strain of cells that allows replication of a specified infecting virus with consequent cell lysis. 3 (in **genetics**) describing a condition, e.g. temperature, or set of conditions that permits the growth of a conditional lethal mutant (see **conditional lethal mutation**). — **permissiveness n.**; **permissively adv.**

permissive temperature see **temperature-sensitive mutation**.

permittivity see **relative permittivity**.

permselective selectively permeable to certain (classes of) molecular entities.

permutation 1 the process or act of permuting. 2 any of the possible arrangements of the members of a specified set of conditions, numbers, residues, etc.

permute to change the sequence or arrangement of.

Permutit a proprietary name for ion-exchange materials and equipment utilizing **permutites**.

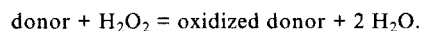
permutite any of a class of artificial zeolite ion-exchangers.

pernicious anemia or (esp. *Brit.*) **pernicious anaemia** a form of anemia characterized by defective production of erythrocytes and the presence of megaloblasts in the bone marrow.

and sometimes accompanied by neurological changes. It is caused by a deficiency of vitamin B_{12} due either to a dietary deficiency of the vitamin or to a failure of production of **intrinsic factor**.

per os *Latin* by mouth; used especially to denote the administration of drugs, etc. by the oral route.

peroxidase 1 any oxidoreductase enzyme of the sub-subclass EC 1.11.1. These act on hydrogen peroxide as (electron) acceptor with the formation of water. 2 the recommended name of the enzyme EC 1.11.1.7, a hemoprotein that catalyses the reaction of hydrogen peroxide with a wide variety of donors, i.e.



Apart from removing H_2O_2 , it participates in the oxidation of toxic reductants and in lignin metabolism. In plants there are isoforms in different tissues. Example, peroxidase C2 precursor from horseradish; database code PER2_ARMRU, 347 amino acids (37.99 kDa); the enzyme consists of the region from amino acid 25 to the end. — **peroxidatic adj.**

peroxidase-antiperoxidase *abbr.*: PAP; a soluble complex of horseradish peroxidase (def. 2) and an antibody to this peroxidase, used in immunochemical procedures in conjunction with specific primary and bridging second antibodies to give a sensitive method of detecting complementary antigens enzymatically.

peroxide 1 (in **inorganic chemistry**) a the O_2^{2-} ion. b any member of a class of metallic oxides, such as sodium peroxide, Na_2O_2 , that contain the peroxide ion and generate hydrogen peroxide on acidification. The name is sometimes used erroneously for certain higher oxides that do not contain the O_2^{2-} ion; e.g. lead peroxide (lead(IV) oxide, PbO_2), manganese peroxide (manganese(IV) oxide, MnO_2), nitrogen peroxide (nitrogen(IV) oxide, NO_2 or N_2O_4). c short for hydrogen peroxide, H_2O_2 ; an oxidizing agent. 2 (in **organic chemistry**) a any compound of general formula $\text{R}'\text{—O—O—R}''$, where the groups R' and R'' are two chains, two rings, or a ring and a chain, and including cases when $\text{R}' = \text{R}''$; e.g. ethyl phenyl peroxide, benzoyl (strictly, dibenzoyl) peroxide. In hydrogen peroxide, $\text{R}' = \text{R}'' = \text{H}$. b see **hydroperoxide**.

peroxisomal bifunctional enzyme a monomeric peroxisomal enzyme (mainly in kidney and liver) with the activities of three steps in the **beta-oxidation system**. Example from human: database code ECHP_HUMAN, 722 amino acids (79.12 kDa). The domain comprising amino acids 1–128 has two enzyme activities: **enoyl-CoA hydratase** and **3,2-trans-enoyl-CoA isomerase**; the domain 280–571 has **3-hydroxyacyl-CoA dehydrogenase** activity, and 720–722 is the microbody targeting signal.

peroxisome a type of **microbody** that is rich in enzymes that act on or generate hydrogen peroxide, especially the enzymes catalase, (S)-2-hydroxy-acid oxidase, and D-amino-acid oxidase. Peroxisomes have been found in mammalian kidney and liver, and in yeasts and certain protozoans. They are particularly numerous in leaf photosynthetic cells of plants photosynthesizing via the Calvin (C_3) pathway; in plants they are closely associated with chloroplasts and mitochondria, which is consistent with their function in **photorespiration**. — **peroxisomal adj.**

peroxo+ prefix (in **inorganic chemical nomenclature**) indicating substitution (in the corresponding acid) of an —OH group by the —O—OH group; e.g. peroxomonosulfuric acid, $\text{HO—SO}_2\text{—O—OH}$.

peroxoacid (in **inorganic chemistry**) any acid that contains the —O—OH group. Compare **peroxy acid**.

peroxy+ prefix or **infix** (in **carboxylic-acid nomenclature**) indicating substitution of the —OH group in the carboxyl group by the —O—OH group; e.g. peroxypropionic acid, $\text{CH}_3\text{—CH}_2\text{—CO—O—OH}$. See also **per+** (def. 2c).

peroxy acid (in **organic chemistry**) any acid that contains the —CO—O—OH group. Compare **peroxoacid**.

peroxyformic acid see **performic acid**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentd.com.

perpendicular dichroism

perpendicular dichroism *see* **dichroic ratio**.

Perrin, Jean Baptiste (1870–1942), French physicist; Nobel Laureate in Physics (1926) ‘for his work on the discontinuous structure of matter, and especially for his discovery of sedimentation equilibrium’.

persalt any salt of a **peroxoacid** or a **peroxy acid**.

perseitol D-glycero-D-galacto-heptitol; a substance found in avocado and the alligator pear.

perspective formula a type of **projection formula** in which the bonds of (a part of) a molecule tend to be depicted with increasing boldness according to their nearness to the viewpoint, and vice versa, in order to give an impression of perspective. *See* **Haworth representation**, **Mills perspective representation**.

Perspex a proprietary name for **poly(methyl methacrylate)**.

pertactin other name: outer membrane protein P68; an agglutinin of *Bordetella* spp. that binds to eukaryotic cells, and is involved in the pathogenicity in whooping cough. It contains the **RGD** sequence motif. Example (precursor) from *Bordetella bronchiseptica*: database code PERT_BORBR, 911 amino acids (93.89 kDa).

perturb to cause a selective and measurable change in the state of equilibrium of a system, or in its associated properties, through the action or influence of a specific external agent; to undergo perturbation. —**perturbable** *adj.*

perturbant an agent (capable of) causing perturbation.

perturbation 1 the action or process of perturbing; an instance of this. 2 the condition of being perturbed. 3 a cause or factor of disturbance; perturbant.

pertussis toxin a mixture of proteins produced by *Bordetella pertussis*, the bacterium responsible for whooping cough in young children. The toxin causes long-lasting activation of G_i protein (*see* **G-protein**) through catalysing ADP-ribosylation of the α subunit. It contains the NAD-dependent ADP ribosyltransferase (EC 2.4.2.-). The toxin enzyme contains five chains (S1–S5) organized in two functional subunits, A and B. Subunit A comprises S1, which has the enzymic activity, and irreversibly uncouples the GTP-binding proteins G_i and **transducin** from their membrane receptors. Subunit B binds to the membrane receptors; it has the composition S2S3S5S4₂, dimers of S2S4 and S3S4 being held together by S5. The database codes of the five chains, S1–S5, are as follows: TOX1_BORPE, 269 amino acids (29.94 kDa); TOX2_BORPE, 226 amino acids (24.80 kDa); TOX3_BORPE, 227 amino acids (24.96 kDa); TOX4_BORPE, 152 amino acids (16.52 kDa); TOX5_BORPE, 133 amino acids (14.48 kDa).

Perutz, Max Ferdinand (1914–), Austrian-born British molecular biologist; Nobel Laureate in Chemistry (1962), the prize being shared with J. C. Kendrew ‘for their studies on the structures of globular proteins’ (actually hemoglobin and myoglobin).

pervaporate to vaporize a liquid from one side of a semipermeable membrane with which it is in contact on the other side, as a means of concentrating a solution or suspension in that liquid. —**pervaporation** *n.*

pervious permeable.

PEST hypothesis the hypothesis that the presence of one or more regions rich in proline (P), glutamic acid (E), serine (S), and threonine (T) in the primary structure of a protein confers susceptibility to rapid intracellular proteolysis.

pesticide any substance, natural or artificial, used to kill pests.

PET *abbr.* for positron emission tomography; *see* **tomography**.

peta+ symbol: P; SI prefix denoting 10^{15} times.

PETG *abbr.* for polyethyleneterephthalate glycol; a plastic developed as an alternative to PVC for use in packaging; in contrast to PVC it does not produce chloride fumes when burned, giving off only CO_2 and H_2O . Flexible, clear, robust, and light, it is also suitable for contact with food, since no plasticizers or stabilizers are used in its manufacture.

petite 1 describing a strain of yeast (or other) cells derived from a spontaneous mutant in which certain respiratory enzymes are defective or deficient, with consequent very slow

growth and formation of unusually small colonies. 2 describing such a colony or mutant.

Petri dish or **Petri plate** a shallow, flat-bottomed, circular dish, usually of glass or transparent plastic and 5–15 cm in diameter, with a loosely fitting, overlapping cover. It is used especially for plate cultures of bacteria and moulds on solid (gelled) media, and for tissue culture. [After Richard Julius Petri (1852–1921), German bacteriologist.]

petroselaidic acid *see* **octadecenoic acids**.

petroselenate numerical symbol: 18:1(6); the trivial name for Z-octadec-6-enoate; a component of the fatty-acid hydrolyse of umbelliferous seed oils.

petroselenic acid *see* **octadecenoic acids**.

pFc' fragment a protein fragment obtained (together with **F(ab')₂ fragment**) by pepsin hydrolysis of an immunoglobulin molecule. That from human IgG is a 27 kDa protein, which is a noncovalently bonded dimer consisting of the C-terminal moieties of the heavy chains, and resembles the **Fc fragment** without the residues containing the linking disulfide bond.

PFP *abbr.* for pore-forming peptide or pore-forming protein; the term embraces cytotoxic peptides such as staphylococcal δ -toxin, **alamethicin**, **magainins**, and **cecropins**; pore-forming proteins include staphylococcal α -toxin, **streptolysin-O**, cnidarian toxins, **perforin**, and **aerolysin**.

PFK *abbr.* for phosphofructokinase.

pg symbol for picogram (i.e. 10^{-12} gram).

pG symbol for titration pK.

PG *abbr.* for 1 phosphatidylglycerol. 2 prostaglandin. 3 protein.

PGA *abbr.* for 1 phosphoglyceric acid. 2 pteroylglutamic acid. 3 prostaglandin A; *see* **prostaglandin**.

PGAH₄ *abbr.* for tetrahydropteroylglutamic acid.

PGAM *abbr.* for phosphoglycerate mutase.

PGB *abbr.* for prostaglandin B; *see* **prostaglandin**.

PGD *abbr.* for 1 phosphogluconate dehydrogenase (EC 1.1.1.43). 2 prostaglandin D (*see* **prostaglandin**).

PGE *abbr.* for prostaglandin E; *see* **prostaglandin**.

PGF *abbr.* for prostaglandin F; *see* **prostaglandin**.

PGF synthase *abbr.* for prostaglandin-F synthase.

PGG *abbr.* for prostaglandin G (*see* **prostaglandin**).

PGH *abbr.* for prostaglandin H (*see* **prostaglandin**).

PGH synthase *see* **prostaglandin-endoperoxide synthase**.

PGI *abbr.* for prostaglandin I (i.e. **prostacyclin**).

PGK *abbr.* for phosphoglycerate kinase.

P glycoprotein or **MDR protein** 1 a glycoprotein associated with drug resistance; it is a transmembrane ATP-dependent transporter that pumps hydrophobic substances out of cells. Drug-resistant cells demonstrate increased expression of the P-glycoprotein gene as one mechanism for their resistance to, e.g., chemotherapeutic anticancer drugs. *See also* **MDR**.

PGT (*in clinical chemistry*) *abbr.* for phosphoglucomutase.

pH a dimensionless quantity notionally defined as the negative decadic logarithm of the molal activity of hydrogen ions in a solution. Since the activity of a single species of ion is not measurable, pH is defined operationally according to the equation:

$$pH(X) = pH(S) + (E_s - E_x)/(RT \ln 10)$$

where pH(S) and pH(X) are the pH values of a standard solution, S, and of the test solution, X, respectively; F is the Faraday constant, R the gas constant, and T the thermodynamic temperature; and E_s and E_x are, respectively, the electromotive forces of the two galvanic cells:

reference electrode | KCl (conc.) || soln. S | $H_2(g)$ | Pt,
and

reference electrode | KCl (conc.) || soln. X | $H_2(g)$ | Pt,

both cells being at the same temperature, with identical reference electrodes and bridge solutions, KCl (aq, $m \geq 3.5$ mol kg^{-1}). A **pH-standard solution**, S_s is chosen from a set of five

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



covering the range pH 3.5 to pH 9, an assigned value of pH(S) at the required temperature being obtained from published tables. To a good approximation, the hydrogen electrodes in the two cells may be replaced by other hydrogen-ion-responsive electrodes, such as glass or quinhydrone electrodes. Originally pH was considered as the negative decadic logarithm of the molar concentration of hydrogen ions, a practice retained to the present day as representing a convenient approximation.

pH* *a notation commonly used to denote the apparent pH of a solution containing $^2\text{H}_2\text{O}$ (i.e. D_2O), as measured electrometrically using a glass electrode without correction for the isotope effect on the electrode.*

Ph *symbol for the phenyl group.*

PH-30 or **PH30** a sperm surface protein with **disintegrin** and metalloprotease domains. Example from *Cavia cobaya* consists of two subunits, α and β ; α subunit: database code CCPH30ASP, 804 amino acids (86.81 kDa); β subunit: database code CCPH30BSP, 735 amino acids (81.81 kDa).

PHA *abbr. for phytohemagglutinin (def. 1).*

pH-activity profile or **pH profile** a plot of enzyme activity against pH, over a range in which the activity is reversibly affected by pH.

pheo+ *an alternative spelling (esp. Brit.) of pheo+.*

pheochromocytoma *an alternative spelling (esp. Brit.) of pheochromocytoma.*

pheofarnesin *an alternative spelling (esp. Brit.) of pheofarnesin.*

pheophorbide *an alternative spelling (esp. Brit.) of pheophorbide.*

pheophytin *an alternative spelling (esp. Brit.) of pheophytin.*

phage *abbr. for bacteriophage.*

phage conversion *an alternative name for bacteriophage conversion.*

phage display library a special example of a gene library encoding fusion proteins of a foreign polypeptide sequence and a coat protein of phage M13 (or more usually) the closely related phage fd. When cloned, the phage is then said to display the foreign protein (in the coat proteins). There are two common versions of the technique. In one, the library consists of random oligonucleotides (e.g. 7-mers). Clones expressing oligopeptides that react with a specific antibody can then be selected using the antibody. In the other version, the library consists of oligonucleotides encoding a range of single-chain variable-domain antibodies. This approach is used where an antibody is required for an oligopeptide, which can thus be used to detect appropriate clones.

phage immunity the immunity conferred on a **lysogenic bacterium** by the presence of a **prophage** and a consequence of the expression of a gene such as *cI* in **lambda phage**.

phage induction or **prophage induction** the stimulation of prophage in lysogenic bacteria to enter the vegetative state. It is accomplished by the action of UV light, X-rays, nitrogen mustards, etc.

phage lambda *see lambda phage.*

phagemid a chimeric phage-plasmid **cloning vector** containing origins of both double-stranded and single-stranded replication. *See also phasmid.*

phage T4 an icosahedral DNA-containing **bacteriophage** that infects *Escherichia coli*. Its single molecule of double-stranded DNA is 174 kb in length. Various enzymes prepared from this phage are commonly used in genetic engineering; these include ligase, polymerase, and polynucleotide kinase.

phage typing a procedure for classifying bacteria according to their susceptibility to various bacteriophages.

phagocyte any cell that characteristically engulfs particles from its surroundings into its cytoplasm by **phagocytosis**. — **phagocytic** *adj.*

phagocytosis a type of endocytosis whereby certain cells — phagocytes — can engulf external particulate material by extension and fusion of pseudopods around each particle. The particles are initially contained within phagocytic vacuoles

(**phagosomes**), which then fuse with primary lysosomes to effect digestion of the particles. Phagocytosis is employed by certain cells of the immune system, notably macrophages and neutrophils, to engulf and destroy target cells, tissue debris, and other particles, and by amoeboid protozoans. *Compare pinocytosis.*

phagolysosome an organelle formed by the fusion of a **phagosome** and a **lysosome**. Immediately after fusion there is a brief rise in the pH of the phagolysosome and neutral proteases and cationic proteins are active. Subsequently the pH falls and acid proteases become active. Indigestible substances remain in the phagolysosomes forming residual bodies.

phagosome an inclusion in a eukaryotic cell that arises from the ingestion of particulate material by **phagocytosis**. The phagosomes subsequently fuse with **lysosomes** to form a **phagolysosome**.

phalloidin *see phallotoxin.*

phallotoxin any member of a family of closely related bicyclic heptapeptides present in the highly poisonous agaric fungus, *Amanita phalloides*, commonly called the death-cap fungus or deadly agaric, and in some related species; **phalloidin** is the best-known member. All phallotoxins but one are toxic, though not as intensely so as the congeneric and much more slowly acting **amatoxins**; high doses of phalloidin in mice or rats cause death in a few hours. The common elements of the molecular structure of phallotoxins include residues of *cis*-4-hydroxy-L-proline, a D-amino acid (which is either threonine or β -hydroxyaspartic acid), and the bisfunctional amino acid, L-tryptathionine (which forms a thioether bridge across the molecule). The majority of phallotoxins bind specifically to F actin and so prevent its depolymerization to G actin; by this means the turnover of microfilaments is blocked and cell movement thereby inhibited.

pharmaceutic *an alternative name for pharmaceutical (def. 2).*

pharmaceutical 1 of, or pertaining to, drugs or pharmacy. 2 or **pharmaceutic** any medicinal substance, mixture, or formulation. — **pharmaceutically** *adv.*

pharmaceutics *an alternative name for pharmacy (def. 1).*

pharmaco+ *comb. form* indicating a drug or drugs.

pharmacodynamics *functioning as sing.* the branch of pharmacology dealing with the effects of drugs on the body, i.e. with the physiological, therapeutic, and toxicological responses to drugs with particular regard to the extent and time course of such effects. — **pharmacodynamic** *adj.*; **pharmacodynamically** *adv.*

pharmacogenetics *functioning as sing.* the branch of pharmacology dealing with the genetic mechanisms underlying individual differences in responses to drugs. — **pharmacogenetic** *adj.*; **pharmacogenetically** *adv.*

pharmacognosy the study of crude drugs of natural origins. — **pharmacognostic** *adj.*

pharmacokinetics *functioning as sing.* the branch of pharmacology dealing quantitatively with the movement of drugs within the body, i.e. with the absorption, distribution, metabolism, and elimination of drugs. — **pharmacokinetic** *adj.*; **pharmacokinetically** *adv.*

pharmacological or (esp. US) **pharmacologic** 1 of, pertaining to, or used in pharmacology. 2 (of an effect of a naturally occurring agonist) caused by an unphysiological concentration; (of a concentration or dose of a naturally occurring agonist) having an unphysiological effect. — **pharmacologically** *adv.*

pharmacology the science or study of drugs — their origin, characteristics, identification, biological effects, and modes of action. *Distinguish from pharmacy.*

pharmacopoeia or (US, sometimes) **pharmacopeia** 1 an authoritative book, often an official one, containing descriptions of medicinal drugs with their uses, preparation, methods of assay, formulae, dosages, etc. 2 the whole range of medicinal drugs in use or available. — **pharmacopoeial** or **pharmacopoeic** *adj.*

pharmacy 1 or **pharmaceutics** the practice of preparing and



phase

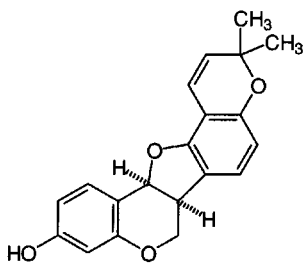
dispensing drugs. **2** a dispensary (for drugs). *Distinguish from pharmacology.*

phase **1** any distinct or characteristic period or stage in a sequence of events, occurrences, or processes, whether cyclical or unidirectional; e.g. the phases in the life cycle of an organism or the phases of the **cell-division cycle** (see also **anaphase**, **gap phase**, **metaphase**, **prophase**, **telophase**). **2** (in physical science) **a** the totality of those parts of a heterogeneous material system that are identical in chemical composition and physical state, and that are separated by an interface from the rest of the system; e.g. solid, liquid, or vapour phases, or the discrete phases into which immiscible liquids separate after mixing, etc. See also **epiphase**, **hypophase**, **liquid-crystalline phase**, **phase transition**. **b** a particular stage in the cycle of a periodic phenomenon, process, or quantity; the fraction of such a cycle that has been completed at a specific reference time, expressed in circular measure (or its equivalent in angular degrees). Two or more waveforms (or two or more similar sequences of events, etc.) that are in corresponding phases at the same time are described as 'in phase', while ones not exhibiting such correspondence are described as 'out of phase'. —**phasic** *adj.*

phase-contrast microscopy or **phase microscopy** a form of light microscopy in which a colourless object having a different refractive index from that of the medium is viewed by converting differences in refractive index into visible differences in light intensity. The technique takes advantage of the quarter-wave retardation of the illuminating light that occurs on its diffraction by the object, imposes a similar change of phase on the undiffracted light, and forms an image by interference between the two beams of light.

phaseolin **1** a crystallizable globulin found in seeds of the kidney bean, or French bean, *Phaseolus vulgaris*, and the lima bean, *Phaseolus lunatus*. It is a major seed-storage protein. α and β types are known. Example, α type precursor from *P. vulgaris*: database code PHSA_PHAVU, 436 amino acids (49.27 kDa). **2** a former name for **phaseollin**. **3** an alternative name for **legumain**.

phaseollin or (formerly) **phaseolin** a fungitoxic **phytoalexin** produced by plants of the kidney (or French) bean, *Phaseolus vulgaris*, and some other species of bean. It is a phenolic fused heterocyclic compound, $C_{20}H_{18}O_4$.



phase rule a generalization, valuable in the study of equilibria between **phases** (def. 2), expressed by the equation:

$$F = C - P + 2$$

where C is the number of chemically distinguishable components present, P is the number of phases in equilibrium, and F is the number of degrees of freedom of the system, i.e. the number of independently variable intensive properties that must be specified in order to describe completely the state of the system.

phaseshift mutation an alternative term for **frameshift mutation**.

phase transition (of lipids) the transition of a lipid between any of the various phases it can adopt. The transition is dependent on factors such as temperature, purity of the lipid, state of hydration, and, if not pure, the composition of the mixture. In the nonhydrated state, a crystalline polar lipid can

on heating, prior to a complete melting, adopt a **liquid-crystalline phase**, in which the polar groups are still associated in sheets, so that the overall structure consists of lipid bilayers, but the fatty-acyl chains undergo increased thermal vibrations leading to considerable disorder; the temperature at which this transition occurs is known as the **phase-transition temperature**. In the presence of water, below the phase-transition temperature, phospholipids adopt a gel phase, in which water penetrates between the sheets of polar headgroups, but the fatty-acyl chains are highly ordered. On increasing the temperature through the phase-transition temperature, the lipid adopts a hydrated liquid-crystalline phase with disordered fatty-acyl chains. The term **fluidity** has been used to provide a qualitative description of these changes, the bilayer being said to become more fluid as the temperature increases. See also **hexagonal phase**.

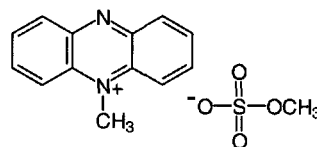
phase variation (of flagellin) see **flagellin**.

phasmid **1** a gene-cloning vector consisting of an artificial combination of a plasmid with a phage such that its genome contains functional origins of replication of both; it may thus be propagated either as a plasmid or as a phage in appropriate host strains. A phasmid is formed by insertion of one or more plasmid molecules into a phage genome. Compare **cosmid**, **phagemid**. **2** either of a pair of lateral caudal organs characteristic of parasitic nematodes of the class Phasmidia; they are thought to function as chemoreceptors. **3** any plant-eating insect of the order Phasmida, which includes leaf insects and stick insects.

PH domain *abbr. for* pleckstrin homology domain.

Phe *symbol for a residue of the α -amino acid L-phenylalanine (alternative to F).*

phenazine methosulfate *abbr.:* PMS; *N*-methylphenazonium methosulfate; 5-methylphenazonium methyl sulfate; $[C_{13}H_{11}N_2]^+ [CH_3SO_3]^-$; an artificial electron carrier, $E^0 = +0.080$ V, useful in the study of redox enzyme systems.



phencyclidine 1-(1-phenylcyclohexyl)piperidine (*abbr.:* PCP); a potent psychotomimetic agent, known to drug users as angel dust, with analgesic and anesthetic actions. It acts as a ligand for the **sigma receptor** and is a blocker of the NMDA-receptor ionophore (see **glutamate receptor**). A controlled substance in the US Code of Federal Regulations.

phone any phenotypic character under genetic control.

phenelzine β -phenylethylhydrazine, (2-phenethyl)hydrazine; a nonselective monoamine oxidase inhibitor used as an antidepressant.

phenobarbital 5-ethyl-5-phenylbarbiturate; a major **barbiturate** drug.

phenocopy a nonhereditary modification of the phenotype of an organism, induced by nutritional or other environmental factors, that mimics a phenotype controlled by a specific gene.

phenol **1** the common and trivial name for hydroxybenzene, C_6H_5OH . **2** any compound containing one or more hydroxyl groups directly attached to an aromatic carbocycle. Unlike hydroxyalkyl compounds, phenols are weakly acidic.

phenolase **1** an alternative name for **catechol oxidase** (EC 1.10.3.1). **2** an alternative name for **monophenol monooxidase** (EC 1.14.18.1). **3** more loosely, any of various other enzymes of the sub-subclass 1.10.3. Such enzymes cause browning of damaged plant tissues when they come into contact with flavonoids, as **catechins** are converted into polymerized brown products.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

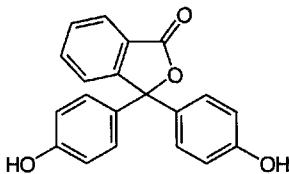


phenolic

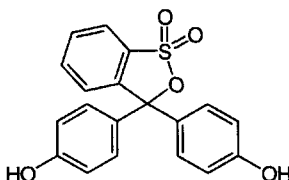
phenolic 1 of, containing, derived from, or having the character of phenol or a phenol. 2 describing a hydroxyl group linked to an aromatic carbocycle.

phenol 2-monooxygenase *see* monooxygenase.

phenolphthalein an indicator dye that changes from colourless to red over the pH range 8.3–10.0. The β -D-glucuronide ($\lambda_{\text{max}} = 420 \text{ nm}$) is used as a substrate for the assay of β -glucuronidase, hydrolysis leading to release of phenolphthalein, which can be measured in alkaline solution by its red colour ($\lambda_{\text{max}} = 550 \text{ nm}$).



Phenol Red an indicator dye that changes colour from yellow to red over the pH range 6.8–8.4.



phenome the whole of the phenotypic characteristics of an organism. *Compare* genome. —**phenomic** *adj.*

phenomenology a branch of a science dealing with the description and classification of natural phenomena rather than with their explanation or cause. —**phenomenological** *adj.*

phenon any set of organisms grouped together by the methods of numerical taxonomy.

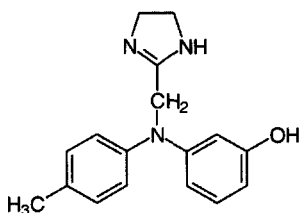
phenothiazine 1 a chemical structure that forms the basis, by *N*-substitution, of a group of neuroleptics, such as chlorpromazine. 2 any of the group of drugs based on the phenothiazine structure. The neuroleptic effects are those now most commonly exploited, but antiemetic, antihistamine and antihelmintic effects have also been of use.

phenotype 1 the totality of the observable functional and structural characteristics of an organism as determined by interaction of the genotype of the organism with the environment in which it exists. 2 any particular characteristic or set of characteristics of an organism so determined. 3 a group of organisms exhibiting the same set of such characteristics. *Compare* genotype. —**phenotypic** *adj.*

phenoxide 1 the anion, $\text{C}_6\text{H}_5\text{O}^-$, derived from phenol by the loss of a proton. 2 any salt of phenol.

phenoxyl *symbol:* OPh or PhO; the $\text{C}_6\text{H}_5\text{O}^-$ group, derived from phenol by the loss of a hydrogen atom.

phentolamine an antagonist for both α_1 and α_2 adrenoceptors. It is an antihypertensive agent and vasodilator.

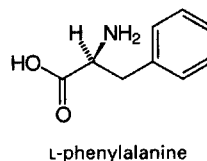


phenyl *symbol:* Ph; the aromatic carbocyclic group C_6H_5^- , derived from benzene.

phenylacetaldehyde dehydrogenase EC 1.2.1.39; an enzyme of the pathway of phenylalanine degradation. It catalyses the oxidation by NAD^+ of phenylacetaldehyde to phenylacetate with formation of NADH.

phenylalaninate 1 phenylalanine anion; the anion, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}^-$, derived from phenylalanine. 2 any salt containing phenylalanine anion. 3 any ester of phenylalanine.

phenylalanine the trivial name for β -phenylalanine; α -aminohydrocinnamic acid; α -amino- β -phenylpropionic acid; 2-amino-3-phenylpropanoic acid; $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$; a chiral α -amino acid. L-phenylalanine (*symbol:* F or Phe), (*S*)-2-amino-3-phenylpropanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: UUC or UUU. In mammals it is an essential dietary amino acid and is both glucogenic and ketogenic. Residues of D-phenylalanine (*symbol:* D-Phe or DPhe), (*R*)-2-amino-3-phenylpropanoic acid, occur in a number of peptide antibiotics, e.g. bacitracin A, gramicidin S, polymyxins B₁ and B₂, and tyrocidins A, B, and C.



L-phenylalanine ammonia-lyase EC 4.3.1.5; *systematic name:* L-phenylalanine ammonia-lyase; an enzyme, widely distributed in plants, that deaminates L-phenylalanine to trans-cinnamic acid, which is rapidly converted to lignin, a major polymer of the plant cell wall.

phenylalanine hydroxylase *see* phenylalanine 4-monooxygenase.

phenylalanine 4-monooxygenase EC 1.14.16.1; *systematic name:* L-phenylalanine, tetrahydrobiopterin: oxygen oxidoreductase (4-hydroxylating); *other names:* phenylalanine 4-hydroxylase; phenylalaninase. An enzyme that catalyses a reaction between L-phenylalanine, tetrahydrobiopterin, and dioxygen to form L-tyrosine, dihydrobiopterin, and H_2O . It is thus of major importance for the provision of L-tyrosine. Absence of the enzyme in humans leads to phenylketonuria. It is an iron protein. Example from human: database code DHPR_HUMAN, 244 amino acids (25.77 kDa).

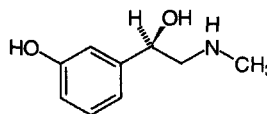
phenylalaninium phenylalanine cation; the cation, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{NH}_3^+)\text{COOH}$, derived from phenylalanine.

phenylalanino the alkylamino group, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{COOH})\text{NH}-$, derived from phenylalanine.

phenylalanyl the acyl group, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{NH}_2)\text{CO}-$, derived from phenylalanine.

phenylalanyl chain (*sometimes*) the B chain of any of those species of insulin whose amino-acid sequences were the first to be determined. It was so called by virtue of the N-terminal phenylalanine residue occurring in those instances.

phenylephrine (*R*)-(-)-1-(3-hydroxyphenyl)-2-methylaminoethanol; a highly selective α_1 adrenergic agonist. It is a vasoconstrictor, used in treating hypotension; it also acts as a decongestant.



R-(-)-enantiomer

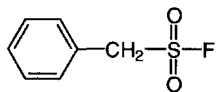
phenylethanolamine *N*-methyl transferase

phenylethanolamine *N*-methyl transferase *abbr.*: PNMT; EC 2.1.1.28; an enzyme catalysing *N*-methylation of norepinephrine in the synthesis of epinephrine. *S*-adenosyl methionine is the methyl donor.

phenylhydrazone any condensation product of the type $R'R''C=N-NHC_6H_5$, formed between carbonyl compounds of the general formula $R'R''C=O$ and phenylhydrazine. Phenylhydrazones, in particular 2,4-dinitrophenylhydrazones, are useful in the characterization of aldehydes and ketones. *See also phenylosazone*.

phenylketonuria (*abbr.*: PKU) or **phenylpyruvic oligophrenia** an autosomal recessive inherited disorder of phenylalanine metabolism characterized by the presence of phenylpyruvate in the urine and excessively high levels of phenylalanine and its metabolites in the blood and tissues. It can be treated by restricting the amount of phenylalanine in the diet. If untreated in young children, severe mental retardation occurs. This form, **classical phenylketonuria**, is due to the absence or deficiency of **phenylalanine 4-monooxygenase**, an enzyme that uses tetrahydrobiopterin as second donor. Other types of hyperphenylalaninemia have been described due to, e.g., deficiency in dihydropteridine reductase (EC 1.6.99.7) or phenylalanine(histidine) aminotransferase (EC 2.6.1.58).

phenylmethylsulfonyl fluoride or **phenylmethanesulfonyl fluoride** *abbr.*: PMSF; a reagent used as an inhibitor of serine proteases; it is highly toxic, but less so than diisopropylphosphorofluoridate. PMSF is both insoluble and unstable in aqueous solutions, so is usually added to such solutions by first dissolving it in, e.g., dioxan. It acts as an irreversible inhibitor of trypsin and chymotrypsin, sulfonylating the active-site histidine; it does not inhibit cholinesterase. *See also aminoethylbenzenesulfonyl fluoride*.



phenylosazone any 1,2-bis(phenylhydrazone); i.e. any condensation product of the type $C_6H_5NH-N=CR'-CR''=N-NH-C_6H_5$ formed between an α -dicarbonyl compound of the general formula $R'-CO-CO-R''$ and phenylhydrazine; α -hydroxy- and α -amino-carbonyl compounds will also form phenylosazones through oxidation of the corresponding [mono]phenylhydrazones initially formed. Phenylosazones, in particular 2,4-dinitrophenylosazones, are useful in the characterization of aldehydes and ketones, especially monosaccharides. *See also phenylhydrazone*.

phenylpropanoid any of a large and heterogeneous group of substances in plants derived from phenylpropane units variously linked and interlinked to form polymers of high molecular mass, especially **lignins**.

phenylpyruvate decarboxylase EC 4.1.1.43; an enzyme of the pathway of phenylalanine degradation. It catalyses the formation of phenylacetaldehyde from phenylpyruvate with release of CO_2 .

phenylpyruvic oligophrenia an alternative name for **phenylketonuria**.

pH 5 enzyme the provisional name given to a protein fraction from rat liver, soluble at pH 7 and isolated by precipitation at pH 5, that was used in early studies of protein biosynthesis *in vitro*. It is now known to be a mixture of amino acid-tRNA ligases.

pheo+ or (*esp. Brit.*) **phaeo+** *comb. form* denoting dark-coloured.

pheochromocytoma or (*esp. Brit.*) **phaeochromocytoma** a tumour of tissues derived from embryological neuroectoderm, often of the adrenal medulla. It is characterized by its ability to synthesize and secrete catecholamines, often in excessive

amounts. The tumour cells commonly show a strong **chromaffin** (or **pheochrome**) reaction.

pheofarnesin or (*esp. Brit.*) **phaeofarnesin** any pigment derived by demetallation of a chlorophyll in which the 17-propionic-acid group is esterified with farnesol.

pheophorbide or (*esp. Brit.*) **phaeophorbide** any pigment derived by removal of metal from a chlorophyll in which the 17-acrylic-acid or 17-propionic-acid group is unesterified or has been de-esterified.

pheophytin or (*esp. Brit.*) **phaeophytin** any pigment derived by removal of metal from a chlorophyll in which the 17-propionic-acid group is esterified with phytol.

pheromone a substance, or characteristic mixture of substances, that is secreted and released by an organism and detected by a second organism of the same or a closely related species, in which it causes a specific reaction, such as a definite behavioural reaction or a developmental process; a type of **ectocrine**. The mating pheromones of fungi and insects are examples. *Compare allomone, kairomone*.

pheromone-binding protein any of a group of insect proteins that bind to and thus solubilize **pheromones**, which are typically hydrophobic. Example (precursor) from the moth *Antheraea polyphemus*: database code PBP_ANTPO, 163 amino acids (18.12 kDa) (first 21 amino acids are the signal).

pheromone receptor a fungal transmembrane G-protein-linked receptor. Example from *Ustilago maydis* (smut fungus), receptor for the A1 mating pheromone: database code PRA2_USTMA, 346 amino acids (38.74 kDa).

phi *symbol*: ϕ or ϕ (lower case) or Φ (upper case); the twenty-first letter of the Greek alphabet. For uses *see Appendix A*.

PHI or **PHI-27** *abbr. for* peptide HI.

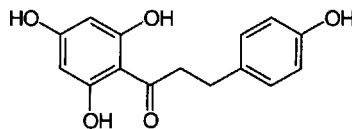
Philadelphia chromosome (Ph¹) a specific chromosomal abnormality in the blood cells of patients with chronic myeloid leukemia. It usually results from a fusion of **BCR** on chromosome 22 with *c-myc* (*see MYC*) on chromosome 9 leading to expression of the latter. The Philadelphia chromosome is a very small chromosome consisting of the short arm and a small piece of the long arm of chromosome 22 fused to a short piece from the end of the long arm of chromosome 9 (the modified chromosome 9 that results being called 9q⁺). It is named after the city in which it was first recorded. *See also abl*.

pH indicator *see indicator* (def. 3), **mixed indicator**.

phi X174 or **φX174** a single-stranded DNA bacteriophage that infects *Escherichia coli*.

phloem the main food-conducting tissue of plants, via which nutrients and photosynthate are conveyed between different sites. Phloem comprises two types of cells: sieve-tube elements, which are joined end-to-end to form sieve tubes; and adjacent companion cells. *Compare xylem*.

phloretin 3-(4-hydroxyphenyl)-1-(2,4,6-trihydroxyphenyl)-1-propanone; the aglycon of **phlorizin**, with which it shares many biochemical and pharmacological properties.



phlorizin or **phlorhizin** or **phloridzin** or **phlorrhizin** phloretin-2'- β -glucoside; a dihydrochalcone glucoside occurring in all parts of the apple tree except the mature fruit. It is an inhibitor of intestinal and renal glucose- Na^+ cotransporters, of the mediated transport of glucose into erythrocytes and hepatocytes, of the glucose 6-phosphate transporter in hepatocytes, and of photophosphorylation. It is used experimentally to produce glucosuria.

phloroglucinol 1,3,5-trihydroxybenzene; 1,3,5-benzenetriol; a compound that is useful as a reagent for the qualitative detection of

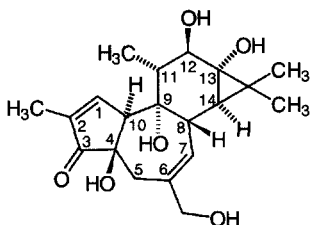


tion of pentoses and pentosans; when these are boiled with a solution of phloroglucinol in hydrochloric acid, a cherry-red colour is produced. Galactose also reacts in this way.

phlorrhizin a variant spelling of **phlorizin**.

pH meter any instrument for determining pH potentiometrically, usually with a glass electrode plus a calomel reference electrode, or with a combination electrode.

phorbol a complex tetracyclic compound, $C_{20}H_{28}O_6$, based on a cyclopropabenzazulene skeleton. Various **phorbol esters** occur in croton oil (from *Croton tiglium*, of the family Euphorbiaceae, hence the name) and act as potent **cocarcinogens**.



phorbol ester any esterified form of **phorbol**. A number of phorbol esters possess activity as tumour promoters and activate the mechanisms associated with cell growth. These actions have been widely attributed to the ability to activate protein kinase C, probably by substituting for diacylglycerol. A much-studied ester is phorbol 12-myristate 13-acetate (PMA); other active esters include phorbol 12,13-dibutyrate (PDBu) and phorbol 12,13-didecanoate (PDD). Activity depends on the β -configuration of the hydroxyl at C-4, and a number of 4α -phorbol esters have been utilized in control studies.

+phore *noun suffix* denoting something, especially a chemical substance or group, that carries or produces; e.g. ionophore, chromophore.

+phoresis *noun suffix* denoting carriage of molecular entities by some agency; e.g. electrophoresis, ionophoresis.

phosducin a part of a complex with the β and γ subunits of **transducin** in outer and inner segments of retinal rod cells. Light-induced changes in cyclic nucleotide levels modulate the phosphorylation of phosducin by cyclic AMP kinase. Example (bovine): database code PHOS_BOVIN, 245 amino acids (28.20 kDa).

+phosph+ a variant form of **+phospho+** (sometimes before a vowel).

phosphacan a chondroitin sulfate proteoglycan of brain that interacts with neurons and neural cell-adhesion molecules. It is an extracellular variant of a receptor-type protein tyrosine phosphatase. Example from rat: database code RNU04998, 1616 amino acids (175.65 kDa).

phosphagen any of a group of guanidine phosphates that occur in muscle and can be used to regenerate ATP from ADP during muscular contraction by a reaction catalysed by **creatine kinase** or by an analogous enzyme of sub-subclass EC 2.7.3. Included in the group are phosphoagmatine (see **agmatine**), **phosphoarginine**, **phosphocreatine**, **phosphoglycocyamine** (see **glycocyamine**), **phospholombricine** (see **lombricine**), **phosphooctopine** (see **octopine**), and **phosphotaurocyamine** (see **taurocyamine**).

phosphataemia a variant spelling (esp. Brit.) of **phosphatemia**.

phosphatase or **phosphomonoesterase** any phosphoric monoester hydrolase of sub-subclass EC 3.1.3. Some phosphatases are very specific while others are relatively unspecific with regard to substrate. See also **acid phosphatase**, **acylphosphatase**, **alkaline phosphatase**, **fructose-bisphosphatase**, **glucose-1-phosphatase**, **glucose-6-phosphatase**, **inorganic pyrophosphatase**, **protein phosphatase**.

phosphatased a jargon term indicating dephosphorylated by treatment with a **phosphatase**.

phosphatase inhibitor protein an inhibitor of protein phosphatase 1 (see **protein phosphatase**) involved in the regulation of glycolysis; it is itself a phosphoprotein. Example, inhibitor 1 from rat: database code IPP1_RAT, 171 amino acids (18.71 kDa).

phosphate 1 the trivalent anion, PO_4^{3-} , derived from phosphoric acid; orthophosphoric acid; tetraoxophosphoric(v) acid; $PO(OH)_3$. 2 any mixture of [ortho]phosphoric acid and its anions. Once widely used in the laboratory as a buffer in the physiological range of pH values; pK_{a2} (37 °C, 0.05 mol L^{-1}) = 6.84. 3 any (partial or full) salt or ester of [ortho]phosphoric acid. Phosphate salts and esters play key roles in the structure and function of all forms of life. 4 any anion or salt of a mixed anhydride of [ortho]phosphoric acid with another acid; e.g. acetyl phosphate (i.e. acetic phosphoric anhydride). 5 the univalent group $-O-P(O)(OH)_2$ (irrespective of its state of ionization). See also **diphosphate**, **inorganic phosphate**, **metaphosphate**, **pyrophosphate**, **triphosphate**.

phosphate bond 1 an ester linkage in a phospho[mono]ester, phosphodiester, or phosphotriester. 2 an anhydride linkage in a phosphoric anhydride. 3 the connecting phosphoric residue in a phosphodiester.

phosphate bond energy a term formerly and incorrectly used for the **bond energy** (def. 2) of a **phosphate bond** (def. 1, 2).

phosphatemia or (esp. Brit.) **phosphataemia** the level of **phosphate** (def. 2) in the blood; in human plasma its normal range is 0.8–1.4 mmol L^{-1} . The level is regulated by **calcitriol**, which stimulates absorption of dietary phosphate as well as calcium, and by **parathyrin**, which decreases phosphate resorption by the kidney. Calcitriol also stimulates bone resorption, which tends to lower the plasma phosphate level, but in states of hypervitaminosis D the effect on the gut predominates and the plasma level is increased. See also **hyperphosphatemia**, **hypophosphatemia**.

phosphate potential the concentration of ATP in any system divided by the product of the concentrations of ADP and inorganic phosphate in that system. Compare **adenylate energy charge**.

phosphatidal a former term for **plasmeyl**.

phosphatidate any anion, salt, or ester of a **phosphatidic acid**.

phosphatidate cytidyltransferase EC 2.7.7.41; *systematic name*: CTP:phosphatidate cytidyl transferase; *other names*: CDPdiglyceride synthetase; CDPdiglyceride pyrophosphorylase. An enzyme that catalyses a reaction between CTP and phosphatidate to form pyrophosphate and CDPdiacylglycerol. It is an endoplasmic reticulum membrane-bound enzyme of phospholipid biosynthesis; it forms the intermediate involved in the synthesis of, e.g., phosphatidylinositol, by reaction with inositol, or phosphatidylserine, by reaction with serine. Example from *Escherichia coli*: database code CDSA_ECOLI, 249 amino acids (27.54 kDa).

phosphatidate phosphatase EC 3.1.3.4; *systematic name*: 3-*sn*-phosphatidate phosphohydrolase; an enzyme that removes phosphate from phosphatidic acid to yield an *sn*-1,2-diacylglycerol.

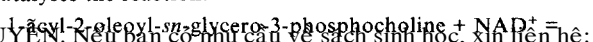
phosphatide an old term, no longer recommended for **glycerophospholipid**.

phosphatidic acid any derivative of glycerol phosphate in which both the remaining hydroxyl groups of the glycerol moiety are esterified with fatty acids; the commonest are derivatives of *sn*-glycerol 3-phosphate.

phosphatidyl *symbol*: Ptd; an acyl group derived from a **phosphatidic acid**.

phosphatidylcholine *abbr.*: PC; any of a class of glycerophospholipids in which a phosphatidyl group is esterified to the hydroxyl group of **choline**. They are important constituents of cell membranes and lipoproteins.

phosphatidylcholine desaturase EC 1.3.1.35; *other names*: oleate desaturase; linoleate synthase; an enzyme in plants that catalyses the reaction:





phosphatidylcholine–dolichol

1-acyl-2-linoleoyl-*sn*-glycero-3-phosphocholine + NADH.

phosphatidylcholine–dolichol O-acyltransferase *see* **lecithin–cholesterol acyl transferase**.

phosphatidylcholine–sterol O-acyltransferase EC 2.3.1.43; *other names*: lecithin–cholesterol acyltransferase (*abbr.*: LCAT); phospholipid–cholesterol acyltransferase; a plasma enzyme that catalyses the transfer of a fatty-acyl group from phosphatidylcholine to a sterol to form a sterol ester and 1-acylglycerophosphocholine. In mammals it is important in plasma lipoprotein metabolism, particularly in esterifying cholesterol, which is then transferred to very-low-density lipoprotein as cholesteryl ester. Example from human: database code LCAT_HUMAN, 440 amino acids (49.52 kDa). *Compare sterol O-acyltransferase*.

phosphatidylethanolamine *abbr.*: PE; any of a class of glycerophospholipids in which a phosphatidyl group is esterified to the hydroxyl group of **ethanolamine**. They are important constituents of cell membranes.

phosphatidylinositol *abbr.*: PI; *symbol*: PtdIns; 1-(3-*sn*-phosphatidyl)-1D-*myo*-inositol; *the recommended name* for any glycerophospholipid in which its *sn*-glycerol 3-phosphate residue is esterified to the 1-hydroxyl group of 1D-*myo*-inositol. The older term **phosphoinositide** is now used as a generic term for any inositol-containing glycerophospholipid. The corresponding names for the monophosphorylated and diphosphorylated derivatives of phosphatidylinositol are **phosphatidylinositol 4-phosphate** and **phosphatidylinositol 4,5-bisphosphate**, respectively. The older names diphosphoinositide and triphosphoinositide for these derivatives are no longer used.

phosphatidylinositol 4,5-bisphosphate *or* (formerly) **triphosphoinositide** *abbr.*: PIP₂ *or* PI-4,5-P₂; *symbol*: PtdIns(4,5)P₂ *or* PtdIns-4,5-P₂; 1-(3-*sn*-phosphatidyl)-1D-*myo*-inositol 4,5-bisphosphate; any of a class of glycerophospholipids in which a phosphatidyl group is esterified to the 1-hydroxyl group of 1D-*myo*-inositol 4,5-bisphosphate.

1-phosphatidylinositol-4,5-bisphosphate phosphodiesterase *see* **phospholipase**.

phosphatidylinositol cycle *or* (formerly) **phosphoinositide cycle** a metabolic cycle that mediates the actions of various hormones and other effectors at the membranes of mammalian cells. The cycle may be summarized as follows. Binding of an effector to its specific receptor at the cell membrane brings about a G-protein-coupled stimulation of a phospholipase C, of 1-phosphatidylinositol-4,5-bisphosphate phosphodiesterase. This action brings about hydrolysis of phosphatidylinositol 4,5-bisphosphate, I, to produce the two intracellular messengers inositol 1,4,5-trisphosphate, II, which causes release of calcium ions from intracellular stores, and *sn*-1,2-diacylglycerol, III, which activates protein kinase C. III is then converted via phosphatidate (under the action of **diacylglycerolkinase** and **phosphatidate cytidyltransferase**) to **CDPdiacylglycerol**, IV, while II is converted via inositol bisphosphate and inositol monophosphate to inositol. Inositol and IV then react together (enzyme is **CDPdiacylglycerol-inositol 3-phosphatidyltransferase**) to form phosphatidylinositol, from which I is regenerated by stepwise phosphorylation by specific kinases to complete the cycle. Alternatively, II may be converted to another messenger, inositol 1,3,4,5-tetrakisphosphate, V, which may be implicated in the influx of extracellular calcium ions; V is then degraded to inositol via inositol 1,3,4-trisphosphate, which is inactive.

phosphatidylinositol glycan protein *abbr.*: PIG protein; *N*-acetylglucosaminyl-phosphatidylinositol biosynthetic protein; a protein required for synthesis of *N*-acetylglucosaminyl-phosphatidylinositol, the very early intermediate in the biosynthesis of the **GPI anchor**. Example, PIG-A (human): database code PIGA_HUMAN, 484 amino acids (54.06 kDa). Another protein required for this synthesis is PIG-F: database code PIGF_HUMAN, 219 amino acids (24.86 kDa).

1-phosphatidylinositol 3-kinase EC 2.7.1.137; *systematic*

name: ATP:1-phosphatidyl-1D-*myo*-inositol 3-phosphotransferase. An enzyme that catalyses the phosphorylation by ATP of 1-phosphatidyl-1D-*myo*-inositol to form 1-phosphatidyl-1D-*myo*-inositol 3-phosphate with release of ADP. It forms part of the complex that associates on activation of certain growth factor (e.g. PDGF) receptors, binding via SH2 domains (*see* **SH domain**). The molecule is a heterodimer of a catalytic subunit (p110) and a regulatory subunit (p85: two forms, α and β , in mammals). p85 acts as an adaptor to mediate association of p110 to the plasma membrane. Examples, (bovine) p110: database code P11A_BOVIN, 1068 amino acids (124.18 kDa); p85(β): database code P85B_BOVIN, 724 amino acids (80.97 kDa).

1-phosphatidylinositol 4-kinase EC 2.7.1.67; an enzyme of the pathway for the biosynthesis of membrane **phosphoinositides**. It catalyses the phosphorylation by ATP of 1-phosphatidyl-1D-*myo*-inositol to form 1-phosphatidyl-1D-*myo*-inositol 4-phosphate with release of ADP. Example, α form (human): database code A55404, 854 amino acids (96.87 kDa).

phosphatidylinositol-3-phosphatase EC 3.1.3.64; an enzyme that catalyses the hydrolysis of the 3-phosphate from phosphatidylinositol-3-phosphate, forming phosphatidylinositol and orthophosphate.

phosphatidylinositol 4-phosphate *or* (formerly) **diphosphoinositide** *abbr.*: PIP *or* PI 4-P; *symbol*: PtdIns4P *or* PtdIns-4-P; 1-(3-*sn*-phosphatidyl)-1D-*myo*-inositol 4-phosphate; any of a class of glycerophospholipids in which a phosphatidyl group is esterified to the 1-hydroxyl group of 1D-*myo*-inositol 4-phosphate.

1-phosphatidylinositol-4-phosphate kinase EC 2.7.1.68; *other name*: diphosphoinositide kinase; an enzyme of the pathway for the biosynthesis of membrane **phosphoinositides**. It catalyses the phosphorylation by ATP of 1-phosphatidyl-1D-*myo*-inositol 4-phosphate to form 1-phosphatidyl-1D-*myo*-inositol 4,5-bisphosphate with release of ADP.

1-phosphatidylinositol phosphodiesterase *see* **phospholipase**.

phosphatidylinositol synthase *see* **CDP-diacylglycerol-inositol 3-phosphatidyltransferase**.

phosphatidylserine any of a class of glycerophospholipids in which a phosphatidyl group is esterified to the hydroxyl group of L-serine. They are important constituents of cell membranes.

phosphatidylserine decarboxylase EC 4.1.1.65; an enzyme that converts phosphatidylserine to phosphatidylethanolamine, of major importance in the synthesis of that phospholipid. It exists as a heterodimer of α and β chains, formed by cleavage of a proenzyme. Example, proenzyme from *Haemophilus influenzae*: database code DPSD_HAEIN, 290 amino acids (33.20 kDa); residues 1–256 form the β chain, 257–290 the α chain.

phosphatidylserine synthase *an alternative name for* **CDP-diacylglycerol-serine O-phosphatidyltransferase**.

phosphite **1** any anionic form of phosphorous acid; trioxophosphoric(III) acid; P(OH)₃. **2** any mixture of phosphorous acid and its ionized forms. **3** any (partial or full) salt or ester of phosphorous acid.

phosphite-triester method a method for the chemical synthesis of oligonucleotides in which a suitably protected phosphite diester of general formula Nuc-3'-O-P(OR)₂X, where R is a methyl or another alkyl group, X is a reactive group such as chloro, dimethylamino, or morpholino, and Nuc is a residue of an appropriate nucleoside, is first condensed with the 5'-hydroxyl of a protected nucleoside (usually anchored through its 3'-hydroxyl to an insoluble support) to form a phosphite triester. Sequential addition of further nucleoside units is then carried out in the same way, the phosphite triester formed at each stage being oxidized to the corresponding phosphate triester before the next cycle of reactions. Protecting groups are removed by hydrolysis at the completion of the synthesis.

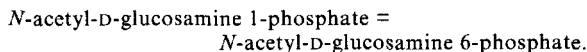
+phospho+ *or* (sometimes before a vowel) **+phosph-** (*in chemical*



phosphoacetylglucosamine mutase

cal nomenclature) prefix or infix denoting a phosphoric residue linked to one, two, or three heteroatoms in an organic compound. *See also* **phosphono-**.

phosphoacetylglucosamine mutase EC 5.4.2.3; *systematic name*: *N*-acetyl-D-glucosamine 1,6-phosphomutase; *other name*: acetylglucosamine phosphomutase. An enzyme that catalyses the reaction:

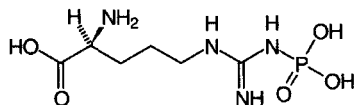


It participates in the pathway for the formation of UDP-*N*-acetylglucosamine: glucosamine 6-phosphate $\rightarrow$ *N*-acetylglucosamine-6-phosphate $\rightarrow$ *N*-acetylglucosamine-1-phosphate, thence by reaction with UTP to UDP-*N*-acetylglucosamine + pyrophosphate, thence also to sialic acid. Example from *Schizosaccharomyces pombe*: database code PCM1_SCHPO, 518 amino acids (56.35 kDa).

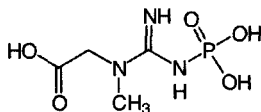
3'-phosphoadenosine 5'-phosphosulfate *an alternative recommended name for adenosine 3'-phosphate 5'-phosphosulfate*.

phosphoamino acid any phosphorylated amino acid. The term often refers to *O*-phosphate esters of residues in proteins, e.g. those of serine, threonine, or tyrosine, that are released as phosphoamino acids on hydrolysis. Phosphohistidine can be formed as part of an enzyme mechanism (e.g. **phosphoglycerate mutase**), and phosphohomoserine is an intermediate in threonine biosynthesis. The guanidine group can also be phosphorylated to form a phosphono derivative (*see* **phosphagen**).

phosphoarginine *the recommended trivial name for N^ω-phosphonoarginine*; the commonest **phosphagen** in invertebrates. The older name, arginine phosphate, is not recommended.



phosphocreatine *abbr.*: PC; *the recommended trivial name for N^ω-phosphocreatine*; the commonest **phosphagen** in vertebrates. The older name, creatine phosphate, is not recommended.



phosphodiester any diester of phosphoric acid; i.e. any phosphoric ester containing the bridging group -PO(OH)- between two oxygen atoms in other groups (which may be the same or different) or at different positions in the same group. Examples include cyclic nucleotides, glycerophospholipids, and nucleic acids.

phosphodiesterase any esterase that catalyses the hydrolysis of either (but not both) of the ester linkages in a phosphodiester, each linkage being hydrolysed only by an enzyme specific for that linkage. Such enzymes are classified in various sub-subclasses within the subclass EC 3.1, hydrolases, especially in the sub-subclass EC 3.1.4, phosphoric diester hydrolases. The term is often applied to **nucleases** without sugar specificity. *See also* **cyclic AMP phosphodiesterase**, **phosphodiesterase I**, **phospholipase C**. —**phosphodiesteratic adj.**

phosphodiesterase I *the recommended name for an enzyme*, EC 3.1.4.1, that catalyses the hydrolytic removal of 5'-nucleotides successively from the (nonphosphorylated) 3' end of an oligonucleotide; *other name*: 5'-exonuclease.

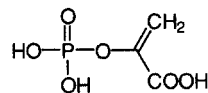
phosphodiesterase II *the former recommended name for an*

phosphoenolpyruvate-dependent :

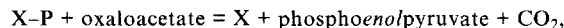
enzyme, now named **spleen exonuclease** (EC 3.1.16.1; previously EC 3.1.4.18), that catalyses the hydrolytic removal of 3'-nucleotides successively from the (nonphosphorylated) 5' end of a (preferably single-stranded) ribonucleic or deoxyribonucleic acid; *other names*: 3'-exonuclease; spleen phosphodiesterase.

phosphodiester method a method for the chemical synthesis of oligonucleotides in which, essentially, a 5'-phosphate of an appropriate [mono]nucleotide, after suitable protection of other functional groups, is condensed with the 3'-hydroxyl of a protected nucleoside, [mono]nucleotide, or previously formed oligonucleotide, using dicyclohexylcarbodiimide or an arylsulfonyl chloride as a coupling agent. The reaction sequence is then repeated with another nucleotide 5'-phosphate. This method is being superseded by the **phosphite-triester method** and **phosphotriester method**.

phosphoenolpyruvate *abbr.*: PEP or *P-enolpyruvate*; 2-hydroxy-2-propenoate phosphate (ester); 2-(phosphonoxy)-2-propenoate; the phosphate ester formed from the hydroxy group of the enol tautomer of pyruvate. It is a so-called high-energy compound, hydrolysis to pyruvate giving a particularly large negative change in standard free energy ($-\Delta G^\circ = 62 \text{ kJ mol}^{-1}$ at pH 7). It is a key intermediate in glycolysis and gluconeogenesis, and in the biosynthesis of aromatic amino acids.



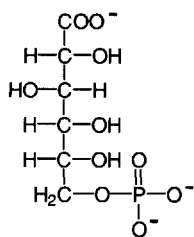
phosphoenolpyruvate carboxykinase *abbr.*: PEPCK; any of several enzymes that catalyse the reaction:



where X-P, depending on source, can be GTP or ITP (in EC 4.1.1.32), ATP (in EC 4.1.1.49), or pyrophosphate (in EC 4.1.1.38). (1) **Phosphoenolpyruvate carboxykinase (GTP)**, EC 4.1.1.32; *systematic name*: GTP:oxaloacetate carboxy-lyase (transphosphorylating); *other names*: phosphoenolpyruvate carboxylase; phosphopyruvate carboxylase; the rate-limiting gluconeogenic enzyme in eukaryotes. It has different forms in cytoplasm and mitochondria. It catalyses a reaction between GTP and oxaloacetate to form GDP, phosphoenolpyruvate, and CO₂. Example, mitochondrial precursor from chicken: database code PPCC_CHICK, 622 amino acids (69.45 kDa). (2) **Phosphoenolpyruvate carboxykinase (ATP)**, EC 4.1.1.49; *systematic name*: ATP:oxaloacetate carboxy-lyase (transphosphorylating); *other names*: phosphoenolpyruvate carboxylase; phosphopyruvate carboxylase (ATP); it catalyses a similar reaction to (1) but with ATP/ADP not GTP/GDP. In many bacteria it is a Ca²⁺-regulated rate-limiting gluconeogenic enzyme. Example from *Escherichia coli*: database code PPCK_ECOLI, 464 amino acids (51.25 kDa). *See also* **gluconeogenesis**.

phosphoenolpyruvate carboxylase EC 4.1.1.31; *systematic name*: orthophosphate:oxaloacetate carboxy-lyase (phosphorylating); an enzyme that catalyses a reaction between orthophosphate and oxaloacetate to form phosphoenolpyruvate, H₂O, and CO₂. It is an allosteric enzyme associated with the **tricarboxylic-acid cycle** in plants; effectors include fatty acids, acetyl-CoA, and fructose 1,6-bisphosphate. It forms oxaloacetate for metabolism by the tricarboxylic-acid cycle in plants. The protein is a homotetramer. Example from *Escherichia coli*: database code CAPP_ECOLI, 883 amino acids (98.95 kDa).

phosphoenolpyruvate-dependent sugar phosphotransferase system *abbr.*: PTS; a system for bacterial sugar transport; the full system comprises two main components:



phosphogluconate

protein-*N*-phosphohistidine-sugar phosphotransferase and **phosphoenolpyruvate-protein phosphotransferase**. The system functions also as a chemoreceptor monitoring the environment for changes in sugar concentration.

phosphoenolpyruvate mutase EC 5.4.2.9; *systematic name*: phosphoenolpyruvate 2,3-phosphonmutase; *other name*: phosphoenolpyruvate phosphomutase. An enzyme that catalyses the conversion of phosphoenolpyruvate to 3-phosphonopyruvate. Phosphoenolpyruvate mutase is an enzyme of the pathway that synthesizes **bialaphos** forming the C-P bond in the phosphinothricin moiety. Example from *Streptomyces hygroscopicus*: database code BCPB_STRHY, 313 amino acids (33.66 kDa).

phosphoenolpyruvate-protein phosphotransferase EC 2.7.3.9; *systematic name*: phosphoenolpyruvate:protein-L-histidine *N*^ε-phosphotransferase; *other name* enzyme I of the **phosphoenolpyruvate-dependent sugar phosphotransferase system**; a soluble bacterial enzyme, part of a system for the transport of hexoses across the cell membrane, that catalyses the phosphorylation by phosphoenolpyruvate of a low-*M_r*, heat-stable protein, **HPr**. Example from *Escherichia coli*: database code PT1A_ECOLI, 711 amino acids (78.30 kDa).

phosphoester any [mono]ester of phosphoric acid; i.e. any phosphoric ester containing the group -PO(OH)₂ linked to an oxygen atom in a single other group.

phosphofructokinase 1 *abbr.*: PFK; EC 2.7.1.11; *recommended name*: 6-phosphofructokinase; *other names*: phosphofructokinase 1; phosphohexokinase. An enzyme that catalyses the phosphorylation of D-fructose 6-phosphate to D-fructose 1,6-bisphosphate at the expense of ATP, CTP, ITP, or UTP.

Phosphofructokinase 1 is an allosteric glycolytic enzyme; the molecule is a homotetramer. Example from *Escherichia coli*: database code K6P1_ECOLI, 320 amino acids (34.72 kDa); 3-D structure known; this is isoenzyme I; *E. coli* has a second entirely unrelated isoenzyme (II), presumably of independent evolution.

2 or phosphofructokinase 2 EC 2.7.1.105; *recommended name*: 6-phosphofructo-2-kinase; an enzyme that catalyses the phosphorylation of D-fructose 6-phosphate to D-fructose 2,6-bisphosphate at the expense of ATP. Phosphofructokinase 2 is the same protein as **fructose-2,6-bisphosphatase**; **phosphorylation** inhibits the kinase activity. Example from human (liver isoform; homodimer): database code F26L_HUMAN, 471 amino acids (54.64 kDa).

3 or fructose-1-phosphate kinase EC 2.7.1.56; *recommended name*: 1-phosphofructokinase; an enzyme that catalyses the phosphorylation of D-fructose 1-phosphate to D-fructose 1,6-bisphosphate at the expense of ATP, GTP, ITP, or UTP.

phosphoglucoisomerase an alternative name for **glucose-6-phosphate isomerase**.

phosphoglucomutase *abbr.* (in clinical chemistry): PGT; EC 5.4.2.2; *systematic name*: α-D-glucose 1,6-phosphomutase; *other name*: glucose phosphomutase. A **phosphotransferase** enzyme that catalyses the interconversion of α-D-glucose 1-phosphate and α-D-glucose 6-phosphate; α-D-glucose 1,6-bisphosphate is an intermediate, formed by phosphoryl transfer from the enzyme. It is a key enzyme linking the metabolism of polysaccharides to glycolysis. Example from *Saccharomyces cerevisiae*: database code PGMU_YEAST, 570 amino acids (63.04 kDa).

phosphogluconate 6-phospho-D-gluconate; an intermediate in the **Entner-Doudoroff pathway** and in the **pentose phosphate pathway**. It is a product of the combined action of **glucose-6-phosphate dehydrogenase** and **6-phosphogluconolactonase**, and is a substrate for **phosphogluconate dehydrogenase (decarboxylating)**. (Illustrated opposite.)

phosphogluconate dehydrogenase (decarboxylating) *abbr.*: 6PGD; EC 1.1.1.44; *systematic name*: 6-phospho-D-gluconate:NADP⁺ 2-oxidoreductase (decarboxylating); *other names*: phosphogluconic acid dehydrogenase; 6-phosphogluconic dehydrogenase; 6-phosphogluconic carboxylase. An enzyme that catalyses a reaction between 6-phospho-D-gluconate and NADP⁺ to form D-ribulose 5-phosphate, CO₂, and NADPH. It is an important enzyme in the **pentose phosphate pathway**, and is significant in maintaining NADP⁺ in the reduced state. Example from *Escherichia coli*: database code 6PGD_ECOLI, 468 amino acids (51.48 kDa).

phosphogluconate (oxidative) pathway an alternative name for **pentose phosphate pathway**.

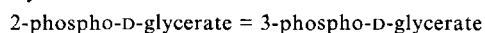
6-phosphogluconolactonase EC 3.1.1.31; *systematic name*: 6-phospho-D-glucono-1,5-lactone lactonohydrolase; an enzyme of the **pentose phosphate pathway** that accelerates the rate of hydrolysis of 6-phosphogluconolactone to 6-phosphogluconic acid.

phosphoglycerate 1 2-phospho-D-glycerate; D-glycerate 2-phosphate; (R)-2,3-dihydroxypropanoate 2-(dihydrogen phosphate). **2** 3-phospho-D-glycerate; D-glycerate 3-phosphate; (R)-2,3-dihydroxypropanoate 3-(dihydrogen phosphate). These phosphoglycerates are important intermediates in glycolysis. 3-phosphoglycerate is a precursor in serine biosynthesis. Compare **glycerophosphate**, **phosphoglyceride**. See also **phosphoglycerate mutase**.

phosphoglycerate dehydrogenase EC 1.1.1.95; *systematic name*: 3-phosphoglycerate:NAD⁺ 2-oxidoreductase; an enzyme that converts 3-phospho-D-glycerate to 3-phosphohydroxypyruvate in the presence of NAD⁺. An enzyme in the pathway for the synthesis of serine from 3-phosphoglycerate. Example from *Bacillus subtilis*: database code SERA_BACSU, 419 amino acids (45.60 kDa).

phosphoglycerate kinase *abbr.*: PGK; EC 2.7.2.3; *systematic name*: ATP:3-phospho-D-glycerate 1-phosphotransferase; an enzyme that catalyses the phosphorylation of 3-phospho-D-glycerate by ATP to 3-phospho-D-glyceroyl phosphate with release of ADP. Phosphoglycerate kinase is important in the glycolytic pathway, and exists in tissue-specific or (in protozoa, etc.) organelle-specific isoforms. Example from wheat chloroplast: database code PGKY_WHEAT, 401 amino acids (42.07 kDa).

phosphoglycerate mutase *abbr.*: PGAM; EC 5.4.2.1; *systematic name*: D-phosphoglycerate 2,3-phosphomutase; *other names*: phosphoglycerate phosphomutase; phosphoglyceromutase. An enzyme of the glycolytic and gluconeogenic pathways that catalyses the reaction:



(the forward direction, as written, being the direction in which the enzyme functions in gluconeogenesis). A phosphotransferase enzyme for which 2,3-bisphospho-D-glycerate acts as a reaction primer. The active enzyme has a phosphoryl group active site (histidine is phosphorylated as part of the mechanism) and transfers phosphoryl to the substrate. Example



phosphoglycerate phosphomutase

from *Saccharomyces cerevisiae*: database code PMGY_YEAST, 246 amino acids (27.45 kDa). *See also* **phosphohistidine**.

phosphoglycerate phosphomutase another name for **phosphoglycerate mutase**.

3-phospho-D-glyceric phosphoric monoanhydride *see* **3-phospho-D-glyceroyl phosphate**.

phosphoglyceride another (less correct) name for **glycerophospholipid**.

3-phospho-D-glyceroyl phosphate *abbr.*: D-glyceric-1,3- P_2 ; one of the recommended names for (R)-[2,3-dihydroxypropanoyl dihydrogen phosphate 3-(dihydrogen phosphate)], a key intermediate in the glycolytic pathway, commonly but incorrectly known as 1,3-diphosphoglycerate or 1,3-bisphosphoglycerate. *Alternative recommended names*: 3-phospho-D-glyceric phosphoric monoanhydride; (D-glyceroyl phosphate) 3-phosphate. This compound should be distinguished from **2,3-bisphospho-D-glycerate**.

phosphohexose isomerase an alternative name for **glucose-6-phosphate isomerase**.

phosphohistidine N^{γ} -phosphohistidine; it is formed as a residue in certain proteins; being an intermediate in the **phosphoglycerate mutase** reaction and the **phosphoenolpyruvate-dependent sugar phosphotransferase system**.

phosphoinositide another old term, no longer recommended, for **phosphatidylinositol** but now much used to denote all phosphatidylinositol derivatives as a class.

phosphoinositide cycle another term for **phosphatidylinositol cycle**.

phosphoketolase EC 4.1.2.9; a thiamine diphosphate-protein of the phosphoketolase pathway that catalyses phosphorylation of D-xylulose 5-phosphate and orthophosphate to acetyl phosphate and D-glyceraldehyde 3-phosphate.

phosphoketolase pathway any metabolic pathway, usually in a microorganism, that involves the enzyme **phosphoketolase**, e.g. in **heterolactic fermentation** or in the fermentation of glucose by *Bifidobacterium* spp.

phosphokinase a former term for **kinase**.

phospholamban a homopentameric sarcoplasmic reticulum membrane protein that regulates the calcium pump of cardiac, slow-twitch skeletal, and smooth muscle. It is phosphorylated by protein kinase A in response to β -adrenergic stimulation. The phosphorylated protein increases the rate of the sarcoplasmic reticulum Ca^{2+} -ATPase. Example from human: database code PPLA_HUMAN, 52 amino acids (6.10 kDa).

phospholemman a 72-residue cell-membrane protein that forms a chloride-selective ion channel. It is a substrate for both protein kinase C and cyclic AMP-dependent protein kinase. It is rapidly phosphorylated after β -adrenergic and α -adrenergic stimulation of the heart and in response to insulin in muscle. It has a single transmembrane domain. The structure of a related protein, Mat-8, expressed in human breast tumours, is known: database code A55571, 87 amino acids.

phospholipase any enzyme that is a lipase and that catalyses the hydrolysis of a glycerophospholipid. Phospholipases have been subdivided as follows.

Phospholipase A₁ EC 3.1.1.32; *the recommended name* for any carboxylic ester hydrolase that removes the acyl residue at C-1 of the glycerol moiety.

Phospholipase A₂ *the recommended name* for any carboxylic ester hydrolase that removes the acyl residue at C-2 of the glycerol moiety; e.g. EC 3.1.1.4; *systematic name*: phosphatidylcholine 2-acylhydrolase; *other names*: lecithinase A; phosphatidase; phosphatidolipase. The enzyme requires calcium. Snake-venom enzymes have been much studied and have broad specificity for phosphoglyceride classes. There are two such mammalian enzymes, the pancreatic type and the membrane-bound type. The pancreatic enzyme is released from the exocrine pancreas on stimulation by cholecystokinin, has broad specificity for phosphoglycerides, and hydrolyses phospholipids of digested food in the intestine in conjunction with the action of bile salts to form micelles. The membrane-bound

enzyme is instrumental in releasing arachidonate for prostanoic synthesis, and is in many cases stimulated by activated hormone and growth-factor receptors. Example, precursor from human (pancreatic enzyme): database code PA2M_HUMAN, 144 amino acids (16.06 kDa), 3-D structure known. **Phospholipase B** EC 3.1.1.5; *recommended name*: lysophospholipase; *other names*: lecithinase B; lysolecithinase. These are enzymes that remove the acyl group from a lysoglycerophospholipid.

Phospholipase C *abbr.*: PLC; *the recommended name* for any phosphoric diester hydrolase that splits the bond between the phosphorus atom and the oxygen atom at C-1 of the glycerol moiety; *other names*: lipophosphodiesterase I; lecithinase C; *Clostridium welchii* α -toxin. Two major types have been studied; one is a bacterial type, which acts on phosphatidylcholine, releasing diacylglycerol (EC 3.1.4.3); there is also a mammalian enzyme of this type. The other type, **1-phosphatidylinositol phosphodiesterase** (EC 3.1.4.10) acts on phosphoinositides; it is found in bacteria and animals, and catalyses the conversion of 1-phosphatidyl-D-myo-inositol to D-myo-inositol 1,2-cyclic phosphate (which is hydrolysed in animals but not bacteria to inositol 1-phosphate) and diacylglycerol; **1-phosphatidylinositol-4,5-bisphosphate phosphodiesterase** (EC 3.1.4.11; *other names*: triphosphoinositide phosphodiesterase; phosphoinositide-specific phospholipase C) catalyses the hydrolysis of 1-phosphatidyl-D-myo-inositol 4,5-bisphosphate to D-myo-inositol 1,4,5-trisphosphate and diacylglycerol. These enzymes play a role in signal transduction. There are at least six forms of the latter, which is required for production of the second messenger molecules **diacylglycerol** and **inositol 1,4,5-trisphosphate**. The activation involves a G_{α} **G-protein** subunit. Example, PLC β_2 from human: database code PIP2_HUMAN, 1181 amino acids (133.53 kDa).

Phospholipase D EC 3.1.4.4; *the recommended name* for any phosphoric diester hydrolase that splits the bond between the phosphorus atom of a glycerophospholipid and the oxygen atom of the nitrogenous base; *other names*: lipophosphodiesterase II; lecithinase D; choline phosphatase.

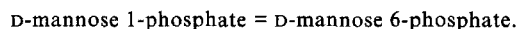
phospholipid or (formerly) **phospholipide** any lipid containing phosphoric acid as mono- or diester.

phospholipid-cholesterol acyltransferase *see* **phosphatidylcholine-sterol O-acyltransferase**.

phospholipid exchange protein any of a number of cytosolic proteins that bind phospholipids and transport them within the cell. Each type of protein exhibits a degree of specificity for the phospholipid bound, and each protein molecule carries one or two phospholipid molecules. These proteins function to exchange phospholipid classes between different intracellular membrane sites.

phospholipid translocase a type of protein that accelerates the transfer of phospholipid (*see* **flip-flop**) between the inner and outer lipid leaflets of the plasma membrane bilayer.

phosphomannomutase EC 5.4.2.8; *systematic name*: D-mannose 1,6-phosphomutase; an enzyme that catalyses the reaction:



One of its functions is to convert mannose 6-phosphate to mannose 1-phosphate, from which GDPmannose is synthesized; this is a precursor of **fucose**, and is an intermediate in the synthesis of the **mannose** core of oligosaccharides. Example from *Escherichia coli*: database code CPSG_ECOLI, 456 amino acids (50.40 kDa).

phosphomonoester any monoester of phosphoric acid.

phosphomonoesterase 1 an alternative name for **phosphatase 2** an alternative name for either **acid phosphatase** or **alkaline phosphatase**.

phosphonic acid *see* **phosphono+**.

phosphono+ *comb. form* denoting a phosphonic residue, $-PO(OH)_2$, derived from **phosphonic acid**, $HPO(OH)_2$. In biochemical usage this name is usually reserved for the $-PO(OH)_2$

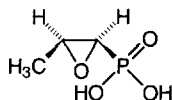


group attached to a carbon atom, as in **phosphonoacetate**, (HO)₂OP-CH₂COO⁻; and **2-phosphonoethylamine**, or **ciliatine**, (HO)₂OP-CH₂CH₂NH₂; the group is commonly termed phospho⁺ when linked to a heteroatom, as in phosphocreatine and phosphoglucanate.

N^w-phosphonoarginine *see* **phosphoarginine**.

N^w-phosphonocreatine *see* **phosphocreatine**.

phosphonomycin or **fosfomycin** (1*R*,2*S*)-(1,2-epoxypropyl)phosphonate; an antibiotic produced by *Streptomyces* and *Pseudomonas* strains. It binds irreversibly to UDP-*N*-acetylglucosamine 1-carboxyvinyl transferase (EC 2.5.1.7), an enzyme catalysing the first stage in the synthesis of bacterial cell-wall peptidoglycan from UDP-*N*-acetylglucosamine, which it thereby inhibits. It penetrates the cell by utilizing the L- α -glycerophosphate transport system. Loss of this system is a primary reason that resistant strains emerge.



phosphonopeptide any of a series of oligopeptide derivatives, designed as inhibitors of bacterial cell-wall biosynthesis, that have a C-terminal residue of L-1-aminoethylphosphonic acid (a phosphorus-containing analogue of L-alanine).

phosphopentomutase EC 5.4.2.7; *systematic name*: D-ribose 1,5-phosphomutase; *other name*: phosphodeoxyribomutase. An enzyme that catalyses the reaction:

D-ribose 1-phosphate = D-ribose 5-phosphate.

The partial sequence is known from *Escherichia coli*: database code DEOB_ECOLI.

phosphoprotein any protein containing one or more phosphoric residues directly attached, usually by ester linkage, to amino-acid residues.

phosphoprotein phosphatase 1 EC 3.1.3.16; *systematic name*: phosphoprotein phosphohydrolase; any enzyme that removes a phosphate group from a protein by hydrolysis. Many different enzymes have been identified with varying specificity with respect to the primary structure of the substrate protein. *See* **protein phosphatase**. 2 *another name for* **serine/threonine-specific protein phosphatase**.

phosphor any substance (capable of) exhibiting **phosphorescence** (def. 1).

phosphor+ a variant form of **phosphoro+** (before a vowel).

phosphoramidate or **amidophosphate** or **phosphoric amide** 1 any ion, salt, or ester of phosphoramidic acid; amidophosphoric acid; (OH)₂PO-NH₂. 2 any compound containing the group (OH)₂PO-NH- or (OH)₂PO-N<.

phosphoramidite any compound having the general structure R₃R₄NP(R₂)-O-R₁. Phosphoramidites are useful in the synthesis of oligonucleotides or for the addition of groups to the 5'-phosphate of nucleotides or oligonucleotides. R₁ is often cyanoethyl-; R₃ and R₄ are often isopropyl with R₂ in 3'-nucleotide linkage.

phosphoramidon *N*-(α -L-rhamnopyranosyl-oxyphospho)-L-leucyl-L-tryptophan; a specific inhibitor of **thermolysin**. It is an inhibitor of the metalloendopeptidase, EC 3.4.24.11, that is responsible for cleaving and inactivating **natriuretic peptide**, with corresponding pharmacological effects.

phosphoresce to exhibit **phosphorescence** (def. 1).

phosphorescence 1 a type of luminescence consisting of the emission by a substance of electromagnetic radiation, especially visible light, as a result of the absorption of energy derived from exciting radiation of shorter wavelength or from incident subatomic particles (especially electrons or alpha particles), and that persists for more than approximately 10 ns after excitation ceases, the persistence being a consequence of the excited electrons decaying via a **metastable** (def. 2) state.

the property of emitting such radiation. 2 the radiation so emitted. 3 a common but erroneous term for **bioluminescence** (which occurs in the absence of exciting radiation). *Compare* **fluorescence**. —**phosphorescent** *adj.*

phosphorescence spectrophotometry measurement of the wavelengths and the intensities of the light emitted from a phosphorescent sample that is excited by monochromatic (or nearly so) exciting radiation.

phosphori+ a variant form of **phosphoro+** (def. 3).

phosphoriboisomerase *see* **ribose-5-phosphate epimerase**.

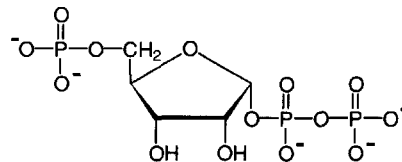
phosphoribosylamine-glycine ligase EC 6.3.4.13; *other names*: phosphoribosylglycinamide synthetase; glycylamide ribonucleotide synthetase (*abbr.*: GARS); an enzyme of the pathway for *de novo* purine biosynthesis that catalyses the reaction between ATP, 5'-phosphoribosylamine, and glycine to form 5'-phosphoribosylglycinamide (glycinamide ribotide, *abbr.*: GAR) with release of ADP and orthophosphate. Example from *Bacillus subtilis*: database code PUR2_BACSU, 422 amino acids (45.23 kDa). Example from human: database code PUR2_HUMAN, 1010 amino acids (107.64 kDa). This multifunctional protein has domains for three proteins required for purine biosynthesis (amino-acid numbers in brackets): GARS (1-433), **phosphoribosylformylglycinamide cycloligase** (434-809), and **phosphoribosylglycinamideformyl-transferase** (810-1010), although variations with only one or two of the domains occur through alternative splicing.

phosphoribosylaminoimidazolecarboxamide formyltransferase EC 2.1.2.3; *other name*: AICAR transformylase; inosinicase; IMP synthetase; an enzyme of the pathway for *de novo* purine biosynthesis that catalyses the formation of 5'-phosphoribosyl-5-formamido-4-imidazolecarboxamide from 10-formyltetrahydrofolate and 5'-phosphoribosyl-5-amino-4-imidazolecarboxamide (5-aminoimidazole-4-carboxamide ribotide; *abbr.*: AICAR) forming also tetrahydrofolate. Example, multifunctional protein from chicken having activities for AICAR transformylase and **IMP cyclohydrolase**: database code PUR9_CHICK, 593 amino acids (64.34 kDa). The IMP cyclohydrolase activity resides in the C-terminal region.

phosphoribosylaminoimidazole carboxylase EC 4.1.1.21; *other name*: AIR carboxylase; an enzyme of the pathway for *de novo* purine biosynthesis that catalyses the formation of 1-(5'-phosphoribosyl)-5-amino-4-imidazolecarboxylate from 1-(5'-phosphoribosyl)-5-aminoimidazole (5'-aminoimidazole ribotide, *abbr.*: AIR) and CO₂. The reaction requires neither biotin nor ATP. For an example, *see* **multifunctional protein ade2**.

phosphoribosylaminoimidazolesuccinocarboxamide synthase EC 6.3.2.6; *other name*: SAICAR synthetase; an enzyme that catalyses the formation of 1-(5'-phosphoribosyl)-4-(*N*-succinocarboxamide)-5-aminoimidazole (*N*-succinyl-5-aminoimidazole-4-carboxamide ribonucleotide; *abbr.*: SAICAR) from ATP, 1-(5'-phosphoribosyl)-4-carboxy-5-aminoimidazole and L-aspartate with release of ADP and orthophosphate. The reaction forms part of the pathway for the *de novo* synthesis of purines. For an example, *see* **multifunctional protein ade2**.

(5-)-phospho-(α -D-)-ribose diphosphate *symbol*: PPRibP or PRibPP; 5-phospho- α -D-ribofuranosyl diphosphate; α -D-ribose 1-diphosphate 5-phosphate; a compound that is essential in the biosynthesis of various nitrogen-containing ring compounds or their derivatives, especially of histidine, and of purines and pyrimidines as their nucleotides. It is commonly but incorrectly known as phosphoribosyl pyrophosphate (*abbr.* (not recommended): PRPP).



Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



phosphoribosylformylglycinamide cyclo-ligase

phosphoribosylformylglycinamide cyclo-ligase EC 6.3.3.1; *other name*: phosphoribosyliminoamidazole synthetase; an enzyme that catalyses the reaction between ATP and 2-formamido-*N*¹-(5-phosphoribosyl)acetamide to form 1-(5-phosphoribosyl)-5-aminoimidazole with release of ADP and orthophosphate. Example from rabbit: database code A53688, 201 amino acids (22.71 kDa).

phosphoribosylformylglycinamide synthase EC 6.3.5.3; *other names*: phosphoribosylformylglycinamide synthetase; FGAM synthase; an enzyme of the pathway for *de novo* purine biosynthesis that catalyses the formation of 5'-phosphoribosylformylglycinamide from ATP, 5'-phosphoribosylformylglycinamide, L-glutamine, and H₂O with release of ADP, orthophosphate, and L-glutamate. Example from yeast: database code PUR4_YEAST, 1360 amino acids (149.15 kDa).

phosphoribosylglycinamide formyltransferase EC 2.1.2.2; *other names*: 5'-phosphoribosylglycinamide transferase; GAR transferase (*abbr.*: GART); an enzyme of the pathway for *de novo* purine biosynthesis that catalyses the formation of 5'-phosphoribosyl-*N*-formylglycinamide (formylglycinamide ribotide; *abbr.*: FGAR) from 10-formyltetrahydrofolate and 5'-phosphoribosylglycinamide (glycinamide ribotide, *abbr.*: GAR) yielding also tetrahydrofolate. Example, multifunctional protein from chicken for **phosphoribosylamine-glycine ligase** (GARS)/**phosphoribosylformylglycinamide cyclo-ligase** (AIRS)/GART: database code PUR2_CHICK, 1003 amino acids (106.42 kDa).

phosphoribosyl pyrophosphate *abbr.*: PRPP; a common name for (5)-phospho- α -D-ribose diphosphate.

phosphoribosyl pyrophosphate synthetase (*abbr.*: PRS) *see* ribose-phosphate pyrophosphokinase.

phosphoribosyltransferase EC 2.4.2.7–2.4.2.12, 2.4.2.17; any of a group of enzymes that catalyse the biosynthesis of a mononucleotide from 5-phospho- α -D-ribose diphosphate (PRPP) in an alternative (or salvage) pathway for nucleoside biosynthesis. They catalyse a reaction between a base and 5-phospho- α -D-ribose diphosphate to form a D-ribonucleotide and pyrophosphate; the base may be (number indicates no. in sub-subclass EC 2.4.2) adenine (7), hypoxanthine (8), uracil (9), orotate (10), nicotinate (11), nicotinamide (12), or ATP (17). Examples: (1) hypoxanthine/guanine phosphoribosyltransferase (HGPRT) from human: database code HPRT_HUMAN, 217 amino acids (24.42 kDa); (2) ATP phosphoribosyltransferase, involved in histidine biosynthesis, from *Escherichia coli*: database code HIS1_ECOLI, 299 amino acids (33.33 kDa). Other phosphoribosyltransferases of sub-subclass EC 2.4.2 involve the following (no. in sub-subclass EC 2.4.2): anthranilate (18), dioxotetrahydropyrimidine (20), and xanthine (22).

phosphoric *symbol*: P or (in the one-letter convention for nucleotides) p; *the general name* for a residue of [ortho]phosphoric acid – i.e. tetraoxophosphoric(v) acid, PO(OH)₃ – whether singly, doubly, or triply attached to other groups and whether or not any residual hydroxyl groups are deprotonated.

phosphorimetry the measurement of phosphorescence.

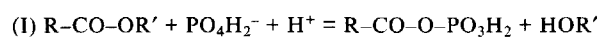
phosphoro+ or (*before a vowel*) **phosphor+** *comb. form* 1 phosphorus. 2 [ortho]phosphoric acid. 3 or **phosphori+** phosphorescence.

phosphoroclastic describing a chemical reaction in which a carbon-carbon bond is split apparently by the action of inorganic phosphate with attachment of a phosphoric group to one of the resulting fragments.

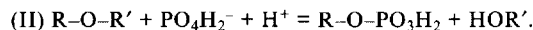
phosphorolysis any reversible lytic process or reaction undergone by an acyl compound (I) or glycosyl compound (II) in which nucleophilic displacement by phosphate (anion) occurs, with formation of the corresponding acyl or sugar phosphate and uptake of a hydron by the other moiety of the original compound, as in the generalized equations:

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



or



phosphorolytic of, or relating to, phosphorolysis; capable of undergoing phosphorolysis.

phosphorous 1 of, containing, or resembling phosphorus. 2 a residue of phosphorous acid – i.e. trioxophosphoric(III) acid, P(OH)₃ – whether singly or doubly attached to other groups and whether or not any residual hydroxyl groups are deprotonated.

phosphorus *symbol*: P; a nonmetallic element of group 15 of the (IUPAC) **periodic table**; atomic number 15; relative atomic mass 30.974; it exists naturally as a single stable nuclide, **phosphorus-31**. Phosphorus occurs only in the combined state, almost exclusively as phosphates. Its commonest covalencies are III and V. Phosphorus compounds are vital components of all living materials. *See also* **phosphorus-32**, **phosphorus-33**.

phosphorus-31 the stable nuclide ³¹P; its relative abundance in natural phosphorus is 100 atom percent.

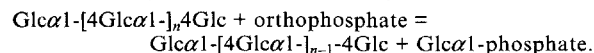
phosphorus-32 the artificial radioactive nuclide ³²P. It emits a relatively high-energy electron or β^- particle (1.709 MeV max.) and no gamma radiation. It has a half-life of 14.3 days.

phosphorus-33 the artificial radioactive nuclide ³³P. It emits a moderate-energy electron or β^- particle (0.249 MeV max.) and no gamma radiation. It has a half-life of 25.4 days.

phosphorus:oxygen ratio *abbr.*: P:O ratio; the number of atoms of inorganic phosphorus (as phosphate) incorporated into organic compounds (usually ATP) per atom of dioxygen consumed during **oxidative phosphorylation**. The value observed depends on the nature of the substrate being oxidized and the state of preservation of mitochondrial function. If the latter is high, a P:O ratio near 3 is observed during the oxidation of NADH, near 2 for succinate, fatty acyl CoA, 3-phosphoglycerate. *See also* **chemiosmotic coupling hypothesis**, **tight coupling**.

phosphoryl 1 the trivalent group –P(O)= derived from phosphoric acid. 2 a name used frequently for the univalent group –P(O)(OH)₂; this usage is not recommended; a phosphoric ester may be named, e.g., choline *O*-(dihydrogenphosphate) or *O*-phosphonocholine. However, phosphoryl is accepted except in derived terms such as the names of enzymes (e.g. phosphorylase) or processes (e.g. phosphorylation). *See also* **phosphono+**.

phosphorylase 1 the name recommended for any of various enzymes, classified together as EC 2.4.1.1, that catalyse the phosphorolytic removal of the nonreducing terminal glucose residue from a glucan according to the general reaction:



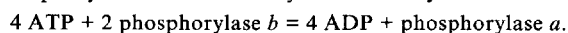
These enzymes vary somewhat in regard to their preferred substrate; the name should be qualified in each instance by prefixing to it the name of the natural substrate (e.g. starch phosphorylase). Glycogen phosphorylase exists in two forms: **phosphorylase a**, the more active, phosphorylated form; and **phosphorylase b** (or dephosphophosphorylase), the less active, unphosphorylated form. The conversion of *b* to *a* is effected with ATP by the enzyme **phosphorylase kinase**, and the conversion of *a* to *b* is brought about by the enzyme **phosphorylase phosphatase** (EC 3.1.3.17). Example from *Rattus norvegicus*, glycogen phosphorylase liver form: database code PHS1_RAT, 850 amino acids (97.48 kDa). 2 a generic name recommended for those glycosyltransferase enzymes within subclass EC 2.4 that utilize orthophosphate as acceptor. Some appear to be specific, e.g. maltose phosphorylase (EC 2.4.1.8) or uridine phosphorylase (EC 2.4.2.3), others less so, e.g. 1,3- β -oligoglucan phosphorylase (EC 2.4.1.30). *See also* **α -glucan phosphorylase**.

phosphorylase kinase EC 2.7.1.38; *systematic name*: ATP-



phosphorylase-limit dextrin

phosphorylase-*b* phosphotransferase; *other name*: dephosphophosphorylase kinase. An enzyme that catalyses the reaction:



Its activity is regulated by Ca^{2+} /calmodulin and by covalent modification, as a result of phosphorylation by **cyclic AMP-dependent protein kinase** or dephosphorylation by **protein phosphatase 1**. The enzyme has the subunit structure $\alpha_4\beta_4\gamma_4\delta_4$; of these γ is the catalytic chain, α and β are regulatory, and δ is a synonym for **calmodulin**. Example, γ chain (human): database code KPBH_HUMAN, 406 amino acids (46.39 kDa).

phosphorylase-limit dextrin the polysaccharide produced by the action of starch phosphorylase on the branched polysaccharide **amylopectin**.

phosphorylase phosphatase EC 3.1.3.17; an enzyme that catalyses the hydrolysis of phosphorylase *a* to two molecules of phosphorylase *b* and four of orthophosphate, thereby converting the phosphorylase to a less active form. *See also* **protein phosphatase**.

phosphorylate to effect **phosphorylation**.

phosphorylation the act or process of introducing a **phosphoric** group into a molecule, usually with the formation of a phosphoric ester, a phosphoric anhydride, or a phosphoric amide. Biochemical phosphorylation reactions are of importance in the trapping of energy and the formation of biosynthetic intermediates during metabolic processes and in the control of the activity of numerous enzymes and other proteins. *See also* **oxidative phosphorylation**, **photophosphorylation**, **substrate-level phosphorylation**.

phosphorylation state ratio *symbol*: R_p ; a measure of the phosphorylating power of the adenylate system within cells. It is given by:

$$R_p = [\text{ATP}]/[\text{ADP}][\text{P}_i].$$

Compare (adenylate) **energy charge**.

phosphorylcholine-glyceride transferase *see* **diacylglycerol cholinephosphotransferase**.

phosphorylcholine transferase *see* **cholinephosphate cytidylyltransferase**.

phosphorylethanolamine transferase *see* **ethanolamine-phosphate cytidylyltransferase**.

phosphoryn the major phosphate-rich protein of **dentine**. It contains about 40 residue % of phosphoserine and 35 residue % of aspartic acid. It is isolated from incisors and is involved in bone mineralization. Soluble forms and a form complexed with collagen in dentine are known.

phosphoserine *O*-phospho-L-serine; a phosphoamino acid found in peptide linkage in proteins. Phosphorylation of serine and threonine residues is catalysed by protein kinases specific for serine or threonine residues, and phosphorylation and dephosphorylation mechanisms have a regulatory role in the protein function. *See also* **protein kinase**, **protein phosphatase**.

3-phosphoshikimate 1-carboxyvinyltransferase EC 2.5.1.19; *systematic name*: phosphoenolpyruvate:3-phosphoshikimate 5-*O*-(1-carboxyvinyl)-transferase; *other name*: 5-enolpyruvyl shikimate-3-phosphate (EPSP) synthase; an enzyme of the **shikimate pathway** that catalyses the formation of *O*⁵-(1-carboxyvinyl)-3-phosphoshikimate (5-enolpyruvyl shikimate 5-phosphate) from phosphoenolpyruvate and 3-phosphoshikimate with release of orthophosphate. *See* **pentafunctional arom polypeptide**.

phosphosphingolipid *see* **sphingolipid**.

phosphothreonine *O*-phospho-L-threonine; a phosphoamino acid found in peptide linkage in proteins. Phosphorylation of serine and threonine residues is catalysed by protein kinases specific for serine or threonine residues, and phosphorylation and dephosphorylation mechanisms have a regulatory role in the protein function. *See also* **protein kinase**, **protein phosphatase**.

phosphotransferase any of the enzymes of sub-subclasses EC 2.7.1 to 2.7.4, and EC 2.7.9; they catalyse transfer of a phosphate group from a donor to an acceptor, and are classified

according to whether the transfer is to an alcohol group (EC 2.7.1), a carboxyl group (EC 2.7.2), a nitrogenous group (EC 2.7.3), a phosphate group (EC 2.7.4) or to paired acceptors (EC 2.7.9 – only two of these are currently classified, transferring phosphate from ATP to both pyruvate and orthophosphate in one case and to pyruvate and water in the other, yielding AMP in both instances). Enzymes of sub-subclass EC 2.7.6 that transfer diphosphate are classified as **diphosphotransferases**.

phosphotransferase system a system for transferring substrates into bacteria (*see* **phosphoenolpyruvate-dependent sugar phosphotransferase system**).

phosphotriester any triester of phosphoric acid.

phosphotriester method (of oligonucleotide synthesis) a method for the chemical synthesis of oligonucleotides in which a suitably protected phosphodiester of general formula Nuc-3'-O-P(O)(OR)OH, where R is an aryl group and Nuc is a residue of an appropriate nucleoside, is first condensed with the 5'-hydroxyl group of another nucleoside (usually anchored through its 3'-hydroxyl group to an insoluble support) to form a phosphotriester. Sequential addition of further nucleoside units is then carried out in the same way, the 5'-protecting group being removed at each stage before the next cycle of reactions. All remaining protecting groups are removed at the completion of the synthesis.

phosphotyrosine *O*-phospho-L-tyrosine; a phosphoamino acid found in peptide linkage in proteins, especially in proteins involved in effector mechanisms in growth stimulation, including growth-factor factor receptors. *See also* **protein kinase**, **protein phosphatase**.

phosphovitin or **phosvitin** a phosphoprotein (subunit size 25 kDa) that, together with **lipovitellin**, constitutes the major part of the mass of the egg-yolk proteins in oviparous vertebrates. It has the highest phosphorus content of all known phosphoproteins (10% P) and approximately half of its amino-acid residues are serine, all of which are *O*-phosphorylated. Phosphovitin and lipovitellin are derived from a common precursor protein, **vitellogenin**.

phot the cgs unit of illuminance, equal to one lumen per square centimetre; 1 phot = 10^4 lux.

photo+ a variant form of **photo+** (sometimes before a vowel).

photagogikon (*pl.* **photagogika**) the light-emitting compound of any luminescent system regardless of binding of the compound or of its state of ionization. *See also* **photogen**.

photic 1 of, or relating to, light. 2 of, or relating to, the reaction to light of, or the production of light by, living organisms. 3 describing the uppermost zone in a body of water that receives sufficient light to permit photosynthesis. —**photically** *adv.*

photo+ or (sometimes before a vowel) **phot+** *comb. form* denoting 1 of, pertaining to, produced by, or responding to light. 2 involving a photographic process.

photoactivate to render active or reactive by means of light. —**photoactivation** *n.*; **photoactivatable** or **photoactivable** *adj.*

photoactive (*of a substance*) capable of giving a physical or chemical response to light or other electromagnetic radiation.

photoaddition any chemical addition reaction promoted by light or other electromagnetic radiation.

photoaffinity label or **photolabel** an **affinity label** carrying a separate photoactivatable group that on illumination reacts with the protein being labelled to form a covalent link.

photoautotroph or **photolithotroph** any organism that is both an **autotroph** and a **phototroph**. —**photoautotrophic** *adj.*; **photoautotrophy** *n.*

photobiochemistry the photochemistry of biochemical processes. —**photobiochemical** *adj.*; **photobiochemically** *adv.*; **photobiochemist** *n.*

photobiology the study of the effects of radiation, mainly of visible and ultraviolet light, on living organisms. —**photobiological** *adj.*; **photobiologist** *n.*

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



photobleaching

photobleaching the loss of colour or fluorescence through the action of incident visible or near-ultraviolet radiation.

photocatalysis 1 the acceleration of a chemical reaction by light. 2 the catalysis of a photochemical reaction. —**photocatalytic adj.**; **photocatalytically adv.**

photocathode (*in an electronic device*) a cathode from which electrons are emitted when exposed to electromagnetic radiation of suitable frequency.

photocell or **photoelectric cell** any type of transducer capable of converting incident electromagnetic radiation to an electric signal of corresponding magnitude. It may function by generating a potential difference or a current, or by changing its conductance.

photochemical action spectrum the relationship between the efficiency of a **photochemical reaction** and the frequency of the illuminating light.

photochemical reaction any chemical reaction that is initiated or accelerated by the absorption of visible or near-ultraviolet radiation.

photochemistry the branch of chemistry concerned with the effect of electromagnetic, especially visible and ultraviolet, radiation on chemical processes and with the uptake or production of photons during chemical reactions. —**photochemical adj.**; **photochemically adv.**

photochromism or **phototropy** the property possessed by certain substances of changing colour reversibly when exposed to electromagnetic radiation of appropriate frequency. *Compare electrochromism, thermochromism.* —**photochromic adj.**

photoconductive describing an electrical device that is capable of changing its conductance on exposure to light or other electromagnetic radiation. —**photoconductivity n.**

photodamage deleterious chemical change caused by the action of light, especially ultraviolet light.

photodensitometer *see* **densitometer** (def. 1).

photodetector a device, e.g. a **photocell**, that responds to incident light.

photodimerization or **photodimerisation** the formation of dimers by a photochemical reaction. It is of particular importance in the formation of **pyrimidine dimers** in DNA. —**photodimerize** or **photodimerise vb.**

photodiode any semiconductor **diode** (def. 1) that generates a potential difference or changes its conductivity when exposed to light.

photodissociation the splitting of a molecule into smaller molecules, radicals, or atoms on absorption of a quantum of electromagnetic radiation.

photodynamic involving, causing, or promoting a deleterious reaction to light, particularly ultraviolet light. Such reactions usually involve oxidation. Damage is caused by **singlet oxygen** produced, e.g., by illumination of chlorophyll in chloroplasts. Normally the singlet oxygen is quenched by the carotenoid pigments coexisting with the chlorophylls in chloroplasts. Hence, any mutation that deletes carotenoids from chloroplasts is lethal, because the chlorophylls are destroyed by singlet oxygen.

photoelectric of, pertaining to, or concerned with electric or electronic effects caused by electromagnetic radiation, especially light.

photoelectric cell *an alternative name for* **photocell**.

photoelectron an electron that is ejected from an atom on interaction of the atom with a photon of sufficient energy.

photoelectron micrograph any photograph of an object taken by **photoelectron microscopy**.

photoelectron microscopy an electron-microscopic technique in which **photoelectrons** emitted from a specimen exposed to ultraviolet radiation are accelerated and focused to produce an image. It is the counterpart in the electron microscope of fluorescence microscopy in the light microscope, the resolution being correspondingly enhanced.

photoelectron spectroscopy a technique in which a specimen (a gas sample or the surface of a solid or a liquid) is exposed to monochromatic electromagnetic radiation in the ultraviolet or X-ray regions and the energy spectrum of the emitted **photoelectrons** recorded. This provides information about the atomic composition of the specimen and also about the types of orbitals to which the constituent electrons of its atoms and molecules are assigned.

photoemission the emission of electrons from a surface as a result of bombardment by photons.

photoexcitation the act or process of raising an atom or molecule to a higher energy state by interaction with a photon.

photogen 1 *an older term for* **photagogikon**. 2 *a term proposed for the specific biochemical precursor of a photagogikon.*

photographic emulsion or **emulsion** any medium that is sensitive to light or other electromagnetic radiation, used for preparing photographic film, paper, etc., and consisting usually of a suspension of a silver salt (especially the bromide or chloride) in gelatin; a coating of such a suspension applied to a substrate.

photoheterotroph or **photoorganotroph** any organism that is both a **heterotroph** and a **phototroph**. —**photoheterotrophic adj.**; **photoheterotrophy n.**

photoinactivation the irreversible abolition of specific biological activity by the action of light.

photoinhibition the reversible abolition of specific biological activity by the action of light.

photoionization or **photoionisation** the ionization of a molecular entity by the loss of an electron on interaction of the entity with a photon of sufficient energy.

photokinesis any change in the undirected movement of an organism in response to variation in light intensity. —**photokinetic adj.**

photolabel *an alternative term for* **photoaffinity label**.

photolabile (of a substance) unstable in the presence of light.

photolithotroph *an alternative term for* **photoautotroph**. —**photolithotrophic adj.**; **photolithotrophy n.**

photoluminescence luminescence as a result of absorption of electromagnetic radiation. The emitted light is always of lower frequency than that of the incident radiation. —**photoluminescent adj.**; **photoluminesce vb.**

photolyse or (*sometimes, US*) **photolyze** to subject to or to undergo **photolysis**.

photolysis the cleavage of one or more covalent bonds in a molecular entity resulting from absorption of energy from light or other electromagnetic radiation; any photochemical process in which such a cleavage is an essential part, e.g. laser photolysis; flash photolysis. —**photolytic adj.**; **photolytically adv.**

photolyse *a US spelling of* **photolysis**.

photometer any instrument used to measure the intensity of light or other electromagnetic radiation; includes **spectrophotometer**.

photometry the measurement of the intensity of light or other electromagnetic radiation; it includes spectrophotometry (*see* **spectrophotometer**). —**photometric adj.**; **photometrically adv.**

photomicrograph a photograph taken through a light microscope; it is sometimes, incorrectly, termed a microphotograph. —**photomicrographic adj.**; **photomicrographically adv.**; **photomicrography n.**

photomultiplier a photocell consisting of a **photocathode** linked to an **electron multiplier**. Photons striking the photocathode cause the emission of electrons, which are then amplified by the electron multiplier to give an enhanced signal.

photon *symbol (in nuclear reactions, etc.): γ* ; a quantum of light (or other electromagnetic radiation) considered as a particle with zero rest mass, zero charge, and energy $h\nu$ joule, where h is the Planck constant and ν the frequency of the radiation in hertz.

photon counter a device, consisting of a photomultiplier and an associated scaler, that is capable of responding to and recording individual photons.

photon *symbol (in nuclear reactions, etc.): γ* ; a quantum of light (or other electromagnetic radiation) considered as a particle with zero rest mass, zero charge, and energy $h\nu$ joule, where h is the Planck constant and ν the frequency of the radiation in hertz.

photon counter a device, consisting of a photomultiplier and an associated scaler, that is capable of responding to and recording individual photons.

photon *symbol (in nuclear reactions, etc.): γ* ; a quantum of light (or other electromagnetic radiation) considered as a particle with zero rest mass, zero charge, and energy $h\nu$ joule, where h is the Planck constant and ν the frequency of the radiation in hertz.

photon counter a device, consisting of a photomultiplier and an associated scaler, that is capable of responding to and recording individual photons.

photon *symbol (in nuclear reactions, etc.): γ* ; a quantum of light (or other electromagnetic radiation) considered as a particle with zero rest mass, zero charge, and energy $h\nu$ joule, where h is the Planck constant and ν the frequency of the radiation in hertz.

photon counter a device, consisting of a photomultiplier and an associated scaler, that is capable of responding to and recording individual photons.

photon *symbol (in nuclear reactions, etc.): γ* ; a quantum of light (or other electromagnetic radiation) considered as a particle with zero rest mass, zero charge, and energy $h\nu$ joule, where h is the Planck constant and ν the frequency of the radiation in hertz.

photon counter a device, consisting of a photomultiplier and an associated scaler, that is capable of responding to and recording individual photons.

photon *symbol (in nuclear reactions, etc.): γ* ; a quantum of light (or other electromagnetic radiation) considered as a particle with zero rest mass, zero charge, and energy $h\nu$ joule, where h is the Planck constant and ν the frequency of the radiation in hertz.

photon counter a device, consisting of a photomultiplier and an associated scaler, that is capable of responding to and recording individual photons.

photon *symbol (in nuclear reactions, etc.): γ* ; a quantum of light (or other electromagnetic radiation) considered as a particle with zero rest mass, zero charge, and energy $h\nu$ joule, where h is the Planck constant and ν the frequency of the radiation in hertz.

photon counter a device, consisting of a photomultiplier and an associated scaler, that is capable of responding to and recording individual photons.



photonicking

photonicking photolysis of a nucleic-acid molecule with production of a nick.

photoorganotroph an alternative term for **photoheterotroph**. — **photoorganotrophic adj.**; **photoorganotrophy n.**

photooxidize or **photooxidise** to oxidize by the use of, or as a result of, radiation in the visible and ultraviolet wavelengths. — **photooxidizable** or **photooxidisable adj.**

photoperiod 1 the length of daylight in a 24-hour period. 2 the duration of daily exposure of an organism to illumination; the length of this period that favours optimum functioning. — **photoperiodic adj.**; **photoperiodically adv.**

photoperiodism or **photoperiodicity** the response of an organism to the relative durations of light and dark in a 24-hour period.

photophore any light-emitting organ of an animal.

photophosphorylation or **photosynthetic photophosphorylation** the metabolic processes by which photosynthetic organisms use light energy to convert ADP to ATP without the concomitant reduction of dioxygen to water that occurs in oxidative phosphorylation (see **photosynthesis**). There are two distinct electron transport mechanisms: in **noncyclic photophosphorylation**, which involves both photosystems I and II, ATP synthesis is linked to the transport of electrons from water to NADP⁺, with production of NADPH and dioxygen; in **cyclic photophosphorylation**, which involves only photosystem I, ATP synthesis is driven by a proton gradient generated across the thylakoid membrane.

photopia adjustment of the eye to vision in light of moderate or high intensity, permitting perception of colour; it is considered to involve mainly the **cone cells** of the retina. Compare **scotopia**. — **photopic adj.**

photopigment any pigment whose chemical state depends on the intensity of the light falling on it.

photopolymerize or **photopolymerise** to effect polymerization by the action of light. — **photopolymerization** or **photopolymerisation n.**

(6-4)photoproduct a type of **pyrimidine dimer**.

photoprotein any protein actively concerned in the emission of light by a living organism.

photopsin an **opsin** that is the apoprotein of an **iodopsin**.

photoreactivating enzyme see **DNA photolyase**, **photoreactivation**.

photoreactivation the process by which genetic damage caused to organisms by ultraviolet radiation can be reversed by subsequent illumination with visible or near-ultraviolet light. The pyrimidine dimers formed in DNA by the action of ultraviolet light bind **DNA photolyase**, which absorbs light, maximally at 380 nm, and catalyses the restoration of two pyrimidine residues in the DNA.

photoreactive (of a substance or reagent) reactive in a photochemical reaction. — **photoreactivity n.**

photoreceptor any biological structure that responds to incident electromagnetic radiation, particularly visible light. It especially refers to a cell that, on absorbing light, generates a nerve impulse or impulses.

photoreduction 1 chemical reduction brought about by electromagnetic radiation, especially visible light. 2 a process occurring at low light intensities in certain algae whereby dihydrogen functions in place of water as an electron donor for the reduction of carbon dioxide and oxygen is not evolved; at higher light intensities normal photosynthesis takes place. 3 reduction in size photographically.

photorespiration a light-dependent catabolic process occurring concomitantly with photosynthesis in plants (especially **C₃ plants**) whereby dioxygen is consumed and carbon dioxide is evolved. The substrate is glycolate formed in large quantities in chloroplasts from 2-phosphoglycolate generated from ribulose 1,5-bisphosphate by the action of **ribulose-bisphosphate carboxylase**; the glycolate enters the peroxisomes where it is converted by **glycolate oxidase** to glyoxylate which undergoes transamination to glycine. This then passes into the mitochondria.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

dria where it is decarboxylated. **C₃ plants** may lose 20–40% of photosynthetically fixed carbon by photorespiration, whereas **C₄ plants** lose relatively little. **C₄ plants** defeat photorespiration in two ways. First, they recapture released CO₂ in the cytoplasm using **phosphoenolpyruvate carboxylase** which has a very high affinity for CO₂. Although this does not directly affect photorespiration, it prevents loss of CO₂. Second, they concentrate CO₂ near ribulose-bisphosphate carboxylase, and this increases the CO₂:O₂ ratio, which has the effect of inhibiting the oxygenase activity. — **photorespiratory adj.**

photosensitivity 1 the property of being reactive or unusually reactive to incident photons, especially to those of visible light. 2 (*in medicine*) an abnormal, local or generalized reaction to sunlight, due to the accumulation in the skin of **photodynamic** substances, either endogenous or exogenous. — **photosensitive adj.**

photosensitize or **photosensitise** 1 to render photosensitive; see **photosensitivity**. 2 to render a target molecule more reactive (e.g. more easily oxidized) by the addition to it of a second type of molecule, often a dye, that absorbs light energy and transfers it to the target molecule. — **photosensitization** or **photosensitisation n.**

photosensitizer or **photosensitiser** any substance that photosensitizes or is photodynamic.

photosuicide inhibitor a type of **suicide inhibitor** that, after reaction with an enzyme or other appropriate protein, is capable of being selectively photodecomposed to a highly reactive species that can irreversibly modify the protein in the neighbourhood of its active site.

photosynthate any product of the dark reactions of photosynthesis.

photosynthesis 1 (*in biology*) the synthesis by organisms of organic chemical compounds, esp. carbohydrates, from carbon dioxide using energy obtained from light rather than from the oxidation of chemical compounds (compare **chemosynthesis**). In higher plants, green algae, and cyanobacteria, water is the electron donor and dioxygen is evolved; in other photosynthetic bacteria, however, various other simple substances (e.g. H₂S) may act as electron donors, dioxygen is not formed, and the photosynthetic pathway is simpler. Photosynthesis comprises two separate processes: the light reactions and the dark reactions of photosynthesis. The system in higher plants, green algae, and cyanobacteria brings about an overall reaction in which electrons are released from water molecules and are transferred through complexes of chlorophylls, cytochromes, proteins, and small molecules to NADP⁺, reducing it to NADPH. Protons are released from water and form a chemiosmotic gradient that results in the formation of ATP by a process similar to that operating in mitochondria (see **chemiosmotic coupling hypothesis**). These are the **light reactions**. The NADPH and ATP formed by these reactions then drive the **dark reactions** i.e. the **reductive pentose phosphate cycle**. The electrons and protons are released from water by the oxygen-evolving complex (*abbr.*: OEC), and dioxygen is formed in the process. The sequence of reactions involves the OEC, from which electrons pass to photosystem II (see **photosystem**), thence to **plastoquinone** reducing it to a quinol which then reduces the cytochrome *b₆-f* complex. This in turn reduces **plastocyanin** which passes electrons to photosystem I (see **photosystem**) which then reduces NADP⁺ to NADPH. In photosynthetic bacteria, a simpler system exists, in which water is not split, dioxygen is not evolved, and NADP⁺ is not reduced directly. In this system, photons excite molecules of bacteriochlorophyll *b* in a **photosynthetic reaction centre** to transfer electrons through a cyclic electron-transfer system.

Cyclic electron flow can also occur under some conditions in plants, and involves only photosystem I, electrons being released by photon activation from photosystem I and returned to it through cytochrome *b₆-f* and plastocyanin, with pumping of protons to generate ATP. Under these conditions there is no reduction of NADP⁺. 2 (*in chemistry*) any synthesis by a



photosynthetic phosphorylation

photochemical reaction. —**photosynthetic adj.**; **photosynthetically adv.**

photosynthetic phosphorylation an alternative term for photophosphorylation.

photosynthetic quotient a ratio of the rate of dioxygen evolution by a cell or organism capable of photosynthesis to that of carbon dioxide uptake. It is an index of photosynthetic activity relative to respiration (including photorespiration).

photosynthetic reaction centre the photosynthetic complex in bacteria, consisting essentially of a transmembrane protein of three subunits, denoted L, M, and H. Collectively the three subunits generate 11 transmembrane helices. Although the L and M helices (which have binding sites for the many prosthetic groups) are different, the groups themselves are arranged with near twofold symmetry because of similarities in the folds of the subunits. The L and M subunits bear bacteriochlorophyll, bacteriopheophytin, ubiquinone, and menaquinone molecules. The bacteriochlorophylls will be either *a* or *b* types, as will be the bacteriopheophytins. Here we describe a system with *b* types. Of the four bacteriochlorophyll *b* molecules, two constitute a 'special pair' with an Mg^{2+} – Mg^{2+} distance of 7 Å and in hydrophobic environments; other prosthetic groups are bacteriopheophytin *b* (two molecules), one nonheme Fe(II), one ubiquinone, and one menaquinone. There is cyclic electron flow through a cytochrome *bc₁* complex, cytochrome *c*, and the bacteriochlorophyll–bacteriopheophytin complex, protons being pumped from the bacterial cytosol to the periplasmic space, and then returning to the cytosol through an ATP synthase with synthesis of ATP. The system differs from that in plants in that there is only one type of reaction centre, no splitting of water, and no direct reduction of $NADP^+$. Its essential features are that photons excite electrons in the 'special pair' and these excited electrons then flow through the bacteriochlorophyll–bacteriopheophytin complex, ubiquinone, and menaquinone, dissipating energy in bringing about protein pumping to form the chemiosmotic gradient that results in ATP formation. They then return to the 'special pair' via cytochromes *bc₁* and *c*. Examples from *Rhodobacter sphaeroides*: L subunit: database code NRL_4RCRL, 266 amino acids (29.74 kDa); M subunit: database code NRL_4RCRM, 296 amino acids (33.15 kDa); H subunit: database code NRL_4RCRH, 237 amino acids (25.63 kDa). This (and the complex from *Rhodospseudomonas viridis*) represents the first transmembrane protein to have its 3-D structure determined in atomic detail by X-ray crystallography (by Deisenhofer, Michel, and Huber). See also P870, P960, photosystem (def. 2).

photosystem 1 either of the two functionally distinct but co-operating systems, designated photosystem I (abbr.: PSI) and photosystem II (abbr.: PSII), between which are distributed the various molecules that effect the light reactions of **photosynthesis**; photosystems are located in the **thylakoid** membranes of the chloroplasts of higher plants and green algae or in the free thylakoid membranes of cyanobacteria. Photosystem I consists of a transmembrane multisubunit complex, of about 10 subunits in all, consisting of a heterodimer complex of PsaA and PsaB which binds P700, together with the chlorophyll acceptor A_0 , the phyloquinone acceptor A_1 , and Fx, a 4Fe–4S centre. PsaC, which is bound to this complex, contains two additional 4Fe–4S centres required for ferredoxin reduction. Other subunits are involved in binding soluble ferredoxin, ferredoxin–NADP oxidoreductase and regulation of electron flow. Approximately 100 chlorophyll *a* molecules are bound to the minimal heterodimer unit. Examples (all from pea), heterodimer of two similar subunits, A_1 : database code PSAA_PEA, 761 amino acids (84.01 kDa) and A_2 : database code PSAB_PEA, 734 amino acids (83.32 kDa); PsaC: database code PSAC_PEA, 80 amino acids (8.84 kDa) is similar to bacterial-type 4Fe–4S **ferredoxins**. Photosystem II is formed from a heterodimer of the D₁ and D₂ proteins which bind P680, together with the pheophytin acceptor, two plasto-

quinone acceptors Q_a and Q_b , and a non-heme iron atom. The minimal purified unit contains about six chlorophyll *a* molecules. **Cytochrome *b₅₅₉*** is always associated with PSII. A large number of other subunits including light-harvesting chlorophyll-protein complexes, proteins involved in stabilizing the Mn complex, and regulatory subunits are required to form a functional photosystem II. Examples (from pea) of proteins of photosystem II are D₁ precursor: database code PSBA_PEA, 353 amino acids (38.92 kDa); and D₂ precursor: database code PSBC_PEA, 353 amino acids (39.51 kDa). 2 any functionally analogous but simpler single photosystem present in photosynthetic bacteria. That from *Rhodospseudomonas viridis* contains three hydrophobic subunits, a special pair of bacteriochlorophyll *b* (the reaction-centre bacteriochlorophyll P960), two molecules of bacteriopheophytin *b* (see **bacteriopheophytin**), a bound iron atom and one molecule each of **ubiquinone** and a **menaquinone**. There are four heme *c*-type cytochromes in this case (but not in all). The complete structure has been determined by X-ray crystallography. See also **photosynthetic reaction centre**.

phototaxis the movement of a motile cell or organism towards (**positive phototaxis**) or away from (**negative phototaxis**) a source of light. —**phototactic adj.**

phototoxic able to cause physiological damage by making the skin abnormally sensitive to light. —**phototoxicity n.**

phototransistor a type of junction transistor that responds to incident light or other electromagnetic radiation by the generation of an electric current, which is then amplified.

phototroph or **phototrophe** any organism whose principal or only source of primary energy is sunlight. The great majority of such organisms are **photoautotrophs** (or **photolithotrophs**), which use light energy to synthesize their organic requirements solely from inorganic precursors; these include all green plants and algae, cyanobacteria, and green and purple sulfur bacteria. Relatively few phototrophs are **photoheterotrophs** (or **photoorganotrophs**), which require a supply of organic precursors to manufacture their own organic components; examples include certain purple nonsulfur bacteria. Compare **chemotroph**. —**phototrophic adj.**; **phototrophy n.**

phototropic 1 involving or showing **phototropism**. 2 involving or showing phototropy (i.e. **photochromism**).

phototropism the orientation or growth of a cell or (a part of) an organism in relation to a source of light. The term is used especially of plants or plant organs.

phototropy an alternative term for **photochromism**.

phototube a type of photocell consisting of a vacuum tube or gas-filled tube in which a photocathode responds to incident light or other electromagnetic radiation by emission of electrons.

pH paper absorbent paper, usually in strip form, that is impregnated with a colour-changing pH indicator dye. It is useful for estimating the approximate pH of a solution.

pH profile see **pH-activity profile**.

phragmoplast a structure formed in the developing plant cell wall from the endoplasmic reticulum and spindle fibres. It is observed during **cytokinesis** in plant cells as a cylindrical structure, composed of microtubules, between the two sets of daughter chromosomes. The microtubules are those of the two sets of polar microtubules of the **spindle**. These microtubules transport small Golgi-derived vesicles to the equatorial (or central) region, which fuse there to form the early **cell plate**, which then grows out to join up with the cell wall.

phrenosin a **cerebroside** in which the acyl constituent is cerebronate. See **lignocerate**, of which cerebronate is the 2-hydroxy derivative.

pH standard any of a set of six standard solutions which may be used as primary pH standards; they are: potassium hydrogen tartrate, saturated at 25 °C, pH 3.557 at 25 °C; 0.1 mol/kg potassium dihydrogen citrate, pH 3.776 at 25 °C; 0.025 mol/kg KH_2PO_4 + 0.025 mol/kg Na_2HPO_4 , pH 6.865 at 25 °C;



pH-stat

0.008695 mol/kg KH_2PO_4 + 0.03043 mol/kg Na_2HPO_4 , pH 7.413 at 25 °C; 0.01 mol/kg disodium tetraborate, pH 9.180 at 25 °C; 0.025 mol/kg NaHCO_3 + 0.025 mol/kg Na_2CO_3 , pH 10.012 at 24 °C.

pH-stat an instrument for maintaining a constant pH in an aqueous liquid mixture in which hydrogen ions are being liberated or taken up. It is frequently arranged to measure the amount of reagent, as a function of time, that has to be added to maintain a predetermined pH.

Pht< or -Pht- symbol for the (divalent) phthaloyl group.

Pht- symbol for the (univalent) phthalyl group.

phthalate **1** the trivial name for 1,2-benzene dicarboxylate, the dianion of phthalic acid. **2** any mixture of phthalic acid and its mono- and dianions. **3** any salt or ester of phthalic acid.

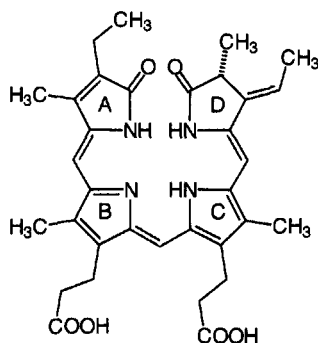
phthaloyl symbol: Pht< or -Pht-; the divalent acyl group, $-\text{OC}-\text{C}_6\text{H}_4-\text{CO}-$, derived from phthalic acid.

phthalyl symbol: Pht-; the univalent acyl group, $\text{HOOC}-\text{C}_6\text{H}_4-\text{CO}-$, derived from phthalic acid.

pH unit the difference between any two successive integral values of pH.

phyco+ comb. form denoting seaweed or other algae.

phycobilin any bilin found in **phycobiliproteins**. The two most common are **phycocyanobilin** and **phycoerythrobilin**. They are linear tetrapyrroles or **open tetrapyrroles**, i.e. chains of four pyrroles, the terminal pyrroles not being linked to each other to form the cyclic tetrapyrrole structure. The structure of phycocyanobilin is shown; phycoerythrobilin is similar except that the ethyl group on ring A is $-\text{CH}=\text{CH}_2$ and the methyne group between rings A and B is a methylene group.



phycocyanobilin

phycobiliprotein or **biliprotein** any phycobilin-protein conjugate found in the algae of the phyla Cryptophyta and Rhodophyta, and in cyanobacteria. Phycobiliproteins are intensely coloured and fluorescent water-soluble globular proteins. The two most common types, **phycocyanin** and **phycoerythrin**, are composed of two kinds of subunits, α (10–20 kDa) and β (14–30 kDa), each of which carries about three molecules of covalently bound **phycobilin** (phycocyanobilin in phycocyanin, phycoerythrobilin in phycoerythrin) and is present *in vivo* most commonly as aggregates of formula $(\alpha, \beta)_n$, where n is 3 or 6. Phycocyanin (blue: A_{max} at 618 nm) predominates in Cyanophyta, and phycoerythrin (red: A_{max} at 545 nm) predominates in Rhodophyta. A third type, **allophycocyanin** (pale blue: A_{max} at 650 nm), present in small amounts in both Cyanophyta and Rhodophyta, has only one type of protein subunit (≈ 15.5 kDa), carries one molecule of covalently bound phycocyanobilin, and is present *in vivo* as aggregates of six subunits. It is a heterodimer of α and β chains. Example from *Mastigocladus laminosus* (β chain only available): database code PHBB_MASLA, 169 amino acids (18.84 kDa). See also **phycobilisome**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

phycobilisome any of the granules, approximately 32 nm $\times$ 48 nm and consisting of highly aggregated **phycobiliproteins**, that are attached in arrays to the external face of a **thylakoid** membrane in algae of the phyla Cyanophyta and Rhodophyta, where they function as light-harvesting devices in photosynthesis. Excitation energy in the phycobilisome flows in the sequence:

phycoerythrin $\rightarrow$ phycocyanin $\rightarrow$ allophycocyanin,

whence it passes to the antenna chlorophyll of photosystem II. In Cyanophyta it is also known as a **cyanosome**.

phycocyanin a blue phycobilin; see **phycobiliprotein**.

phycocyanobilin a type of **phycobilin**; see also **phycobiliprotein**.

phycoerythrin a red phycobilin; see **phycobiliprotein**.

phycoerythrobilin a type of **phycobilin**; see also **phycobiliprotein**.

phycofluor any artificial conjugate of a **phycobiliprotein** with a molecule conferring biological specificity (e.g. avidin, immunoglobulin, protein A). Such conjugates are useful in fluorescence immunoassay and fluorescence-activated cell sorting by virtue of the distinctive and intense fluorescence of the phycobiliprotein moiety.

phycotoxin any poisonous substance produced by algae.

phyl+ a variant form of **phyl-** (sometimes before a vowel).

phyla the plural of **phylum**.

PHYLLIP a suite of computer programs used for determining phylogeny and/or cladistic relationships from protein and/or nucleic-acid sequences, using, in part, **parsimony**.

phyllo+ or (before a vowel) **phyll+** comb. form denoting leaf.

phyllocerulein or (esp. Brit.) **phyllocaerulein** see **cerulein**.

phylloquinone abbr.: K; the recommended name for vitamin K₁; 2-methyl-3-phytyl-1,4-naphthoquinone. See **vitamin K**.

phyl+ or (sometimes before a vowel) **phyll+** comb. form denoting race or tribe.

phylogenetic tree a graphical representation of the (putative) evolutionary relationships of a group of organisms.

phylogeny the evolutionary history of an organism or group of related organisms. In biochemistry, such evolutionary relationships are based on similarities and differences in nucleic acid or protein sequences. — **phylogenetic** or **phylogenetic adj.**

phylum (pl. **phyla**) a primary taxon of animals or plants (in plants also termed division). It consists of a number of classes (or sometimes a single class).

physalaemin Glp-Ala-Asp-Pro-Asn-Lys-Phe-Tyr-Gly-Leu-Met-NH₂; an 11-residue peptide amide **tachykinin** from the skin of the amphibian *Physalaemus fuscumaculatus*. It is more potent than **eledoisin** and **substance P** in some assay systems, and is useful in the study of receptors for substance P.

physeteric acid see **tetradecenoic acid**.

physical **1** of, or pertaining to, matter and energy; of material things. **2** of, or relating to, natural science, especially of non-living systems. Compare **biological**. **3** having reference to or used in physics. — **physically** adv.

physical adsorption an alternative term for **physisorption**.

physical biochemistry the application of physical chemistry to biochemical substances, structures, and systems. Often used synonymously with **biophysical chemistry**.

physical chemistry the branch of chemistry concerned with the relationship between the physical properties of substances and their chemical properties, reactions, and structures.

physical map a map which identifies the physical location of genes on a chromosome in contrast to a **linkage map** which is generated by measuring recombination frequencies. Physical length may be measured in base pairs, kilobases, or megabases. Such maps are obtained, e.g. in **pulsed field gel electrophoresis**, **fluorescence in situ hybridization**, and **contig mapping**.

physico+ comb. form denoting physical or of physics.

physicochemical of, pertaining to, or using the methods of physical chemistry. — **physicochemically** adv.

physics functioning as sing. the branch of science concerned



physiological

with the properties of matter and of energy, and with the interactions and interconversions between matter and energy.

physiological 1 of, pertaining to, or used in physiology. 2 normal (def. 4); not **pathological** (def. 2) or **pharmacological** (def. 2). —**physiologically** *adv.*

physiological antagonist a drug that produces the opposite effect to a specified agonist by an action that is unrelated to that of the agonist.

physiological chemistry an older term for **biochemistry** (especially that of humans and higher animals).

physiological conditions environmental conditions simulating those under which the (normal) functions of a cell, organ, or tissue can be expressed.

physiological saline or (sometimes) **normal saline** a (sterile) solution of 0.9% w/v (0.154 mol L⁻¹) sodium chloride in water. It is approximately isotonic with mammalian blood and lymph, and is useful in the temporary maintenance of living cells and tissues, and as a vehicle for injections.

physiology the science dealing with the functioning of cells, tissues, organs, and organisms, and with the chemical and physical phenomena concerned.

physisorption or **physical adsorption** or **van der Waals adsorption** the type of **adsorption** (def. 1) in which only van der Waals forces are involved and hence in which no significant changes occur in the electronic orbital patterns of the interacting substances. *Compare chemisorption.*

physostigmine or **eserine** a toxic alkaloid extracted from Calabar beans (*Physostigma venenosum*). It inhibits **cholinesterase** by binding to the anionic site of the enzyme, forming a carbamoyl ester of the serine hydroxyl, which is only slowly hydrolysed. It has been used in West Africa in witch ordeals.

phyt+ a variant form of **phyto+** (before a vowel).

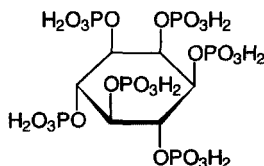
phytanic acid 3,7,11,15-tetramethylhexadecanoic acid; a branched chain fatty acid that is formed metabolically from **phytol**, a constituent of chlorophyll, by bacteria in the ruminant stomach. Its oxidation in animals depends on the **alpha oxidation** pathway, since the methyl group on C-3 (the beta carbon) blocks beta oxidation. After alpha oxidation this methyl group is on C-2 of the product, so that beta oxidation can occur, releasing propionyl-CoA as a product..

phytase either of the two enzymes, designated 3-phytase (EC 3.1.3.8) and 6-phytase (EC 3.1.3.26), that catalyse the hydrolytic removal of a phosphoric residue from **phytate** at the 3-position or the 6-position, respectively; most commonly it signifies 6-phytase.

phytate any anion of **phytic acid**, any mixture of free phytic acid and its ionized forms, or any salt of phytic acid. It is the major phosphorus compound in plants and is important for phosphate storage in cereal grains, especially as its calcium-magnesium salt, **phytin**. In higher animals phytin binds dietary Ca²⁺ and Mg²⁺, making them less available.

+**phyte** *noun suffix* denoting 1 plant (of the habitat or type specified). 2 pathological outgrowth. —+**phytic** *adj.*

phytic acid the trivial name for myo-inositol hexakis(dihydrogen phosphate). See **phytate**.



phytin see **phytate**.

phyto+ or (before a vowel) **phyt+** *comb. form* denoting plant.

phytoagglutinin or (rarely) **phytohemagglutinin** any plant **lectin**. The majority are glycoproteins, although some, e.g. **concanavalin A**, contain no carbohydrate. They accumulate mainly in seeds, in cytoplasm of cotyledons and embryonic cells, although they are synthesized in leaves.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

phytoalexin any substance with antibiotic activity that is produced by plant tissues in response to infection, especially with a fungus. A variety of different types of substance are regarded as phytoalexins. They include compounds as divergent as **phaseollin**, some sesquiterpenes and long-chain acetylenic alcohols.

phytochelatin any of a group of peptides that bind heavy metals (Cd, Zn, Cu, Pb, Hg) in thiolate coordination complexes, and can be isolated from plant cell suspension cultures. The structure is of the type (γ-glutamyl-cysteinyln)-glycine, where n is 2 to 11.

phytochemistry the branch of chemistry concerned with plants, especially with **secondary metabolites**.

phytochrome a chromoprotein present in trace amounts in all higher plants and involved in regulating light-dependent processes such as leaf development and flowering. It consists of a **phycobillin**-related chromophore covalently linked to a polypeptide of ≈124 kDa. Phytochrome exists in two forms, P_R (A_{max} at 666 nm) and P_{FR} (A_{max} at 730 nm), which are indefinitely interconvertible on absorption of light in the red and far-red regions of the spectrum, respectively. The spectral changes are due to *cis-trans* isomerization in the phycobillin. The active form, P_{FR}, plays a central role in light-promoted modulation of gene expression, which is of importance in plant growth and development in every phase of the life cycle; conversion of P_{FR} to P_R cancels these responses. The phytochromes are homodimeric proteins with a linear tetrapyrrole chromophore covalently linked to a polypeptide. Examples, phytochrome A (vegetable marrow): database code PHYA_CUCPE, 1124 amino acids (124.94 kDa); phytochrome B (tobacco): database code PHYB_TOBAC, 1132 amino acids (125.67 kDa).

phytyecdysone (sometimes) any member of the group of **ecdysteroids** found in plants.

phytoglycogen plant glycogen; a highly branched polymer of D-glucose, with a structure similar to that of animal **glycogen**, found in fungi, corn (maize), and rice. A similar substance is found in bacteria and cyanobacteria. In contrast to the major branched chain glucan, amylopectin, which is found only as a component of starch, plant glycogen is found as an independent molecule.

phytohemagglutinin or (esp. Brit.) **phytohaemagglutinin** 1 *abbr.*: PHA; or **phytomitogen** the **lectin** from the red kidney bean, *Phaseolus vulgaris*; a potent mitogen. Example, leucoagglutinating phytohemagglutinin precursor: database code PHAL_PHAVU, 272 amino acids (29.56 kDa). 2 a rare alternative name for **phytoagglutinin**.

phytohormone or **plant hormone** any plant growth factor or similar substance that regulates plant development. The group is diverse and includes **abscisic acid**, **auxins**, **ethylene**, **cytokinins**, and **gibberellins**.

phytokinin an alternative term for **cytokinin**.

phytol a trivial name for *trans*- (or *E*-) (7*R*,11*R*)-3,7,11,15-tetramethyl-2-hexadecen-1-ol; the pure substance is a colourless oily liquid. Phytol occurs in combined form in chlorophyll and vitamin K₁. See also **phytyl**.

phytolysosome a **lysosome** of plant cells.

phytomitogen an alternative name for **phytohemagglutinin** (def. 1).

phytoplankton plant plankton; consisting of diatoms, dinoflagellates, and other microscopic algae.

phytoporphyrin any plant **porphyrin**.

phytosphingosine 2*S*,3*S*,4*R*-2-amino-1,3,4-octadecanetriol; a constituent of many plant sphingolipids.

phytosterol any sterol occurring in higher plants; they differ from animal sterols in having C₁ or C₂ residues at C-24 and/or a double bond at C-22.

phytotoxic toxic to a plant.

phytotoxin 1 any toxin produced by a plant. 2 any substance that is toxic to a plant or plants.

phytotron

phytotron a laboratory where plants can be grown and studied under a variety of controlled conditions.

phytyl the alkyl group, $(\text{CH}_3)_2\text{CH}-[\text{CH}_2]_3-\text{CH}(\text{CH}_3)-[\text{CH}_2]_3-\text{CH}(\text{CH}_3)-[\text{CH}_2]_3-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_2-$, (*E*)-(7*R*,11*R*)-3,7,11,15-tetramethyl-2-hexadecenyl, derived from **phytol**.

pi 1 *symbol*: π (lower case) or Π (upper case); the sixteenth letter of the Greek alphabet. For uses see **Appendix A. 2** (in *mathematics*) *symbol*: π ; a transcendental number that is the ratio of the circumference of a circle to its diameter. It has the approximate value 3.141 592 653 59.

pi *symbol* for isoelectric point.

PI *abbr.* for phosphatidylinositol.

pi adduct or π **adduct** any adduct formed by coordination and involving donation of an electron pair from a **pi orbital** into a **sigma orbital**, from a sigma orbital into a pi orbital, or from a pi orbital into a pi orbital. It is also known, less correctly, as a **pi complex**.

pi bond or π **bond** the type of covalent bond formed between atoms by electrons occupying a **pi orbital**.

picein (wax) an unreactive thermoplastic material, composed of a mixture of hydrocarbons of very low vapour pressure, that adheres well to glass and metal. It is useful as a sealing substance in the assembly of vacuum systems.

Pichia see **methylotrophic yeasts**.

picket-fence (porphyrin) complex any synthetic iron-porphyrin complex in which the porphyrin has been derivatized with four large groups attached to one side of the porphyrin ring, to each of the inter-pyrrole carbon atoms, so as to form a protective enclosure for the binding of a dioxygen molecule as in **myoglobin** or **hemoglobin**.

pico+ *symbol*: p; SI prefix denoting 10^{-12} times.

picrogram *symbol*: pg; a unit of mass equal to 10^{-15} kilogram or 10^{-12} gram.

picometre or (*US*) **picometer** *symbol*: pm; a unit of length equal to 10^{-12} metre.

picomolar *symbol*: pM or pM; describing a solution containing one picomole of solute per litre of solution, or a specified multiple or fraction thereof; used also of a substance concentration similarly expressed.

picomole *symbol*: pmol; a unit of amount of substance equal to 10^{-12} mole.

pi complex or π **complex** a less suitable name for **pi adduct**.

picoplankton plankton having cell dimensions of between 0.2 and 2 μm and including bacteria, cyanobacteria, some algae, and certain microscopic protozoans. Picoplankton appears to be ubiquitous in open waters of lakes and seas and probably exceeds in total biomass any other group of pelagic organisms.

picornavirus any of a group of RNA animal viruses consisting of naked, icosahedral 27 nm capsids with single-stranded infectious RNA (plus strand) of 2.7 MDa. The group includes human poliovirus, some common-cold viruses, and foot-and-mouth disease virus (of cattle and other cloven-footed mammals).

picosecond *symbol*: ps; a unit of time equal to 10^{-12} second.

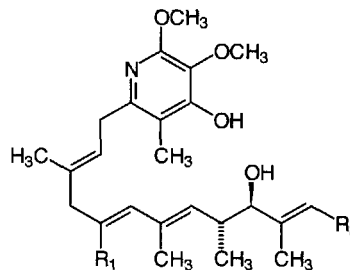
4-pi counter any type of radiation counter that detects radiation emitted through a solid angle of 4π steradians, i.e. in all directions from a source.

picrotoxin or **cocculin** a preparation containing an equimolar mixture of **picrotoxinin** ($\text{C}_{15}\text{H}_{16}\text{O}_6$) and **picrotin** ($\text{C}_{15}\text{H}_{18}\text{O}_7$) obtained from the seeds of *Anamirta cocculus* (fishberries). It contains a potent ligand for the chloride ionophore of the GABA_A receptor, binding to the channel protein and inhibiting Cl^- transport without displacing GABA; this endows it with potent convulsant activity. It can thus also act as an antidote to barbiturates, which enhance GABA binding. It is displaced from its binding site by anticonvulsants.

pi electron or π **electron** an electron in a **pi orbital**.

piericidin A an antibiotic that is a structural analogue of **ubiquinone**. It is a classical inhibitor of mitochondrial electron transport similar in action to **rotenone**. It is a potent inhibitor

of NADH:ubiquinone oxidoreductase (see **NADH dehydrogenase**), acting at or near the ubiquinone catalytic site.



pig 1 any nonruminant, omnivorous ungulate mammal of the family Suidae, especially the domestic pig, *Sus scrofa*. 2 informal short for **guinea pig**.

PI-G *abbr.* for phosphatidylinositolglycan.

PIG-A *abbr.* for phosphatidylinositol glycan protein class A.

pigeon any stout-bodied bird of the family Columbidae, especially the domesticated or feral pigeon, *Columba livia*. Pigeon-breast muscle has been extensively used as a convenient tissue for the study of oxidative metabolism.

pigment 1 any general or particular colouring matter in living organisms. 2 any natural or artificial substance used to impart colour. 3 to colour with pigment.

pigment 680 see **P680**.

pigment 700 see **P700**.

pigmentation 1 natural coloration of organisms. 2 an increase in such coloration.

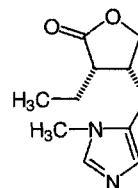
pigment system an alternative name for **photosystem**.

PI-G-tailed proteins proteins that are anchored to the membrane through linkage to phosphatidylinositol-glycan.

pili the plural of **pilus**.

pilin any protein subunit of a bacterial **pilus**.

pilocarpine an alkaloid extracted from *Pilocarpus jaborandi*; it has muscarinic cholinomimetic activity.



pilot plant a prototype apparatus set up to develop and scale up a new manufacturing process.

pilus (*pl.* **pili**) a hairlike appendage, about 25 nm in diameter and 10–12 μm long, present sometimes on cells of certain Gram-negative bacteria, especially members of the Enterobacteriaceae and *Pseudomonas* spp., in addition to any flagella. The two main types, distinguishable morphologically and antigenically, are **common pilus** and **sex pilus**, both of which may be present on the same cell. Up to several hundred common pili per cell may be present, either distributed over the entire surface or concentrated at a pole; their function is not known. Only one or two sex pili commonly are present; they appear to be required for bacterial **conjugation**. Both types of pili are thought to be tubes or tubelike structures formed from one or two helically wound strands, each of which consists of a linear arrangement of molecules of the protein pilin. The term pilus is often used interchangeably with **fimbria**, but may be used specifically to mean sex pilus (in which case fimbria then denotes common pilus).

PIM a gene family encoding serine/threonine kinases of un-

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



pl marker

known function named from proviral integration site MuLV-induced murine T-cell lymphomas. They are highly conserved between human and mouse, and are expressed at high levels in hemopoietic tissues, testes and ovaries, and embryonic stem cells. Example of gene product from mouse: database code KPIM_MOUSE, 313 amino acids (35.54 kDa).

pl marker an **isoelectric point marker**; *see* **isoelectric point**.

pineal body or **pineal gland** a small, pine-cone-shaped structure situated deep between the cerebral hemispheres of the mammalian brain. It elaborates and secretes **melatonin**.

pineal hormone an older name for **melatonin**.

ping-pong mechanism an alternative name for **double-displacement (enzyme) mechanism**.

pinocytic an alternative term for **pinocytotic** (*see* **pinocytosis**).

pinocytic vesicle an alternative term for **pinocytotic vesicle**.

pinocytosis the process whereby cells take in liquid material from their external environment; literally 'cell drinking'. Liquid is enclosed in vesicles, formed by invagination of the plasma membrane. These vesicles then move into the cell and pass their contents to endosomes. Pinocytosis is the type of **endocytosis** occurring in most types of cells; hence the term is often applied interchangeably with **endocytosis** to such cells (*compare* **phagocytosis**). —**pinocytotic** or **pinocytic adj.**

pinocytotic vesicle or **pinocytic vesicle** a cellular vesicle serving **pinocytosis**; an alternative term to **endocytotic vesicle** when containing nonparticulate material.

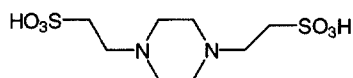
pinosome an alternative name for **endosome** (def. 1) (when containing nonparticulate material).

pi orbital or π orbital a bonding molecular orbital formed by the in-phase, sideways-on overlap of parallel p atomic orbitals, one from each of two atoms, to produce a charge-density distribution that is mainly between the atomic nuclei but maximal in two zones, one on either side of and parallel to the internuclear axis. When occupied by two electrons of opposed spins it becomes the second bond in a double covalent bond (two such occupied orbitals, mutually orthogonal, providing the second and third bonds of a triple bond).

PIP or **PI 4-P** *abbr.* for phosphatidylinositol 4-phosphate.

PIP₂ or **PI 4,5-P₂** *abbr.* for phosphatidylinositol 4,5-bisphosphate.

Pipes or **PIPES** *abbr.* for 1,4-piperazinediethanesulfonic acid; a Good buffer substance; pK_a (20 °C) = 6.8.



pipette or (US) **pipet** 1 any device for transferring or delivering (usually measured) volumes of liquid and commonly consisting of a slender, calibrated glass or plastic tube with a finely drawn-out delivery tip and (for larger volumes) a centrally placed bulb. It may be filled by applied suction or by capillarity (according to its bore), and emptied by drainage or by applied pressure. 2 to transfer or deliver a (usually measured) volume of liquid with a **pipette** (def. 1). —**pipettable adj.**

pipettor any device for the rapid, repetitive dispensing of a predetermined volume of a liquid.

pipsyl *symbol:* Ips; *p*-iodophenylsulfonyl; $I-C_6H_4-SO_2$; a chemical group with which free amino groups may be derivatized by reaction with pipsyl chloride, *p*-iodophenylsulfonyl chloride. *Compare* **tosyl**.

pipsylate to convert (an amino compound) to a **pipsyl** derivative. —**pipsylation n.**

Pirani gauge an electric gauge for low gas pressures that functions over the range 10^{-2} to 10^{-5} mm Hg. It depends on the principle that an electrically heated wire loses heat by conductivity, and thus varies in electric resistance, to an extent empirically dependent on the pressure of the surrounding gas.

piscine of, relating to, or being a fish.

pitch height (of a helix) *symbol:* p ; the linear displacement

along the axis of a helix corresponding to one complete turn of the helix.

PIT a gene encoding the pituitary-specific positive transcription factors, Pit. It activates the genes for somatotropin and prolactin, binding specifically to the consensus sequence 5'-TAAAT-3'. It contains a **POU domain** and contributes to the naming of the domain. Example Pit-1 from human: database code HMP1_HUMAN, 291 amino acids (32.91 kDa).

Pitocin a proprietary name for (a solution of) **oxytocin**.

Pitressin a proprietary name for (a solution of) **vasopressin**.

pituitary 1 pertaining to or produced by the **pituitary gland**. 2 the **pituitary gland**.

pituitary function *see* **combined test of pituitary function**.

pituitary gland or **hypophysis (cerebri)** the most important endocrine gland of vertebrates. It is situated below the brain, to which it is attached by a stalk, and measures about 8 mm by 12 mm in an adult human. It consists of two parts: the anterior lobe, or **adenohypophysis**, and the posterior lobe, or **neurohypophysis**; both lobes secrete a range of hormones.

Pituitrin a proprietary name for an aqueous extract of bovine neurohypophysis; it contains both **oxytocin** and **vasopressin**.

PIVKA *abbr.* for protein induced by vitamin K absence; examples of such substances are the abnormal proteins produced in vitamin K-deficient animals, corresponding to coagulation factors II, VII, IX, and X (*see* **blood coagulation**) in which some glutamate residues have not been carboxylated to γ -carboxyglutamate residues.

pixel one of the very small units that make up the image on a cathode-ray tube. [From 'pix' (i.e. pictures) + el(ement).]

pK *symbol* for the decadic logarithm of the reciprocal of a (thermodynamic) dissociation constant, K ; i.e.

$$pK = \log 1/K = -\log_{10} K.$$

The type of constant may be indicated by a subscript: e.g. pK_a , a dissociation constant of an acid; pK_b , a dissociation constant of a base; pK_2 , a second dissociation constant. For the dissociation of an acid, HA, to its ions, H^+ and A^- ,

$$K = [H^+][A^-]/[HA].$$

Thus when $[A^-] = [HA]$, i.e. when the acid is half-dissociated, $K = [H^+]$, and hence pK is then equal to pH. Formerly symbolized as p_x . *See also* **group pK**, **Henderson-Hasselbalch equation**, **molecular pK**, **titration pK**.

pK' *symbol* for the decadic logarithm of the reciprocal of an apparent (concentration) dissociation constant, K' .

pK₂ *see* **pK**.

pK_a *see* **pK**.

pK_b *see* **pK**.

pK_w *symbol* for the decadic logarithm of the reciprocal of the ion product of water, K_w .

PK (in clinical chemistry) *abbr.* for **pyruvate kinase**.

PKC *abbr.* for **protein kinase C**.

PKU *abbr.* for **phenylketonuria**.

PL *abbr.* for 1 placental lactogen (i.e. **choriomamototropin**). 2 pyridoxal.

placebo (*pl.* **placebos** or **placeboes**) a tablet, capsule, or liquid medicine devoid of any specific pharmacological activity, but appearing identical to a preparation containing a biologically active substance. Placebos are used in human medicine for their psychological effects, and as controls in experimental therapeutics for elucidating the clinical actions of a drug.

placenta (*pl.* **placentas** or **placentae**) a highly vascularized organ within the uterus of a pregnant animal by which the embryo is attached to the uterine wall and through which the embryo can exchange solutes with the maternal circulation. The placenta also functions as an endocrine gland, secreting *inter alia* **choriogonadotropin**, **choriomamototropin**, and certain steroid hormones. —**placental adj.**

placental lactogen *abbr.*: PL; an alternative name for **choriomamototropin**.

plakoglobin or **desmoplakin III** a protein of **cell junctions**. It is



thought to play a central role in the function of desmosomes and intermediate junctions. Example from human: database PLAK_HUMAN, 743 amino acids (81.40 kDa).

planar 1 of, or pertaining to, a **plane**. 2 lying in one plane; flat.

planchet or **planchette** a small, shallow dish used to contain a specimen for determination of its radioactivity with an end-window or windowless counter.

Planck, Max Karl Ernst Ludwig (1858–1947), German physicist; Nobel Laureate in Physics (1918) 'in recognition of the services he rendered to the advancement of Physics by his discovery of energy quanta'.

Planck constant *symbol:* h ; a fundamental constant of proportionality between the frequency, ν , of electromagnetic radiation and the energy, E , of a quantum of the radiation; i.e. $h = E/\nu$. It has a value of $6.626\,075\,5(40) \times 10^{-34}$ J s.

plane 1 a surface in which a straight line joining any two contained points lies entirely within the surface; a flat surface. 2 an imaginary surface of such a kind within a real structure or lying conceptually in space. 3 lying wholly in one plane; flat or level; planar.

plane of polarization or **plane of polarisation** (of linearly polarized light or other electromagnetic radiation) the plane containing the electric vectors of the vibrations.

plane of symmetry a plane through a three-dimensional structure, e.g. a crystal or molecule, that divides the object into two parts that are mirror images of each other.

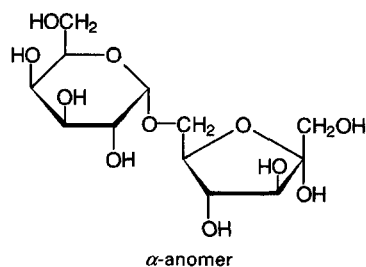
plane-polarized or **plane-polarised** an alternative term for linearly polarized. *See* polarized light.

planimeter a mechanical integrating instrument for measuring the area of an irregular plane figure. It consists of a moveable tracing arm, which is made to follow the boundary of the figure, and a dial on which the arm's motion is recorded. — **planimetry** *n*.

plankton the collective term for the small or microscopic plants (phytoplankton) and animals (zooplankton), that drift freely in the surface waters of lakes, seas, and oceans.

plant any organism of the kingdom Plantae. Plants are characterized by their ability to effect photosynthesis and the possession of rigid cell walls that contain cellulose.

planteobiose the disaccharide, 6-*O*- α -D-galactopyranosyl-D-fructose; a unit of the trisaccharide **planteose**.



planteose *O*- α -D-galactopyranosyl-(1 $\rightarrow$ 6)-*O*- β -D-fructofuranosyl-(2 $\rightarrow$ 1)- α -D-glucopyranoside; a trisaccharide isolated from defatted seeds of *Plantago ovata*.

plant hormone an alternative term for **phytohormone**.

plantibody a human antibody produced by genetically engineered tobacco plants.

plaque 1 any small disk-like object, patch, or zone. 2 a macroscopic or microscopic rounded clear zone in a layer of cells or bacterial lawn that results from the killing or lysis of adjacent cells by the action of a virus or other agent. 3 **a** (a patch of) fibrous or lipid material on the inner surface of an artery. **b** an area of tissue degeneration with distinctive histological features characteristic of certain (especially neurological) diseases, e.g. **kuru**, **multiple sclerosis**. 4 (a patch of) deposit closely adherent to the surface of a tooth that contains a mixed micro-

bial flora and is composed largely of extracellular bacterial polysaccharide.

plaque-lift a technique used in recombinant DNA technology for screening actively expressing clones. Candidate clones for expression of a particular protein are plated out onto a dish. After incubation a nitrocellulose (or other) membrane is pressed against the surface of the plate, thereby adsorbing any protein 'plaques' formed by actively expressing clones. The membrane can then be screened, and the active clones identified.

+plasia or **+plasy** *comb. form* indicating development or growth. — **+plastic** *adj.*

plasm+ a variant form of **plasm-** (before a vowel).

+plasm *comb. form* denoting the colloidal material of which living cells are composed. — **+plasmic** *adj.*

plasma 1 the proteinaceous fluid in which the cells of blood or lymph are suspended; the meaning is sometimes extended to include also the analogous fluid in which the fat droplets of milk are suspended. *See also* **blood plasma**. 2 a less common word for **protoplast**. 3 (*in physics*) any region of highly ionized gas (as in a gas-discharge tube, a hot flame, or a thermonuclear reaction) containing approximately equal numbers of electrons and positively charged ions. It differs from ordinary gas in being a good conductor of electricity and being affected by a magnetic field. — **plasmatic** or **plasmic** *adj.*

plasma albumin an alternative name for **serum albumin**.

plasma cell or **plasmacyte** or **plasmocyte** a fully mature antibody-secreting B lymphocyte, found in lymphoid tissue and in connective tissue liable to encounter foreign material. It has an eccentrically placed nucleus, basophilic cytoplasm, and an unusually large amount of rough endoplasmic reticulum.

plasma cholesterol (*in medical biochemistry*) the concentration of cholesterol in blood plasma. In human blood, cholesterol occurs as a component of all plasma lipoproteins, but especially **very-low-density lipoprotein**, **low-density lipoprotein**, and **high-density lipoprotein**. A correlation exists between the level of plasma cholesterol and the occurrence of **atheroma** and heart attacks; this concept is associated with the role of low-density lipoprotein, which has the highest concentration of cholesterol and is thought to be the lipoprotein that distributes cholesterol to the tissues. However, the ratio of high-density lipoprotein cholesterol to low-density lipoprotein cholesterol must be taken into account, as high-density lipoprotein is thought to have a protective function in cycling cholesterol from peripheral tissues back to the liver. The circulating level of cholesterol may be altered by disturbances in carbohydrate and endocrine metabolism (e.g. diabetes or thyroid hormone imbalance), diet, and drugs. It is thought that the level of circulating total cholesterol should desirably be maintained below about 5 mmol L⁻¹.

plasma clearance *see* **clearance** (def. 2).

plasmacyte an alternative name for **plasma cell**.

plasmacytoma a tumour of plasma cells; most such tumours secrete myeloma proteins belonging to the same category as monoclonal antibodies.

plasma kallikrein *see* **kallikrein**.

plasmalemma an alternative term for **cell membrane**.

plasmalogen any glycerophospholipid in which the glycerol moiety bears an *O*-(1-alkenyl) group at position 1. The term embraces **plasmenic acid**.

plasmalogen synthase EC 2.3.1.25; an enzyme that catalyses the formation of plasmenylcholine from acyl-CoA and 1-*O*-alk-1-enyl-*sn*-glycero-3-phosphocholine with release of CoA.

plasma membrane an alternative term for **cell membrane**.

plasmanic acid a derivative of *sn*-glycerol 3-phosphate in which the glycerol moiety is etherified with an alkyl (or alkenyl other than 1-alkenyl) group at position 1 and esterified with an acyl group at position 2; i.e. any 2-acyl-1-*O*-*R*-*sn*-glycerol 3-phosphate where *R* = alkyl or *n*-alkenyl when *n* > 1. *Compare* **plasmenic acid**.

plasmanyl the trivial name for any phosphoric acyl group de-



plasmapheresis

rived from a **plasmanic acid**. The term is useful for naming phosphoric diesters of plasmanic acid, e.g. plasmanylethanolamine, 2-acyl-1-*O*-alkyl-*sn*-glycerol 3-phosphoethanolamine. *Compare plasmanyl*.

plasmapheresis or **plasmaphoresis** or **plasmapharesis** or **plasmaphaeresis** a technique for decreasing the protein content of blood plasma circulating in the body by withdrawing blood, separating the cells by centrifugation, and then returning them to the circulation as a suspension in a suitable protein-free medium.

plasma protein 1 any of the numerous proteins present in or derived from blood plasma. **2** such proteins collectively. In mammalian plasma the concentration of plasma protein is 60–80 g L⁻¹; of this, ≈60% is serum albumin, ≈18% is immunoglobulin G, ≈9% is β₁-lipoprotein, and ≈4% is fibrinogen.

plasma thromboplastin antecedent an alternative name for factor XI; *see blood coagulation*.

plasmenic acid a derivative of *sn*-glycerol 3-phosphate in which the glycerol moiety is etherified with a 1-alkenyl group at position 1 and esterified with an acyl group at position 2; i.e. any 2-acyl-1-*O*-(1-alkenyl)-*sn*-glycerol 3-phosphate. *Compare plasmanic acid, plasmalogen*. [Contraction of plasmalogenic acid.]

plasmenyl or (formerly) **phosphatidal** the trivial name for any phosphoric acyl group derived from a **plasmenic acid**. The term is useful for naming phosphoric diesters of plasmenic acid, e.g. plasmenylethanolamine, 2-acyl-1-*O*-(1-alkenyl)-*sn*-glycerol 3-phosphoethanolamine. *Compare plasmanyl*.

plasmid strictly any extrachromosomal genetic element; however, mitochondrial and chloroplast DNA, yeast killer, and other cases are commonly excluded, hence the term plasmid generally refers to an extrachromosomal covalently continuous double-stranded DNA molecule that occurs in bacteria and (rarely) other microorganisms. Natural plasmids vary from ≈5 kb to over 90 kb in size. They are classified in a number of different ways: the **copy number** (between 1 and 50) describes the ratio of plasmid/chromosome molecules; the **host range** describes the variety of species that can act as host (*see also incompatibility*). Plasmids capable of transfer are said to be conjugal; wide host-range, conjugal plasmids are described as promiscuous. Plasmid genes include drug resistance (**resistance plasmids**), genes for catabolic enzymes (catabolic plasmids), genes for antibiotic synthesis, and genes for toxin production. Plasmids are popular choices for **cloning vectors**.

plasmid chimera a hybrid DNA molecule that has been constructed *in vitro* by joining fragments of separate plasmids and that forms a new, biologically functional replicon when inserted into a cell.

plasmid curing the elimination of a plasmid from a cell culture by treatment with **Acridine Orange** at a concentration insufficient to inhibit chromosome replication but sufficient to inhibit plasmid replication.

plasmid engineering genetic engineering when a plasmid is used as a cloning vector.

plasmid incompatibility *see incompatibility*.

plasmid segregation *see segregation* (def. 3).

plasmid transconjugant *see transconjugant*.

plasmid vector *see cloning vector*.

plasmin or **fibrinolysin** or **fibrinase** EC 3.4.21.7; a serine proteinase that converts the insoluble fibrin of a blood clot into soluble products. It is formed from **plasminogen** by proteolysis, which results in a two-chain protein linked by a single disulfide bond; the heavier A chain is derived from the N-terminal part of plasminogen and the lighter B chain contains the enzymically active site. It exhibits preferential cleavage at Lys-|Xaa > Arg-|Xaa bonds, with a higher selectivity than trypsin. *See also blood coagulation*.

plasminogen or **profibrinolysin** a ≈90 kDa, single-polypeptide glycoprotein found in blood plasma. It is the inactive precursor of **plasmin**, into which it may be converted by hydrolysis of one peptide bond (Arg⁵⁶⁰-|Val⁵⁶¹) brought about, e.g., by the

action of **plasminogen activator** or **streptokinase**. Example from *Mus musculus* (precursor): database code PLMN_MOUSE, 812 amino acids (90.35 kDa). 3-D structure for plasminogen Kringle 4 chain a: database code NRL_IPMKA, 78 amino acids (8.90 kDa); chain b: database code NRL_IPMKB, 78 amino acids (8.90 kDa). *See also blood coagulation*.

plasminogen activator any serine proteinase that converts **plasminogen** into plasmin. Numerous variants occur in different tissues; they may be classified immunologically into **urokinase-type** and **tissue-type** plasminogen activators (*abbr.*: UPA and TPA respectively). Example 1, urokinase-type (EC 3.4.21.73); this catalyses specific cleavage of the Arg⁵⁶⁰-|Val⁵⁶¹ bond in plasminogen to form plasmin. It is found in urine, and is used clinically for therapy of thrombotic disorders. Example from human: database code UROK_HUMAN, 431 amino acids (48.47 kDa). Example 2, tissue-type (EC 3.4.21.68); this catalyses specific cleavage of the Arg⁵⁶⁰-|Val⁵⁶¹ bond in plasminogen to form plasmin. Cleavage after Arg³¹⁰ by plasmin or trypsin results in a two-chain form of the molecule; the two halves are held together by one or more disulfide bonds. TPA is active in tissue remodelling and destruction, and particularly also in fibrinolysis, and in cell migration. Example from human: database code UROT_HUMAN, 562 amino acids (62.85 kDa). Example 3, found in certain pathogens, is the activator EC 3.4.21.—, *also known as* coagulase/fibrinolysin precursor; it is a homopentamer occurring in the outer membrane. Example from *Yersinia pestis*, the plague bacterium, in which it is involved in transmission by the insect vector (flea): database code COLY_YERPE, 312 amino acids (34.57 kDa). *See also urokinase*.

plasma+ or (before a vowel) **plasm+** *comb. form* denoting of, pertaining to, related to, or derived from plasma.

plasmocyte an alternative name for **plasma cell**.

plasmodesma (*pl. plasmodesmata* or **plasmodesmas**) a fine cytoplasmic channel, found in all higher plants, that connects the cytoplasm of one cell to that of an adjacent cell. It is a roughly cylindrical, membrane-lined channel of diameter 20–40 nm, through the centre of which runs a narrower cylindrical **desmotubule**; the latter appears to be continuous with elements of the endoplasmic reticulum membrane of each of the connected cells. The function of plasmodesmata appears to be similar to that of **gap junctions** between animal cells. They permit passage of molecules of *M_r* less than about 800.

plasmodium (*pl. plasmodia*) **1** any member of the protozoan genus *Plasmodium*. Certain species parasitize humans causing malaria, while others parasitize other mammals. Their life cycle is complex, requiring, at different stages, an insect host and a mammalian host. **2** the motile multinucleate mass of protoplasm formed by the true **slime moulds** (Myxomycetes).

plasmolysis temporary shrinkage of the protoplasm of a plant or bacterial cell away from the cell wall, caused by the withdrawal of water from the cell. —**plasmolyse** *vb*.

plasmon the sum total of the extrachromosomal hereditary determinants in a cell.

plasmoptysis the localized extrusion of cytoplasm through the cell wall of a bacterium.

plasmosome an alternative name for **nucleolus**.

+plast *comb. form* denoting an organized subcellular particle from a living organism.

plastein a gel that may be formed on treating a partial hydrolysate of a protein with an endopeptidase. It was once thought to be a demonstration of the synthetic capability of proteinases.

plastic **1** any of a large and diverse group of polymeric materials, either synthetic or semisynthetic, that may be obtained following moulding, extrusion, etc. at a stage in their manufacture when soft or liquid. **2** made of or containing plastic; of the nature of plastic. **3** (of a material) capable of being formed or moulded. **4** (*in biology*) of or pertaining to an ability to adapt to the environment; able to change, develop, or grow. —**plasticity** *n*.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

+plastic

+plastic *see* **+plasia**.

plasticizer or **plasticiser** any substance added to another to render the latter less brittle and/or more mouldable and flexible.

plastid any member of a family of organelles found in the cytoplasm of eukaryotic plants, all of which contain DNA, are bounded by a double membrane, and develop from a common type, the **proplastid**. Plastids also contain pigments and/or storage materials. They include **aleuroplasts**, **amyloplasts**, **chloroplasts**, **chromoplasts**, **elaioplasts**, and **etioplasts**.

plastidome the total complement of **plastids** of a cell.

plastocyanin a soluble protein of eukaryotic plants that contains one atom of copper per molecule and is blue in its oxidized form. It occurs in the chloroplast at a molar concentration equal to that of P700, where it functions as a one-electron carrier ($E'_0 = +0.43$ V) between the cytochrome b_6-f and P700 of photosystem I. It is loosely bound to the inner thylakoid membrane surface. Example from pea: database code PLAS_PEA, 168 amino acids (17.14 kDa).

plastogene any of the hereditary determinants of plastids.

plastoglobuli droplets, 50–220 nm in diameter and without a bounding membrane, that occur close to thylakoid membranes in chloroplasts. Rich in lipids, particularly **plastoquinone**, they are particularly numerous when thylakoids are disrupting, and appear to be liberated therefrom.

plastome or **plastom** the total complement of the **plastogenes** of a cell.

plastoquinol *abbr.*: PQH₂; any hydroquinone derived from the corresponding plastoquinone by a two-electron reduction.

plastoquinone *abbr.*: PQ; any 2,3-dimethyl-5-multiprenylbenzoquinone; plastoquinones occur mainly in plants; the number of intact prenyl units in the side chain (commonly nine in the plastoquinone of higher plants) may be designated by a suffix numeral, e.g. plastoquinone-9 (*abbr.*: PQ-9). The plastoquinone–plastoquinol redox system ($E'_0 \approx +0.1$ V) functions in the electron-transport system, transferring electrons from photosystem II to the cytochrome b_6-f complex.

plate 1 (*in biology*) a Petri dish or other shallow vessel containing so-called solid or semisolid culture medium; a culture grown on this. **2** (*in chemistry*) any one of the horizontal perforated plates that comprise the fractionating component of a **plate column** for fractional distillation, or the theoretical equivalent of such a plate. *See also* **theoretical plate**. **3** (*in biology*) or **plate out** to inoculate a **plate** (def. 1) with microbial or other cells, or with material (suspected of) containing such cells. **4** to coat (a metal or other substrate) with a thin layer of (other) metal.

plateau (*pl. plateaux or plateaus*) **1** a part of a graph in which the value of the ordinate shows little or no change with increasing value of the abscissa. **2** a period of (relative) stability in the course of a progression. **3** to reach or pass through a plateau.

plate column *see* **plate** (def. 2).

plate height *see* **height equivalent to a theoretical plate**.

platelet or **thrombocyte** the smallest of the blood cells. It is an anucleate biconvex disk, 2–3 μ m in diameter, formed by division of the cytoplasm of a megakaryocyte and having an exterior coat rich in glycoprotein, a normal cell membrane, and a well-developed system of microtubules some of which are arranged as an equatorial ring. The main functions of platelets are in hemostasis, by (1) adhering to damaged blood vessels and aggregating to form a plug; (2) releasing various vasoconstrictive agonists, e.g. **epinephrine**, **serotonin**, and **thromboxane A₂**; (3) activating or binding certain coagulation factors; and (4) releasing factor XIII, **platelet-derived growth factor**, and **platelet factor IV**: blood clotting then occurs around the aggregated platelets. *See* **blood coagulation**.

platelet-activating factor *abbr.*: PAF or PAF-acether; 1-*O*-alkyl-2-acetyl-*sn*-glycerol 3-phosphocholine, where 'alkyl' may be hexadecyl or octadecyl; a potent general inflammatory mediator, active at concentrations in the range 10^{-12} to 10^{-9} .

platelet-derived growth factor receptor

M. It is released from a variety of cells in response to a number of stimuli, e.g. by IgE-mediated release from basophils and by the interaction of tissue-bound IgG, complement, and neutrophils. It is so named because of its high activity in causing aggregation of platelets; other effects of PAF include activation of polymorphonuclear leukocytes, monocytes, and macrophages, mediation of increased vascular permeation, hypotension, decreased cardiac output, and stimulation of uterine contraction. It is synthesized in the cell in either of two ways: (1) from 1-*O*-alkyl-*sn*-glycerol-3-phosphate by addition of an acetyl group and release of phosphate, forming 1-alkyl-2-acetyl-*sn*-glycerol, which then reacts with CDPcholine to form PAF and CMP; or (2) by an alternative remodelling pathway, in which the 2-fatty acyl group is removed from a phosphatidylcholine by phospholipase A₂ and replaced with acetyl. A number of analogues act as PAF agonists, e.g. the stable analogues in which the 2-acetyl group is replaced by an ethyl, methyl, or benzyl ether group; replacement of the choline moiety by (*N*-methylpyrrolidino)-ethanolamine greatly increases its effect. PAF antagonists include gossypol, hexanolamino-PAF, *bis*(methylthio)gliotoxin, and 2,5-*bis*(3,4,5-trimethoxyphenyl)-1,3-dioxolane.

platelet basic protein precursor *abbr.*: PBP; the precursor of several peptides generated by alternative processing. Example from human: database code PF4L_HUMAN, 128 amino acids (13.88 kDa); this contains a signal (residues 1–34), platelet basic protein (35–128), **LA-PF4** (44–128), **β -thromboglobulin** (48–128), and **NAP-2** (59–128).

platelet-derived endothelial cell growth factor *abbr.*: PD-ECGF; a **cytokine** produced by platelets, fibroblasts, and smooth muscle cells. It is a monomer of 471 amino acids. A related protein, **gliostatin**, which also exists as a dimer, is produced by astrocytes. Both proteins stimulate growth of endothelial cells *in vitro* and angiogenesis *in vivo*, and promote survival and differentiation of CNS neurons; they inhibit the growth of astrocytes and glial cells. These proteins may be identical with thymidine phosphorylase.

platelet-derived growth factor *abbr.*: PDGF; a **cytokine**, of ≈ 30 kDa, composed of two disulfide-bonded polypeptide chains denoted A and B, which are nonidentical but homologous. It is believed that human PDGF is an AB heterodimer but pig PDGF is a homodimer of B chains. PDGF is a potent mitogen for cells of mesenchymal origin (endothelial cells, epithelial cells, some fibroblasts, smooth muscle cells) and for glial cells; it has chemotactic properties. It is released by platelets on wounding and stimulates nearby cells to grow and repair the wound; it is also produced by endothelial cells, fibroblasts, monocytes, macrophages, smooth muscle cells, and neurons. PDGF-like proteins are produced by numerous human tumour cell lines and the B chain is almost identical to part of the transforming protein produced by simian sarcoma virus (*see* **sis**). Homodimers of A and B chains (both can bind to the receptor) are implicated in transformation processes. Examples from human (precursors); A chain (two forms, derived by alternative splicing; data refer to long form): database code PDGA_HUMAN, 211 amino acids (24.01 kDa); B chain: database code PDGB_HUMAN, 241 amino acids (27.25 kDa). *See also* **platelet-derived growth factor receptor**.

platelet-derived growth factor receptor or **PDGF receptor** a **tyrosine kinase** dimer, of $\alpha\alpha$, $\beta\beta$, or $\alpha\beta$ subunits, belonging to the immunoglobulin superfamily. It is a transmembrane cell surface receptor; the binding of platelet-derived growth factor (PDGF) stimulates tyrosine-kinase activity, which leads to phosphorylation of tyrosine residues in the receptor and in some cytoplasmic proteins. It belongs to a class of receptor tyrosine kinases, subclass III, that have a non-kinase insert in the cytoplasmic kinase region. Example from human (precursor); α subunit: database code PGDS_HUMAN, 1089 amino acids (122.53 kDa); β subunit: database code PGDR_HUMAN, 1106 amino acids (123.83 kDa).



platelet endothelial cell adhesion molecule

platelet endothelial cell adhesion molecule *see* **PECAM**.

platelet factor IV a 70-residue polypeptide secreted by platelets when they adhere at a site of injury. It is a **chemokine** that binds heparin strongly and is a powerful attractant for neutrophils and monocytes. Example from *Bos taurus*: database code PLF4_BOVIN, 88 amino acids (9.52 kDa).

plating *see* **plate** (def. 3, 4).

platysome (sometimes) the core particle of a **nucleosome**.

PLC *abbr.* for 1 preparative-layer chromatography. 2 phospholipase C; *see* **phospholipase**.

pleated sheet *see* **beta-pleated sheet**.

pleckstrin *other name*: platelet p47 protein; a platelet protein that is the major substrate for Ser/Thr phosphorylation by protein kinase C. Its function is not known. Example from human: database code P47_HUMAN, 350 amino acids (40.08 kDa). *See also* **pleckstrin homology domain**.

pleckstrin homology domain *abbr.*: PH domain; a domain first noted in the platelet protein **pleckstrin**, consisting of about 100 residues. Its function is unknown, but it has since been found in many (more than 60) proteins, especially those associated with intracellular signal transduction downstream of cell-surface receptors. It binds inositol-1,4,5-triphosphate. The 3-D solution structure of the domain from **dynamitin** is known.

plectonemic coiling a type of coiling of a double-stranded helix in which the two strands are intertwined, i.e. they may not be separated without uncoiling.

pleiotropin or **pleiotrophin** *other name*: heparin-binding growth factor 8; a heparin-binding mitogenic protein with neurite extension activity for osteoblasts and brain cells. Example (precursor) from human: database code PTN_HUMAN, 168 amino acids (18.92 kDa).

pleiotropism or **pleiotropy** the phenomenon in which a single gene is responsible for producing multiple, distinct, apparently unrelated phenotypic effects; an instance of such a phenomenon. The meaning is sometimes extended to include multiple, distinct, unrelated actions or effects of a single polypeptide agonist. —**pleiotropic** *adj.*

pleiotypic relating to the process by which a single stimulus to a living cell elicits multiple, unrelated responses.

pleomorphism or **pleomorphy** the quality of occurring in, or exhibiting more than one distinct form. *Compare* **polymorphism**. —**pleomorphic** *adj.*

pleuropneumonia-like organism *abbr.*: PPLO; an obsolete name for **mycoplasma** or **mollicute**.

Plenixglas or **Plenixglass** a proprietary name for **poly(methyl methacrylate)**.

PLI *abbr.* for pulsed laser interferometry.

+ploid *adj.* and *n. comb. form* indicating the specific multiple of homologous sets of chromosomes in cell. —**+ploidy** *n.*

ploidy the number of homologous sets of chromosomes in a cell, or in each cell of an organism.

plot 1 a diagram showing the relationship between two or more variables. 2 to construct such a plot.

plotter an instrument for automatically making a **plot** (def. 1).

PLP *abbr.* for 1 pyridoxal phosphate. 2 proteolipid protein; *see* **myelin proteins**.

plug flow a type of flow of fluids in which there is no mixing in the direction of flow.

plumper a disposable, nonscratching plastic device for adding a small volume (5–50 μ L) of a reagent to, and mixing it with, a liquid in a small container such as a spectrophotometer cuvette.

pluripoietin granulocyte colony stimulating factor (G-CSF) precursor. Pluripoietin α is a second human hemopoietic colony stimulating factor produced by the human bladder carcinoma cell line 5637. *See* **colony stimulating factor**.

pluripotent 1 (of a biomolecule) capable of being converted (biologically) into more than one active product. 2 (of a cell) capable of differentiating into more than one alternative type of mature cell.

plus and minus screening a technique for isolating genes for gene products that are present in one type of cell and absent in another. It involves comparing the results of **colony hybridization** of a cDNA clone library from the first cell type with cDNAs derived from the mRNAs of both cell types and then picking off the desired colonies from the reference plate.

plus end the end of a microtubule or actin filament at which addition of monomers occurs most readily; the 'fast-growing' end of a microtubule or actin filament. The plus end of an actin filament is also known as the **barbed end**.

plus strand 1 the parental polynucleotide strand in bacteriophages whose genomes consist of duplex RNA. It is the template used for the production of the complementary minus strand. 2 virus-specified messenger RNA, or any given strand of nucleic acid (RNA or DNA) whose base sequence is equivalent to that of virus-specified mRNA. Any strand having a base sequence complementary to virus-specified mRNA is described as a minus strand. 3 (sometimes) an alternative term for **coding strand** (of genomic duplex DNA). *Compare* **minus strand**.

pM or **pM** symbol for picomolar.

Pm symbol for promethium.

PMA *abbr.* for phorbol 12-myristate 13-acetate; *see* **phorbol ester**.

PMF *abbr.* for protonmotive force, or sometimes (facetiously), after its discoverer, Peter Mitchell force.

PMN *abbr.* for 1 pyridine mononucleotide. 2 polymorphonuclear (leukocyte).

pmol symbol for picomole.

PMR or **pmr** *abbr.* for proton magnetic resonance; *see* **nuclear magnetic resonance spectroscopy**.

PMS *abbr.* for 1 phenazine methosulfate. 2 pregnant mares' serum.

PMSF *abbr.* for phenylmethylsulfonyl fluoride.

PMSG *abbr.* for pregnant mares' serum gonadotropin.

PN-1 *see* **nexin 1**.

PNA *abbr.* for pentose nucleic acid (i.e. ribonucleic acid).

PNAD *abbr.* for protein NH₂-terminal arginine deamidase; an enzyme, discovered in the pig, that catalyses deamidation of N-terminal arginine residues in peptides.

pneumo+ or **pneumono+** or (before a vowel) **pneum+** or **pneumon+** *comb. form* of or pertaining to a lung or lungs; respiratory.

pneumococcus (*pl.* **pneumococci**) any strain of bacterium belonging to the species *Streptococcus pneumoniae*. —**pneumococcal** *adj.*

pneumococcus capsule swelling reaction an alternative name for **Quellung reaction**.

p[NH]ppA symbol for adenosine 5'-[β , γ -imido]triphosphate (alternative to **AdoPP[NH]P**).

PNK *abbr.* for polynucleotide kinase.

PNMT *abbr.* for phenylethanolamine N-methyltransferase.

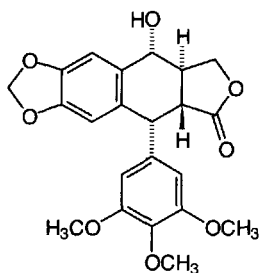
PNP *abbr.* for 1 para-nitrophenol. 2 purine-nucleoside phosphorylase.

Po symbol for polonium.

pocket (in molecular biology) a cavity or hollow in a three-dimensional structure, e.g. a substrate pocket of an enzyme.

podocalyxin a sialated glycoprotein of the kidney glomerulus. It is the major sialoprotein of the podocyte foot process glycolyx. It is prominently expressed on renal glomerular epithelial cells and found in patients with glomerulonephritis.

podophyllotoxin a **lignan** found in the May apple, or American mandrake (*Podophyllum peltatum*). It is known in folk medicine as a cathartic and poison, and is useful in topical treatment of warts, including venereal warts; it may have antiviral action. In mammalian cells it inhibits assembly of tubulin into microtubules, and also nucleoside transport. It is used in the synthesis of **Etoposide** and **Teniposide**.



podophyllotoxin

+poiesis *noun comb.* form indicating an act or process of making (something specified), e.g. erythropoiesis, hemopoiesis. — **+poietic** *adj.*

poikilo+ *comb. form* indicating various, variable.

poikilocyte an erythrocyte that is distorted in shape and usually larger than normal, found in the blood of persons with pernicious anemia and some other anemias.

poikilosmotic describing an organism whose internal osmotic pressure varies with that of its environment. *Compare* homoiosmotic.

poikilothermic or **poikilothermal** describing an organism whose body temperature varies with that of the environment. *Compare* homoiothermic. — **poikilotherm** *n.*

point **1** (*in mathematics*) a any dimensionless geometric element whose position in space is defined by its coordinates. **b** a location or position, especially on a diagram, graph, etc. **2** a decimal point. **3** a **point of inflection**.

point group **1** a notation used in describing the geometrical arrangement of the symmetry of globular oligomeric proteins. Such molecules may belong to any of the three point groups having **cyclic symmetry**, **dihedral symmetry**, or **cubic symmetry**. **2** (*in crystallography*) a group of symmetry operations that leave unmoved at least one point within the object to which they apply. It can describe single molecules or actual crystals.

point ligand any of two or more individual atoms or groups linked covalently to a specified carbon atom in an organic compound. Point ligands may be identical or nonidentical, but they may not occupy the same position in space.

point mutation any **mutation** (def. 1) involving a single base change. *See also* transition, transversion.

point of inflection any point on a graph where the direction of curvature changes.

poise *symbol:* P; the cgs unit of dynamic viscosity; it has the dimensions of $\text{g s}^{-1} \text{cm}^{-1}$; $1 \text{ P} = 0.1 \text{ N s m}^{-2}$. [After Jean-Louis Marie Poiseuille (1799–1869), French physician and physicist.]

Poiseuille's equation an equation describing the behaviour of a liquid flowing under pressure through a capillary tube of length l and radius r . The volume flowing per second, dV/dt , is given by:

$$dV/dt = \pi Pr^4/8\eta l,$$

where η is the viscosity of the liquid and P is the pressure difference of the liquid along the tube over a distance l . [After Jean-Louis Marie Poiseuille (*see* poise).]

poison **1** (*in toxicology*) any substance that, in relatively small amounts, can impair the functioning of, or cause structural damage to a cell or organism. **2** (*in chemistry*) any substance that, at relatively low concentrations, inhibits a chemical reaction or inactivates a catalyst. **3** to add or administer poison to (a cell, organism, catalyst, etc.).

pokeweed mitogen *abbr.:* PWM; a plant lectin (phytoagglutinin) extractable from the roots of pokeweed, *Phytolacca americana*, a North American plant. It is one of the lectins most commonly used in immunology and cell biology. It has an affinity for *N*-acetyl- β -D-glucosamine oligomers, and ag-

glutinates leukocytes and erythrocytes (of all ABO blood groups). It also has mitogenic properties, which appear to reside in five (or more) separable components; one of these activates both T and B lymphocytes, and the others are specific for T lymphocytes.

poky mutant a mutant strain of *Neurospora* with a lower rate of growth than that of wild-type strains, due to absence or deficiency of certain components of the respiratory chain. The *poky* gene appears to occur in mitochondrial DNA and is transmitted to sexually derived progeny only when the female parent exhibits the poky phenotype.

pol **1** a retroviral gene that encodes a polypeptide containing **retropepsin**, **reverse transcriptase**, and **ribonuclease H** (*see* **ribonuclease**). **2** as **polA**, **polB**, **polC**, genes that encode DNA polymerases I to III.

Pol the gene product of **polA**.

polA the structural gene in *Escherichia coli* for DNA polymerase I (Pol or Pol I; *see* **DNA polymerase**), which serves *in vivo* as a repair polymerase and as an exonuclease in the generation of **Okazaki fragments**. The enzyme belongs to family A of DNA polymerases and has three characteristic domains: 5'-exonuclease (amino acids 1–323), 3'-exonuclease (amino acids 324–517), and polymerase (amino acids 521–928). Klenow polymerase (amino acids 324–928) contains the latter two domains. *See* **Klenow enzyme**.

polar **1** pertaining to, or possessing, a **pole** or poles. **2** (*in chemistry*) (of a group or a molecular entity) having an uneven distribution of electrons and hence a permanent dipole moment. *See also* **polarity**.

polar body (*in biology*) either of the two small cells that form as a result of the division of a primary oocyte during its development to a mature ovum. One is formed in the first division of meiosis and the other in the second division of meiosis. At each division, the cytoplasm divides unequally, the polar body being of much smaller size than the developing oocyte. At the second division in which a polar body is formed, the polar body and the developing oocyte each contains a haploid set of chromosomes. The polar bodies eventually degenerate.

polar bond **1** a covalent bond in which the electron pair(s) of the bond are held with greater force by one of the two bonded atoms leading to the existence of a permanent dipole moment in the group or molecular entity in question. **2** an ionizable bond between two oppositely charged ions of a salt.

polar coordinate (*in mathematics*) either of two numbers that locate a point in a plane, one being the distance along a line to the point from the origin and the other being the angle this line makes with a fixed line passing through the origin.

polar group *see* **polar** (def. 2).

polarimeter **1** an instrument for measuring the degree of polarization of light. **2** an instrument for measuring the rotation of the plane of vibration of linearly polarized light as a result of passing through a given length of optically active material. — **polarimetric** *adj.*; **polarimetry** *n.*

polarity **1** the state, or condition, of having **poles**. **2** having a differential distribution of a property or properties along an axis; *see* **polar**. **3** having opposing physical properties at different points, e.g. north and south magnetic poles, positive and negative electric charges, hydrophobic and hydrophilic regions. **4** the state of being electrically either positive or negative. **5** (of a solvent) **ionizing power**. **6** (of an organism) the tendency to develop differentially along an axis; e.g. plants exhibit polarity in developing roots downwards and stems upwards.

polarity gradient (*in genetics*) the quantitative effect of a **polar mutation** in one gene on the expression of later genes in the operon.

polarity mutant a mutant gene formed by a **polar mutation**.

polarizability or **polarisability** the quality of being able to be polarized.

Bạn đang đọc một quyển sách được tặng bởi HUỲ NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



polarization

polarization or **polarisation** 1 inducing polarity. 2 the quality of being or becoming polarized.

polarization microscope or **polarisation microscope** an alternative name for **polarizing microscope**.

polarize or **polarise** 1 to acquire or cause to acquire **polarity**. 2 to cause the polarization of light or other electromagnetic radiation; see **polarized light**. 3 to cause the electrical resistance of an electric cell to increase through the accumulation of electrolysis products at an electrode.

polarized light or **polarised light** light (or other electromagnetic radiation) in which the angular orientations of the electric vectors of its transverse vibrations are nonrandom. In **linearly** (or **plane-**) **polarized light** the electric vectors all lie in a given plane containing the light beam; to an observer looking along the light beam the vectors appear as a straight line through the axis of propagation. In **circularly polarized light** the angular orientations of the electric vectors are such that a given point on the vectors generates a (left-handed or right-handed) helix whose axis is the direction of the beam; to an observer looking along the light beam this point appears to describe the circumference of a circle, the light being defined as right or left circularly polarized depending on whether the point appears to the observer looking towards the source to be rotating clockwise or anticlockwise, respectively. Circularly polarized light may be considered to consist of two superimposed linearly polarized waves of equal amplitude but differing in phase by an odd number of quarter-wavelengths and having their electric vectors mutually at right angles; conversely, if right and left circularly polarized waves of equal amplitude are superimposed the resultant is a linearly polarized wave. In **elliptically polarized light** the angular orientations of the electric vectors are such that a given point on the vectors generates a (left-handed or right-handed) flattened helix whose axis is the direction of the beam; to an observer looking along the light beam this point appears to describe the perimeter of an ellipse, the degree of such polarization being characterized by the **ellipticity** and by the angular orientation of the major axis of the ellipse. Elliptically polarized light may be considered to result from the superimposition of two linearly polarized waves of equal amplitude and having their electric vectors mutually at right angles but where the phase difference is neither zero nor exactly an even or odd number of quarter-wavelengths; alternatively, it may be considered to result from the superimposition of right and left circularly polarized waves of differing amplitude.

polarizer or **polariser** an optical device for producing linearly polarized light, e.g. a **Nicol prism** or a sheet of **Polaroid** (def. 1).

polarizing microscope or **polarization microscope** a microscope that is useful for examining the birefringence properties of small objects, or for visualizing objects by virtue of their birefringence properties. Also: polarising microscope, polarisation microscope.

polar microtubule see **mitotic spindle**, **prometaphase**.

polar molecule a molecule in which there is a polarized distribution of positive and negative charges due to an uneven distribution of electrons. Polar molecules are likely to be soluble in water.

polar mutation a mutation in one gene that reduces the expression of genes further from the promoter in the same operon.

polarogram the current-voltage diagram obtained by **polarography**.

polarograph 1 an instrument used for **polarography**. 2 to subject (a sample) to analysis by polarography.

polarography a method for quantitative or qualitative determination of small concentrations of reducible or oxidizable substances based on the nature of the current-voltage curve obtained when a test solution is subjected to electrolysis with a steadily increasing voltage between, e.g., a dropping mercury cathode and a pool of mercury serving as anode. The current-voltage curve obtained (called a **polarographic wave**) is

sigmoidal; the potential at the midpoint (called the **half-wave potential**) is characteristic of the substance being oxidized or reduced and the limiting current is related to its concentration. —**polarographic** *adj.*

Polaroid a *proprietary name* for 1 a thin, transparent material that linearly polarizes light transmitted by it (see **polarized light**). It consists of oriented, doubly refracting microcrystals of herapathite (quinine iodosulfate) in a matrix of cellulose acetate. 2 a photographic process or camera producing a nearly instantaneous, permanent photograph.

polar solvent any solvent of relatively high **ionizing power**.

polar zipper a motif in proteins that is considered to promote binding between protein subunits or between proteins as a result of interactions between polar residues, which are aligned in the form of a 'zip fastener'. The polar groups may consist of repeats of glutamines, Asp-Arg or Glu-Arg repeats, or repeats of glutamates and either lysines or histidines. The concept derives from the previously coined term **leucine zipper**.

pole 1 either end of an axis of rotation of a sphere or spheroid body, especially of the Earth. 2 (*in biology*) a either of the differentiated zones at opposite ends of an axis in a cell, organ, or organism. b (*in mitosis or meiosis*) either of the opposite ends of the **spindle**. 3 (*in physics*) a any of the zones in a magnetized body, commonly two in number and near its extremities, where the lines of magnetic flux appear to originate (**north pole**) or end (**south pole**). b either of the (positively or negatively charged) terminals of an electric battery, or of the corresponding terminals of a d.c. electric generator; either of two points, positions, or regions carrying opposite electric charges in a piece of material, system, or (polar or polarized) molecular entity. Compare **dipole**. See also **polar**.

policeman a device used for detaching adherent particles of precipitate, cells, etc. from the wall of a vessel and consisting commonly of a glass stirring-rod with a short piece of rubber tubing fitting tightly over one end, but sometimes of a small snipe feather cemented into the end of a capillary tube.

poly+ *comb. form* 1 denoting more than one; many or much; more than the usual number or amount; often used interchangeably with **multi+**. 2 indicating heterogeneity of a physical attribute e.g. polychromatic, polydisperse. Compare **mono+** (def. 2). 3 (*in chemistry and biochemistry*) a denoting more than one of a specified component or substituent, e.g. polyamine, polyene, polyester; in names of large molecules, usually denoting more than 10–20 of the component specified, especially in contrast to **oligo+** (def. 2), e.g. polynucleotide, polypeptide, polysaccharide. Compare **mono+** (def. 3). b denoting polymer of a specified substance, e.g. polyglycine, polystyrene. c indicating more than one function of the kind specified, e.g. polyacidic, polybasic, polydentate, polyhydric. Compare **mono+** (def. 5).

poly(A) *abbr.* for polyadenylate.

polyacidic (of a **base** (def. 1)) able to accept more than one proton per molecule in solution.

polyacrylamide (gel) a common name for any hydrogel containing a three-dimensional matrix of randomly cross-linked polymers of acrylamide (2-propenamide; $\text{CH}_2=\text{CH}-\text{CO}-\text{NH}_2$). The cross-linking agent commonly used is *N,N'*-methylene bisacrylamide, $(\text{CH}_2=\text{CH}-\text{CO}-\text{NH})_2\text{CH}_2$, polymerization being initiated by various alternative methods for generating free radicals. The pore size and rigidity of the gel depend on the concentration of acrylamide and the proportion of cross-linking agent used.

polyacrylamide gel electrophoresis *abbr.*: PAGE; a technique of **zone electrophoresis** that uses polyacrylamide gel as supporting medium. It is especially useful for the separation of biopolymers and fragments thereof. A two-dimensional electrophoresis has been used to resolve as many as 1000 proteins: in the first step the polypeptides are separated by **isoelectric focusing electrophoresis** and in the second step by **SDS (polyacrylamide) gel electrophoresis**. The 2-D process was devised by P. H. O'Farrell after whom it is often named.



polyacrylamide gradient gel electrophoresis

polyacrylamide gradient gel electrophoresis *an alternative name for gradient gel electrophoresis.*

polyadenylate **1** *abbr.*: poly(A); any homopolymer of adenylate (i.e. adenosine 5'-monophosphate). Most eukaryotic mRNAs are terminated at the 3' end by a tract of poly(A) 40–200 residues long (see **poly(A) tail**). **2** to add a stretch of polyadenylate to a molecule. —**polyadenylation** *n.*

poly(ADP-ribosyl) a structure found attached to histones H1 and H2B, to other eukaryotic chromosomal proteins, and to some mitochondrial proteins. It consists of a polymer of 20–50 ADP-D-ribosyl units linked by glycosidic bonds; linkage is formed by transfer of the ADP-ribosyl unit of NAD⁺, linking its ribosyl C-1 position to the C-2' position of the ribosyl of the terminal ADP-ribose of the growing chain. The attachment to protein appears, at least in some instances, to be through an ester linkage between the terminal ribose group and certain free carboxyl groups in the protein. See also **ADP-ribosylation**; **NAD(P)⁺–arginine ADP-ribosyltransferase**.

poly(ADP-ribosyl) transferase EC 2.4.2.30; *recommended name*: NAD⁺ ADP-ribosyltransferase; *systematic name*: NAD⁺:poly(adenine-diphosphate-D-ribosyl)-acceptor ADP-D-ribosyl-transferase; *other names*: poly(ADP) polymerase; poly(adenosine diphosphate ribose) polymerase; ADP-ribosyl-transferase (polymerizing); poly(ADP-ribose) synthetase. An enzyme responsible for ADP-ribosylation of acceptors; it catalyses a reaction between NAD⁺ and (ADP-D-ribosyl)_n-acceptor to form nicotinamide and (ADP-D-ribosyl)_{n+1}-acceptor. It often has zinc as cofactor (as in example); the acceptor is a histone or the enzyme itself. The modification is dependent on DNA and is involved in the regulation of cellular processes such as differentiation, proliferation, and tumour transformation and also in the regulation of the molecular events involved in the recovery of a cell from DNA damage. Example from human: database code PPOL_HUMAN, 1013 amino acids (112.82 kDa). See also **NAD(P)⁺–arginine ADP-ribosyltransferase**. Compare **ADP-ribosylation**.

polyamide any polymer in which the constitutional units are joined by amide or thioamide links. Compare **nylon**, **polypeptide**.

polyamine any organic compound containing two or more amino groups. **Putrescine**, **spermine**, and **spermidine** are important animal polyamines. Polyamines are synthesized by various routes. In animals, **putrescine** is the decarboxylation product of **ornithine**. Decarboxylated S-adenosylmethionine (see **S-adenosylmethionine decarboxylase**) is a substrate in a reaction in which propylamine is transferred to putrescine to form spermidine, and again to spermidine to form spermine. See also **ornithine decarboxylase**.

poly(amino acid) any homopolymer, or mixture of homopolymers, of a particular (specified) amino acid.

polyampholyte any amphoteric **polyelectrolyte**.

polyanion any macromolecular ion that carries multiple negative charges.

poly(A) polymerase *abbr.*: PAP; *the recommended name for polynucleotide adenyllyltransferase*.

poly(A) tail a sequence of adenyl residues at the 3' end of eukaryotic mRNA. Almost all mature eukaryotic mRNAs have 3'-poly(A) tails of 40–200 nucleotides, those of histones being a notable exception. The poly(A) tail is added enzymically to the primary transcript, which is first cleaved 10–30 nucleotides past a highly conserved AAUAAA sequence. The poly(A) tail is then generated from ATP through the action of **polynucleotide adenyllyltransferase**. In practical terms the poly(A) tail on mRNA has facilitated its ready isolation from total cellular RNA by affinity chromatography on oligo(dT)-cellulose.

polyatonic (of a molecular entity) composed of two or more atoms.

polybasic (of an acid (def. 1)) able to give up more than one proton per molecule in solution.

poly(C) *abbr.* for polycytidylate.

polycarbonate any polymer in which the structural units are

linked by carbonate ester groups, usually formed by the polymerization of diphenylols with phosgene (i.e. carbonyl chloride). Such polymers are tough thermoplastic resins.

polycation any macromolecular ion that carries multiple positive charges.

polychromatic **1** relating to, or possessing, various or varying colours; consisting of, relating to, or emitting light (or other electromagnetic radiation) of two or more frequencies or wavelengths. **2** of, relating to, or showing **polychromatophilia**.

polychromatophilia the quality of being stainable by more than one type of dye, especially by both acid and basic dyes.

polycistronic messenger RNA or **multicistronic messenger RNA** or **polygenic messenger RNA** any messenger RNA molecule that carries more than one initiation site and encodes the synthesis of two or more (usually related) gene products. It is formed by transcription of a single operon and is characteristic of prokaryotes.

polyclonal of, pertaining to, being formed by, or derived from two or more clones (def. 1).

polyclonal antibody an antibody that arises from the immune response to an immunogen. Each **epitope** on the immunogen has the potential for causing the formation of a clone of **B lymphocytes** each capable of synthesizing an antibody specific to the particular epitope.

polycyclic having more than one ring of atoms in the molecule; used especially of molecules with condensed ring systems.

polydentate *an alternative term for multidentate.*

polydeoxynucleotide or **polydeoxyribonucleotide** any **polynucleotide** composed solely of **deoxynucleotide** residues.

polydisperse describing a colloidal system whose disperse phase consists of particles of many different sizes. —**polydispersity** *n.*

polyelectrolyte any macromolecular substance that, on dissolution in water or another ionizing solvent, dissociates to give polyanions or polycations together with an equivalent of oppositely charged ions, which may be simple ions or themselves polyanions.

polyene any unsaturated organic compound containing two or more carbon-carbon double bonds.

polyene antibiotic any member of a group of structurally related antibiotics produced by *Streptomyces* spp. They possess a large lactone ring containing 4–7 conjugated double bonds (generally all-trans) and substituted by aliphatic groups, by a residue of **mycosamine** or another amino sugar, and by numerous hydroxyl groups. The polyene antibiotics include **amphotericin B** and **nystatin**; they interact with sterols in cell membranes and hence have antifungal but not antibacterial activity and tend to be toxic to mammals. Compare **macrolide**.

polyenoic acid any alkenyl carboxylic acid containing two or more double bonds per molecule.

polyester any synthetic polymer in which the constitutional units are joined by ester bonds; proprietary materials include Dacron and Mylar. Polyesters are extensively used in textiles, ropes, etc.

polyethylene *an alternative name (esp. US) for polythene. See also polyethylene glycol.*

polyethylene glycol *abbr.*: PEG; any polymer of general formula H(OCH₂CH₂)_nOH, where *n* ≥ 4; a suffixed number may be used as a guide to the average relative molecular mass, e.g. polyethylene glycol 4000. Polyethylene glycols are clear viscous liquids or white solids, useful as embedding media for microscopy, as bases for ointments, and in radioimmunoassays.

poly(ethyleneimine) a water-soluble synthetic linear polymer of ethyleneimine (i.e. aziridine) of 30–40 kDa. It is useful for the selective precipitation of proteins and nucleic acids. It is also used as a substituent in hydroxylated chromatographic support matrices, as in PEI-cellulose, to form anion exchangers. *Proprietary name*: Polymin P.

polyfructosan *former term for fructan.*

poly(G) *abbr.* for polyguanylate.

Bạn đang đọc một quyển sách được tặng bởi HUỠNG NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



polygalactan

polygalactan *former term for galactan.*

polygalacturonase EC 3.2.1.15; *other name:* pectinase; an enzyme that hydrolyses 1,4- α -D-galactosiduronate linkages in pectin and other galacturonans in a random fashion. It participates in cell-wall degradation in plants. Example, polygalacturonase I precursor from *Aspergillus niger*: database code PGL1_ASPNG, 368 amino acids (38.11 kDa).

polygene any one of a group of genes, each of which individually exerts a small effect on the phenotype but that together control a quantitative character such as stature.

polygenic describing a character or trait that is controlled by polygenes. *Compare* **multigenic**.

polyglucosan *a former term for glucan.*

polygon any closed plane figure bounded by three or more (nonintersecting) straight lines. —**polygonal** *adj.*

polyhedron (*pl.* **polyhedrons** or **polyhedra**) any solid figure bounded by four or more plane faces, each of which is a polygon. —**polyhedral** *adj.*

polyhydric describing any chemical compound containing two or more hydroxyl groups per molecule; used especially of alcohols.

poly(hydroxybutyrate) a linear polymer of approximately 1500 (R)-3-hydroxybutyrate residues held together in ester linkage. It is found in the cells of a number of prokaryotic genera where it occurs in membrane-bound granules and acts as a storage material.

poly(I) *abbr. for polyinosinate.*

polyion any macromolecular ion that carries multiple charges.

polykaryocyte a multinucleate cell. —**polykaryotic** *adj.*

polyketide or (*sometimes*) **acetogenin** any of a diverse group of natural products synthesized via linear poly- β -ketones, which are themselves formed by repetitive head-to-tail addition of acetyl (or substituted acetyl) units indirectly derived from acetate (or a substituted acetate) by a mechanism similar to that for fatty-acid biosynthesis but without the intermediate reductive steps. In many cases, acetyl-CoA functions as the starter unit and malonyl-CoA as the extending unit. Various molecules other than acetyl-CoA may be used as starter, often with methylmalonyl-CoA as the extending unit. The poly- β -ketones so formed may undergo a variety of further types of reaction, which include alkylation, cyclization, glycosylation, oxidation, and reduction. The classes of product formed – and their corresponding starter substances – comprise *inter alia*: conine (of hemlock) and orsellinate (of lichens) – acetyl-CoA; flavanoids and stilbenes – cinnamoyl-CoA; tetracyclines – amide of malonyl-CoA; urushiols (of poison-ivy) – palmitoleoyl-CoA; erythronolides – propionyl-CoA and methylmalonyl-CoA as extender.

polylabelling labelling where more than one atom of a molecule has been isotopically substituted.

polylinker any linker containing multiple restriction sites.

polymannan *a former term for mannan.*

polymer **1** (*in chemistry and biochemistry*) any substance that is composed of molecules containing a large number of constitutional units (or 'mers') that are in repetitive covalent linkage and that may be of one or more than one species. Polymers are generally considered to comprise at least ten mers, although sometimes the term is taken to imply simply 'more than one' mer. Hence 'polymer' may or may not embrace **oligomer**, depending on the branch of chemistry or biochemistry concerned. **2** (*in molecular biology and enzymology*) (*sometimes*) *an alternative term for multimer.* See also **heteropolymer**, **homopolymer**. —**polymeric** *adj.*; **polymerize** or **polymerise** *vb.*; **polymerization** or **polymerisation** *n.*

polymerase *a general name for any of various enzymes (transferases) catalysing the formation of biological polymers (def. 1), especially of polynucleotides. See DNA polymerase, RNA polymerase.*

polymerase accessory protein any of a group of proteins that are associated with **DNA polymerases**. Examples (1) DNA polymerase α binding factor from yeast; database code

CTF4_YEAST, 926 amino acids (104.18 kDa); (2) phage T4 accessory proteins: protein 44, database code DPA4_BPT4, 319 amino acids (35.74 kDa); protein 45, database code DPA5_BPT4, 227 amino acids (24.80 kDa); protein 62, database code DPA6_BPT4, 187 amino acids (21.34 kDa).

polymerase chain reaction *abbr.:* PCR; a method whereby a specific sequence of nucleotides within a double-stranded DNA is amplified. The sequence is identified by the use of short synthetic oligonucleotides that are complementary to the two terminal regions of the DNA sequence to be amplified; these oligonucleotides are extended by the thermostable **Taq DNA polymerase** on the DNA template. The effect is that the new chains span the region delimited by the two chosen termini. The nascent chains are heat denatured and the process is repeated. Up to 100 000-fold **amplification** of the DNA stretch required may be obtained. The method was first described by Kary B. Mullis in 1984. The use of the thermostable **Taq** polymerase from a thermophilic bacterium avoids the need to add more polymerase at each cycle because the enzyme is not inactivated at the temperature of denaturation of the DNA. PCR amplification has become an indispensable tool in a great variety of applications. Clinically it may be used for the rapid diagnosis of infectious diseases and the detection of rare mutations associated with cancer. Forensically, the DNA from even a single hair or sperm can be used to identify the donor. RNA may be amplified by first converting it to cDNA through the use of reverse transcriptase.

polymetaphosphate (*sometimes*) *an alternative name for polyphosphate.*

poly(methyl methacrylate) a clear, easily worked, relatively nontoxic thermoplastic, useful as a substitute for glass and in the construction of many pieces of scientific apparatus. *Proprietary names:* Lucite, Perspex, Plexiglass, etc.

Polymix P *a proprietary name for poly(ethyleneimine).*

polymorph *see polymorphonuclear leukocyte.*

polymorphic or **polymorphous** having, assuming, or passing through various different forms or styles; exhibiting or undergoing **polymorphism**.

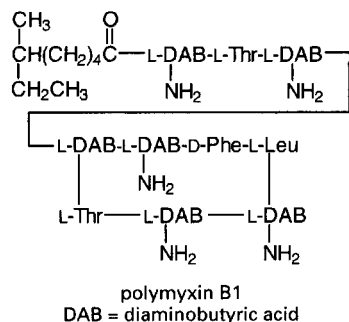
polymorphism **1** (*in biology*) **a** the occurrence of two or more distinct kinds of individual (**morphs**) belonging to the same species within a population that is interbreeding freely. **b** **genetic polymorphism.** **2** (*in chemistry*) the existence of the same substance in two or more different crystalline forms. See also **enantiomorph**.

polymorphonuclear **1** describing a cell whose nucleus is multipartite or **polymorphic.** **2** a **polymorphonuclear leukocyte**.

polymorphonuclear leukocyte or **polymorphonuclear** or **polymorph** *abbr.:* PMN; any white blood cell of the myeloid series that in its mature form has a lobed and variably shaped nucleus and granular cytoplasm; included are basophil leukocyte, eosinophil leukocyte, and neutrophil leukocyte. The term is often used erroneously for neutrophil leukocyte solely.

polymorphous *a variant form of polymorphic.*

polymyxin any of a group of related antibiotics isolated from cultures of *Bacillus polymyxa* and active against most Gram-negative bacteria. Individual polymyxins are designated by suffixed letters, e.g. polymyxin B. All are decapeptides containing five or six residues of L-2,4-diaminobutyric acid. Their biosynthesis does not involve ribosomes (*see* **protein and peptide biosynthesis**). The sequence of seven residues at the C-terminal end is formed into a heterodetic ring through an isopeptide link to one of the diaminobutyric acid residues, while the N-terminal residue is acylated with a 6-methylheptanoyl or 6-methyloctanoyl group. Polymyxins bind to the cytoplasmic membrane of sensitive organisms, causing leakage of cytoplasmic components; however, they also attack the cell membranes of eukaryotic cells and therefore have limited clinical usefulness; they are used in antibiotic ointments. Polymyxin B inhibits protein kinase C. See also **colistin**.

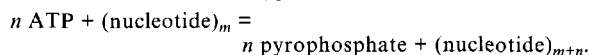


polynucleate or **polynuclear** (of a cell) containing several or many nuclei; **multinucleate**.

polynucleotidase *an alternative name for polynucleotide phosphatase.*

polynucleotide any one-stranded homo- or heteropolymer of at least about ten **nucleotide units** connected by phosphodiester linkages between (usually) the 3' position on the glycoside moiety of one nucleotide unit and the 5' position on the glycoside moiety of the adjacent one, or any two-stranded molecule comprised of two such one-stranded molecules held together by hydrogen bonds. Comprehends the term **nucleic acid**.

polynucleotide adenyllyltransferase EC 2.7.7.19; *other names*: NTP polymerase; RNA adenylating enzyme; poly(A) polymerase; an enzyme that adds the **poly(A) tail** to mRNA. It catalyses a reaction of the type:



PAP I and II are long forms of the enzyme that contain four functional domains: a ribonucleoprotein-type RNA-binding domain, a catalytic region, two nuclear localization signals, and a C-terminal serine/threonine-rich region. Clones of a number of truncated forms have been obtained, some of which are expressed in tissues. Examples from vaccinia virus subunits PAP1 (VP55) and PAP2 (VP39): database codes PAP1_VACCC, 479 amino acids (55.52 kDa), and PAP2_VACCC, 333 amino acids (38.87 kDa). Example from yeast: database code PAP_YEAST, 568 amino acids (64.48 kDa). The enzyme from *Escherichia coli* (also known as **plasmid copy number protein**): database code PCNB_ECOLI, 468 amino acids (53.81 kDa). *See also* **polyadenylate**.

polynucleotide kinase *abbr.*: PNK; any of a number of enzymes belonging to subclass EC.2.7 that phosphorylate the 5' end of nucleic acids. The reaction is between ATP and the 5'-OH of RNA or DNA to form ADP and RNA or DNA-5'-phosphate. Example, phage T4 enzyme: database code KIPN_BPT4.301 amino acids (34.58 kDa).

polynucleotide ligase *an alternative name for* DNA ligase (ATP); *see* **DNA ligase**.

polynucleotide ligase (NAD⁺) *an alternative name for DNA ligase (NAD⁺); see DNA ligase.*

polynucleotide phosphatase or **polynucleotidase** either of the enzymes polynucleotide 3'-phosphatase (EC 3.1.3.32) or polynucleotide 5'-phosphatase (EC 3.1.3.33).

polynucleotide phosphorylase *an alternative name for*
polyribonucleotide nucleotidyltransferase (EC 2.7.7.8).

polyol any polyhydric alcohol.

polyol (metabolic) pathway (*sometimes*) a metabolic pathway by which nonphosphorylated sugars and sugar alcohols are interconverted.

polyomavirus any member of a genus (*Polyomavirus*) of mouse viruses of the Papovaviridae family. They are widely distributed and may be oncogenic under appropriate conditions. Ban đầu đọc, một quyển sách được tặng bởi HUY

polyose *an obsolete term for any polymer consisting of an unknown or unspecified number of covalently linked monose residues.*

polypeptide **1** any peptide comprising more than about 10–20 amino-acid residues connected by peptide linkages. Unlike a **protein** it usually lacks appreciable tertiary structure in solution and is not liable to irreversible denaturation. **2** (*loosely*) an alternative term for **protein**. **3** describing another entity of a polypeptide nature, e.g. polypeptide chain, polypeptide hormone.

polypeptide hormone *an alternative term for peptide hormone.*

polyphase or **polyphasic** 1 (of matter) consisting of two or more phases. 2 or **multiphase** (of an electrical system or device) having, generating, or using two or more alternating electric voltages of the same frequency but differing from each other in phase. 3 (of a process) consisting of or occurring in a number of separate stages.

polyphenol oxidase EC 1.10.3.1; *recommended name* catechol oxidase; any of a family of copper-containing oxidoreductases that catalyse the oxidation of mono and *o*-diphenols to *o*-diquinones. These enzymes are found in plants, localized in membranes of plastids, but their normal function is unknown. However, they are responsible for brown discoloration ('browning') in fruit and vegetables on storage, as their products give rise, probably nonenzymatically, to polyphenolic melanin-like compounds. Sequences of polyphenol oxidase genes from different plants are very similar, but when compared with related bacterial, fungal, or animal enzymes (*see* **monophenol monooxygenase**) only the putative copper-binding sites are conserved. Example from *Lycopersicon esculentum* (tomato), polyphenol oxidase B *Lycopersicon*: database code PPOB_LYCES, 596 amino acids (67.23 kDa).

polyphosphate or (sometimes) **polymetaphosphate** 1 any ionic form of polyphosphoric acid, $H_{n+2}P_nO_{3n+1}$ (where n is an unknown or unspecified integer greater than 1), or any mixture of polyphosphoric acid and its ionized forms. Polyphosphate occurs in the **volutin granules** observed in many microorganisms, where it is synthesized by the action of the enzyme **polyphosphate kinase** and is believed to serve as a phosphate store. 2 any (partial or full) salt or ester of polyphosphoric acid. *See also* **metaphosphate**.

polyphosphate kinase EC 2.7.4.1; a microbial enzyme that catalyses the synthesis of **polyphosphate** by transfer of the terminal phosphoric residue of ATP to diphosphate (i.e. pyrophosphate) or to a polyphosphate one residue smaller than the product. Example from *Klebsiella aerogenes*: database code PPK_KLEAE, 684 amino acids (80.03 kDa).

polyphosphoinositide or **polyphosphatidylinositide** *a former inappropriate general name for any [mono]phosphate, bisphosphate, triphosphate, or tetrakisphosphate of phosphatidylinositol; use not recommended.*

polyploid describing a cell (or organism) that has three, four, five, or more times the haploid number of chromosomes in its nucleus. —**polyploidy** *n*.

polyprenol *a recommended term for any prenol in which there are more than four isoprene residues; the isoprene units may be all-trans or a mixture of cis and trans. The number of such units in a particular polyprenol may be indicated by a multiplicative prefix, e.g. undecaprenol. Polyprenols and their derivatives are widely distributed in living organisms. See also dolichol.*

polyprenyl the univalent alkyl group derived from a **polyprenol** by loss of its hydroxyl group. The number of isoprene residues in a particular polyprenyl derivative may be indicated by a multiplicative prefix, e.g. undecaprenyl diphosphate.

polyprotein any polypeptide that on cleavage gives rise to two or more biologically active proteins, e.g. **proopiomelanocortin**.

polyribonucleotide any polynucleotide composed solely of ribonucleotide residues.

polyribonucleotide nucleotidyltransferase EC 2.7.7.8;
other name: polynucleotide phosphorylase. An enzyme that is



polyribonucleotide synthase (ATP)

involved in the degradation of mRNA, hydrolysing single nucleotides successively from the 3' end of the RNA, to yield a nucleoside diphosphate and a shortened mRNA. Example from *Escherichia coli*: database code PNP_ECOLI, 711 amino acids (77.10 kDa).

polyribonucleotide synthase (ATP) *see* RNA ligase (ATP).

polyribosome *an alternative name for polysome.*

polysaccharide *an alternative name for glycan*; i.e. any linear or branched polymer consisting of monosaccharide residues. Important polysaccharides include glycogen, starch, hyaluronic acid, and cellulose.

polysome or **ergosome** or **polyribosome** any structure consisting of two or more ribosomes attached to different points on the same strand of messenger RNA. Hence, a polysome is a complex involved in the cellular synthesis of the polypeptide specified in the message. Polysomes effect the synthesis of all proteins, but in the case of secretory proteins they become attached to the rough endoplasmic reticulum.

polysomic 1 describing a cell or organism that has one or more normal chromosomes in excess of the usual diploid or polyploid complement; being such a chromosome. 2 a polysomic organism. —**polysomy** *n.*

polystyrene phenylethene homopolymer; a clear, brittle, rigid thermoplastic with high electrical resistivity. A proportion of divinylbenzene may be added as a cross-linking agent during polymerization in order to produce polystyrene resins of various porosities for use in the preparation of polystyrene ion-exchange resins.

polysubstituted having more than one chemical substituent. —**polysubstitution** *n.*

poly(T) *abbr. for polythymidylate.*

polytene chromosome a type of chromosome in a polyploid cell, formed when multiple copies of homologous chromosomes are aligned side by side to give a giant chromosome in which distinct chromosome bands are readily visible.

polyteny *an alternative name for endomitosis.* —**polytene** *adj.*

polytetrafluoroethylene *abbr.:* PTFE; tetrafluoroethene homopolymer; a polymer consisting of chains of (≈2000) linked $-\text{CF}_2-$ units prepared by polymerization of tetrafluoroethylene, $\text{F}_2\text{C}=\text{CF}_2$. It is a nonflammable thermoplastic, extremely chemically resistant, of high electrical resistivity, and with an extremely low coefficient of friction. It is useful as tubing and sheeting in chemical laboratory work, as a lining to reaction vessels, as gasket material, and as packing for pumps and bearings. *Proprietary names:* Fluon, Teflon, etc.

polythene or (*esp. US*) **polyethylene** ethene homopolymer; a versatile thermoplastic polymer of ≈1.5–100 kDa, prepared by polymerization of ethylene. **High-density polyethylene**, a linear polymer, is fairly rigid and crystalline, while **low-density polyethylene**, a branched polymer, is softer and more flexible; both forms are chemically resistant and have high electrical resistivity.

polytopic describing an integral membrane protein that completely traverses the membrane two or more times.

polytrioxophosphate(v) *the recommended name for metaphosphate.*

poly(U) *abbr. for polyuridyate.*

polyubiquitin the form in which ubiquitin (def. 1) is synthesized, a precursor with a number of head-to-tail repeats.

polyunsaturated describing a substance containing more than one carbon-carbon multiple bond per molecular entity or residue; applied especially to a fatty acid containing more than one multiple bond in the hydrocarbon chain of its molecular structure, and to a fat or oil derived from one or more such fatty acids. *See also* **polyunsaturated fatty acid**. —**polyunsaturation** *n.*

polyunsaturated fatty acid *abbr.:* PUFA; any fatty acid that is **polyunsaturated**; the multiple bonds are most frequently all ethylenic and usually methylene-interrupted, the configuration of their substituents generally being *Z* (i.e. *cis*); more rarely is an acetylenic bond present. In animal metabolism,

polyunsaturated fatty acids belong to families denoted by the position of the double bond nearest to the methyl group and often also by the name of the fatty acid from which the remainder can be derived; hence **linoleic family**, **linolenic family**, **oleic family**, and **palmitoleic family**. In recent years, emphasis has been placed on a role of some polyunsaturated fatty acids in lowering plasma cholesterol, and potentially reducing the risk of heart disease. Foremost among these have been those of the *n*-6 family such as linoleic acid. A recommendation has been made that the diet should contain these polyunsaturated fats in a ratio of 2:1 to saturated fats. *See also* **essential fatty acid**, **fatty-acid nomenclature**, **fish oil**.

polyuria the passing of large volumes of urine.

polyuronide *an old term for glycuronan.*

polyvalent or **multivalent** 1 (*in chemistry*) a having a numerical valency greater than unity. b capable of exhibiting more than one valency value. 2 (*in immunology*) a (of an antibody molecule or fragment) having at least two combining sites. b (of an antiserum or other antibody preparation) capable of combining with two or more different antigens. c (of a vaccine) capable of eliciting immunity to more than one (strain of) microorganism, toxin, or other antigenic agent.

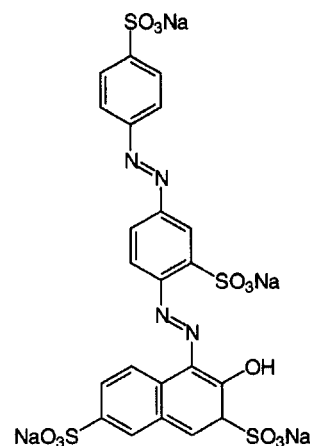
polyvinyl acetate *abbr.* PVA; ethenoic acid homopolymer; a synthetic polymer obtained from ethyne and acetic acid, used in adhesives and paints and in the manufacture of polyvinyl alcohol.

polyvinyl alcohol *abbr.:* PVA; ethenol homopolymer; a polymer prepared from polyvinyl acetates by replacement of the acetate groups by hydroxyl groups; the commercial products have differing contents of residual acetyl groups and hence differing physical characteristics. Polyvinyl alcohols are used in adhesives and can also be compounded into elastomers, which are useful in the manufacture of artificial sponges, hydrocarbon-resistant tubing, etc.

polyvinyl chloride *abbr.:* PVC; chloroethene homopolymer; a plastic solid, soluble in many organic solvents, that discolours on exposure to light or heat unless containing stabilizers. It is useful for the manufacture of rubber substitutes and electric wire coverings.

polyvinyl pyrrolidone *abbr.:* PVP; 1-ethenyl-2-pyrrolidinone homopolymer; commercial preparations are ≈10 kDa to ≈700 kDa; it is soluble in water to give colloidal solutions. PVP is used in the preparation of enzymes from plants; it removes phenolic compounds that can destroy enzyme activity.

polyxenic *see* **synxenic**.



**pomace**

pomace the pulpy residue remaining after extraction of juice from fruits such as apples, or after extraction of oil from seeds, nuts, or similar materials.

POMC *abbr. for proopiomelanocortin.*

Pompe's disease an alternative name for type II glycogen (storage) disease; *see* **glycogen disease**.

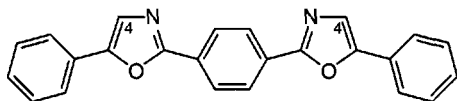
Ponceau S or **Fast Ponceau 2B** a dye used for staining protein, especially on cellulose acetate strips. (Illustrated on p. 523.)

pontal (of an atom or group) bridging.

ponticulin an F-actin-binding transmembrane glycoprotein of plasma membranes of the cellular slime mould *Dictyostelium discoideum*. It is a major membrane protein of this organism and a protein of the major high-affinity actin-membrane link. It is directly involved in actin polymerization. Example: database code A54793, 143 amino acids (14.75 kDa).

pool *see* **metabolic pool**.

POPOP 1,4-bis(5-phenyloxazol-2-yl)benzene; a secondary fluor used in liquid scintillation counting, originally as a wavelength shifter and more recently to decrease sensitivity to the action of quenching agents.



population 1 (*in biology*) any group of individual organisms of the same species inhabiting a given area; the number of such individual organisms. **2** (*in statistics*) or **universe** any finite or infinite collection of individuals or items from which a sample has been drawn.

population standard deviation *see* **standard deviation**.

P:O ratio *abbr. for phosphorus:oxygen ratio.*

porcine *abbr. (in compound abbrs.) p or (formerly) P;* of, characteristic of, or relating to a pig or pigs; resembling a pig.

pore any very small opening, e.g. in a (cell) membrane or in the integument of an organism, that allows the passage of gases and/or liquids.

porin 1 any member of a class of similar proteins of ≈ 15 kDa (monomer) that can be isolated from the outer membranes of certain Gram-negative bacteria. Trimers of these proteins are thought to make these membranes porous to hydrophilic substances of ≤ 600 Da by forming in them water-filled channels that allow transmembrane diffusion of uncharged small solutes. The proteins are usually homotrimers with several characteristic motifs; they are typically beta barrels with short α -helical domains. Examples: (1) from *Rhodobacter capsulatus*: database code NRL_2POR, 301 amino acids (31.50 kDa); 3-D structure known; (2) from *Escherichia coli* (*Neisseria* protein is similar): database code PHOE_ECOLI, 351 amino acids (38.88 kDa). **2** any of the transmembrane proteins typically of the mitochondrial outer membrane in eukaryotes that form aqueous channels through the lipid bilayer, allowing passage of small molecules. These constitute a structural family, characterized by four motifs and much β structure. Human mitochondrial membrane porin (voltage-dependent anion-selective channel protein) allows diffusion of small hydrophilic molecules; the channel adopts an open conformation at low or zero membrane potential and a closed conformation at potentials above 30–40 mV. Example: database code PORI_HUMAN, 282 amino acids (30.61 kDa).

porosity 1 the state or quality of being **porous**. **2** the degree to which a material or substance is porous.

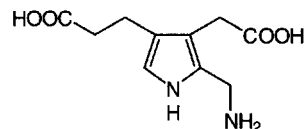
porosome nuclear pore complex.

porous having pores; allowing the passage or uptake of gases and/or liquids, especially by diffusion rather than by bulk flow.

porphin or **porphine** an alternative name for **porphyrin** (def. 1).

porphobilinogen *abbr.:* PBG; *see* (aminomethyl)-4-(carboxy

methyl)-1H-pyrrole-3-propanoic acid; the biosynthetic precursor of all porphyrins. It is excreted in relatively large amounts in the urine in some porphyrias.



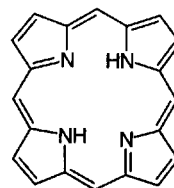
porphobilinogen deaminase *see* **hydroxymethylbilane synthase**.

porphobilinogen synthase EC 4.2.1.24; *systematic name:* 5-aminolevulinate hydro-lyase (adding 5-aminolevulinate and cyclizing); *other names:* δ -aminolevulinic acid dehydratase; aminolevulinate dehydratase. An enzyme that catalyses the conversion of two moles of 5-aminolevulinate to one mole of porphobilinogen + 2 H₂O in the pathway for the synthesis of **porphyrins**; zinc is a cofactor. Example from *Escherichia coli*: database code HEM2_ECOLI, 323 amino acids (35.45 kDa).

porphyr+ a variant form of **porphyr+** (before a vowel).

porphyria any of a group of metabolic disorders characterized by the excretion of abnormally large amounts of various porphyrins and/or their biosynthetic precursors. Porphyrins are currently classified as primary (if inherited) or secondary (if acquired). The primary disorders may be further classified into two broad categories: (a) the neurological/psychiatric forms and (b) the forms associated with cutaneous photosensitivity, though these two types of symptoms may in some cases overlap. The acquired disorders are much more common than the inherited conditions. There is no clear-cut association of enzyme defects with specific porphyrias. Elevated blood levels of intermediates in porphyrin biosynthesis are normally present during attacks. Elevated intermediates that are found, together with (in parentheses) associated enzymes and conditions are (1) aminolevulinic acid (or ALA) (porphobilinogen (or PBG) synthase; PBG synthase deficiency porphyria); (2) ALA and PBG (PBG deaminase; acute intermittent porphyria, variegated porphyria); (3) uroporphyrin (uroporphyrinogen decarboxylase; porphyria cutanea tarda, acute intermittent porphyria) (4) coproporphyrin (coproporphyrinogen oxidase; variegated porphyria, hereditary coproporphyrin). Acquired porphyria may be due to lead exposure, some malignancies, and chronic ethanol exposure among other causes. It has been suggested that the British king, George III, suffered from a form of porphyria leading to psychiatric symptoms.

porphyrin 1 the *systematic name* for the fundamental skeleton, C₂₀H₁₄N₄, of the macrocyclic **tetrapyrroles**; *often known also as porphin or porphine*. It consists of a ring of four pyrrole nuclei linked each to the next at their α positions (i.e. those adjacent to their nitrogen atoms) through a methine group, -CH= (*compare* **corrin**).



The structure is aromatic and tautomeric, two hydrogen atoms being associated with any two of its four nitrogen atoms (though for the purposes of nomenclature the name 'porphyrin' implies that the saturated nitrogen atoms are at positions 21 and 23 in the molecule unless there is a specific indication otherwise). Many di- or trivalent metal ions are chelated by porphyrin through these four central nitrogen



atoms. [Note: It was recommended in 1986 that the widely used Fischer system of (incomplete) numeration for the porphyrin ring be discontinued in favour of one in which all the atoms are numbered, the C atoms 1 to 20 and the N atoms 21 to 24.] The systems for numbering tetrapyrrole carbons are shown at **bacteriochlorophyll**. Porphyrin itself is a synthetic substance, deep purple in colour. **2** any member of a large group of naturally occurring or synthetic derivatives or analogues of porphyrin (def. 1), of which **protoporphyrin IX** and its derivatives, especially the chlorophylls and hemes, are the most important.

porphyrinogen **1** 5,10,15,20,22,24-hexahydroporphyrin; *see* **porphyrin**. **2** a general name for the corresponding hexahydro derivative of any **porphyrin** (def. 2).

porphyrinuria the excretion of abnormally large amounts of porphyrins in the urine. *See also* **porphyria**.

porphyro+ or (before a vowel) **porphy+** *comb. form* denoting purple.

porphyropsin a purple, light-sensitive, visual pigment in the rod cells of the retinas of freshwater and migratory fishes and some amphibians, consisting of a rod-type **opsin** combined (as a Schiff's base) with 3,4-didehydro-11-*cis*-retinal. It has an absorption maximum at ≈ 522 nm and is analogous in function to **rhodopsin** in other vertebrates.

portable (*in molecular biology*) describing DNA sequences that can be inserted into recombinant molecules for the purposes of gene expression; e.g. portable promoter, portable Shine-Dalgarno sequence, etc.

portal **1** relating to a blood vessel, or system of blood vessels, that carries blood from one capillary bed to another, especially the hepatic portal vein. **2** relating to a porta, such as the hepatic porta (porta hepatis). **3** the route of entry into the body of a drug, microorganism, beam of radiation, etc. (or sometimes of its exit therefrom).

porter any transporter of solutes between aqueous phases on either side of a lipid membrane. The term includes **antiporter**, **symporter**, and **uniporter** (*see* **uniport**).

Porter **1** George, Baron Porter of Luddenham (1920–), British physical chemist; Nobel Laureate in Chemistry (1967) jointly with R. G. W. Norrish, the prize being shared with M. Eigen 'for their studies of extremely fast chemical reactions, effected by disturbing the equilibrium by means of very short pulses of energy'. **2** Rodney Robert (1917–85), British biochemist and immunochemist noted particularly for his isolation of the constituent polypeptide chains of antibody molecules and for formulating the four-chain structure of such molecules; Nobel Laureate in Physiology or Medicine (1972) jointly with G. M. Edelman 'for their discoveries concerning the chemical structure of antibodies'.

positional cloning a strategy for the identification of a gene of unknown function responsible for a genetic disease. The phenotype is correlated with a chromosomal site, the DNA of that region is cloned and compared with that in normal individuals. If a single gene is involved, it is sequenced, and the protein structure deduced. An example is the discovery of the genetic basis of cystic fibrosis and the identification of the **cystic fibrosis transmembrane conductance regulator** (CFTR).

positional information information supplied to or possessed by cells according to their position in a multicellular organism.

positional isomer any of two or more constitutional isomers that differ in the position of a particular substituent or group.

positive **1** having a value greater than zero. **2** (*in mathematics*) of the same value but of opposite sense to that regarded as negative. **3** (of the results of a test) indicating the presence of the material or condition tested for. **4** (of an electric charge) having the opposite polarity to the charge of an electron. **5** (of a molecular entity) having a positive electric charge; having a deficiency of electrons. **6** (of a point in an electric circuit) having a higher potential than some other point to which a potential of zero is assigned. **7** (*in biology*) signifying growth or

movement towards a (specified) stimulus. **8** (*in photography*) an exposed and developed photographic film, plate, or print showing an image in colours or tones corresponding to those of the object photographed. *Compare* **negative**. —**positiveness** *n.*; **positivity** *n.*

positive chromatography (*sometimes*) any method of purification of substances by affinity chromatography in which a substance of interest in a sample interacts with and is selectively retained by the adsorbent, especially as opposed to **negative chromatography**.

positive control the initiation or enhancement of some biological activity by the presence of a specific molecular entity.

positive cooperativity cooperative ligand binding in which binding of one ligand to one site on a macromolecule increases the affinity at other sites for the subsequent binding of the same or ligands.

positive correlation *see* **correlation** (def. 2).

positive effector *see* **effector**.

positive electron an alternative name for **positron**.

positive feedback *see* **feedback** (def. 1).

positive staining a technique used for preparing specimens for electron microscopic examination in which charged biological macromolecules or structures are rendered electron-dense by allowing heavy-metal ions of opposite charge to bind to them. *Compare* **negative-contrast technique**.

positron or **positive electron** symbol: β^+ ; an elementary particle that is the antiparticle of the electron. It has the same mass as an electron and a numerically equal but positive electric charge.

positron emission tomography *see* **tomography**.

post+ or (*sometimes*) **post-** *prefix* denoting after in time or sequence; behind, posterior to.

postabsorptive describing the period immediately after absorption of food.

posterior after in position, sequence, or time; behind; following.

posterior lobe the posterior lobe of the pituitary gland; i.e. the **neurohypophysis**.

postlysosome a degenerate **telolysosome** that has lost its enzymes.

postmitochondrial fraction or **postmitochondrial supernatant** or **supernatant fraction** an operational term for the part of a homogenate of eukaryotic cells or tissue remaining after sedimentation of the mitochondrial fraction by centrifugation. It consists essentially of the **microsomal fraction** and the cytosol.

post-mortem **1** occurring after death. **2** a post-mortem examination; necropsy, or **autopsy**.

postnatal describing the period after birth or giving birth.

postpartum describing the period immediately after birth or giving birth.

postprandial describing the period immediately after intake of food.

postreplication repair *see* **recombinational repair**, **mismatch repair**.

postreplicative describing the period following replication of a cell or of a nucleic-acid molecule.

postribosomal fraction or **postribosomal supernatant** or **supernatant fraction** an operational term for the part of a homogenate of eukaryotic cells or tissue remaining after sedimentation of the ribosomal (microsomal) fraction by centrifugation. It consists essentially of the cell sap.

postsynaptic **1** describing a neuron or a neuronal membrane that is on the efferent side of a synapse. **2** describing the membrane of a muscle cell at a motor end-plate.

postsynaptic membrane *see* **synapse**.

post-transcriptional describing any phenomenon or process occurring after completion of transcription of genomic nucleic acid. —**post-transcriptionally** *adv.*

post-translational describing any phenomenon or process occurring after completion of translation of messenger RNA into polypeptide; *postsynthetic*. —**post-translationally** *adv.*

Bản đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.



post-translational modification

post-translational modification or **post-translational processing** any enzyme-catalysed change to a protein made after it is synthesized by the translation of messenger RNA. Examples are proteolytic cleavage, **glycosylation**, **methylation**, **phosphorylation**, and **prenylation**.

potash 1 the common name for potassium hydroxide or potassium carbonate. 2 the name for potassium in the common names of certain of its salts.

potassium or (in scientific and medical Latin) **kalium** symbol: K; a soft, silvery white, highly reactive, metallic element of group 1 of the (IUPAC) **periodic table**; atomic number 19; relative atomic mass 39.0983. Natural potassium is a mixture of stable nuclides of mass numbers 39 (93.2 atom percent) and 41 (6.7 atom percent) together with a minute proportion of the radionuclide of mass number 40 (0.012 atom percent). Potassium is widely distributed in all living organisms, besides occurring in various minerals; it is one of the most abundant elements of the Earth's crust. It forms a monovalent cation, K^+ . Several artificial radioactive isotopes are known. In the body, potassium is a major intracellular cation, with little in the extracellular compartment; the range in normal human plasma is 3.6–5.0 mmol L^{-1} , with serum 0.2–0.3 mmol L^{-1} higher due to release from platelets on clotting. The cellular uptake of potassium is stimulated by insulin, and its excretion by the kidney is indirectly controlled by aldosterone, which stimulates sodium reabsorption in the distal tubules, potassium being excreted by passive transfer due to the electrochemical gradient thereby created. It is actively pumped into cells by **sodium/potassium-ATPase**. See also **hyperkalemia**, **hypokalemia**.

potassium-40 the naturally occurring radioactive nuclide $^{40}_{19}K$. It has a half-life of 1.28×10^9 years, emitting beta particles (i.e. electrons) of 1.34 MeV max. and gamma radiation of 1.46 MeV.

potassium-42 the artificial radioactive nuclide $^{42}_{19}K$. It has a half-life of 12.36 h, emitting high-energy beta particles (i.e. electrons) of 3.52 MeV max. (82%) and 1.97 MeV max. (18%), together with gamma radiation of 1.52 MeV (18%) and 0.31 MeV (0.15%).

potassium channel or **K^+ channel** any one of a number of types of structure that permit controlled (gated) passage of potassium ions, predominantly efflux, through membranes; these channels exhibit considerable diversity, but in excitable cells their role is to set the resting potential, participate in restoring it after depolarization, and attenuate and regulate potentials generated by other excitatory inputs. The main subdivisions of potassium ion channel are: (1) those primarily responding to membrane polarization, termed voltage-dependent, voltage-gated, or delayed rectifier (see **rectification**) K^+ channels; transmembrane (typically six transmembrane domains, designated S1 to S6) proteins; example, channel protein KV1.3 (human): database code CIK3_HUMAN, 523 amino acids (58.18 kDa); there are six motifs; (2) Ca^{2+} -activated K^+ channels, which are activated by increase of intracellular Ca^{2+} concentration; (3) receptor-coupled channels, which may be activated or inhibited by receptor occupancy; these themselves fall into two classes – those which have an intrinsic ion channel as part of the receptor, and those which act through a G-protein that activates an independent channel; example of the latter from human: database code IRK7_HUMAN, 423 amino acids (48.51 kDa); and (4) ATP-sensitive K^+ channels that close in response to increased intracellular ATP/ADP ratios generated by increased metabolic flux; example, heart K_{ATP} channel from human: database code IRK5_HUMAN, 419 amino acids (47.67 kDa). Types 3 and 4 are known as inward rectifier channels, in that they favour inward movement of K^+ . Inward rectifier channels have two transmembrane domains rather than the six of voltage-gated channels. However, all K^+ channels have a loop known as the H5 or P region, which is thought to line the conducting pore. In voltage-gated channels, this lies between segments S5 and

S6, and in inward rectifier channels between the two transmembrane domains.

potency (in pharmacology) an expression of the activity of a drug or other agent, in terms of either the concentration or the amount needed to produce a defined effect. It is an imprecise term that should always be further defined. Compare **EC₅₀**, **IC₅₀**.

potential (in physics) 1 the work done in bringing unit positive charge, unit positive pole, or unit mass from infinity to a specified point in an electric, magnetic, or gravitational field, respectively. 2 **electric potential**. 3 **electric potential difference**.

potential difference *abbr.*: p.d.; 1 generally, the difference in potential (def. 1) between any two points. 2 see **electric potential difference**.

potential divider a chain of resistors (or inductors or capacitors) connected in series, across which an electric potential may be applied and from which a desired fraction of the potential may then be obtained from a contact suitably placed in the chain. See also **potentiometer** (def. 3).

potential energy *abbr.*: PE; symbol: E_p or V or Φ ; the energy of a body or system consequent upon its physical state or its position in an electric, magnetic, or gravitational field. It is expressed as the work necessary to bring it to its present state or position from a reference state or position, to which a potential energy value of zero is assigned.

potential energy barrier or **energy barrier** 1 the difference between the bond energy at which a molecule dissociates and the energy of that molecule in the ground state. 2 the difference between the energy of the ground state of an enzymically activated reaction complex and the sum of the energies of the ground states of the reactants forming the activated complex.

potential gradient the potential difference per unit distance between a point of higher and one of lower (electric) potential; the algebraic differential of this at a particular point.

potentiate to cause to be (more) potent, especially to increase the potency of a drug or other agent. —**potentiation** *n*.

potentiometer 1 a precision instrument for measuring an electric potential difference of constant polarity without drawing current from the circuit being examined. In its simplest form, it consists of a length of wire of uniform resistance and across which a constant potential is applied. By means of a sliding contact the proportion of this applied potential is found that will balance the potential being measured as determined by a null reading on a galvanometer. 2 a direct-reading electronic device for measuring electric potential difference or electromotive force without drawing appreciable current. 3 or (informal) **pot** a development of a potential divider consisting of a resistor with a movable contact and used, especially in electronics, either as a potential divider or as a variable resistor. —**potentiometry** *n*.; **potentiometric** *adj.*; **potentiometrically** *adv.*

potentiometric titration or (sometimes) **electrometric titration** titration in which an electrode reversible to one of the ionic components of the analyte or titrant is immersed in the analyte solution and the electric potential of the electrode (relative to that of an inert reference electrode) is measured during addition of titrant. Because the potential of an electrode is a logarithmic function of the activity of the ion to which the electrode is reversible, an abrupt change in its potential is observable as an equivalence point is approached.

potocytosis a mechanism for small molecule transport across eukaryotic cell membranes that uses **caveolae** rather than clathrin-coated pits.

Potter-Elvehjem homogenizer a device for preparing small quantities of cell or tissue homogenates. It consists of a cylindrical pestle (of glass, Perspex, or Teflon) that fits closely into a hard-glass test tube. Cells or small pieces of tissue in a suspending medium are placed in the tube and the pestle is rotated by means of an electric motor while the tube is slowly moved up and down by hand, a shear thereby being imparted to the contents of the tube. (After van Rensselaer, Potter



POU domain

(1911–), US biochemist, and Conrad Arnold Elvehjem (1901–62).]

POU domain a feature found in many **transcription factors**. Its name derives from the fact that it was first recognized after comparison of cDNAs encoding three families of transcriptional regulators, Pit (see **PIT**), Oct, and Unc (see, e.g., **unc-86**). POU domain genes now also include the Brn family and others. The POU domain is a bipartite DNA-binding domain, i.e. it has two prominent protein domains that confer sequence-specific DNA binding. One of these domains is homologous to the domain found in other transcription factors and is called the POU homeodomain, whereas the other is specific to these factors and is called the POU-specific domain. The recognition sequences are A/T-rich, including the highly conserved octamer sequence 5'-ATGCAAAT-3'. The domain can also confer special protein-protein binding properties. Example i-POU (Pou inhibitory) protein of *Drosophila melanogaster*: database code IPOU_DROME, 367 amino acids (39.83 kDa).

pound *symbol*: lb; the avoirdupois unit of mass, now defined as 0.453 592 338 kg exactly.

powder pattern or **powder diagram** a diffraction pattern obtained from a specimen of a crystalline material in finely divided form. The specimen is rotated in a parallel beam of monochromatic X-rays, electrons, neutrons, etc. so that all planes of the crystals assume all possible orientations with respect to the incident beam.

power **1** *symbol*: P; a measure of the rate of doing work, expressed as energy divided by time. The derived SI unit of power is the **watt**. **2** (in optics) a measure of the ability of a lens or mirror to magnify, expressed as the reciprocal of the focal length of the lens or mirror in metres. **3** (in mathematics) **a** an alternative name for **exponent** or **index**; e.g. 2 to the power 3 = 2³. **b** one more than the cardinal number of times that a number or expression has to be multiplied by itself to give a required number; e.g. 2 to the power 3 = 2 × 2 × 2. **c** one more than the ordinal number of times that a particular number has to be multiplied by itself to give a required number; e.g. the third power of 2 is 8 = 2 × 2 × 2.

poxvirus any member of the Poxviridae, a family of large, double-stranded DNA viruses, in class I of the **Baltimore classification**. They consist of a single molecule of DNA of 130–240 MDa together with 30 or more structural proteins, a number of viral enzymes, and lipid. Their hosts range from insects to mammals; examples include the viruses of variola (smallpox) and vaccinia (cowpox) – but not of varicella (chickenpox) – in humans, and the virus of myxomatosis in rabbits.

Pp *symbol* for the propionyl group.

PP *abbr.* for pancreatic polypeptide.

PP_i *symbol* for inorganic pyrophosphate.

P particle a liberated, mature **kappa particle**.

ppb or **p.p.b.** *abbr.* for parts per billion, i.e. parts per 10⁹.

PP cell or (formerly) **F cell** a type of cell containing granules that store pancreatic polypeptide (**pancreatic hormone**). In adult humans such cells are located exclusively in the pancreas (79% in the islets, 19% in the acini, and 2% in the ducts).

pp[CH₂]₃pA *symbol* for adenosine 5'-[α,β-methylene]triphosphate; an analogue of ATP in which the α,β-bridging oxygen atom has been replaced by a methylene group.

PPF or **PP factor** *abbr.* for pellagra-preventative factor; an early name for **nicotinic acid** or **nicotinamide**.

ppGpp *symbol* for guanosine 3'-diphosphate 5'-diphosphate (also known as **magic spot I**). See **relA**, **spoT**.

PPase *abbr.* for peptidylprolyl isomerase.

PPLO *abbr.* for pleuropneumonia-like organism (i.e. **mycoplasma**).

ppm or **p.p.m.** *abbr.* for parts per million, i.e. parts per 10⁶.

PPO *abbr.* for 2,5-diphenyloxazole; a substance useful as a primary fluor in liquid scintillation counting.

pppGpp *symbol* for guanosine 3'-diphosphate 5'-triphosphate (also known as **magic spot II**). See **relA**.

PPRibP *symbol* for (p-ribosephosphoryl)-p-ribose diphosphate

PPS (in clinical chemistry) *abbr.* for pepsin A.

ppt **1** or **ppt.**; *abbr.* for precipitate. **2** or **p.p.t.** *abbr.* for parts per trillion, i.e. parts per 10¹².

PPT *abbr.* for preprotachykinin (see **protachykinin β precursor**, **tachykinin**).

PQ *abbr.* for plastoquinone.

PQH₂ *abbr.* for dihydroplastoquinone (i.e. **plastoquinol**).

PQQ *abbr.* for pyrroloquinoline quinone (i.e. **methoxatin**).

Pr *symbol* for 1 praseodymium. **2** the propyl group.

Prⁱ *symbol* for the isopropyl group, (CH₃)₂CH–.

prazosin or **furazosin** a quinazoline antagonist of α₁ adrenoceptors; it has anti-hypertensive properties.

pre+ or **pre-** *prefix* **1** before in time, position, sequence, rank, etc.; *compare* **pro+** (def. 1). **2** denoting a substance that moves faster (or further) than a specified substance in a separative procedure (e.g. prealbumin). Usage in this sense for naming proteins is now deprecated in order to avoid confusion with def. 3. **3** denoting a (usually metabolic) precursor, whether endogenous (e.g. prehormone, prekalikrein, pre-messenger RNA, preprotein, pretyrosine) or exogenous (e.g. precarcinogen), of the (class of) substance specified; *Compare* **pro+** (def. 3); *see also* **prepro+**.

PRE *abbr.* for proton relaxation enhancement.

prealbumin an operational (and confusing) term for either of two protein-containing zones observed to migrate more rapidly towards the anode than does serum albumin on zone electrophoresis of serum at pH 8.6 by some techniques. The faster-moving of these zones is now known to be **transferrin** and the other to be acidic α₁-glycoprotein (i.e. **orosomucoid**).

prebiotic of or relating to the period before the appearance of living organisms on Earth.

precancerous describing a tissue or condition that is not yet malignant but is expected to form a cancer.

precarcinogen (sometimes) a chemical carcinogen that can be converted by metabolism into another of no less potency. *Compare* **proximate carcinogen**.

precarcinomatous describing a tissue that is expected to form a carcinoma.

P receptor *see* **purinoceptor**.

precerebellin a cerebellum-specific protein containing a 16-residue peptide known as cerebellin, flanked by Val-Arg and Glu-Pro. Precerebellin is the precursor of cerebellin, which is a protein of the postsynaptic structures of Purkinje cells. Concentrations of cerebellin are low at birth and (in humans) peak at days 5–15. It has a conserved sequence in mammals and birds. Example from human: database code CERB_HUMAN, 193 amino acids (21.07 kDa).

precess to undergo, or cause to undergo, **precession**.

precession (slow) gyration of the axis of rotation of a spinning body caused by application of a torque such that the axis of rotation generates a cone.

precipitant an agent or a reagent that induces the formation of a precipitate.

precipitate **1** to (cause to) come out of solution or suspension. **2** substance or material that is precipitated. —**precipitation** *n*.

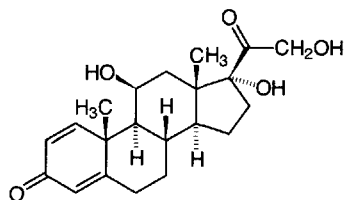
precipitating antibody an alternative name for **precipitin**.

precipitin or **precipitating antibody** an antibody that is capable of reacting with a soluble antigen with the formation of an insoluble antigen-antibody complex.

precipitin reaction the reaction together of antigen and antibody near their equivalence point to form a cross-linked precipitate. Quantitative analysis of this interaction gives both the antibody content of an immune serum and an indication of the valency of the antigen. If the reaction occurs in a gel then the reactants form precipitin arcs.

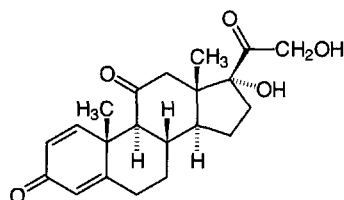
precision-bore (of tubing or a tubular object) having a lumen whose diameter varies only within extremely narrow limits.

prednisolone 1,4-pregnadiene-11β,17α,21-triol-3,20-dione; a semisynthetic steroid with glucocorticoid action. *Compare* **prednisone**.



prednisolone

prednisone 1,4-pregnadiene-17 α ,21-diol-3,11,20-trione; a synthetic steroid with glucocorticoid action. Compare **prednisolone**.



preelectrophoresis a preliminary treatment of an electrophoresis gel in which a potential difference is applied in order to remove undesirable substances such as excess polymerization reagents.

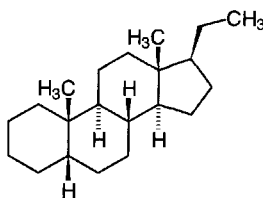
preflashing the hypersensitization of photographic emulsion by a preliminary exposure to a brief (≈ 1 ms) flash of light in order to increase its sensitivity to low levels of radiation and to render its response more nearly linear. It is useful especially in fluorography and (auto)radiography.

Pregl, Fritz (1869–1930), Austrian chemist; Nobel Laureate in Chemistry (1923) 'for his invention of the method of microanalysis of organic substances'.

pregnancy-associated α_2 -glycoprotein *see* **pregnancy zone protein**.

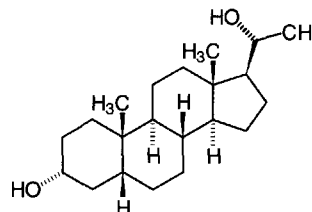
pregnancy zone protein *abbr.*: PZP; *other names*: pregnancy-associated α_2 -glycoprotein; α_2 -pregnoglobulin; a homotetrameric disulfide-linked proteinase inhibitor of the α_2 -macroglobulin family that inhibits all four classes of proteinase by a 'bait-and-trapping' mechanism (*see* **bait region**). One of the major human pregnancy-associated plasma proteins, it may reach levels of 1.0–1.4 g L⁻¹ just before term. It has extensive homology with human α_2 -macroglobulin. In the sequence (example below) amino acids 1–24 form the signal, 26–1482 are the pregnancy zone protein, and 685–735 the bait region. Example (precursor) from human: database code PZP_HUMAN, 1482 amino acids (163.65 kDa).

pregnane a C₂₁ steroid to which **pregnanediol** is related.

5 β -pregnane

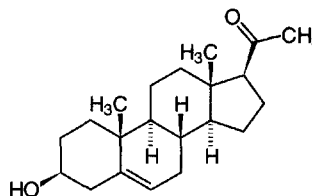
pregnanediol a common name for the steroid 5 β -pregnane

3 α ,20 α -diol; (20S)-5 β -pregnane-3 α ,20-diol; the principal urinary metabolite of progesterone, occurring mainly as its 3-glucuronide. Its level in the urine increases during the luteal phase of the menstrual cycle and especially in the latter part of pregnancy, the excretion rate being a convenient (if approximate) index of the functional state of the corpus luteum and placenta.



pregnant mares' serum gonadotropin *abbr.*: PMSG; a clinically useful impure gonadotropin preparation extracted from the serum of pregnant mares – such serum has high activities of both follicle-stimulating hormone and luteinizing hormone. Compare **urogonadotropin**.

pregnenolone a common name for the steroid Δ^5 -pregnen-3 β -ol-20-one; 3 β -hydroxypregn-5-en-20-one; a key intermediate on the biosynthetic route from cholesterol to progesterone. It has glucocorticoid activity and has been used in the treatment of rheumatoid arthritis. It is the product of **cholesterol monooxygenase (side-chain-cleaving)**.



α_2 -pregnoglobulin *see* **pregnancy zone protein**.

preincubate to bring a reaction mixture or a cell or tissue preparation to (chemical, metabolic, thermal, etc.) equilibrium before starting the reaction to be investigated. —**preincubation** *n*.

prekallikrein a plasma glycoprotein of 88 kDa (bovine), and a **blood coagulation** factor. It is the precursor of the proteinase **kallikrein**, to which it is converted by the action of factor XIIa. The protein contains four **apple domains**. The active site contains His, Asp, and Ser residues characteristic of a **charge-relay system**. Example from human (precursor): database code KAL_HUMAN, 638 amino acids (71.37 kDa).

Prelog, Vladimir (1906–98), Bosnian-born Swiss organic chemist and stereochemist; Nobel Laureate in Chemistry (1975) 'for his research into the stereochemistry of organic molecules and reactions' [prize shared with J. W. Cornforth].

prelysosomal vacuole or **prelysosome** a former name for **endosome** (def. 1).

pre-messenger RNA *abbr.*: pre-mRNA; the class of primary gene transcripts from which the corresponding messenger RNA molecules are subsequently formed by post-transcriptional processing; any member of this class. It comprises a substantial part of **heterogeneous nuclear RNA**.

pre-mRNA *abbr.* for pre-messenger RNA.

pre-mRNA processing protein (*abbr.*: PRP protein) or **pre-mRNA splicing factor** any of the proteins involved in processing the primary transcript RNA. Many of them function as components of a **spliceosome**. Most are components of snRNP particles, which are numbered U1, U2, etc. these proteins are

pre-murein-leader peptidase

usually numbered PRP2, etc.; the examples given are from yeast (*Saccharomyces cerevisiae*) unless stated to the contrary. PRP2 is an ATP-binding RNA helicase required for the first cleavage ligation reaction: database code PR02_YEAST, 876 amino acids (99.70 kDa). PRP5 is an ATP-binding RNA helicase required for unwinding of pre-mRNA and is part of the spliceosome, a so-called **DEAD-box** helicase: database code PR05_YEAST, 849 amino acids (96.25 kDa). PRP6: database code PRP6_YEAST, 899 amino acids (104.11 kDa). PRP8 is a U5 snRNP protein: database code PR08_YEAST, 2413 amino acids (279.18 kDa). PRP9 is required for U2 snRNP to bind to pre-mRNA: database code PR09_YEAST, 530 amino acids (62.96 kDa). PRP15 is an RNA-dependent ATPase required for branch-point recognition. PRP16 is an RNA helicase (another DEAD-box helicase): database code PR16_YEAST, 1071 amino acids (121.51 kDa). PRP18 is a component of the U4/U5/U6 snRNP; a homodimer: database code PR18_YEAST, 251 amino acids (28.35 kDa). PRP19 is a U2-binding protein of unknown function: database code PR19_YEAST, 502 amino acids (56.41 kDa). PRP21 is probably involved in retention of pre-mRNA in the nucleus: database code PR21_YEAST, 280 amino acids (33.02 kDa); PRP21 is a DEAD-box helicase involved in release of spliced mRNA from the spliceosome: database code PR22_YEAST, 1145 amino acids (129.86 kDa). PRP28 is a DEAD-box helicase involved in U4/U5/U6 snRNP: database code PR28_YEAST, 588 amino acids (66.62 kDa). PRP38 is involved in maintenance of U6 snRNA: database code PR38_YEAST, 242 amino acids (27.94 kDa). PRP39 facilitates interaction between U1 snRNP and the 5'-splice site: database code PR39_YEAST, 629 amino acids (74.55 kDa).

pre-murein-leader peptidase EC 3.4.99.35; *other names*: lipoprotein-signal peptidase; signal peptidase II; SPase II. An enzyme that cleaves a single bond between Gly²⁰ and alkylated Cys²¹ in a bacterial lipoprotein, detaching the leader peptide. Example from *Escherichia coli*: database code LSPA_ECOLI, 164 amino acids (18.14 kDa).

prenol the recommended name for any **isoprenoid** of general formula $H-[CH_2-C(CH_3)=CH-CH_2]_n-OH$; i.e. any primary monohydroxy alcohol whose carbon skeleton consists of two or more isoprene residues linked head to tail. *See also* **polyprenol**.

prenyl the univalent alkyl group derived from a **prenol** by loss of its hydroxyl group. *See also* **polyprenyl**. *Compare* **isopentenyl**.

prenylation the enzymic addition of **prenyl** moieties to proteins as a post-translational modification; **geranyl**, **farnesyl**, or **geranylgeranyl** groups may be added. The process involves reaction of a prenyl diphosphate with a cysteinyl sulfhydryl group near the C terminus of the protein, to give a prenyl-S-Cys moiety. Characteristically a Cys-ali-ali-Xaa sequence is recognized by the transferase that mediates the reaction, where Cys is cysteine, ali is an aliphatic amino acid, and Xaa is any amino acid. When Xaa is serine, alanine, or methionine the protein is farnesylated; when Xaa is leucine it is geranylgeranylated, but this is variable; for example, in the case of Rab 3A (*see* **RAB**), the Cys-Xaa-Cys motif leads to geranylgeranylation on both cysteine residues and methyl esterification of the terminal carboxyl, while in other members of the Rab family Cys-Cys directs geranylgeranylation without esterification. The function of the process appears often to be regulation of protein-protein and protein-membrane interaction.

prenyltransferase *see* **dimethylallyltransferase**.

PR-enzyme an old name for **1 phosphorylase phosphatase** (EC 3.1.3.17). [PR stood originally for prosthetic-group removing, later for phosphorylase rupturing, and later still for phosphate removing.] **2 deoxyribodipyrimidine photo-lyase** (EC 4.1.99.3). [PR stood for photoreactivating.]

preparative ultracentrifuge any ultracentrifuge designed for preparation or purification of tangible quantities of macromolecular substances, subcellular fractions, etc.

pre-part or **pre-peptide** an alternative name for **signal peptide**.

prephenate a trivial name for 2-oxo-3(1-carboxylato-4-hydroxy-2,5-cyclohexadiene-1)-propanoate; a nonaromatic intermediate in the branched biosynthetic pathway (the **prephenate pathway**) from chorismate to phenylalanine or tyrosine in most autotrophic organisms. *See also* **arogenate**.

prephenate dehydratase EC 4.2.1.51; *systematic name*: prephenate hydro-lyase (decarboxylating); an enzyme that catalyses the formation of phenylpyruvate from prephenate with release of H₂O and CO₂. This is a reaction in one of the pathways for the biosynthesis of phenylalanine. Example from *Bacillus subtilis*: database code PHEA_BACSU, 285 amino acids (31.86 kDa). In other cases (e.g. *Escherichia coli*) the activity is one domain of a bifunctional enzyme that also contains **chorismate mutase** (another enzyme of the phenylalanine biosynthetic pathway). Example from *E. coli*: database code PHEA_ECOLI, 386 amino acids (43.06 kDa). The mutase domain is in the N-terminal part of the database code, the dehydratase in the C-terminal part. *See also* **arogenate**.

prephenate dehydrogenase any of several enzymes that convert prephenate to 4-hydroxyphenylpyruvate with release of CO₂ and reduction of either NAD⁺ or NADP⁺. They function at a branch point in the pathway for the biosynthesis of aromatic amino acids, at which they initiate the branch of the pathway leading to tyrosine biosynthesis. They include **prephenate dehydrogenase**, EC 1.3.1.12; *systematic name*: prephenate:NAD⁺ oxidoreductase (decarboxylating); *other name*: hydroxyphenylpyruvate synthase; an enzyme that also has **chorismate mutase** activity, so can convert chorismate to hydroxyphenylpyruvate. Example from *Escherichia coli*: database code TYRA_ECOLI, 373 amino acids (42.04 kDa). There is also **prephenate dehydrogenase (NADP⁺)**, EC 1.3.1.13; *systematic name*: prephenate:NADP⁺ oxidoreductase (decarboxylating). Example from *Saccharomyces cerevisiae*: TYR1_YEAST, 452 amino acids (50.92 kDa).

prephenate pathway a pathway for the synthesis of phenylalanine and tyrosine from chorismate, utilizing prephenate as an intermediate. Prephenate is formed from chorismate by **chorismate mutase**. It is converted by **prephenate dehydrogenase** to 4-hydroxyphenylpyruvate, from which tyrosine is formed by transamination with glutamate as the amino-group donor. For the synthesis of phenylalanine, phenylpyruvate is formed by **prephenate dehydratase**, and is then converted to phenylalanine by transamination, glutamate again acting as amino-group donor. *See also* **arogenate**.

pre-piece an alternative name for **signal peptide**.

prepolypeptide or **presecretory polypeptide** the transient, primary translation product of a eukaryotic messenger RNA for any polypeptide destined to be secreted by a cell, irrespective of whether or not an intermediary **propolypeptide** is formed; the polypeptide counterpart of **preprotein**. *See also* **prepropolypeptide**.

prepro+ or **prepro-** compound prefix denoting preprotein of the proprotein of a specified protein, or prepolypeptide of the propolypeptide of a polypeptide.

preprocessing a fanciful name for enzymic conversion of a preprotein (or prepropolypeptide) to a proprotein (or propolypeptide).

preprohormone the prohormone of any prohormone of a protein or (poly)peptide hormone. Examples are: preproglucagon, preproinsulin, preproparathyrin, prepro-VIP.

preprophase band a dense band of microtubules, 1–3 μm wide, that appears just beneath the cell membrane before the start of cell division in the cells of higher plants. It precedes the onset of prophase and then disappears as mitosis begins, yet it somehow determines the plane of orientation of the new cell plate forming in late telophase and marks the zone of the parental cell wall where fusion with the growing cell plate ultimately occurs.

prepropolypeptide the prepolypeptide of any propolypeptide. Examples are: preproenkephalin and prepromelanin.

preprotein the proprotein of any proprotein. Examples



preprotachykinin

are: preproalbumin (serum) and prepro-von Willebrand factor. Note that preprolactin is not a preproprotein since prolactin is not a proprotein.

preprotachykinin *abbr.*: PPT; *see* **protachykinin β precursor**.

preprotein the transient, primary translation product of a eukaryotic messenger RNA for any protein destined to be secreted by a cell, irrespective of whether or not an intermediary **proprotein** is formed. It is detectable only in a cell-free protein-synthesis system since in an intact cell its N-terminal presequence (i.e. **signal peptide**) is cleaved off before translation has been completed. *See also* **prepolypeptide**, **preproprotein**.

presecreatory polypeptide an alternative name for **prepolypeptide**.

presequence an alternative name for **1 signal peptide**. **2** transport sequence (of a protein imported into mitochondria), especially in those instances where the sequence is not removed by proteolytic cleavage prior to translocation through the membrane but is removed at its final destination.

pressor (*in physiology*) **1** of, relating to, or bringing about an increase in arterial blood pressure. **2** any drug or other agent that has a pressor effect.

pressure *symbol*: p or P ; the force per unit area acting on a surface; i.e. $p = F/A$. The pressure at a point in a fluid is equal to the force per unit area on an infinitesimal plane situated at that point, and in a fluid at rest it is the same in all directions. The SI derived unit of pressure is the **pascal**; the **bar** is also acceptable (*see* **standard atmosphere**). *See also* **millimetre of mercury**, **torr**.

pressure dialysis dialysis in which a hydrostatic pressure is applied to the material being dialysed in order to minimize its initial dilution by solvent entering by osmosis.

pressure homogenization or **nitrogen cavitation** a method of disrupting cells that depends on equilibrating a cell suspension with a gas, e.g. nitrogen, at high pressure (up to 7 MPa) and then suddenly reducing the pressure, whereupon the gas dissolved in the cytoplasm is explosively released.

pressure-jump method a relaxation technique in which change of pressure is the disturbing factor. *See* **relaxation kinetics**.

pre-steady-state kinetics or **transient-phase kinetics** the kinetics of a catalysed reaction during the phase in which the concentration of the catalyst-reactant complex rises from zero to its steady-state value. For an enzyme-catalysed reaction this phase usually occupies a very brief period of time (from a second to a microsecond or less) and special techniques are required for its study. *See also* **rapid reaction kinetics**.

pre-swollen describing a commercial gel preparation that has already been equilibrated with water or another liquid phase.

presynaptic **1** (*in anatomy*) a designating a neuron that transmits an impulse directly to a synapse. **2** of or pertaining to that part of such a neuron close to the synapse. **2** (*in cytology*) prior to meiotic **synapsis**.

presynaptic membrane *see* **synapse**.

presynaptic nerve ending *see* **synapse**.

pre-transfer RNA *abbr.*: pre-tRNA; the class of primary gene transcripts from which the corresponding transfer RNA molecules are subsequently formed by post-transcriptional processing; any member of this class.

pretransition temperature the specific temperature at which a lipid bilayer being heated shows the first signs of **melting** (def. 2), before it reaches its transition temperature. Special techniques (e.g. employing spin-labelled or fluorescent probes) are required for its detection and study.

pretyrosine an obsolete name for **arogenate**.

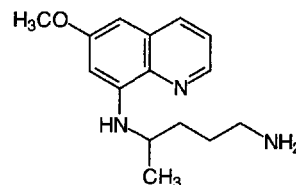
previtamin a less common word for **provitamin**.

Pribnow box a promoter site in *Escherichia coli* and phage genes that specifically binds **RNA polymerase**. The Pribnow box has the consensus sequence TATAAT and is centred at -10 (ten nucleotides on the 5' side of the first nucleotide transcribed, which is designated +1).

PRibPP *symbol* for (5- α -phospho- α -D-ribose) diphosphate.

primaevial soup a variant spelling of primeval soup; *see* **primordial soup**.

primaquine one of several 8-aminoquinoline antimalarial drugs. It is active against the exo-erythrocytic form of the parasite, and can effect radical cures as well as serving as a preventive medicine.



primary (*in chemistry*) **1** a describing an alkyl compound (e.g. an alcohol) in which the functional group (e.g. a hydroxyl group) is attached to a carbon atom linked to only one other. **2** describing the carbon atom bearing the functional group in such a compound. **2** describing an amide or an amine in which a single appropriate group is attached to a nitrogen atom. **3** describing a salt formed by replacing only one of the ionizable hydrogen atoms of a tribasic acid by another cation. *Compare* **secondary**, **tertiary**.

primary cell wall the outer layer of the wall of a plant cell. It is produced by deposition of cellulose microfibrils against the cell plate separating two daughter cells newly formed by division. When cell extension is complete, it is supplemented by deposition of the secondary cell wall (*see* **secondary cell wall deposition**).

primary culture a culture of cells (or tissue) that have been obtained directly from their place of origin.

primary electron an electron incident on a surface that causes **secondary electrons** to be emitted.

primary F'-containing strain *see* **F' plasmid**.

primary fluor or **primary scintillator** or **primary solute** (*in liquid scintillation counting*) the fluorescent substance to which radiation energy is directly transferred from excited molecules of the solvent. *Compare* **secondary fluor**.

primary granule an alternative name for **azurophil granule**.

primary hormone a term proposed to embrace any hormone that acts rapidly and uses a cyclic nucleotide as second messenger.

primary immune response the response produced in an animal when it is first exposed to an antigen. It consists characteristically of a slow production of antibody molecules and the priming of lymphoid tissue in readiness for the production of a **secondary immune response** on subsequent challenge with the same antigen.

primary kinetic isotope effect an isotope effect on a reaction rate that is attributable to isotopic substitution of an atom to which a bond is made or broken in the rate-limiting step or in a pre-equilibrium step of a specified reaction.

primary lysosome or **protolysosome** a **lysosome** containing newly synthesized acid hydrolases that have not as yet reacted with their ultimate substrates. A primary lysosome may be considered as a vesicle for transport between the site of enzyme synthesis and the digestive body or **secondary lysosome**.

primary metabolism the ensemble of metabolic activities that are common to most if not all living cells and are necessary for growth, maintenance, and survival. *Compare* **secondary metabolism**.

primary metabolite any intermediate in, or product of, **primary metabolism**.

primary response *see* **primary immune response**.

primary scintillator an alternative name for **primary fluor**.

primary solute an alternative name for **primary fluor**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

primary solvent

primary solvent (in liquid scintillation counting using mixed solvents) the solvent to which energy of the radiation emitted by the specimen is first transferred as excitation energy.

primary standard 1 any standard of measurement of nationally or internationally agreed and specified value that is used as the basis of a unit of measurement. 2 a specimen of a particular substance that is of extremely high purity and stability and that serves as an ultimate reference to which may be compared the purity of any secondary standard used in analysis.

primary structure the first order of complexity of structural organization exhibited by polypeptide and protein molecules and by polynucleotide and nucleic-acid molecules. When applied to a segment of a polypeptide chain or to a polypeptide or protein molecule, it refers to the linear sequence of the amino-acid residues of the polypeptide chain(s) without regard to spatial arrangement apart from configuration at the α -carbon atoms (and excluding the positions of any disulfide bonds). When applied to a segment of a polynucleotide chain or to a polynucleotide or nucleic-acid molecule, it refers to the linear sequence of the nucleotide residues of the polynucleotide chain(s) without regard to spatial arrangement. Compare **secondary structure**, **tertiary structure**, **quaternary structure**.

primary transcript the RNA as synthesized by RNA polymerase directly from a gene, unaltered by any further modification, such as **splicing**, that it may undergo.

primase 1 any RNA polymerase that synthesizes an RNA primer needed for the initiation of DNA synthesis. Such enzymes vary greatly in structure, specificity, and mode of regulation. 2 or **DNA primase** the DNA-directed **RNA polymerase** (EC 2.7.7.-) of the bacterial **primosome**. In *Escherichia coli*, primase is the product of the *dnaG* gene while in T4 phage the primase is the product of gene 61. In either case, the primase is only active in the presence of other proteins which create a complex called a **primosome**. The complex participates in the unwinding of parental DNA strands ahead of the replication fork, and synthesizes short RNA primers for the **Okazaki fragments**. Example from *E. coli*: database code PRIM_ECOLI, 581 amino acids (65.49 kDa).

primate any member of the Primates, an order of placental mammals. Primates are characterized by flexible pentadactyl limbs, opposable first digits, good stereoscopic vision, and, in the higher apes, highly developed brains. Included are lemurs, tarsiers, monkeys, apes, and humans.

prime 1 the printer's character, ', used (singly, doubly, etc.) as a mark to discriminate between characters or symbols representing things that are or would otherwise be indistinguishable. In chemical nomenclature it is used to distinguish between (1) identical locants referring to different moieties of the same molecule; or (2) two or more unspecified but differing univalent groups represented by the same symbol R. In physical chemistry it is used to differentiate the symbol for a quantity measured under arbitrary conditions from that for the corresponding quantity referred to the (thermodynamic) standard state; e.g. K' , apparent equilibrium constant, from K , thermodynamic equilibrium constant; E_o' , standard redox potential at a specified pH, from E_o , standard redox potential at a hydron activity of unity. 2 (in biochemistry) to provide a metabolic reaction with a primer; to bring about activation of a process or substrate. 3 (in immunology) to induce a **primary immune response** in an animal; to activate a cell specifically with respect to a given antigen. 4 (in mathematics) a having no other integral factor than itself and one. b **prime number**.

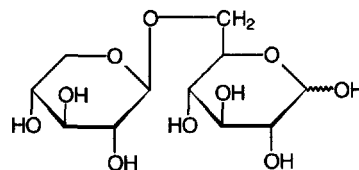
prime number any integer that possesses no integral factors and hence is divisible only by itself and 1.

primer 1 something used to **prime** (def. 2, 3). 2 an oligonucleotide required as the starting point for the stepwise synthesis of a polynucleotide from mononucleotides by the action of a nucleotidyltransferase. 3 a short-chain fatty acyl-CoA, e.g. acetyl-CoA or butyryl-CoA, required for the initiation of the

stepwise enzymic synthesis, from two-carbon units, of long-chain fatty acids as their CoA derivatives. See also **glycogenin**.

primeval soup or (esp. Brit.) **primaevial soup** an alternative name for **primordial soup**.

primeverose 6-O- β -D-xylosyl-D-glucose; a disaccharide occurring in madder.



primordia the plural of **primordium**.

primordial existing at or from the beginning, primeval; earliest formed, primitive.

primordial soup or **primordial slime** or **primeval soup** or (esp. Brit.) **primaevial soup** the aqueous solution of organic compounds that is considered to have been the environment in which the synthesis of macromolecules first occurred and cellular life originated. See also **coacervate droplet**.

primordium (pl. **primordia**) an organ or part at the earliest stage of its development; a rudiment.

primosome a protein complex that is essential for the activity of the **primase** responsible for the synthesis of RNA primers in the replication of DNA in *Escherichia coli*. See **replication complex**, **replication of DNA**.

PRINTS a computer database of protein sequence motifs (**signatures**) used in the compilation of this dictionary and elsewhere to identify amino-acid sequences (typically sequences of about 10 residues) defining structural and functional features of protein sequences. See also **Appendix E**.

prion a term originally introduced to emphasize the proteinaceous and infective nature of the pathogen responsible for transmissible encephalopathies. These include **scrapie**, **kuru**, **bovine spongiform encephalopathy**, **Creutzfeld-Jakob disease**, **fatal familial insomnia**, and **Gerstmann-Sträussler syndrome**. The agent is now widely accepted to be a variant of the normal **prion protein**; it is believed that the variant folds abnormally and induces the normal protein also to fold abnormally, typically into beta configurations, which form aggregates. A similar phenomenon has been observed in yeast, hence the term 'yeast prion'. See also **virino**.

prion protein *abbr.*: PrP^C; a membrane protein of 33–35 kDa that is encoded in the mammalian genome; it contains, in humans, 253 amino-acid residues (near the same number in other species) and has a glycosylphosphatidylinositol anchor. A variant, termed PrP^{Sc}, with the same amino-acid sequence, is found in the brains of sheep and other mammals affected by **spongiform encephalopathy**. PrP^C is rich in α helix but devoid of β sheet, while PrP^{Sc} contains much β sheet. PrP^{Sc} associates to form aggregates ('rods') in the central nervous system; these aggregates are seen in many manifestations of susceptibility to spongiform encephalopathy. Such mutations appear to be important in respect of the susceptibility of spongiform encephalopathy. Polymorphism, e.g. at position 171, occurs in several species and in subgroups in one species. Example: database code PRIO_SHEEP, 256 amino acids (27.88 kDa); eight motifs.

prism 1 any polyhedron of which two faces are polygons in parallel planes and the other faces are parallelograms. 2 a crystal form of which three or more faces are parallel to one axis. 3 a solid, bounded in part by two nonparallel plane faces and made of material transparent in the relevant region of the spectrum, that is used in optics to invert an image or to disperse or deviate a beam of light.

PRL *abbr.* for prolactin.

pro+

pro+ or **pro-** prefix **1** before in time or position; anterior; projecting; compare **pre+** (def. 1). **2** for, on behalf of, substituting for, favouring, encouraging, etc. **3** (in biochemistry) denoting (inactive or less active) biosynthetic precursor of the (class of) proteinaceous agent specified, to which it is convertible by proteolytic cleavage in a specific manner. Examples are: proalbumin, prohormone, proinsulin, proprotein, and prothrombin (but not prolactin!). Compare **pre+** (def. 3). See also **+gen**.

Pro symbol for a residue of the α -amino (strictly, cyclic α -alkylamino) acid L-proline (alternative to P).

pro- prefix in symbols designating prochirality; see **pro-R/pro-S** convention.

proaccelerin an alternative name for factor V; see **blood coagulation**.

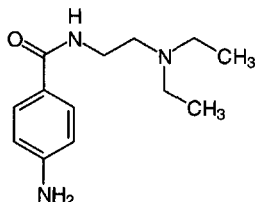
probability (in statistics) symbol: *p* or *P*; a measure of the likelihood of the occurrence of a given event, expressed as the ratio of the number of times it occurs in a series of observations to the total number of observations, or as its decimal equivalent.

probable error an older measure of sampling variability such that one-half of the events, observations, etc. in a normal distribution lie between the mean value minus the probable error and the mean value plus the probable error. The probable error is equal to the standard error multiplied by 0.6745.

proband an alternative term for **propositus**.

probe **1** something that searches into, examines, or tests; e.g. an instrument or tube used for exploring a cavity or sampling a fluid, or an electric conductor introduced into an electric circuit or a cavity resonator to provide electric coupling to an external circuit. **2** (in biochemistry) a chemical species, or a chemical group attached to a carrier molecule, used to investigate the chemical nature of some other particular chemical species or the physical nature of a particular environment, especially within an intact cell; a **reporter group**. Typically some measurable physical property of the probe (e.g. fluorescence, electron spin resonance) alters in a way that can be correlated with changes in the environment or in the macromolecule. Examples are a group having paramagnetic properties, for detection using electron spin resonance (see, e.g., **TEMPO**), and a fluorescent lipid molecule used to report membrane fluidity (see, e.g., **parinaric acid**). **3** (in molecular biology) or **gene probe** or **hybridization probe** an oligo- or polynucleotide that is complementary to an oligonucleotide or nucleic-acid sequence under investigation. The probe is normally labelled in a way that permits its ready detection, e.g. with a radioactive isotope or a fluorescent constituent. Hybridization of the probe to the oligonucleotide sequence under investigation allows that sequence to be detected. The technique finds much use in screening gene libraries, detecting oligonucleotides on blots of electrophoresis gels, and in other associated technologies. **4** to examine with or as with a probe.

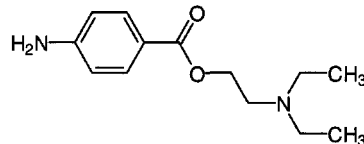
procainamide a local anaesthetic similar in action to **procaine** but having a less easily hydrolysed amide linkage, which gives it longer duration of action.



procainamide azide a local anaesthetic; it binds the 43 kDa postsynaptic protein (μ chain or rapsyn) of the nicotinic cholinergic receptor and is useful as a photoaffinity label.

procaine or **novocaine** the prototype injectable local anaesthetic. It brings about changes in the physical properties of the

lipid bilayer towards a more fluid state, decreasing ordering of the fatty-acyl side chains.



procapsid a precursor of the virion of certain viruses, consisting of a viral capsid without its specific nucleic acid.

procarboxypeptidase component III a protein, resembling elastase but without enzyme activity, found in the pancreatic juice of ruminants in noncovalent association with zymogens, carboxypeptidase A, etc. It probably prevents denaturation of the latter in the acid of the duodenum. Example (bovine): database code CAC3_BOVIN, 240 amino acids (25.83 kDa).

procaryon a variant spelling of **prokaryon**.

procaryote a variant spelling of **prokaryote**.

process (in anatomy) a thin prominence or protuberance of a structure, or an extension or outgrowth of a cell.

processed gene a **pseudogene** that lacks the intervening sequence or sequences of the homologous functional gene. It usually contains a short segment consisting entirely or almost entirely of adenine nucleotides, attached near the 3' end, and is thought to originate by reverse transcription of an mRNA template with incorporation of the cDNA so formed into genomic DNA. A processed gene is sometimes called a **retrogene**.

processing (in molecular biology) **covalent modification** either of the primary transcription product (i.e. RNA) of a DNA or of the primary translation product (i.e. polypeptide) of a messenger RNA. See also **post-translational modification**.

processive (in molecular biology) or **nondistributive** describing (the action of) any enzyme or catalytic complex that progressively synthesizes or degrades a biopolymer by effecting several or many cycles of the same type of reaction or reaction sequence without dissociating from the template or intermediate product (as appropriate) between catalytic events. Compare **distributive**.

processivity (of a nucleic acid polymerase) the average number of nucleotides inserted before the enzyme dissociates from its template. In the case of DNA polymerases, repair enzymes (such as Pol I in *Escherichia coli*) have low processivity replication enzymes (such as Pol II and its associated proteins in *E. coli*) have high processivity.

processosome a hypothetical ribonucleoprotein associated with processing of pre-RNA in the nucleolus.

prochiral describing a molecule, or a centre or axis in a molecule, that in itself is achiral but is capable of becoming **chiral** by a replacement reaction. In general, a prochiral molecule has a structure *Caabc*, where the 'a' groups (or atoms) are identical. Replacement of one 'a' by another group or atom, 'd', leads to a chiral molecule. Thus, in ethanol, replacement of one 'H' of the methylene group by ^2H leads to a chiral structure which can be shown to be optically active. In a prochiral molecule *Caabc*, the chemically like 'a' groups may be differentiated one from another in a reaction with a chiral molecule, especially an enzyme. An achiral molecule containing a double bond or a planar cyclic ring system is also prochiral if separate addition of an achiral reagent to each face leads to two enantiomeric products. See also **meso-carbon atom**, **pro-R/pro-S** convention. — **prochirality** *n*.

procholecystokinin precursor a member of the gastrin/cholecystokinin family, and a precursor for five different **cholecystokinins**. Example from human: database code CCKN_HUMAN, 115 amino acids (12.65 kDa).

Procion the proprietary name for a range of reactive dyes that includes Procion Blue H-B, Procion Red H-EB, as well as a



procoagulant

number of other dyes useful in biochemistry, especially in **dye-ligand chromatography**.

procoagulant (*sometimes*) any of factor V, factor VII, or factor VIII, each of which acts to accelerate the conversion of prothrombin to thrombin during **blood coagulation**.

procollagen a ≈ 150 kDa triple-helical protein that is an intermediate in the biosynthesis of **tropocollagen**, into which it is converted proteolytically by removal of the additional sequences at both the N and the C termini of each of the three peptide chains. Example, procollagen $\alpha 1$ (human): database code CA11_HUMAN, 1464 amino acids (138.73 kDa). *Compare* **protocollagen**.

procollagen N-endopeptidase EC 3.4.24.14; *other name*: procollagen N-proteinase; an enzyme that catalyses the cleavage of the N-propeptide of collagen chain $\alpha 1(I)$ at Pro-[Gln and of chain $\alpha 2(II)$ at Ala-[Gln.

procollagen-lysine 5-dioxygenase EC 1.14.11.4; *other names*: procollagen-lysine,2-oxoglutarate 5-dioxygenase; lysine hydroxylase; lysine,2-oxoglutarate 5-dioxygenase; an enzyme that catalyses the formation of hydroxylysine residues in collagen. The reaction is between dioxygen, procollagen L-lysine, and 2-oxoglutarate to form procollagen 5-hydroxy-L-lysine, succinate, and CO_2 . Iron and **ascorbic acid** are cofactors. Example (precursor) from human: database code LYSH_HUMAN, 727 amino acids (83.49 kDa). *See also* **procollagen-proline dioxygenase**.

procollagen-proline dioxygenase EC 1.14.11.2; *systematic name*: procollagen-L-proline,2-oxoglutarate:oxygen oxidoreductase (4-hydroxylating); *other names*: proline,2-oxoglutarate-4-dioxygenase; protocollagen hydroxylase; proline hydroxylase; prolyl 4-hydroxylase. An enzyme converting L-proline into 4-hydroxy-L-proline; it catalyses a reaction between dioxygen, procollagen L-proline, and 2-oxoglutarate to form procollagen *trans*-4-hydroxy-L-proline, succinate and CO_2 . Hydroxylation only occurs when proline is present in a polypeptide chain elongating on a ribosome. **Ascorbic acid** and Fe^{3+} are also required as cofactors, and the reaction is one of the few well-defined roles for ascorbic acid; it seems likely that scurvy, as a result of vitamin C deficiency, is due to defective activity of this enzyme, which plays a key role in collagen synthesis. The enzyme is a tetramer of two α chains and two β chains; the β chain is the multifunctional **protein disulfide isomerase** and hence has microsomal triacylglycerol transfer protein activity. Subcellular location is the endoplasmic reticulum lumen. There are two forms of α subunit, which are produced by alternative splicing of the same gene. Example from human (α precursor): database code P4HA_HUMAN, 534 amino acids (60.84 kDa). *See also* **procollagen-lysine 5-dioxygenase**.

proconvertin an alternative name for factor VII; *see* **blood coagulation**.

procratocortin or **procortin** an alternative name for **proopiomelanocortin**.

procyclin or **PARP** a major surface antigen of procyclic forms of trypanosomes. It is GPI-anchored. Example B1- α precursor from *Trypanosoma brucei brucei*: database code PAR1_TRYBB, 143 amino acids (14.88 kDa).

produg a drug molecule that is itself inert but has pharmacological effects after **bioactivation**. An example is **ganciclovir**.

produce (*in mathematics*) to extend a line or a plane.

product 1 (*in biochemistry and chemistry*) something formed in a reaction. **2** (*in mathematics*) the result of multiplying together two or more numbers, quantities, or expressions.

product inhibition the inhibition of an enzymic reaction caused by increased concentration of one or more products of that reaction.

productive (capable of) yielding a product, result, or benefit.

productive binding or **eutopic binding** the binding of a substrate in a reactive mode at the active site of an enzyme. *Compare* **nonproductive binding**.

productive complex any enzyme-substrate complex in which the substrate is bound to the enzyme in a manner that renders

catalysis possible so that products can be formed. *Compare* **abortive complex**.

productivity (*in biotechnology*) *symbol*: r ; the mass of product formed per unit reactor volume per unit time; often per unit of enzyme or biomass. It is measured typically in $\text{kg m}^{-3} \text{h}^{-1}$.

proenkephalin a protein precursor for several neuropeptides including various **enkephalins** formed in brain and adrenal medulla. For example, in the human, proenkephalin A precursor yields four copies of [Met]enkephalin (100–104; 107–111; 136–140, 210–214) and single copies of each of [Leu]enkephalin (230–234), [Met]enkephalin-Arg-Gly-Leu (octapeptide) (186–193) and [Met]enkephalin-Arg-Phe (heptapeptide) (261–267), which arise by enzymatic cleavage of the gene product by cleavage at paired basic amino acids: database code PENK_HUMAN, 267 amino acids (30.79 kDa). Another human protein, proenkephalin B precursor (*other name*: β -neoendorphin-dynorphin precursor) yields single copies of α -neoendorphin, dynorphin, [Leu]enkephalin, rimorphin, and leumorphin, also by cleavage at paired basic amino acids: database code NDDB_HUMAN, 254 amino acids (28.38 kDa).

proenzyme an alternative name for **zymogen**.

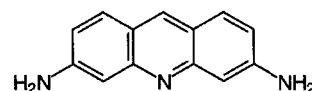
pro-Epro-Z convention if the two chemically-like hydrogen atoms of $abC=CH_2$ are separately replaced by the achiral, d , an achiral stereogenic element is generated and two (diastereoisomeric) alkenes, $abC=CHd$ are formed that are E and Z stereoisomers (*see* **E/Z convention**). The replacement has converted a prostereogenic element into which one which is stereogenic; the situation is comparable to the change, prochiral $\rightarrow$ chiral, although a chiral element is not involved. The two chemically-like hydrogen atoms are designated as *pro-E* and *pro-Z*, and are abbreviated as H_E and H_Z . Replacement of a *pro-E* hydrogen with 2H yields an E diastereoisomer and the similar replacement of a *pro-Z* hydrogen yields the Z form. Example: the two hydrogen atoms of phosphoenolpyruvate; that on the same side of the double bond as the COOH group is H_E , that on the same side as the OPO_3H_2 group is H_Z . Replacement of 1H_E by 2H yields (E)-[3- 2H]phosphoenolpyruvate; the Z diastereoisomer results if 1H_Z is replaced by 2H . A similar situation is found in the side chain of **chorismic acid**.

proestrus *see* **estrous cycle**.

profilinolyisin an alternative name for **plasminogen**.

profilin a protein that *in vitro* prevents the polymerization of actin. In many cells it occurs as a 1:1 complex with monomeric actin, that appears to act as an intracellular storage form of actin. Profilin binds PtdInsP_2 and inhibits the γ isoform of phospholipase C (*see* **phospholipase**), an inhibition that is overcome when the enzyme is activated as a result of receptor kinase-dependent tyrosine phosphorylation. This activation appears in some way to recruit profilin-actin to the site of filament assembly which occurs when cells are stimulated e.g. by nitrogen. Example, profilin I (human): database code PRO1_HUMAN, 139 amino acids (14.91 kDa).

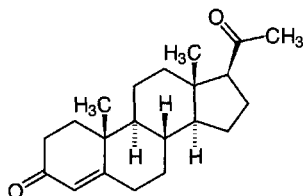
proflavin 3,6-diaminoacridine; an **acridine** dye, molecules of which are capable of intercalating between adjacent base pairs in duplex DNA. It is useful as an antibacterial agent, by virtue of its ability to inhibit the biosynthesis of both DNA and RNA, possibly through intercalation into preexisting DNA, and as a frameshift mutagen for bacteriophages.



progesterone the common name for pregn-4-ene-3,20-dione; the principal steroid hormone of the corpus luteum, from which it is secreted during the latter half of the estrous cycle, acting upon the endometrium to prepare it for embryo im-



plantation. During pregnancy, secretion occurs also from the placenta, the rate of production increasing markedly as pregnancy proceeds, when it acts to maintain the uterus, inhibit further release of ova, and promote proliferation of acini within the mammary gland. Progesterone occurs also in the adrenal cortex and testis. It is synthesized from **pregnenolone** by 3 β -hydroxy- Δ^3 -steroid dehydrogenase, EC 1.1.1.145, and is an intermediate in the formation of androgens, estrogens, glucocorticoids, and mineralocorticoids. It also occurs in some plants, e.g. *Hollatrheia floribunda*.



progesterin 1 an alternative name for **progestogen**. **2** natural **progesterone**, especially in unpurified form.

progestogen or **progestagen** or **progestin** any natural or synthetic compound with the hormonal activity of progesterone. In oral contraception regimes, a progestogen is normally combined with an estrogen. The effect is to inhibit ovulation. A preferred method is to use a gestagen, which acts by an effect on the endometrium; these have fewer side effects than progestogens.

proglucagon the putative prohormone of **glucagon** and **enteroglucagon**. In mammals it appears to be an 18 kDa single-chain polypeptide of some 170 residues that contains the entire 29-residue sequence of glucagon within its N-terminal moiety and in addition the sequences of two glucagon-like peptides, designated GLP-1 and GLP-2, of 37 and 35 residues respectively, within its C-terminal moiety. The 69-residue N-terminal portion of the molecule, containing the glucagon sequence, is identical with **glucicentin**, and thus in mammals proglucagon appears to be the common biogenetic precursor both of pancreatic glucagon and of enteroglucagon and other nonpancreatic forms of glucagon. Angler-fish proglucagon is similar but smaller, and apparently exists in two forms: a 14 kDa form (lacking GLP-2 and a 13-residue connecting peptide) and a 12 kDa form (of unknown sequence).

program 1 a sequence of coded instructions for controlling a computer or other electrical or electromechanical device, especially for enabling a computer to manipulate data in a predetermined way; an equivalent of this in an electronic or magnetic storage medium. **2** to constrain to function or operate in a particular way, especially by providing with a program or other (set of) instructions. **3** to write (and test) a computer program for a specific application. —**programmable** or **sometimes** (*esp. US*) **programmable** *adj.*

programmable messenger a term proposed for a cytosolic protein messenger molecule in a hypothesis advanced to explain the multiple actions on their target cells of hormones whose cellular receptors are linked to oligomeric GTP-regulatory proteins in cell membranes. According to this concept, the activation of GTP-regulatory proteins (by GTP and the respective hormones) causes release of their α subunits from the membranes as cytosolic proteins, which become the primary messengers of hormone action and which thereupon are transformed by various protein-modifying enzymes (e.g. kinases and proteases) to yield new structures with selective affinities for different effectors, each responsive to its specific activating system. A variety of responses of the target cells thus may be elicited: accordingly, these messengers are considered **programmable**.

programmed ribosomal frameshifting a mechanism

whereby different proteins may result from a single mRNA molecule, due to the presence of alternative sites within the mRNA that are recognized for initiation and termination of translation.

progress curve a graph showing the change with time in the concentration or amount of an analyte as a chemical reaction proceeds.

prohormone a natural precursor of a hormone, especially for the proprotein of a (poly)peptide or protein hormone.

prohormone convertase an enzyme activity that cleaves prohormones at **dibasic cleavage sites** to form active polypeptides. As EC 3.4.23.17 its *recommended name* is **proopiomelanocortin converting enzyme**. However, no sequence has been reported. An example of a convertase enzyme is given in **kexin**. See also **neuroendocrine convertase**.

prohormone converting enzyme see **pro-opiomelanocortin converting enzyme**.

proinsulin or (*formerly*) **pro-insuline** the prohormone of **insulin**; it is a single-chain polypeptide of ≈ 9 kDa with three intrachain disulfide bridges, the number of amino-acid residues varying somewhat with species (78 in dog, 81 in cattle and sheep, 84 in cod and pig, 86 in horse, human, and rat). Proinsulin is synthesized in the pancreatic B cell, where it is normally present at about 5% of the concentration of insulin, and from where it is secreted in small amounts into the blood.

projection formula any two-dimensional diagrammatic representation of the (three-dimensional) structural formula of a molecule (or part thereof) as if seen by projection onto a plane surface from a particular viewpoint in space. Bonds with different spatial orientations relative to the observer may be depicted distinctively. The term includes **conformational formula**, **Fischer projection**, **Newman projection**, and **perspective formula**.

prokaryon or **procaryon** (*pl.* **prokarya** or **procarya**) a primitive nucleus; i.e. genomic DNA in a defined structure, but lacking a nuclear membrane or other nuclear inclusions. Such an arrangement is found in bacteria (including actinomycetes) and cyanobacteria. See also **nucleoid** (def. 2). —**prokaryous** or **procaryous** *adj.*; **prokaryosis** or **procaryosis** *n.*

prokaryosis the condition of possessing a prokaryon or prokarya.

prokaryote or **procaryote** any organism in which the genomic DNA is not enclosed by a nuclear membrane within the cells; i.e. any organism whose cells possess a **prokaryon**. The prokaryotes comprise the bacteria, cyanobacteria, and archaea and are classified in their own kingdom, variously named Monera or Procaryotae. Compare **eukaryote**. —**prokaryotic** or **procaryotic** *adj.*

prolactin *abbr.*: PRL; a hormone secreted by the anterior pituitary that stimulates lactation. Human prolactin (hPRL) is a 199-residue protein, 23 kDa in its monomeric form (**little prolactin**), although dimeric (**big prolactin**, 48–56 kDa) and polymeric (>100 kDa) forms circulate in blood. The prolactin gene belongs to a family of genes, along with those encoding placental lactogen and human growth hormone (see **somatotropin**). Secretion is episodic; it is inhibited by dopamine, which binds to lactotrope D₂ receptor (see **dopamine receptor**) to inhibit adenylate cyclase, and is probably stimulated by **thyrotropin-releasing hormone** and **vasoactive intestinal peptide**. Prolactin synthesis is positively regulated by estrogen acting at the level of the prolactin gene. Receptors for prolactin are located in breast, liver, ovary, testis, and prostate, but its main site of action is the mammary gland, which is stimulated to develop by estrogen, progesterone, prolactin, and placental mammatropic hormones. Lactation is stimulated by prolactin but is prevented during pregnancy by the inhibitory action of estrogen and progesterone, the levels of which fall after parturition, allowing prolactin to initiate lactation. Example (precursor) from *Gallus gallus*: database code PRL_CHICK, 229 amino acids (25.81 kDa). See also **prolactin receptor**, **proliferin**.

prolactin receptor a membrane protein of the cytokine receptor family. Example (type 2 precursor) from human: database



prolamin

code PLR2_HUMAN, 622 amino acids (69.43 kDa). The prolactin receptor is associated with a small protein **calcyclin**.

prolamin or (formerly) **prolamine** any of a group of simple proline-rich proteins, found especially in the seeds of cereal plants, that are insoluble in water, neutral salt solutions, and absolute ethanol, but soluble in dilute acid or alkali and in 70–80% aqueous ethanol. The group includes the **gliadins** of wheat and rye, **hordein** of barley, and **zein** of maize. Compare **glutelin**.

prolate **1** having a polar diameter longer than the equatorial diameter. **2** describing an **ellipsoid of rotation** generated by the rotation of an ellipse about its major axis. Compare **oblate**.

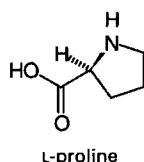
prolidase another name for **X-Pro dipeptidase**. Distinguish from **prolinase**.

proliferin or **mitogen-regulated protein** a prolactin-related protein, the mRNA of which appears in several murine cell lines during growth. One of several proteins of the somatotropin/prolactin family that provide a growth stimulus to target cells in maternal and fetal tissues during the development of the embryo. The mRNA occurs in placenta, which secretes proliferin as a glycoprotein distinct from placental lactogen. Examples: three proliferin precursors from mouse; database code PRO1_MOUSE, 224 amino acids (25.34 kDa); database code PRO2_MOUSE, 224 amino acids (25.28 kDa); and database code PRO3_MOUSE, 224 amino acids (25.31 kDa).

prolinase the recommended name for **Pro-X dipeptidase**. Distinguish from **prolidase**.

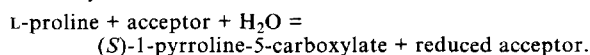
prolinate **1** proline anion; the anion, $(C_4H_7NH)-COO^-$, derived from **proline**. **2** any salt containing proline anion. **3** any ester of proline.

proline the trivial name for pyrrolidine-2-carboxylic acid, $(C_4H_7NH)-COOH$; a chiral cyclic *N*-alkylated α -amino acid. L-proline (symbol: P or Pro), (S)-pyrrolidine-2-carboxylic acid, is a coded amino acid found in peptide linkage in proteins; codon: CCA, CCC, CCG, or CCU; by virtue of its more constrained molecular structure relative to most other α -amino acids it cannot participate in α -helix formation. In mammals it is a nonessential dietary amino acid and is gluconogenic. It is formed from **glutamate 5-semialdehyde** by reduction by NADPH. D-Proline (symbol: D-Pro or DPro), (R)-pyrrolidine-2-carboxylic acid, is not known to occur naturally. See also **hydroxyproline**.



proline/arginine-rich protein abbr.: PARP; a 24 kDa proline- and arginine-rich protein from bovine cartilage, closely related to the N-terminal domain of collagen $\alpha 1$ (XI).

proline dehydrogenase EC 1.5.99.8; a flavoprotein enzyme that catalyses the reaction:



The flavin is FAD. Example, from *Escherichia coli*, a bifunctional enzyme, having the activities of both proline dehydrogenase and **1-pyrroline-5-carboxylate dehydrogenase**; database code PUTA_ECOLI, 1320 amino acids (143.66 kDa).

proline dipeptidase see **X-Pro dipeptidase**.

proline hydroxylase see **procollagen-proline dioxygenase**.

prolinium proline cation; the cation, $(C_4H_7NH_2^+)-COOH$, derived from proline.

proline the pyrrolidino group, $HOOC-(C_4H_7N)-$, derived from proline.

prolyl the acyl group, $(C_4H_7NH)-CO-$, derived from proline.

prolyl dipeptidase see **Pro-X dipeptidase**.

L-prolylglycine dipeptidase see **Pro-X dipeptidase**.

prolyl hydroxylase see **procollagen-proline dioxygenase**.

prolyl oligopeptidase EC 3.4.21.26; other names: prolyl endopeptidase; post-proline cleaving enzyme; post-proline endopeptidase. An enzyme that catalyses the hydrolysis of $Pro\text{-}[-Xaa] \gg Ala\text{-}[-Xaa]$ in oligopeptides. It is a cytoplasmic (in eukaryotes) serine endopeptidase, also found in some bacteria. Example from pig: database code PPCE_PIG, 710 amino acids (80.68 kDa).

prometaphase the stage in **mitosis** or **meiosis**, following **prophase**, during which the nuclear envelope is disrupted and breaks into membrane vesicles, and the spindle microtubules enter the nuclear region. **Kinetochore**s mature on each **centromere** and attach to some of the spindle microtubules; these are then called **kinetochore microtubules** while those to which kinetochores do not attach are called **polar microtubules**; microtubules outside the spindle are called **astral microtubules**. Kinetochore microtubules begin the process of aligning chromosomes in one plane halfway between the poles.

promoter symbol: P; a DNA sequence located 5' to a gene that indicates the site for the initiation of transcription. It may influence the amount of mRNA produced and the tissue specificity. Examples include the **TATA box**. The symbol may be subscripted to indicate particular promoters, e.g. P_{RM} , a promoter for the *cl* gene, RM standing for repressor maintenance.

promoter ligand an additional ligand that is included in the loading buffer in **affinity chromatography**, to promote specific binding to the immobilized ligand.

promutagen any substance that, when acted upon by the cell's metabolism, becomes a **mutagen**.

Pronase a proprietary name for a mixture of various exo- and endo-peptidases, obtained from *Streptomyces griseus*, that is able to hydrolyse virtually any protein almost completely to free amino acids. It is used for hydrolysis of many proteins, for digestion of mucins, in isolation of intact DNA or RNA from cells or viruses, and in dispersal of certain types of mammalian tissue cells, e.g. chondrocytes.

pronucleus (pl. **pronuclei**) the nucleus of either the ovum or the spermatozoon following fertilization. Thus, in the fertilized ovum, there are two pronuclei, one originating from the ovum, the other from the spermatozoon that brought about fertilization; they approach each other, but do not fuse until just before the first cleavage, when each pronucleus loses its membrane to release its contents. —**pronuclear adj.**

proof a standard by which the strength of distilled alcoholic solutions (or vinegars) is judged. In the UK it is the strength of a mixture of alcohol and water having a specific gravity of 0.9184, and containing 0.495 of its weight or 0.5727 of its volume of absolute alcohol. Degrees of proof express the percentage of proof spirit present; hence 70° proof spirit contains $0.7 \times 57.27\%$ alcohol. In the US, proof is a number that is twice the percentage (by volume) of alcohol present; hence 70° proof spirit contains $0.5 \times 70\%$ alcohol.

proofreading (in genetics) verification of the exact nature of a sequence of bases in DNA or amino-acid residues in a protein. Proofreading of DNA sequence occurs in transcription, during the action of DNA polymerase, when incorporation of an incorrect nucleotide causes mismatch in pairing with the complementary base on the template strand; this leads to immediate excision of the incorrect nucleotide, retreat of the enzyme to the previous position, and another attempt to insert the correct nucleotide. During translation, charging of the tRNA with its aminoacyl group is verified by its fit in the binding site; if incorrect, the aminoacyl-tRNA is hydrolysed to tRNA and amino acid and the whole process is repeated until successful. Examples of such proofreading during translation include rejection of threonine by valine-tRNA ligase, and of valine by isoleucine-tRNA ligase. See also **mismatch repair**.

Pro(3-OH) symbol for a residue of the α -amino acid (trans)-3-hydroxy-L-proline (alternative to 3Hyp).

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



Pro(4-OH) symbol for a residue of the α -amino acid (*trans*)-4-hydroxy-L-proline (alternative to 4Hyp).

proopiomelanocortin *abbr.*: POMC; a 241-residue prohormone protein, also called **corticotropin-lipotropin precursor**, synthesized in the intermediate and anterior lobes of the pituitary as the primary gene product of the POMC gene; in humans this is located on chromosome 2. POMC is enzymically cleaved in the anterior pituitary by a **prohormone convertase**, PC1 to yield an N-terminal peptide (residues 1–80), **corticotropin** (residues 110–148) and β **lipotropin** (β -LPH, residues 148–241). The segment 83–107, which is thus released, is known as **joining peptide**. Prohormone convertase, PC2, then cleaves corticotropin to yield α -**melanocyte-stimulating hormone** (α -**MSH**; residues 110–122), and cleaves β -LPH to yield γ -LPH (residues 151–208) and β **endorphin** (residues 211–241). Example from human: database code COL1_HUMAN, 267 amino acids (29.39 kDa) residues 1–26 being the signal peptide. Derived hormones (cleavage points and some sizes differ from porcine above): corticotropin, 39 amino acids; **corticotropin-like intermediate lobe peptide**, 21 amino acids; β **endorphin**, 31 amino acids; lipotropins β -LPH (89 amino acids) and γ -LPH (56 amino acids); melanocyte-stimulating hormones α -MSH (13 amino acids), β -MSH (18 amino acids), and γ -MSH (11 amino acids); [Met]enkephalin (*see* **enkephalin**), 5 amino acids. In the intermediate lobe, POMC synthesis is regulated mainly by dopamine and serotonin; in the anterior lobe by glucocorticoids and corticotropin-releasing hormone. The prohormone convertases are related to **furin** and **kexin**.

proopiomelanocortin converting enzyme EC 3.4.23.17; *the recommended name for* **prohormone convertase**.

propargylamine *see* **monoamine oxidase inhibitors**.

propellant or **propellent** any substance that through pressure or combustion or other means causes propulsion, as, e.g., the use of compressed gases in spray cans.

propeptide that part of a peptide chain that follows the signal sequence, and which is cleaved off in formation of the mature protein.

properdin *see* **complement**.

prophage a form of the genome of a bacteriophage in which the nucleic acid is integrated into the genome of the phage's bacterial host, a state known as **lysogeny**. Activation to produce mature phage can occur, with accompanying lysis of the bacterium. For example, in **lambda phage**, transcription of the prophage is repressed by a gene, *cI*, that codes for a repressor protein that binds to the operator (*O*) genes to prevent use of the promoter (*P*) genes. The product of the *cII* gene, CII, is critical in determining the switch between lysogeny and lysis; if CII is not active (e.g. due to breakdown by host proteases) lysogenization is favoured, since CII competes with the repressor product of *cI* and prevents it binding to *O*.

prophage induction *see* **phage induction**.

prophage-mediated conversion another name for **bacteriophage conversion**.

prophage reactivation the phenomenon in which the survival of UV-irradiated **lambda phage** is higher in a host carrying a homologous heteroimmune prophage than in a host that is nonlysogenic or is lysogenic with a nonhomologous prophage.

prophase the initial stage of **mitosis** or **meiosis** in which the chromosomes are condensed, becoming apparent within the nucleus, but are not yet attached to a mitotic spindle. By early prophase, the single centrosome contains two centriole pairs; at late prophase, the centrosome divides and the two asters move apart. In meiosis, the chromosomes come together in homologous pairs, and undergo crossing-over. This occurs during the first of the two meiotic cell divisions, and is divided into several stages. In the first, zygotene, a complex structure develops between the two sister chromatids of each pair, which may play a part in **recombination**. They remain in the complex during a stage known as pachytene, then separate at the diplotene stage.

propheromone convertase **Y** a protease present in yeast.

with specificity towards paired basic residues (–Lys-Arg-, –Arg-Arg-, or –Arg-Lys-) and no activity towards single basic residues.

propionate the anion, $\text{CH}_3\text{--CH}_2\text{--COO}^-$, derived from propionic acid, in any salt; designation for the **propionyl** radical in naming an ester.

propionibacteria bacteria belonging to the genus *Propionibacterium*. They are characterized by their activity in fermenting glucose or lactic acid to propionic acid; acetic acid is also a major product. *See also* **propionic-acid fermentation**.

propionic acid propanoic acid; $\text{CH}_3\text{--CH}_2\text{--COOH}$; a carboxylic acid, important in the energy metabolism of ruminants, in which it is formed by rumen bacteria. It is the basis of a series of nonsteroidal anti-inflammatory agents, known as substituted propionic acids; *see* **ibuprofen**.

propionic-acid fermentation fermentation of glucose or lactate by *Propionibacterium* spp. (*see* **propionibacteria**) to either propionate or acetate; pyruvate is a common intermediate for both transformations, the pathway then branching to form either propionate or acetate. In propionate formation, pyruvate first reacts with methylmalonyl-CoA to form propionyl-CoA and oxaloacetate. The latter is converted by reactions of the tricarboxylic-acid cycle to succinate, which then reacts with the propionyl-CoA to form propionate and succinyl-CoA. The succinyl-CoA is converted to methylmalonyl-CoA, which is then available to react with a further molecule of pyruvate. Conversion of pyruvate to acetic acid involves oxidative decarboxylation to acetyl-CoA, reaction of this with orthophosphate to form acetyl phosphate, and reaction of the latter with ADP to form ATP and acetic acid. When lactate is the initial substrate, this is first converted to pyruvate by lactate dehydrogenase. This enzyme is a flavoprotein, that is oxidized by a cytochrome *b*.

propionyl symbol: Pp; *the trivial name for* propanoyl, $\text{CH}_3\text{--CH}_2\text{--CO-}$, the univalent acyl group derived from propionic (i.e. propanoic) acid.

propionyl-CoA carboxylase *abbr.*: PCCase; EC 6.4.1.3; an enzyme that catalyses a reaction between ATP, propanoyl-CoA, and HCO_3^- to form (*S*)-methylmalonyl-CoA, ADP, and orthophosphate; biotin is a coenzyme. It is an enzyme in the catabolic pathways of odd-chain fatty acids and isoleucine, occurring in the mitochondrial matrix. Defects cause propionic acidemia, which is inherited as a recessive trait. Example from human (precursor); six α chains (biotin binding): database code PCCA_HUMAN, 702 amino acids (77.01 kDa); and six β chains: database code PCCB_HUMAN, 155 amino acids (16.99 kDa).

proplastid the precursor of a **plastid**.

propolypeptide the inactive form in which some polypeptide hormones or other agonists are produced; it is then activated by removal of part of the sequence by a proteinase.

proportional counter any device for measuring radioactive disintegrations by means of a gas-filled ionization chamber, e.g. Geiger counter, operating in the **proportional region**.

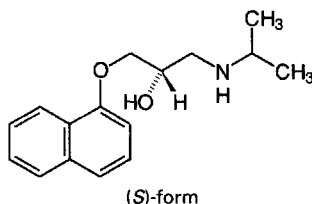
proportional region a range of voltages applied to a gas-filled ionization chamber (e.g. Geiger counter) over which the gas amplification is greater than unity. In radioactive counters of this type, measurement depends on ionization of the gas in the ionization chamber by emitted radiation. Up to a certain applied voltage, ion production and thus the current through the chamber depends solely on collisions between emitted radiation and gas molecules; at higher voltages, the current is enhanced by additional collisions, so that sensitivity is increased, but the current remains proportional (hence the name **proportional region**) to the rate of emission of radiation. *Compare* **Geiger region**.

propositus or (*fem.*) **proposita** (*pl.* **propositi**) or **proband** the first individual to present as a patient, in a series of cases related (usually) to a heritable disorder, and from whom the line of descent is traced.

propranolol an important β -adrenoceptor antagonist; the (*S*)



form is the active structure. It is used therapeutically as an antihypertensive, anti-anginal, and anti-arrhythmic. *One proprietary name:* Inderal (as hydrochloride).



propressophysin the polypeptide precursor of **vasopressin** and **neurophysin** (see **vasopressin-neurophysin 2-copeptin precursor**). A clinical name for the product of the gene associated with familial neurogenic diabetes insipidus.

proprotein an inactive form of a protein, having an additional sequence that is removed by a proteinase to yield the active protein; a classic example is **proinsulin**. A **zymogen** is a form of proprotein, sometimes referred to as a proenzyme.

propyl symbol: Pr; *n*-propyl; the alkyl group, $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--}$, derived from propane. Compare **isopropyl**.

pro-R/pro-S convention stereochemical descriptors for paired, chemically-like ligands at a prochiral centre; they derive from an extension of the **sequence rule** used to assign *R,S* descriptors to a chiral centre. In the prochiral (but achiral) structure, *Caabc*, the chemically-like ligands are the two atoms (or groups of atoms), *a*. For the sake of example assume that the sequence rule gives the priority sequence, $a > b > c$. In the molecular model (or projection formula) arbitrarily label the *a* groups as *a'* and *a''* and arbitrarily assign higher priority to one of them, e.g., $a' > a''$. The modified priority sequence, $a' > a'' > b > c$, is then considered with the model viewed from the side remote from *c*. If this gives a right-handed ordering for $a' \rightarrow a'' \rightarrow b$, thus leading to *R* chirality for the arbitrarily created chiral element, the promoted *a'* is designated the *pro-R* group; conversely, if the arbitrarily created chiral element reveals an *S* ordering, the promoted group is *pro-S*. Example: glycerol, $\text{HOCH}_2\text{--CHOH--CH}_2\text{OH}$, contains two chemically-like CH_2OH groups, and the normal priority sequence is $\text{OH} > \text{CH}_2\text{OH} > \text{H}$. In the **Fischer projection** formula with the central carbon atom having the OH group to the right and H to the left, arbitrarily assign the top $\text{C}'\text{H}_2\text{OH}$ group priority over that at the bottom, $\text{C}''\text{H}_2\text{OH}$, giving the modified sequence, $\text{OH} > \text{C}'\text{H}_2\text{OH}$ (top) $> \text{C}''\text{H}_2\text{OH}$ (bottom) $> \text{H}$. When the model is viewed from the side remote from the lowest priority atom (H) the ordering, $\text{OH} \rightarrow \text{C}'\text{H}_2\text{OH}$ (top) $\rightarrow \text{CH}_2\text{OH}$ (bottom) is right-handed so that the arbitrarily created chiral element has *R* chirality; hence the top $\text{C}'\text{H}_2\text{OH}$ is the *pro-R* group. Subscript letters *R* and *S* can be used to indicate whether an atom or group of atoms is *pro-R* or *pro-S*. This is most commonly done for the two hydrogens of a methylene group, with H_R indicating the *pro-R* hydrogen and H_S the *pro-S* hydrogen. For a methylene group, replacement of $^1\text{H}_R$ by ^2H or ^3H gives a centre with *R* chirality; similar replacement of $^1\text{H}_S$ gives a centre with *S* chirality. See also **enantiotopic**.

pros a designator of position of a nitrogen in the histidine imidazole ring, being that nitrogen nearest to the side chain; often abbreviated to π , as in N^π . Compare **tele**.

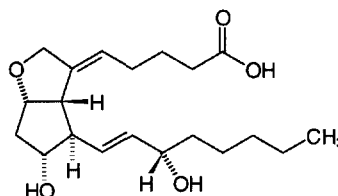
PROSITE a database of all documented protein motifs.

prosolin see **stathmin**.

prosome another name for **proteasome**.

prostacyclin or **prostaglandin I** (abbr.: PGI) any of a group of prostanoids synthesized by **prostaglandin-I synthase** (EC 5.3.99.4) from prostaglandin H – which itself is the product of **prostaglandin-endoperoxide synthase** – as prostaglandin I_2 , I_1 , or I_3 , depending on whether the substrate for the enzyme is arachidonate, dihomo-(6,9,12)-linolenate, or eicosa-

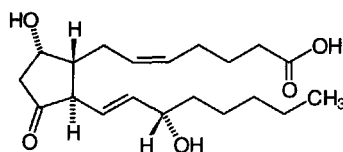
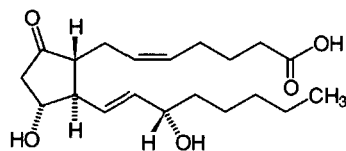
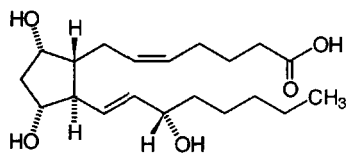
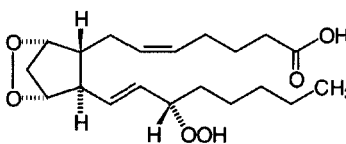
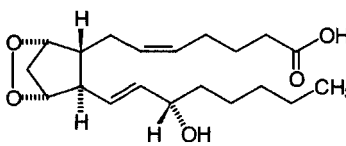
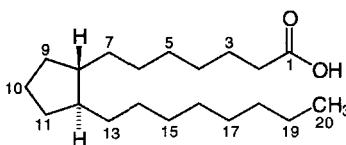
5,8,11,14,17-pentaenoate respectively. PGI_2 is the major product *in vivo*; it is produced by endothelial cells of blood-vessel walls, has potent platelet anti-aggregatory properties resulting from stimulation of adenylate cyclase, and induces vasodilation. It has a normal half-life of 2 min, being degraded by hydrolysis to 6-oxo-prostaglandin $\text{F}_{1\alpha}$; this is then converted to the 6,15-dioxo-13,14-dihydro-PGF $_{1\alpha}$, and follows a similar degradative path to other prostaglandins. More stable but still active analogues have been synthesized, e.g. carbacyclin. (S)-15-hydroperoxyeicosatetraenoate inhibits prostaglandin-I synthase. See also **prostaglandin**.



prostaglandin I_2

prostacyclin synthase see **prostaglandin-I synthase**.

prostaglandin abbr.: PG; any of a group of biologically active metabolites of arachidonate (PG $_2$ series), dihomo- γ -linolenate (PG $_1$ series), or eicosa-5,8,11,14,17-pentaenoate (PG $_3$ series). They characteristically contain a cyclopentane ring due to formation of a bond between C-18 and C-12 of the fatty acid; hydroxy or oxo substitution at positions 9 and 11 is a major distinguishing feature between prostaglandin classes. They were originally discovered as a uterus-contracting activity in seminal plasma, and are now known to possess a protean array of highly potent biological activities that defy simplification. However, to give an indication of these roles, their pharmacological effects include: vasodilation (PGE $_2$ and **prostacyclin**) which is short-acting, and vasoconstriction (PGF $_{2\alpha}$ and **thromboxane A $_2$**); oxytocic activity (PGF $_{2\alpha}$ can be used to terminate pregnancy in the second trimester and PGE $_2$ induces labour at term); bronchodilation (PGE series) and constriction (PGF series). They are involved in inflammatory reactions, and the enzymes that are involved in their synthesis are targets for antiinflammatory agents such as aspirin and non-steroidal antiinflammatory drugs such as ibuprofen. They are also involved in the regulation of cell proliferation in normal and neoplastically transformed cells. Prostaglandins are formed from the precursors prostaglandin G $_1$, G $_2$, or G $_3$, the intermediate products of **prostaglandin-endoperoxide synthase**, depending on the substrate fatty acid; the same enzyme converts these to prostaglandins H, from which other enzymes form prostaglandins D, E, and F, and also **prostacyclins** (or prostaglandins I). These are, from the bioactive aspect, the end products of the system, but they have a half-life of minutes, being rapidly converted to metabolites with much weaker, or often inhibitory properties. The initial metabolites are the 15-oxo-13,14-dihydroprostaglandins, produced by the sequential actions of 15-hydroxyprostaglandin dehydrogenases (e.g. EC 1.1.1.196 and 1.1.1.197) and 15-oxoprostaglandin 13-reductase (EC 1.3.1.48); these are further degraded by beta oxidation and omega oxidation to urinary metabolites such as 7 α -hydroxy-5,11-dioxotetranorprostane-1,16-dioate (from PGE $_2$) and 5,7-dihydroxy-11-oxotetranorprostane-1,16-dioate. The structures of prostanic acid, with its carbon numbering, and of PGG $_2$, PGH $_2$, PGD $_2$, PGE $_2$ and PGF $_{2\alpha}$ are shown

prostaglandin D₂prostaglandin E₂prostaglandin F_{2α}prostaglandin G₂prostaglandin H₂

prostanic acid

prostaglandin-D synthase *abbr.*: PGD synthase; EC 5.3.99.2; *other name*: prostaglandin-H₂ D-isomerase; an enzyme that catalyses the conversion of (5Z,13E)-(15S)-9α,11α-epidioxy-15-hydroxyprosta-5,13-dienoate to (5Z,13E)-(15S)-9α,15-dihydroxy-11-oxoprosta-5,13-dienoate (prostaglandin D). Glutathione is a cofactor in some cases. Example (precursor) from

human, PGD₂ synthase (glutathione-independent PGD synthetase or β-trace protein): database code PGHD_HUMAN, 190 amino acids (21.00 kDa).

prostaglandin-endoperoxide synthase EC 1.14.99.1; *other names*: prostaglandin synthase; prostaglandin G/H synthase. An enzyme whose reaction embraces the activity formerly referred to as **cyclooxygenase**; it catalyses a reaction between dioxygen, arachidonate and two molecules of reduced glutathione (GSH) to form prostaglandin H₂, oxidized glutathione (GSSG), and H₂O. The enzyme acts as both a dioxygenase and as a peroxidase. It catalyses the formation of the precursor, prostaglandin G₂ (PGG₂), for the biosynthesis of all prostanoid compounds, by forming an endoperoxide bridge between C-9 and C-11, simultaneously bonding C-8 and C-12 to form the cyclopentane ring, and also forming the hydroperoxide at C-15. The latter is reduced by a reduced glutathione (GSH)-dependent peroxidase activity of the enzyme to a hydroxyl group to form prostaglandin H₂; these, and their further metabolites, prostaglandins D, E and F, are highly potent bioactive compounds with a great diversity of actions (*see prostaglandin*). The enzyme can also act on dihomo-(6,9,12)-linolenic acid, to form a series of prostaglandins (denoted by a subscript 1, e.g. PGE₁) lacking the C-5 double bond, and on eicosa-5,8,11,14,17-pentaenoic acid, to form the 3 series of prostaglandins (e.g. PGE₃), with the additional double bond at C-17. Two isoenzymes, COX-1, constitutively expressed, and COX-2, inducible in inflammation by cytokines, are highly homologous membrane proteins, but selective inhibition of COX-2 is possible due to its larger inhibitor-binding site. Aspirin acetylates an active-site serine residue in both, the IC₅₀ being 10 to 100 times lower for COX-1 than for COX-2. Its inhibition of COX-1 is the basis of its antithrombotic effects. Examples, both from human, COX-1 (precursor): database code PGH1_HUMAN, 599 amino acids (68.58 kDa); COX-2 (precursor): database code PGH2_HUMAN, 604 amino acids (68.98 kDa). *See also* prostacyclin, thromboxane.

prostaglandin-E₂ 9-reductase EC 1.1.1.189; an enzyme that catalyses the reduction by NADPH of (5Z,13E)-(15S)-11α,15-dihydroxy-9-oxoprosta-5,13-dienoate (PGE₂) to (5Z,13E)-(15S)-9α,11α,15-trihydroxyprosta-5,13-dienoate (PGF_{2α}) with formation of NADP⁺. The monomeric cytoplasmic human enzyme catalyses reduction of many carbonyl compounds. In addition to the activity of prostaglandin-E₂ 9-reductase (also termed prostaglandin 9-ketoreductase) described here two other activities are present. They are that of carbonyl reductase (NADPH) EC 1.1.1.184, and 15-hydroxyprostaglandin dehydrogenase (NADP⁺), EC 1.1.1.197; database code DHCA_HUMAN, 276 amino acids (30.21 kDa).

prostaglandin-E isomerase EC 5.3.99.3; *other names*: prostaglandin-H₂ E-isomerase; endoperoxide isomerase; an enzyme that catalyses the conversion of (5Z,13E)-(15S)-9α,11α-epidioxy-15-hydroxyprosta-5,13-dienoate to (5Z,13E)-(15S)-11α,15-dihydroxy-9-oxoprosta-5,13-dienoate (PGE₂). Glutathione is a cofactor.

prostaglandin-F synthase EC 1.1.1.188; *other name*: prostaglandin-D₂ 11-reductase; an enzyme that catalyses the formation of (5Z,13E)-(15S)-9α,11α,15-trihydroxyprosta-5,13-dienoate and NADP⁺ from (5Z,13E)-(15S)-9α,15-dihydroxy-11-oxoprosta-5,13-dienoate and NADPH. It is a monomeric protein of the prostaglandin biosynthetic pathway. Example of bovine enzyme (precursor): database code PGFS_BOVIN, 322 amino acids (36.55 kDa).

prostaglandin-I synthase EC 5.3.99.4; *other name*: prostacyclin synthase; an enzyme that catalyses the conversion of (5Z,13E)-(15S)-9α,11α-epidioxy-15-hydroxyprosta-5,13-dienoate to (5Z,13E)-(15S)-6,9α-epoxy-11α,15-dihydroxyprosta-5,13-dienoate (PGI₂, prostacyclin). This is a heme-thiolate protein. Example from human: database code JC2231, 500 amino acids (57.04 kDa).

prostanate the anion of prostanic acid, 2-octylcyclopent-



prostanoid

aneheptanoic acid; the 8*S*,12*S* form is the parent compound for prostanoid nomenclature. For structure, see **prostaglandin**.

prostanoid any compound based on or derived from the prostanate structure, including the **prostaglandins**, **prostacyclins**, and **thromboxanes**.

prostasin a serine protease with trypsin-like activity, found in human seminal fluid.

prostate-specific antigen *abbr.*: PSA; a proteinase which is virtually prostate specific. Levels may be elevated in prostate hypertrophy but the release into the serum from cancer is higher than from normal tissue. It is a much better marker than **acid phosphatase**, being elevated at an earlier stage of the tumour. PSA is a serine endopeptidase belonging to the family of glandular **kallikreins** with a His-Asp-Ser **charge-relay system**. Its physiological substrate is the predominant protein of the seminal vesicle coagulum. It can be isolated from prostate epithelial cells and seminal plasma. Example from human (precursor): database code PROS_HUMAN, 261 amino acids (28.74 kDa); residues 1–18 are a signal sequence, 19–24 a propeptide, and 25–261 PSA.

prostatic of, pertaining to, or related to the prostate gland.

prosthetic group a nonprotein group that is combined specifically with a protein, in stoichiometric proportion.

protachykinin β precursor a preproprotein from which certain polypeptide hormones are cleaved, not all by the same processing pathway. Example from human: database code TKNB_HUMAN, 129 amino acids (14.99 kDa). Derived peptides: **substance P**, residues 58–68; **neurokinin A**, residues 98–107; **neuropeptide K**, residues 72–107; residues 68 and 107 are C-amidated in the derived peptides. See also **tachykinin**.

protamine any of a group of small peptides, containing arginine, alanine, and serine, that are isolated from the sperm of fish (e.g. salmon, herring, sturgeon, trout, and mackerel), but present in the sperm of all animals. They have a histone-like function in the chromatin of sperm, compacting sperm DNA into a condensed complex. Example, sperm protamine P1 from *Alouatta seniculus* (howler monkey): database code HSP1_ALOSE, 51 amino acids (6.82 kDa).

protamine sulfate a sulfated derivative of a **protamine**. Protamine sulfates combine with, and inactivate, **heparin**, and are used to prevent bleeding in heparin overdose.

protease a former name for any enzyme that catalyses the hydrolysis of peptide bonds in a protein or polypeptide. The specific names **endopeptidase**, **aminopeptidase**, and **carboxypeptidase** are now recommended. See also **proteinase**.

protease inhibitor 4 see **kallistatin**.

protease nexin 1 see **nexin 1**.

proteasome or **proteasome** EC 3.4.99.46 *recommended name*: multicatalytic endopeptidase complex; *other names*: multicatalytic proteinase (complex); *ingensin*; *macropain*; *prosome*; *lens neutral proteinase*. Proteasomes are large multi-subunit protease complexes that selectively degrade intracellular proteins. Most of the proteins removed by these proteases are tagged for destruction by ubiquitination. Proteasomes play a role in controlling cellular processes, such as metabolism and the cell cycle, through signal-mediated proteolysis of key enzymes and regulatory proteins. They also operate in the stress response, by removing abnormal proteins, and in the immune response, by generating antigenic peptides. The proteolytic activity is due to a 26S protease, which is a 2 MDa complex. The ability of this 26S protease to degrade proteins generally depends on both the ubiquitination of the substrate and the presence of ATP. The core of the 26S protease is a 20S particle that carries the catalytic activity. It is this 20S particle that was first isolated, and was called 'multicatalytic proteinase' because it possessed trypsin-like, chymotrypsin-like and peptidyl-glutamyl activities, with the ability to cleave peptide bonds carboxy-terminal to basic, hydrophobic, and acidic residues. The 20S particle does not require ATP for activity. Following the determination of the crystal structure of this core from the Archaeon *Thermoplasma acidophilum*, the structure has been clarified.

The proteasome itself is inactive and requires two specific subunits of a 19S particle to mediate recognition of ubiquitin-protein conjugates. The proteasome subunits are arranged in four heptameric rings which are stacked together to form a hollow cylinder with the protease activity on the inside. The active site nucleophile is threonine. The size of the channel in the cylinder is such that the protein substrate first must be unfolded and the disulfide bonds reduced. In *Thermoplasma*, the two outer rings of the 20S proteasome are composed of seven α subunits, and each inner ring contains seven β subunits, an organization that is likely also to apply in eukaryotic 20S proteasomes. The enzyme catalyses the cleavage at Xaa-|- bonds in which Xaa carries a hydrophobic, basic, or acidic side chain. Components (all examples are human): c2 (*other names*: macropain subunit c2; proteasome ν chain; multicatalytic endopeptidase complex subunit c2; 30 kDa proteasomal protein): database code PRC2_HUMAN, 263 amino acids (29.52 kDa); c3 (EC 3.4.99.46; *other names*: macropain subunit c3; multicatalytic endopeptidase complex subunit c3): database code PRC3_HUMAN, 233 amino acids (25.74 kDa); c5 (*other names*: macropain subunit c5; proteasome γ chain): database code PRC5_HUMAN, 241 amino acids (26.46 kDa); proteasome chain 7 (*other names*: macropain chain 7; multicatalytic endopeptidase complex chain 7; ring 12 protein): database code PRC7_HUMAN, 219 amino acids (23.24 kDa); proteasome component c8 (*other names*: macropain subunit c8; multicatalytic endopeptidase complex subunit c8): database code PRC8_HUMAN, 254 amino acids (28.27 kDa); proteasome component c9 (*other names*: macropain subunit c9; multicatalytic endopeptidase complex subunit c9): database code PRC9_HUMAN, 261 amino acids (29.45 kDa); proteasome β chain (*other names*: macropain β chain; multicatalytic endopeptidase complex β chain (fragment)): database code PRCB_HUMAN, 28 amino acids (2.79 kDa); component c13 (*other names*: macropain subunit c13; multicatalytic endopeptidase complex subunit c13): database code PRCC_HUMAN, 208 amino acids (23.03 kDa); δ chain (*other names*: macropain δ chain; multicatalytic endopeptidase complex δ chain): database code PRCD_HUMAN, 205 amino acids (21.68 kDa); ϵ chain (*other names*: macropain ϵ chain; multicatalytic endopeptidase complex ϵ chain; proteasome chain 6 (fragment)): database code PRCE_HUMAN, 26 amino acids (2.32 kDa); component mecl-1 (*other names*: macropain subunit mecl-1; multicatalytic endopeptidase complex subunit mecl-1): database code PRCF_HUMAN, 273 amino acids (28.90 kDa); τ chain (*other names*: macropain τ chain; multicatalytic endopeptidase complex τ chain; 27 kDa proteasomal protein (also in rat)): database code PRCL_HUMAN, 246 amino acids (27.37 kDa); ζ chain (*other names*: macropain ζ chain; multicatalytic endopeptidase complex ζ chain): database code PRCZ_HUMAN, 241 amino acids (26.44 kDa).

protectin see **CD59**.

protecting group or **protective group** or **blocking group** any chemical group added to a reactive centre of a molecule to prevent it from reacting during chemical treatments in which it is not intended to participate. Such groups are commonly used during chemical synthesis of macromolecules or shorter polymers of amino acids or nucleotides. Reagents contributing protecting groups should react readily under mild conditions, with high specificity, and lead to addition of a group that can be removed highly specifically under mild conditions. For protein or amino-acid protecting groups see **aminoethyl**, **Boc**, **citronylation**, **DABITC**, **dabsyl**, **dansyl**, **Fmoc**, **maleylation**, **pipsyl**, **silylation**, **succinylation**, **tosyl**, **triphenylmethyl** group.

protegrin *abbr.*: PG; any of a group of leukocyte antimicrobial peptides, denoted PG-1 to PG-5, that are active against *Escherichia coli*, *Listeria monocytogenes*, and *Candida albicans* *in vitro*. Like **tachyplesins** they contain two intramolecular cystine disulfide bonds and show homology to **defensins**. Example, protegrin 1 (precursor) from *Sus scrofa*: database code



protein

PGI_PIG, 149 amino acids (16.67 kDa); protegrin 1 is residues 131–148.

protein any of a large group of organic compounds found as major macromolecular constituents of living organisms. All enzymes are proteins, although catalytic activity is shown by some nucleic-acid molecules (see **ribozyme**). A protein is a linear polymer of **amino acids** linked by peptide bonds in a specific sequence. In the biosynthesis of the polypeptide chain, any of 20 different amino acids may be incorporated, according to the genetic instructions of the cell (see **protein and peptide biosynthesis**). The amino-acid residues may be modified subsequently so that the chains may contain a much wider variety of residues, amounting to nearly 200. The modifications may involve the covalent attachment of various groups such as carbohydrates and phosphate; these are the simple proteins. Other substances may be more loosely associated with the polypeptide chains, such as heme or lipid, giving rise to the **conjugated proteins**. Four hierarchies of structure may be detected in the polypeptide chains: **primary structure** is the order of the specific amino-acid residues; **secondary structure** is the way the amino-acid residues interact within a chain to form structures such as the alpha helix and the beta strand; **tertiary structure** signifies the way in which an entire single polypeptide chain (or chains linked by disulfide bonds) folds to form a three-dimensional structure; and **quaternary structure** is the interaction between intact chains to form multi-subunit proteins. The various ways in which the polypeptide chains fold have been elucidated, in particular with regard to the presence of **alpha helices** and **beta-pleated sheets**. Four types of proteins may be classified: all-alpha; all-beta; alpha/beta; and alpha + beta. Some important protein types are listed separately in this dictionary and many others are listed individually: contractile proteins (**actin**, **myosin**, **tropomyosin**, **tubulin**), **heat-shock proteins**, **hemoproteins** (**cytochromes**, **hemoglobin**, **myoglobin**), **lipoproteins**, **membrane proteins** (**ATPases**, **ion channels**, **rhodopsin**), nonheme proteins (**blood coagulation factors**, **serum albumin**, **serum globulin**), nuclear proteins (**histones**), **plasma proteins**, and structural proteins (**collagen**, **crystallin**, **keratin**). Nutritional aspects of protein metabolism include the fact that higher animals cannot synthesize certain of the amino acids required for protein synthesis; such amino acids are referred to as **essential amino acids**. Plants and most microorganisms can synthesize all of the amino acids required for protein synthesis and many others. The nutritional value of different proteins depends on their composition. Some proteins are of poor nutritional value due to lack of essential amino acids, e.g. gliadin, which lacks lysine, and zein, which lacks lysine and tryptophan. —**proteinaceous adj.**

14-3-3 protein any of a highly conserved family of proteins, widespread in higher eukaryotes, that seem to act as regulators in signal transduction or phosphorylation. The name derives from the migration position on DEAE-cellulose columns, and on starch-gel electrophoresis, of certain brain proteins that were the first members of the family to be studied. Members of this family have been implicated in regulation of **protein kinase C**, transcriptional regulation in plants, exocytosis, cell-cycle regulation, and interaction with **Raf** in the **MAP kinase** cascade. For example, 14-3-3 protein β (*other name*: protein kinase C inhibitor protein-1) from sheep activates tyrosine and tryptophan hydroxylases in the presence of Ca^{2+} /calmodulin-dependent protein kinase II. The bovine equivalent has an identical sequence: it is a homodimer: database code 143B_BOVIN, 244 amino acids (27.82 kDa).

protein A a bacterial cell-wall protein, found in many strains of *Staphylococcus aureus*, that can bind, without masking the antigen-binding site, to the Fc region of IgG. It binds weakly to human IgA, IgE, and IgM but fails to bind to human IgG₃ and IgD; it does not bind to cell Fc receptors. The purified protein or a preparation of cells of *S. aureus* that carry protein A is widely used as an immunochemical reagent, often in an immobilized form on affinity columns, for isolation and purification of immunoglobulins and, by forming ternary com-

plexes, of antigens and immune complexes. It is also used as a binding reagent in enzyme-linked immunoassay, in histochemistry, and in blotting techniques.

protein and peptide biosynthesis the biochemical synthesis of proteins. Biochemists became actively concerned with the elucidation of the mechanism of protein synthesis from about 1950. The classical pathway was expected to involve the utilization of enzymes to synthesize peptide intermediates, which would sequentially be linked together, but it soon became apparent that the number of enzymes involved would be horrendous and, moreover, there was no evidence that such peptide intermediates existed in living cells. Although there was a suggestion that peptide bonds were synthesized by transpeptidation using **proteolytic enzymes**, on theoretical grounds there was increasing support for the concept that the amino acids would first be aligned on a template and then be sequentially linked by peptide bonds. Following the elucidation of one structure for DNA in 1953 it was confidently predicted that the template would be nucleic acid. Shortly after, in 1955, P. C. Zamecnik and his group in Boston demonstrated the synthesis of a polypeptide by the isolated **microsome** fraction from rat liver. This involved the presence of particles (**ribosomes**), a soluble extract of the cell, and ATP and GTP.

As will be described, the synthesis of most peptides in living cells depends on the presence of ribosomes. This includes the small peptide hormones, such as **oxytocin** and **vasopressin**, which are synthesized as **polypeptides**. Other peptides are not synthesized by means of ribosomes. Thus **glutathione** is synthesized by soluble enzymes in two steps. Many biologically active peptides, especially cyclic structures, are products of microorganisms. These include such antibiotic peptides as **gramicidin S**, the **tyrocidines**, and the **polymyxins**. The systems responsible for their synthesis consist of interacting multienzymes in which activated aminoacyl groups are transferred onto -SH groups to form intermediate thioesters. The multienzymes are unusual in size, ranging from 120 to 1700 kDa. The respective genes consist of repeating modules, each encoding the incorporation of one amino-acid residue. The genes are arranged in clusters associated with genes encoding auxiliary proteins for the synthesis of precursors, modifying enzymes, exporting proteins, and regulatory systems. In addition to the amino acids and ATP, *S*-adenosylmethionine is required as methyl group donor if *N*-methylated bonds are involved. Due to their complexity (cyclosporin synthase for example, integrates 40 reactions on one polypeptide chain), the multienzyme systems have low turnover numbers. It has been possible to set up cell-free systems for the production of various peptides and their analogues.

With respect to the synthesis involving ribosomes it was shown that the first step was the activation of the amino acids to form amino-acid **adenylates**. The amino acids were then transferred to an RNA present in the soluble extract of the cell, and so named soluble RNA but later termed **transfer RNA** (tRNA) to which the amino acids became attached by an ester linkage. The template was then found to be **messenger RNA** (mRNA), which associates with the ribosomes themselves. Although the ribosomes themselves contain much RNA it came as a surprise that this did not serve as a template. The **translation** of the triplet codons in mRNA to provide a polypeptide involves three steps: chain initiation, chain elongation (see **elongation factor**), and chain termination (see **release factor**). Chain initiation normally starts with the codon for methionine, **AUG**. For this purpose a unique initiator tRNA is involved. Many protein factors play a part in the three processes and five molecules of ATP and GTP participate. The ribosomes participate in a cycle during which they subdivide into their two subunits only to reassemble. Hence the synthesis of a peptide bond is a surprisingly expensive process in terms of cell physiology. The targeting of the newly synthesized proteins to their final destination is termed **protein kin-**

proteinase

esis. The mechanism of ribosomal protein synthesis is very similar in all types of cell but there are subtle differences between prokaryotes and eukaryotes, which account for the differential action of antibiotics such as chloramphenicol and cycloheximide. Apart from differences in the number and properties of the soluble factors the main difference involves chain initiation: in prokaryotes this is by way of formylmethionine while in eukaryotes, apart from mitochondria, this involves methionine.

proteinase name introduced in 1928 to distinguish a sub-class of **proteases** that catalyse the hydrolysis of peptide bonds more readily in an intact protein than in small peptides. It is still widely used. **Endopeptidase** is now recommended.

proteinase K EC 3.4.21.64; *recommended name*: endopeptidase K; *other name*: *Tritirachium* alkaline proteinase. A serine proteinase enzyme that catalyses the hydrolysis of keratin, and of other proteins with subtilisin-like specificity. Example from the fungus *Tritirachium album*: database code NRL_2PRK, 279 amino acids (28.88 kDa); 3-D structure of this, and with inhibitor bound, is known.

proteinase V8 *see* glutamyl endopeptidase.

protein B a cell-surface protein of group B *Streptococcus* species that binds specifically to the Fc region of human IgA, without affecting its antigen-binding capacity, but not to other immunoglobulins or serum proteins. It is used in enzyme-linked immunoassay and blotting techniques.

protein biosynthesis *see* protein and peptide biosynthesis.

protein-bound iodine *abbr.*: PBI; that fraction of iodinated molecules bound to protein in serum. It is due largely to thyroxine (T_4), 99.97% of which in human serum is present bound to the binding proteins T_4 -binding globulin, T_4 -binding prealbumin and albumin, but also to T_3 , 99.7% of which is bound to the same proteins. Determination of PBI (now replaced by immunoassay) was used for estimation of blood thyroid hormones as a basis for assessment of thyroid function.

protein C a vitamin K-dependent glycoprotein that is the **zymogen** of a serine endopeptidase in normal blood plasma; M_r 62 000 (human), 54 300 (bovine). It is formed from a precursor, autoprothrombin IIA, which is cleaved into a light chain and a heavy chain held together by a disulfide bond to form protein C. Thrombin cleaves a tetradecapeptide from the amino end of the heavy chain; this reaction, which occurs at the surface of endothelial cells, is strongly promoted by thrombomodulin; protein C then becomes the enzyme protein C (activated), EC 3.4.21.69. This activated form specifically hydrolyses factors Va and VIIIa to inactive forms (in a reaction requiring phospholipid, calcium, and **protein S**), thereby acting as a potent anticoagulation factor; it contains 10–12 γ -carboxyglutamyl residues in the N-terminal 40 residues. Example (precursor) from human: database code PRTC_HUMAN, 461 amino acids (52.01 kDa). A congenital deficiency of protein C (usually autosomal dominant) is often not compatible with survival, or leads to hemorrhagic disease of the newborn. Protein C is so named because it eluted from columns as the third of four peaks labelled A, B, C, and D.

protein C23 *see* nucleolin.

protein C-binding protein *see* apolipoprotein H.

protein disulfide-isomerase EC 5.3.4.1; *other name*: disulfide rearrangase; an enzyme that enhances the rate of interchange in the groups attached to two or more intrachain or interchain disulfide bonds in a protein. It was originally described by Anfinsen in 1966 concerning the reversible denaturation of ribonuclease and then called 'rearrangase'. It has since been shown to be a **luminal protein** of the endoplasmic reticulum. It is a homodimer; two of its domains are homologous to **thioredoxin** of *Escherichia coli*. It is also the triacylglycerol transfer protein (MTP) that facilitates the incorporation of lipids into newly synthesized core **lipoproteins** within the endoplasmic reticulum. It serves as the β subunit of **procollagen-proline dioxygenase**. Examples: probable protein disulfide-isomerase er-60 precursor from human (*other names*: erp60; 58

protein-glutamine γ -glutamyltransf

kDa microsomal protein): database code ER60_HUMAN, 505 amino acids (56.64 kDa); precursor (human) of this and also prolyl 4-hydroxylase β subunit (EC 1.14.11.2) and cellular thyroid hormone binding protein (p55): database code PDI_HUMAN, 508 amino acids (57.04 kDa).

protein engineering the process whereby a protein of any desired primary structure can (in principle) be created by the use of recombinant DNA technology. The ability to express DNA in a bacterium such as *Escherichia coli* creates the potential to synthesize a protein with the desired primary structure. Bacteria cannot effect post-translational modifications so if these also are required a eukaryotic host cell such as yeast must be employed. In principle there are no problems with the proteins of bacterial cells but **introns** present in the genome of eukaryotic cells present a potential problem. This can be overcome by preparing complementary DNA, **cDNA**, which is synthesized by the use of the **mRNA** as template and the enzyme **reverse transcriptase**. There are several ways whereby the DNA for the required protein may be prepared by modification of the base sequence of the natural DNA. DNA may be used that contains inserts of synthetic oligonucleotides, or stretches replaced by these. Oligonucleotide-directed mutagenesis may be used to modify a single base to produce a protein with a single amino-acid replacement. This is done by preparing an oligonucleotide primer with the modified sequence which will base pair with the natural DNA around the site to be altered; a single base substitution still permits satisfactory pairing. The primer is elongated by DNA polymerase and the double-stranded circle closed by **DNA ligase**. A specific deletion can be obtained by cleaving a plasmid at two sites with a restriction enzyme and religating to form a smaller circle. A smaller deletion can be obtained by cutting a plasmid at a single site. The ends of the linear DNA are then digested with an exonuclease that removes nucleotides from both strands and the shortened piece of DNA is religated.

protein folding disease any disease that results from the abnormal folding of a protein. **Prion** diseases are examples of such diseases that can be inherited or acquired; **Alzheimer's disease** is a different type of disease, caused by alternative folding of proteins derived from the amyloid precursor.

protein G a bacterial cell-wall protein isolated from group G streptococci. Its properties and uses are similar to those of **protein A**, but it does not bind to IgM, IgD, or IgA; it does bind to all subclasses of human IgG, and animal immunoglobulins with which protein A does not react well.

protein gene product any one of the proteins or protein subunits synthesized biochemically on the basis of information encoded in a genome, whether generated at **translation** or by **post-translational modification**; the pattern of protein gene products formed or capable of being formed by an organism is unique to any one genotype. Individual proteins are often referred to using the symbol p followed by the relative molecular mass, e.g. p53. The protein may be further identified by adding an indication of its origin as a superscript, e.g. p21^{ras}. A protein gene product may also be referred to by the name of the gene that encodes it, but normal font is used instead of italics, with a capital initial followed by lower case characters, e.g. Src protein encoded by *SRC* gene.

protein-glutamate methylesterase a chemotaxis protein encoded by *cheB* (*see che*).

protein-glutamine γ -glutamyltransferase EC 2.3.2.13; *systematic name*: protein-glutamine:amine γ -glutamyltransferase; *other names*: transglutaminase, factor XIIIa; the enzyme activity of **blood coagulation** factor XIIIa. It forms inter- and intramolecular links between a γ -carboxyl group of a glutamine residue of a protein and the ϵ -amino group of a lysine residue, releasing ammonium ion. Its natural substrate is the **fibrin** monomer, and its function is to link these into polymers. The protein is a heterotetramer of two each of the A and B chain,



protein intron

the A chain bearing the catalytic activity. Examples from human, A chain: database code F13A_HUMAN, 731 amino acids (83.14 kDa); residues 1–37 are a propeptide that is cleaved from the A chain by thrombin in the conversion of factor XIII to XIIIa; B chain (precursor): database code F13B_HUMAN, 661 amino acids (75.49 kDa); residues 1–20 are the signal peptide; the molecule contains ten **Sushi domains**.

protein intron *see* **intron**; **splicing** (def. 3).

protein kinase any of a number of enzymes that phosphorylate one or more hydroxyl or phenolic groups in proteins, ATP being the phosphoryl-group donor. Two classes are recognized: those that phosphorylate seryl or threonyl hydroxyls, and those that phosphorylate tyrosyl phenolic groups. The first class contains enzymes that were classically identified as regulating pathways of intermediary metabolism (e.g. glycogen **phosphorylase kinase**); subsequently, protein kinase A (**cyclic-AMP-dependent protein kinase**) and **protein kinase C** were recognized in this class. Enzyme activity of the second class is found in some cytokine receptors such as **platelet-derived growth factor receptor** and **epidermal growth factor receptor** and in oncogene products such as that of *SRC*. Over 110 unique gene products are known to make up the protein kinase superfamily; important motifs include GXGXXG...AHK and APE...DXWSXG, which are common to both serine/threonine- and tyrosine-specific activities. The tyrosine-specific kinase activity associated with the cytoplasmic domains of certain cytokine receptors forms part of an important signal-transduction mechanism implicated in cell growth (*see* **tyrosine kinase**).

protein kinase A *an alternative name for* **cyclic-AMP-dependent protein kinase**.

protein kinase C *abbr.*: PKC; any of a family of **protein kinase** enzymes identified following discovery of a protein kinase requiring anionic phospholipid for activity, and regulated by diacylglycerol and Ca^{2+} . They phosphorylate hydroxyl groups in substrate serine and threonine residues. Several isoforms have now been characterized: α , β_1 , β_2 , γ , δ , ϵ , ζ , θ , η , ι , and μ . Four conserved regions are recognized, C_1 , C_2 , C_3 , and C_4 ; C_3 and C_4 comprise the C-terminal half of the molecule and bear the active site, including the ATP-binding sequence GXGXXG...K. PKC- ϵ is Ca^{2+} -independent, and δ , ϵ , and ζ lack the homologous C_2 region. Activation of PKC occurs when plasma membrane receptors coupled to phospholipase C are themselves activated, releasing diacylglycerol. Phorbol ester tumour promoters can substitute for diacylglycerol in enzyme activation, which may indicate their mechanism of action, and much evidence points to PKC activation being involved in cell-growth stimulation. Inhibitors of PKC attract considerable interest, and include **bisindolylmaleimide**, 1-O-hexadecyl-2-O-methylglycerol, **melittin**, **phloretin**, **polymyxin B**, and **staurosporine**. Examples: α from *Bos taurus*: database code KPCA_BOVIN, 672 amino acids (76.84 kDa); β_2 from *Oryctolagus cuniculus*: database code KPC2_RABIT, 673 amino acids (76.97 kDa); δ from *Mus musculus*: database code KPCD_MOUSE, 674 amino acids (77.59 kDa).

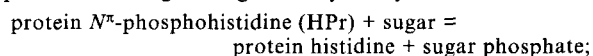
protein kinesis the directed movement of proteins from their site of synthesis to specific targets within cells and the modulation by proteins of interactions among intracellular organelles. It is a factor in generating subcellular compartments and determining the way in which reactions are compartmentalized to control metabolic processes. It embraces the mechanisms for ensuring that proteins reach their site of action and that the amounts and locations of particular proteins are controlled during development, during the cell-division cycle, in combating pathological states of the cell, and during apoptosis.

protein mapping the resolution of a mixture of proteins into a two-dimensional pattern (i.e. map) by successive use of two differing separative procedures, as in isoelectric focusing followed by gradient gel electrophoresis, the pattern being rendered visible by an appropriate detection method.

protein phosphatase any enzyme that hydrolyses phosphate

groups from proteins. Together with protein kinases, these enzymes control the state of phosphorylation of cell proteins and thereby provide an important mechanism for regulating cellular activity. Protein phosphatases are classified into several groups. **Class I enzymes** are mainly specific for seryl- or threonyl-phosphate. Within this class, phosphatase 1 is ATP, Mg^{2+} -dependent (AMD phosphatase); it consists of a catalytic subunit of 38 kDa and a modulator subunit of 23 kDa, which is subject to phosphorylation, thus activating the phosphatase. Phosphatase 2A is polycation-stimulated (PCS), being directly stimulated by protamine, polylysine, or histone H1; it constitutes a subclass of several enzymes activated by different histones and polylysine. These have a very high affinity for the tumour promoter **okadaic acid**. Phosphatase 2B (or **calcineurin**) is a heterodimer; one of the subunits, of 61 kDa, contains the catalytic site, a high-affinity calmodulin-binding site, and a binding site for the other subunit, a 19 kDa, Ca^{2+} -binding protein. Phosphatase 2C is a Mg^{2+} -dependent monomeric 43 kDa protein; it has been purified as myosin light chain phosphatase, but is active against the phosphorylated forms of glycogen synthase, 6-phosphofructo-1-kinase, pyruvate kinase, fructose 1,6-bisphosphatase, and 6-phosphofructo-2-kinase. **Class II enzymes** hydrolyse tyrosylphosphates, and are generally designated 'phosphatase 3'. They are subdivided into those that are cytoplasmic, and others that are membrane-bound and may have receptor properties. Example of phosphatase 2A from human: database code P2AA_HUMAN, catalytic subunit 309 amino acids (35.55 kDa); same in rabbit. This enzyme can modulate the activities of phosphorylase kinase *b*, casein kinase 2, mitogen-stimulated S6 kinase, and MAP-2 kinase. It is a cytoplasmic enzyme that exists in several oligomeric forms, all of which contain a catalytic (C) subunit (36–38 kDa) associated with one or both of two putatively regulatory subunits, the A subunit (61–65 kDa) and the B subunit (51–55 kDa). Example of protein tyrosine phosphatase, leukocyte antigen-related PTPase precursor from human: database code LAR_HUMAN, 1897 amino acids (211.60 kDa).

protein- N^{π} -phosphohistidine-sugar phosphotransferase EC 2.7.1.69; *systematic name*: protein- N^{π} -phosphohistidine:sugar N^{π} -phosphotransferase; *other name* enzyme II; any of a group of related membrane-bound bacterial enzymes, part of the **phosphoenolpyruvate-dependent sugar phosphotransferase system** for the transport of hexoses across the cell membrane. It involves phosphorylation of a protein substrate (9.5 kDa), a phosphocarrier known as HPr. Individual enzymes are specific for a range of sugars. They catalyse the reaction:



example from *Escherichia coli*: database code PT2G_ECOLI, 477 amino acids (50.62 kDa).

protein S a single-chain glycoprotein occurring in blood, M_r 69 000. It is a vitamin K-dependent protein (not a proteinase) that promotes the binding of **protein C** to platelets, and functions as a cofactor for the anticoagulant activity of activated protein C. It is named after Seattle, where it was discovered.

protein splicing *see* **splicing**.

protein synthesis 1 (*in cell biology*) *see* **protein and peptide biosynthesis**. 2 (*in chemistry*) *see* **peptide synthesis** (def. 2).

protein thiotemplate mechanism *see* **thiotemplate mechanism**.

proteinuria the presence of protein in urine. Normal excretion of protein in human urine is less than 0.1 mg every 24 h, about half of which is **Tamm-Horsfall glycoprotein**, and about 20% is albumin. Protein may appear in urine due to extrarenal factors such as fever, strenuous exercise, or burns; direct involvement of the kidney occurs in the nephrotic syndrome, which may arise from several causes, such as diabetic nephropathy, cancer of the kidney, or various forms of glomerulonephritis of unknown etiology.

protein Z a major protein of barley endosperm albumin. It is



proteoclastic

structurally similar to a **serpin**, but its main role is as a storage protein, being especially rich in lysine. Database code PRTZ_HORVU, 399 amino acids (22 lysine), 43.23 kDa.

proteoclastic an obsolete term for **proteolytic**.

proteoglycan any **glycoprotein** in which the carbohydrate units are **glycosaminoglycans**.

proteolipid a term variously applied to describe certain hydrophobic membrane proteins, or any other protein, that has a lipid moiety covalently bound to one or more of its constituent amino acids. Many such proteins are soluble in chloroform-methanol, a property that was significant in their first identification.

proteoliposome a **liposome** into which a specific protein, or group of proteins, has been incorporated.

proteolysis degradation of a protein, usually by hydrolysis at one or more of its peptide bonds. —**proteolytic adj.**

proteolytic enzyme see **peptidase**.

proteome the complete expression profile of the proteins of an organism. See also **proteomics**.

proteomics the study of the **proteome** by the analysis of protein structure and composition.

proteosome a variant spelling of **proteasome**.

prothrombin an alternative name for factor II; see **blood coagulation**.

protide 1 symbol: $^1\text{H}^-$; a **hydride** (def. 1) ion derived from an atom of **protium**. 2 any **hydride** (def. 2) formed from **protium**. 3 or **protid** a generic name embracing protein, peptide, or amino acid, suggested by analogy with glucide and lipid(e). Its use in English is largely restricted to translations or abstracts from French. —**protidic adj.**

protio+ comb. form indicating that one or more atoms of hydrogen in a hydrogen-containing chemical compound consists exclusively of **protium**.

Protista a taxonomic grouping of eukaryotic organisms that are unicellular, coenocytic, or multicellular but that are not classified as fungi, animals, or embryophytes.

protium symbol: ^1H ; hydrogen-1, the nuclide of hydrogen, ^1H , of relative atomic mass 1.008. It is a stable isotope, having a relative abundance in natural hydrogen of 99.98 atom percent. The name is used especially when distinction from **deuterium** or **tritium** is required.

protoalkaloid an alkaloid that does not contain a heterocyclic ring, e.g. **damascenine** in *Nigella* spp.

protobiont one of the postulated first precursors of cells, supposed to have arisen when a boundary or membrane formed around one or more macromolecules possessing catalytic properties. Compare **coacervate droplet**.

protochlorophyllide Mg^{2+} -vinylpheoporphyrin A_5 methyl ether; an intermediate in the synthesis of chlorophyll, it is formed from Mg^{2+} -protoporphyrin, and converted to **chloro-**

phyllide by NADPH:protochlorophyllide oxidoreductase (EC 1.3.1.33). Protochlorophyllides may be suffixed with the letter of the related chlorophyll, e.g. protochlorophyllide *a*.

protochlorophyllide holochrome a pigment protein complex that carries out a light-catalysed reduction of **protochlorophyllide** to **chlorophyllide** with an external hydrogen donor, which is probably strongly bound NADPH. The enzyme involved in regenerating NADPH is **NADPH:protochlorophyllide oxidoreductase**; it is a single polypeptide, M_r 36 000, and is only one component of the complex protein entity of the holochrome.

protocollagen collagen chains in which proline and lysine residues have not been hydroxylated. Protocollagens may be produced by experimental inhibition of the relevant hydroxylases either by use of a chelating agent or by imposed anaerobiosis. Compare **procollagen**.

protoheme or (esp. Brit.) **protohaem** the iron-porphyrin prosthetic group of the **cytochromes b**.

protohemin see **hemin**.

protokaryon an obsolete term for **prokaryon**.

protokaryote an obsolete term for **prokaryote**.

protolysosome see **primary lysosome**.

protomer any of the subunits of an oligomeric protein that are identical. See also **monomer**, **oligomer**. —**protomeric adj.**

proton 1 symbol: $^1\text{H}^+$; the cation derived from an atom of **protium**; a hydrogen ion of nucleon number unity; used in this sense especially when distinction between $^1\text{H}^+$, $^2\text{H}^+$ (i.e. deuterium), and $^3\text{H}^+$ (i.e. triton) is needed; any solvated species is included. 2 symbol: H^+ ; commonly used term for the cation derived from any atom of natural **hydrogen** (of which a small proportion is **deuterium**) by loss of its orbital electron; any solvated species is included. See also **hydrogen ion**, **hydron**. 3 symbol: p ; a nuclear particle of nucleon number unity, having a charge equal and opposite to that of an **electron** and having a mass of 1.007 276 470 (12) Da. —**protonic adj.**

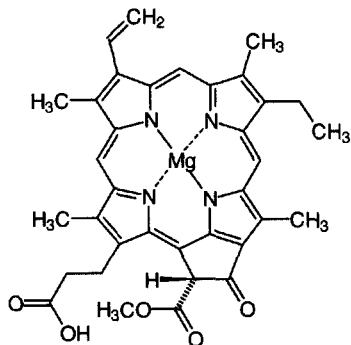
proton magnetic resonance abbr.: PMR or pmr; see **nuclear magnetic resonance spectroscopy**.

proton magnetogyric ratio symbol: γ_p ; a fundamental physical constant, of value $2.675\,221\,28(81) \times 10^8 \text{ s}^{-1} \text{ T}^{-1}$. See also **gyromagnetic ratio**.

protonmotive indicating, or related to an ability to transport protons (i.e. hydrons).

protonmotive Q cycle a hypothetical mechanism put forward to explain the **protonmotive** and **redox** properties of certain components of the **respiratory chain** in the **coupling membrane** of bacteria, mitochondria, and chloroplasts. The **Q cycle** applies to the part of the respiratory chain involving passage of electrons through cytochrome *b*. Essentially, the cycle accepts two hydride ions from a reduced flavoprotein (e.g. of complex I or complex II of the respiratory chain; see **complex I**, **II**, **III**, **IV**) passes electrons singly to cytochrome c_1 and releases the two hydrons to the intermembrane space. **Ubiquinone** is the molecule that accepts the hydride ions, and releases the electrons and hydrons. Cytochromes b_{566} and b_{562} are involved in shuttling electrons between different oxidation states of ubiquinone during the cycle. The property of ubiquinone that it can exist as a **semiquinone** is important in the switch from two-electron to one-electron transfer.

proton number or **atomic number** symbol: Z ; a dimensionless physical quantity, with a unique integral value for each chemical element, that is equal to the number of protons contained in the nucleus of each atom of the element irrespective of the mass of any particular atom. It also is equal to the number of unit positive charges on the nucleus of each atom of a given chemical element, to the number of orbital electrons surrounding the nucleus of an un-ionized atom of that element, and, for a specific nuclide of that element, to the arithmetical difference between the nuclide's **nucleon number**, A , and its **neutron number**, N ; i.e. $Z = N - A$. The proton number of an element determines the position of the element in the periodic



protochlorophyllide a



proton relaxation enhancement

table. It may be attached to the symbol for the element as a left subscript, e.g. ${}^4\text{C}$.

proton pump another term (*esp. in pharmacology*) for **hydrogen-ion pump**.

proton relaxation enhancement *abbr.*: PRE; a technique that yields information concerning (macro)molecular motion. It depends on the enhancement of the solvent water-proton relaxation rates on addition of a macromolecule.

protooncogene a normal gene, usually concerned with the regulation of cell proliferation, that can be converted into a cancer-promoting **oncogene** by mutation. For example, in the case of the Rous sarcoma virus the viral oncogene is *v-src* (*see src*). This is closely related to a normal gene, *c-src*. The appearance of *v-src* is believed to have followed the acquisition of a copy of *c-src* by an ancestor of the Rous sarcoma virus. Through many generations this ancestral gene was mutated and became dominant. Protooncogenes encode proteins involved in all aspects of controlling cell proliferation, including growth factors, growth-factor receptors, nuclear factors regulating gene expression proteins that generate second messengers, and protein kinases.

protoplasm the living contents of a cell; i.e. the matter contained within (and including) the plasma membrane, usually taken to exclude large vacuoles and masses of secretory or ingested material. In eukaryotes it includes the **nucleus** and **cytoplasm**.

protoplast a spherical, osmotically sensitive plant cell without its cell wall but retaining an intact cell membrane. The cell wall is degraded by digestion with different hydrolytic enzymes, as a mixture or sequentially, e.g. cellulase, hemicellulase. Protoplasts are used to create hybrid cells via protoplast fusion. In some cases they are totipotent, and are thus capable of yielding plants. Such protoplasts can be used in plant transformation.

protoporphyrin IX the specific substrate for the enzyme **ferrochelatase**, which catalyses the insertion of iron to form **protoheme**. It is also probably the substrate for the insertion of magnesium in the formation of chlorophylls in plants.

prototroph any strain of a microorganism (alga, bacterium, or fungus) that does not require any substances in its nutrition additional to those required by the wild type. *Compare auxotroph*. —*prototrophic adj.*

Protozoa a phylum or subkingdom comprising single-celled eukaryotic microscopic organisms, usually classified as animals. *Compare Metazoa, Parazoa*.

provirus a virus, e.g. a **retrovirus**, that is integrated into the chromosome of its host cell and can be transmitted through the generations without causing lysis of the host cell.

provitamin a compound which is converted to an active vitamin in the body. In practice the term is only applied to **carotenoids**.

Pro-X dipeptidase EC 3.4.13.8; *recommended name*: prolinase; *other names*: prolyl dipeptidase; iminodipeptidase; prolinase; L-prolylglycine dipeptidase; an enzyme that catalyses the hydrolysis of Pro-[Xaa] dipeptides. It also acts on hydroxy-prolyl analogues.

proximal protein any protein that binds early in the assembly of complex structures, such as ribosomes. Their presence in the partially assembled structure is required for the binding of distal proteins.

proximate carcinogen or **proximate carcinogenic metabolite** any metabolite of a **chemical carcinogen** that is itself carcinogenic; the term may embrace **ultimate carcinogen**.

Prozac *see* **fluoxetine**.

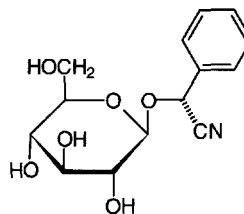
PrP *abbr.* for prion protein.

PRP protein *abbr.* for pre-mRNA processing protein.

PRPP *abbr.* (*not recommended*) for phosphoribosyl pyrophosphate (*see* **(5-)-phospho-(α -D-ribose) diphosphate**).

PRPP synthetase another name for **ribose-phosphate pyrophosphokinase**.

prunasin (–)-(R)-mandelonitrile β -D-glucoside; a **cyanogenic glycoside**.



prunasin

ps symbol for picosecond.

P-selectin *see* **selectin**.

pseudoalkaloid any **alkaloid** in which the skeleton is partly derived from a terpene.

pseudoallele any of two (or more) mutations that are allelic functionally but not structurally. *See* **allele**.

pseudoaxial (*in chemistry*) *abbr.*: a' ; *see* **conformation**.

pseudocholinesterase EC 3.1.1.8; *recommended name*: cholinesterase; *other name*: choline esterase II. An enzyme, found in blood and other tissues, that catalyses the hydrolysis of a variety of choline esters and a few other compounds. *Compare* **acetylcholinesterase**.

pseudoequatorial (*in chemistry*) *abbr.*: e' ; *see* **conformation**.

pseudoxon a portion of an **exon** that is duplicated in an adjacent intron.

pseudogene symbol: ψ ; a sequence in DNA that is related to a functional gene but cannot be transcribed. This may be due to mutational changes that preclude it from being expressed in the form of functional products or the lack of some flanking control region in the promoter. Pseudogenes may or may not contain the introns of the functional gene. Examples include pseudogenes for α , β , and ζ globulins in several species, V_K chains of immunoglobulins, actin, and tubulin. **Processed pseudogenes**, or **LINES** (*see* **LINE**), occur in a genome somewhere other than the normal position of the corresponding functional gene and so cannot be expressed because of alterations in regulatory DNA sequences, or coding sequences, or both.

pseudoglobulin a globulin that is soluble in solutions of low ionic strength or in pure water. *Compare* **euglobulin**.

pseudointron an **intron** that contains 5' and 3' splice sites that are appropriate but not utilized in the excision of the intron.

pseudoisoenzyme any one of the epigenetically derived multiple forms of an enzyme.

pseudo-operator a functional **operator** element within a structural protein-coding region of DNA.

pseudopapsin *see* **pepsinogen**.

pseudoplasmodium *see* **Dictyostelium discoideum**.

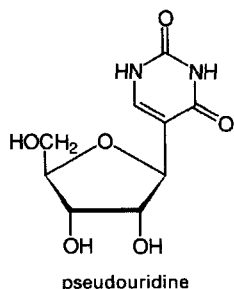
pseudopodium (*pl.* **pseudopodia**) a temporary protrusion or retractile process of a cell, associated with flowing movements of the protoplasm and serving for locomotion and feeding. — *pseudopodial adj.*

pseudouridine symbol: Ψ ; 5- β -D-ribofuranosyluracil; a component of **transfer RNA** (tRNA) containing a C–C bond between the ribose (C-1) and uracil (C-5). Typical of the **minor nucleosides** present in tRNA, it is formed by modification of uridine in the original transcript; the chemistry of this process is unclear. When pseudouridine occurs in the anticodon of transfer RNA it base pairs with adenine. *Compare* **uridine**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

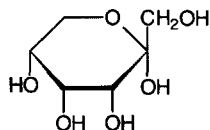


psi 1 *symbol*: ψ (lower case) or Ψ (upper case); the twenty-third letter of the Greek alphabet. For uses see **Appendix A. 2** or **p.s.i.** *abbr.* for pounds per square inch (a unit of pressure in the Imperial and US systems of units).

PSI *abbr.* for photosystem I (see **photosystem**).

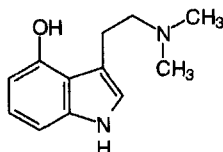
psicofuranine 6-amino-9-D-psicofuranosylpurine; an antibiotic produced by *Streptomyces hygroscopicus*, found to have antitumour activity. See also **D-psicose**.

D-psicose a nonsystematic name for the keto-hexose D-ribo-2-hexulose. The hydroxyl at C-3 has the opposite configuration to that of D-fructose. It is a component of **psicofuranine**.



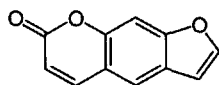
PSII *abbr.* for photosystem II (see **photosystem**).

psilocin 3-[2-(dimethylamino)ethyl]-1H-indol-4-ol; a hallucinogenic alkaloid extracted from the mushroom *Psilocybe mexicana*.



P-site *abbr.* for a peptidyl-tRNA binding site of ribosome; in bacteria this involves the **ribosomal proteins** L2, L14, L18, L24, L27, and L33, and a region close to the 3' end of 16S rRNA.

psoralen 6-hydroxy-5-benzofuranacrylic acid δ -lactone; a furocoumarin present as a **phytoalexin** in many plants. It has photosensitizing and phototoxic effects in humans and animals, and is used in photochemistry, intercalating into DNA and forming cross-links on irradiation. See also **methoxypsoralen**.



PSP *abbr.* for pancreatic stone protein (see **pancreatic thread protein**).

psychosine 1-O- β -D-galactosylsphingosine; it is formed from

UDPgalactose and sphingosine by sphingosine β -galactosyltransferase, EC 2.4.1.23. It is acylated to a cerebroside by reaction with a long-chain acyl-CoA.

psychrometer or **wet-and-dry bulb hygrometer** a type of hygrometer consisting of two thermometers, one with a wet bulb and one with a dry bulb. The cooling effect of evaporation from the wet bulb creates a temperature difference between the two thermometers, from which the relative humidity can be calculated.

psychrophile an organism that grows best at low temperatures (below about 20 °C). —**psychrophilic** *adj.*

P system a system of blood groups in which the antigenic determinants are oligosaccharides. The antigens are detected on erythrocytes, erythroblasts, platelets, megakaryocytes, fibroblasts, and endothelial cells; they are designated P^k, P, and P₁. The common phenotypes are P₁ (antigens P₁ and P positive) and P₂ (P antigen positive); P^k phenotypes are rare. Individuals with the p phenotype lack all of the above determinants, having only the core structure lactosylceramide, and have anti-P and anti-P^k antibodies. The determinants are:

P^k (Gb₃Cer): Gal(α 1-4)Gal(β 1-4)GlcCer

P (Gb₄Cer): GalNAc(β 1-3)Gal(α 1-4)Gal(β 1-4)GlcCer

P₁: Gal(α 1-4)Gal(β 1-4)GlcNAc(β 1-3)Gal(β 1-4)GlcCer.

Pt *symbol* for platinum.

PTA *abbr.* for plasma thromboplastin antecedent (i.e. factor XI); see **blood coagulation**.

PTC *abbr.* for 1 plasma thromboplastin component (i.e. factor IX); see **blood coagulation**. 2 phenylthiocarbamoyl.

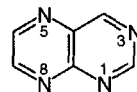
Ptd *symbol* for the phosphatidyl group.

PtdIns *symbol* for phosphatidylinositol.

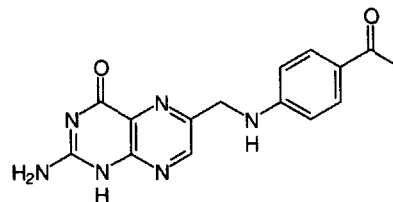
PtdIns4P or **PtdIns-4-P** *symbol* for phosphatidylinositol 4-phosphate.

PtdIns(4,5)P₂ or **PtdIns-4,5-P₂** *symbol* for phosphatidylinositol 4,5-bisphosphate.

pteridine pyrazino[2,3-d]pyrimidine; the parent structure of pterins and the **pteroyl** group. See also **xanthopterin**.



pteroyl 4-[(2-amino-1,4-dihydro-4-oxo-6-pteridiny]methyl]amino]benzoyl; the acyl group of pteric acid, and the structural basis of **folate** and its derivatives.



PTFE *abbr.* for polytetrafluoroethylene.

PTH *abbr.* for 1 parathyroid hormone (i.e. **parathyrin**). 2 phenylthiohydantoin; as in PTH-amino acid (i.e. the phenylhydantoin derivative of an amino acid).

PTI *abbr.* for pancreatic trypsin inhibitor.

PTM *abbr.* for post-translational modification.

PTS *abbr.* for phosphoenolpyruvate-dependent sugar phosphotransferase system.

ptyalin a substance in saliva, identified as α -amylase.

Pu *symbol* for plutonium.

pUC vector any of a series of plasmid cloning vectors having an



ampicillin resistance gene and part of the *lacZ* gene containing multiple cloning sites (MCSs). Different members of the series (e.g. pUC18, pUC19) vary in the restriction enzyme recognition sites included in the MCS. They are high copy number plasmids and so can be purified by rapid, small-scale methods of isolation. A related series of bacteriophage M13 vectors, the M13mp vectors, contain the same MCS and are useful for DNA sequencing. *See also* **Messing vector**.

PUFA *abbr. for* polyunsaturated fatty acid.

puff any discrete, localized expansion occurring at specific sites in polytene chromosomes, e.g. the giant salivary gland chromosomes in Diptera. Typically a few micrometres long, they are believed to be a gene or group of genes that are actively synthesizing RNA, which accounts for a considerable part of the mass of the puff. *See also* **Balbani ring**.

'pull' model a nickname for the **Monod-Wyman-Changeux model** (for ligand-induced changes in protein conformation) because, in this model, ligands influence the equilibrium between enzyme states by sequestering the enzyme in ligand-bound forms. *Compare* **'push' model**.

pullulan an extracellular **glucan** elaborated by *Aureobasidium pullulans* (yeast form) and formed by other fungi. It is a linear structure with both (1→4)- α -D- and (1→6)- α -D-linkages. Pullulan is used as a non-caloric material, a water-retention agent, and as a film and fibre former.

pullulanase **1** isopullulanase EC 3.2.1.57; an enzyme that catalyses the hydrolysis of **pullulan** to isopanose (i.e. 6- α -maltosylglucose). **2** neopullulanase EC 3.2.1.135; an enzyme that catalyses the hydrolysis of pullulan to panose (i.e. 6- α -D-glucosylmaltose). *3* *see* **α -dextrin endo-1,6- α -glucosidase**.

pulsed field gel electrophoresis *abbr.*: PFGE; a type of **gel electrophoresis** in which large fragments of DNA can be separated by continuously altering the angle at which the electric current is applied.

pulse radiolysis a technique whereby reactive chemical species are generated by pulses of high-energy radiation and the subsequent reactions of these species are followed spectrophotometrically as a function of time.

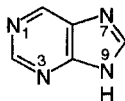
pump (*in cell biology*) any transmembrane protein that drives the active transport of ions and small molecules across the lipid bilayer. *See also* **hydrogen-ion pump**, **sodium pump**.

PUMP-1 *see* **matrilysin**.

pump-1 protease *see* **matrilysin**.

Purcell, Edward Mills (1912–97), US electrical engineer, physicist, and radioastronomer notable for his discovery (independently of F. Bloch) of the phenomenon of nuclear magnetic resonance of certain atomic nuclei and for being the first to report detection of the emission of 1420-Hz microwave radiation by interstellar dihydrogen; Nobel Laureate in Physics (1952) jointly with F. Bloch 'for their development of new methods for nuclear magnetic precision measurements and discoveries in connection therewith'.

purine **1** *1H*-purine; **7H**-imidazo[4,5-*d*]pyrimidine; an organic nitrogenous base, sparingly soluble in water. **2** any of a class of derivatives of *1H*-purine; these form one of the two classes of nitrogen-containing ring compounds found in DNA and RNA, which include **adenine** and **guanine**. *See also* **purine biosynthesis**.



purine biosynthesis the biosynthesis of purine ribonucleotides. The pathway for the synthesis of purine ribonucleotides begins with the formation of 5-phospho- α -D-ribosyl diphosphate (5-phosphoribosyl- α -D-pyrophosphate, *common abbr.*: PRPP) by ribose-phosphate pyrophosphokinase, EC

2.7.6.1. This is converted to β -5-phosphoribosylamine (**I**) by reaction with glutamine, catalysed by amidophosphoribosyltransferase, EC 2.4.2.14; reaction of **I** with glycine, with hydrolysis of ATP to ADP and P_i , catalysed by **phosphoribosylamine-glycine ligase**, EC 6.3.4.13 (glycinamide ribonucleotide synthetase, *common abbr.*: GAR synthetase), brings about the formation of N^1 -(5-phospho-D-ribosyl)glycinamide (**II**, glycinamide ribotide, *common abbr.*: GAR); **II** reacts with 10-formyltetrahydrofolate to form 5'-phosphoribosyl-*N*-formylglycinamide (**III**, *common abbr.*: FGAR), the enzyme being **phosphoribosylglycinamide formyltransferase**, EC 2.1.2.2 (GAR transformylase); **III** is converted to 5'-phosphoribosyl-formylglycinamide (**IV**, formylglycinamide ribotide, *common abbr.*: FGAM) by **phosphoribosylformylglycinamide synthase**, EC 6.3.5.3 (*common abbr.*: FGAM synthetase) by reaction with glutamine with ATP hydrolysis to ADP and P_i . **IV** is converted to 1-(5-phosphoribosyl)-5-aminoimidazole (**V**, 5-aminoimidazole ribotide, *common abbr.*: AIR), with hydrolysis of ATP to ADP and P_i , by **phosphoribosylformylglycinamide cyclo-ligase**, EC 6.3.3.1 (*common abbr.*: AIR synthetase); **V** is carboxylated to 1-(5-phosphoribosyl)-5-amino-4-imidazole-carboxylate (**VI**, carboxyaminoimidazole carboxamide, *common abbr.*: CAIR), by **phosphoribosylaminoimidazole carboxylase**, EC 4.1.1.21 (*common abbr.*: AIR carboxylase); **VI** reacts with L-aspartate, with hydrolysis of ATP to ADP and P_i , to form 1-(5-phosphoribosyl)-5-amino-4-(*N*-succinocarboxamide)-imidazole (**VII**, 4-(*N*-succinylcarboxamide)-aminoimidazole ribotide, *common abbr.*: SCAIR) catalysed by **phosphoribosylaminoimidazolesuccinocarboxamide synthase**, EC 6.3.2.6 (*common abbr.*: SCAIR synthetase); **VII** is converted, with loss of fumarate, to 1-(5-phosphoribosyl)-4-carboxamide-5-aminoimidazole (**VIII**, aminoimidazolecarboxamide ribotide, *common abbr.*: AICAR) by adenylosuccinate lyase, EC 4.3.2.2 (**VII** is an alternative substrate for this enzyme); **VIII** reacts with 10-formyltetrahydrofolate to form 5'-phosphoribosyl-5-formamido-4-imidazolecarboxamide (**IX**, formaminoimidazolecarboxamide ribotide, *common abbr.*: FAICAR), catalysed by **phosphoribosylaminoimidazolecarboxamide formyltransferase**, EC 2.1.2.3 (*common abbr.*: AICAR transformylase); **IX** undergoes ring closure catalysed by IMP cyclohydrolase, EC 3.5.4.10, to form inosine monophosphate (IMP), from which other purine nucleoside monophosphates are formed. Thus it is converted to AMP via adenylosuccinate, in steps catalysed by adenylosuccinate synthase, EC 6.3.4.4, and adenylosuccinate lyase, EC 4.3.2.2. It is converted to GMP via xanthosine monophosphate, in steps catalysed by **IMP dehydrogenase**, EC 1.1.1.205, and **GMP synthase**, EC 6.3.4.1. A **salvage pathway** also exists, that reconverts purine bases released during nucleic-acid degradation to nucleoside monophosphates. The enzymes adenine phosphoribosyltransferase and hypoxanthine-guanine phosphoribosyltransferase are responsible for these reactions.

purine-nucleoside phosphorylase *abbr.*: PNP; EC 2.4.2.1; *systematic name*: purine-nucleoside:orthophosphate ribosyltransferase; *other name*: inosine phosphorylase. An enzyme that catalyses the phosphorylysis of a purine nucleoside to a purine and α -D-ribose 1-phosphate. It functions in the degradation of purines; decreased activity leads to accumulation of dGTP, which affects DNA replication. Enzyme deficiency is associated with moderate immunodeficiency. Example from *Escherichia coli*, homoheptamer; database code DEOD_ECOLI, 238 amino acids (25.79 kDa).

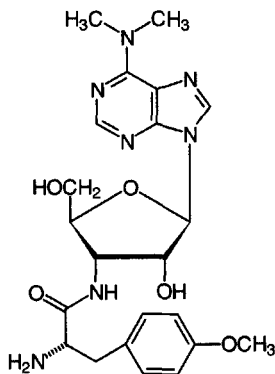
purinergic describing a type of nerve in which synaptic transmission is brought about by purine-like transmitter substances. *See* **purinoceptor**. *Compare* **adrenergic**, **cholinergic**.

purinoceptor or purinergic receptor a family of receptors (P_1 , P_{2X} , P_{2Y} , P_{2Z} , P_{2T} , and P_{2U}) classified according to the relative potency of purine nucleotides – ATP, ADP, AMP, and UTP – in stimulating them and by their response to either adenosine (P_1) or ATP (P_2). P_{2Y} and P_{2U} have seven transmembrane domains and regulate turnover of phosphoinositides; example of



P_{2U} from *Rattus norvegicus*: database code P2UR_RAT, 374 amino acids (42.26 kDa). **P_{2X}**, **P_{2Z}**, and **P_{2T}** modulate the opening of an intrinsic cation channel but **P_{2T}** also stimulates release of intracellular Ca^{2+} and reduces synthesis of cyclic AMP. Example of **P_{2X}** from *R. norvegicus*: database code P2X1_RAT, 472 amino acids (52.62 kDa).

puromycin a nucleoside antibiotic produced by the soil actinomycete *Streptomyces alboniger*. It interrupts protein synthesis by virtue of its structural similarity to the 3' end of aminoacyl-tRNA: it enters the A-site (see **aminoacyl site**) on the ribosome, and is incorporated into the growing polypeptide chain, thereby causing premature release of the puromycinyl polypeptide from the ribosome.



purothionin any of a group of basic peptides, isolated from grain and of $M_r \approx 5000$, that contain a large number of cysteine residues and are toxic to bacteria and fungi by inhibiting DNA transcription and RNA translation. Example purothionin γ -1 from *Triticum aestivum*: database code THG1_WHEAT, 47 amino acids (5.24 kDa).

purple membrane part of the cell membrane of *Halobacterium halobium* containing **bacteriorhodopsin**.

'push' model a nickname for the **Koshland model** (for ligand-induced changes in protein conformation) because, in this model, binding of the ligands results in a conformational change and does not simply select a pre-existing state. Compare **'pull model'**.

putidaredoxin a ferredoxin, isolated from *Pseudomonas putida*, that is an electron carrier in a hydroxylase system acting on camphor and closely related monoterpenes. Compare **adrenodoxin**.

putrescine 1,4-diaminobutane; the polyamine formed by decarboxylation of **ornithine** and the metabolic precursor of **spermidine** and **spermine**.

PVA *abbr.* for 1 polyvinyl alcohol. 2 polyvinyl acetate.

PVC *abbr.* for polyvinyl chloride.

PVP *abbr.* for polyvinyl pyrrolidone.

PWM *abbr.* for pokeweed mitogen.

pycnometer an apparatus for measuring the density of liquids, and also sometimes of solids. It consists of a container whose volume can be accurately measured, and that can be filled with great precision, so that when filled with the liquid in question it can be weighed to determine the density. — **pycnometric** *adj.*

Pyd *symbol* for a residue of an unspecified pyrimidine nucleoside (alternative to Y).

pyr-1 the symbol for the pyrimidine 1 gene, which encodes **CAD protein** in several eukaryotes.

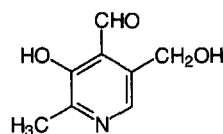
pyran the trivial name for either of the hypothetical isomers α -pyran, 2H-pyran, oxacyclohexa-3,5-diene, or γ -pyran, 4H-pyran, oxacyclohexa-2,5-diene, $\text{C}_5\text{H}_6\text{O}$. These pyrans are not known as such, but are the parent structures of various known pyranoid compounds such as dihydropyran, α - and γ -pyrones, and pyranoses.

pyranoid pyran-like, i.e. describing a molecular structure that consists of or contains a ring of five carbon atoms and one oxygen atom (as in **pyran**).

pyranose *symbol*: p; any monosaccharide or monosaccharide derivative whose molecule contains a **pyranoid** ring. Compare **furanose**, **septanose**.

pyridine-nucleotide coenzyme any of a group of coenzymes containing a nicotinamide nucleotide residue, e.g. nicotinamide-adenine dinucleotide (NAD^+); now known as **nicotinamide-nucleotide coenzymes**.

pyridoxal *abbr.*: PL; 3-hydroxy-5-(hydroxymethyl)-2-methyl-4-pyridine carboxaldehyde; a compound with **vitamin B₆** activity.



pyridoxal kinase EC 2.7.1.35; an enzyme that catalyses the formation of pyridoxal 5'-phosphate from ATP and pyridoxal with release of ADP. Example (fragment) from sheep: database code A61483, 11 amino acids (1.24 kDa).

pyrimidine 1,3-diazine; an organic nitrogenous base, sparingly soluble in water. 2 any of a class of derivatives of pyrimidine (def. 1); these form one of the two categories of nitrogen-containing ring compounds found in DNA and RNA, which include **cytosine**, **thymine**, and **uracil**. See also **pyrimidine biosynthesis**.



pyrimidine biosynthesis or **pyrimidine ribonucleotide biosynthesis** pyrimidines are formed from carbamoyl phosphate (see **carbamoyl-phosphate synthase**) after conversion to carbamoyl aspartate in a reaction with L-aspartate catalysed by **aspartate transcarbamylase**, EC 2.1.3.2. Carbamoyl aspartate is converted to orotate via dihydroorotate, in steps catalysed by **dihydroorotase**, EC 3.5.2.3, and dihydroorotate dehydrogenase, EC 1.3.99.11. Orotate is phosphorylated to oridine monophosphate (OMP) in a reaction with 5-phospho- α -D-ribose diphosphate catalysed by **orotate phosphoribosyltransferase**, EC 2.4.2.10. OMP is converted to uridine monophosphate by **oridine 5'-phosphate decarboxylase**, EC 4.1.1.23.

pyrimidine dimer any structure formed within a strand of DNA from two neighbouring pyrimidine residues through covalent cross-linking induced by ultraviolet irradiation. Two types are known. (1) Addition of one pyrimidine ring to the other at each end of their respective 5,6-double bonds generates a **cyclobutadipyrimidine**, a fully substituted cyclobutane. Such a dimer is formed most readily from two thymine residues (see **thymine dimer**), but may be formed also from two cytosine residues, or from one cytosine and one thymine residue. For each of these varieties there are four possible configurational isomers: if the pyrimidine rings have become linked in a parallel manner, i.e. by 5-5' and 6-6' bonds, the configuration is designated by the prefix *syn*-, but if they are linked in an antiparallel manner, i.e. by 5-6' and 6-5' bonds, it is designated by *anti*-; further, if the pyrimidine rings are both on the same side of the plane of the cyclobutane ring, the dimer is described as being in the **'doghouse configuration'**, designated by the prefix *cis*-, but if the rings are on opposite sides of the cyclobutane plane, the dimer is said to be in the **'chair configuration'**, designated by *trans*-. The presence of a

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



pyrimidine-nucleoside phosphorylase

cyclobutadipyrimidine in a DNA strand causes a local distortion of the DNA duplex and blocks the action of DNA polymerase, thus preventing replication; cell death may ensue unless repair is effected by **photoreactivation**. (2) Alternatively, addition may take place between position 6 of one pyrimidine residue and position 4 of an adjacent one, with creation of a single covalent link and generation of a (6-4)photoproduct. Such a dimer is usually formed between position 6 of a cytosine or thymine residue on the 5' side of a dinucleotide segment and position 4 of a cytosine residue on its 3' side. The (6-4)photoproduct is thought to be the main type of lesion in DNA responsible for mutations; it is not susceptible to photoreactivation, but can be eliminated by other DNA-repair mechanisms, e.g. **excision repair**.

pyrimidine-nucleoside phosphorylase any of several enzymes that function in the catabolism of pyrimidines; examples are: (1) **Uridine phosphorylase**, EC 2.4.2.3; it catalyses the phosphorolysis of uridine to uracil and α -D-ribose 1-phosphate. Example from *Escherichia coli*, homohexamer showing 4-, 6-, and 8-fold symmetry; database code UDP_ECOLI, 252 amino acids (27.00 kDa). (2) **Thymidine phosphorylase**, EC 2.4.2.4; it catalyses the phosphorolysis of thymidine to thymine and 2-deoxy-D-ribose 1-phosphate. Example from *E. coli*, homodimer; database code DEOA_ECOLI, 441 amino acids (47.28 kDa).

pyro+ or (before a vowel) **pyr+** *comb. form* 1 of, relating to, caused, or produced by fire. **2** (in chemistry) indicating material derived by the action of heat; it often indicates an acid formed by loss of a molecule of water in a heating process, and hence also some anhydrides, e.g. pyrophosphate.

pyrocatechol or **pyrocatechin** or (formerly) **catechol** 1,2-benzenediol; *o*-dihydroxybenzene. Compare **catecholamine**, **o**-benzoquinone, **hydroquinone**.

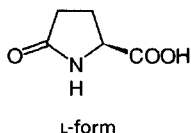
pyrolysis the decomposition or dissociation of a chemical compound by the application of heat. See also **thermolysis**. —**pyrolytic adj.**; **pyrolyse** or (sometimes, US) **pyrolyze vb.**

pyrophosphatase **1** an imprecise name for **inorganic pyrophosphatase**. **2** the generic name for any of a number of the enzymes within sub-subclass EC 3.6.1 (enzymes acting on diphosphate bonds); e.g. ATP pyrophosphatase (EC 3.6.1.8).

pyrophosphate an older name for **diphosphate** (def. 1-5) (use no longer recommended although it is still used as an alternative in such terms as inorganic pyrophosphate, pyrophosphatase, etc.). See also **inorganic diphosphate**.

pyrophosphokinase the general name for any enzyme belonging to the sub-subclass EC 2.7.6, diphosphotransferases, which transfer a diphosphate group from ATP to a named substrate; e.g. GTP pyrophosphokinase (EC 2.7.6.5), which converts guanosine 5'-triphosphate, GTP to guanosine 3'-phosphate 5'-triphosphate, pppGpp.

pyrrolidonecarboxylic acid *symbol*: <Glp; pyroglutamic acid; 5-oxoproline; pyrrolid-2-one-5-carboxylic acid.



1-pyrroline-5-carboxylate dehydrogenase EC 1.5.1.12; an enzyme that catalyses a reaction between 1-pyrroline-5-carboxylate (*abbr.*: P5C), NAD^+ , and H_2O to form L-glutamate and NADH. P5C is an intermediate in the conversion of glutamate, proline, and ornithine. Example from *Escherichia coli*, a bifunctional protein that also has **proline dehydrogenase** activity; database code PUTA_ECOLI, 1320 amino acids (143.66 kDa).

pyrroloquinoline-quinone *abbr.*: PQQ; an alternative name for **methoxatin**.

pyruvate *trivial name* for 2-oxopropanoate, CH_3COCO^- ; it is found in virtually every type of cell and species, being an important intermediate in many pathways including the catabolism of glucose, and the biosynthesis of amino acids and carbohydrates.

pyruvate carboxylase *abbr.*: PCB; EC 6.4.1.1; *systematic name*: pyruvate:carbon dioxide ligase (ADP-forming); *other name*: pyruvic carboxylase; an enzyme that catalyses a reaction between ATP, pyruvate, and HCO_3^- to form oxaloacetate, ADP, and orthophosphate; biotin and manganese or zinc are cofactors. Example from yeast (PCB1) homotetramer; database code PYC1_YEAST, 1178 amino acids (129.95 kDa).

pyruvate decarboxylase EC 4.1.1.1; *systematic name*: 2-oxoacid carboxy-lyase; *other names*: α -carboxylase, pyruvic decarboxylase; α -ketoacid carboxylase. An enzyme that catalyses the decarboxylation of a 2-oxo acid to an aldehyde and CO_2 ; thiamine diphosphate is a coenzyme. Example from yeast (there are three related isozymes), homotetramers, isozyme 1: database code DCP1_YEAST, 563 amino acids (61.44 kDa).

pyruvate decarboxylase kinase or **pyruvate dehydrogenase kinase** a subunit of the mammalian **pyruvate dehydrogenase complex**. It phosphorylates three serine residues on pyruvate dehydrogenase (lipoamide) resulting in loss of activity, and is stimulated by elevated levels of acetyl-CoA and NADH.

pyruvate dehydrogenase complex an enzyme complex that catalyses the overall reaction between pyruvate, NAD^+ , and coenzyme A to form acetyl-CoA, NADH, and CO_2 . The complex consists of three activities: (1) **pyruvate dehydrogenase (lipoamide)**, designated E_1 , catalysing a reaction between pyruvate and lipoamide to form S-acetyldihydrolipoamide and CO_2 ; (2) **dihydrolipoamide S-acetyltransferase**, designated E_2 , catalysing a reaction between S-acetyldihydrolipoamide and coenzyme A to form acetyl-CoA and dihydrolipoamide; and (3) **dihydrolipoamide dehydrogenase**, designated E_3 , which oxidizes dihydrolipoamide to lipoamide. The complex is regulated by the phosphorylation and dephosphorylation of the pyruvate dehydrogenase (lipoamide) component, phosphorylation being catalysed by **pyruvate decarboxylase kinase** and removal of phosphate being catalysed by **[pyruvate dehydrogenase (lipoamide)] phosphatase** (or pyruvate decarboxylase phosphatase), EC 3.1.3.43. Elevated levels of NADH and acetyl-CoA inhibit the complex by stimulating the kinase.

pyruvate dehydrogenase (lipoamide) EC 1.2.4.1; *systematic name*: pyruvate:lipoamide 2-oxidoreductase (decarboxylating and acceptor-acetylating); *other names*: pyruvate decarboxylase; pyruvate dehydrogenase; pyruvic dehydrogenase. A component enzyme of the **pyruvate dehydrogenase complex**. It catalyses a reaction between pyruvate and lipoamide to form S-acetyldihydrolipoamide and CO_2 ; thiamine diphosphate is a coenzyme. Example, pyruvate dehydrogenase E_1 component from *Escherichia coli*, homodimer, flavoprotein; database code ODP1_ECOLI, 886 amino acids (99.50 kDa).

[pyruvate dehydrogenase (lipoamide)]-phosphatase EC 3.1.3.43; an enzyme that catalyses the hydrolysis of phosphate from [pyruvate dehydrogenase (lipoamide)] phosphate. This hydrolysis activates the enzyme activity. Example (precursor) from *Bos taurus*, catalytic subunit; database code PDP_BOVIN, 538 amino acids (61.12 kDa).

pyruvate kinase EC 2.7.1.40; *systematic name*: ATP:pyruvate 2-O-phosphotransferase; *other names*: phosphoenolpyruvate kinase; phosphoenol transphosphorylase. An enzyme that catalyses the formation of pyruvate and ATP from phosphoenolpyruvate and ADP; it requires Mg^{2+} and K^+ . In the glycolytic pathway, it effects the phosphorylation of ADP to ATP. In many cases the enzyme is a homotetramer. Example from yeast; database code KPYK_YEAST, 500 amino acids (54.48 kDa).

[pyruvate kinase]-phosphatase EC 3.1.3.49; an enzyme that

pyruvate-malate cycle

catalyses the hydrolysis of phosphate from [pyruvate kinase] phosphate in order to activate pyruvate kinase.

pyruvate-malate cycle a metabolic cycle involving malate dehydrogenase (oxaloacetate-decarboxylating) (the **malic enzyme**) that occurs during fatty-acid biosynthesis. In this situation, acetyl-CoA is transported into the cytoplasm as citrate, formed by mitochondrial citrate synthase; the citrate is cleaved in the cytoplasmic compartment to acetyl-CoA, which participates in fatty-acid synthesis, and oxaloacetate. No transport system exists for taking this oxaloacetate back into the mitochondrion; however, it can be reduced to malate by

cytoplasmic NAD⁺-linked malate dehydrogenase, and the malate so formed can be converted to pyruvate by malic enzyme, EC 1.1.1.38; the pyruvate can then be transported into the mitochondrion, to be carboxylated to oxaloacetate by **pyruvate carboxylase**.

pyruvic-malic carboxylase *an alternative name for malic enzyme.*

PYY *abbr. for peptide YY.*

PZ *abbr. for pancreozymin.*

PZP *abbr. for pregnancy zone protein.*

Qq

q symbol for quartet (in nuclear magnetic resonance spectroscopy).

Q symbol for 1 heat (alternative to *Q*). 2 partition function (individual entity). 3 specific humidity.

Q symbol for 1 a residue of the α -amino acid L-glutamine (alternative to Gln). 2 a residue of the ribonucleoside **pseudouridine** (replacing Ψ for computer work). 3 coenzyme Q (i.e. **ubiquinone**); not recommended. The number of prenyl residues may be indicated by a suffixed arabic number, e.g. Q-10. 4 queuosine.

Q symbol for 1 heat (alternative to *q*). 2 electric charge. 3 radiant energy (alternative to *W*). 4 or *Q_v* quantity of light. 5 partition function (whole system).

Q₁₀ symbol for temperature coefficient of a reaction.

QCO₂ symbol for metabolic quotient in respect of discharge of carbon dioxide; it is expressed in terms of microlitres CO₂ evolved per milligram dry mass of material per hour.

QO₂ symbol for metabolic quotient in respect of uptake of dioxygen; it is expressed in terms of microlitres O₂ absorbed per milligram dry mass of material per hour.

Q_s symbol for a measure of the activity of an enzyme, especially a respiratory enzyme, defined as the number of microlitres (at stp) of gaseous substrate S used up per hour per milligram enzyme. It could be extended to nongaseous substrates by taking 1 μ mol as equivalent to 22.4 μ L.

QAE abbr. for the quaternized aminoethyl; denoting the group diethyl-(2-hydroxypropyl)aminoethyl; (C₂H₅)₂(CH₃-CHOH-CH₂)N⁺-CH₂-CH₂-; a strongly basic substituent sometimes used in hydroxylated chromatographic support matrices, as in QAE-Sephadex, to form anion exchangers. Compare **DEAE**.

Q cycle an alternative name for **protonmotive Q cycle**.

Q enzyme an alternative name for amylopectin branching enzyme; see **branching enzyme**.

Q gas a mixture of 98.7% helium and 1.3% butane, used as filling for gas-flow Geiger counters.

QH[•] symbol for ubisemiquinone.

QH₂ symbol for ubiquinol.

quad denoting four kinetically important substrates or products in an enzyme mechanism. See **reactancy of enzymes**.

quadr+ or (sometimes) **quadr+** or (before a vowel) **quadr+** comb. form four; fourth or one-fourth; square.

quadruplex DNA another name for **tetraplex DNA**.

quality factor or (formerly) **relative biological effectiveness** an index of the ability of a given type of ionizing radiation to cause biological damage, depending on the density of the ionization produced relative to that produced by gamma radiation. For all X- (and gamma) radiation and beta particles (+ or -) likely to be encountered in the use of radioisotopes it is assigned a value of unity, for thermal neutrons usually a value of 3, and for all alpha particles a value of 10. See also **dose equivalent**.

quanta the plural of **quantum**.

quantal 1 of, pertaining to, or being a **quantum** or something that is quantized; of or pertaining to the quantum theory. 2 of a biological effect or response in which there are only two categories, conditions, or states (e.g. all or none, alive or dead). —**quantally** adv.

quantitative electrophoresis (sometimes) an alternative name for **crossed immunoelectrophoresis**.

quantity or **physical quantity** (in systems of measurement) any property whose value can be expressed as the product of a numerical value and a unit. In the International System of Units (see **SI**) there are seven **base quantities**, physical quantities regarded as being dimensionally independent; these are: length, mass, time, electric current, thermodynamic temperature, amount of substance, and luminous intensity; all are arbitrarily defined.

Derived units are expressed algebraically in terms of the base units. Symbols for physical quantities are printed in italic type; they are generally single Roman or Greek letters, and may be lower case or capital.

quantity of electricity an alternative name for **electric charge**.

quantize or **quantise** to restrict (a physical quantity) to a set of discrete values, characterized by quantum numbers. A quantized physical quantity cannot vary continuously but must change in steps. —**quantization** or **quantisation** *n*.

quantum (*pl.* **quanta**) 1 a (specific) quantity or amount of something. 2 a unit amount of a physical quantity, especially of electromagnetic energy, such that all other amounts are integral multiples of it. According to the **quantum theory**, a quantum of energy released during radiation emission or taken up during radiation absorption is equal to $h\nu$, where h is the Planck constant and ν the frequency of the radiation in hertz. A quantum of light (or other electromagnetic energy) is termed a **photon**.

quantum efficiency 1 (in a luminescent system) the fraction of the molecules in a particular excited state that emit luminescence. 2 (in biochemistry) an alternative term for **quantum yield** (def. 1).

quantum mechanics a mathematical theory developed from the quantum theory (see **quantum**) as a replacement for classical mechanics in order to explain satisfactorily the behaviour of atoms, molecules, and elementary particles in terms of observable quantities such as the intensities and frequencies of spectral lines.

quantum number any of a set of small integral or half-integral numbers that give the various possible values of a quantized property of a system. Quantum numbers are used especially in describing the properties of elementary particles, such as their charge (values of 0, +1, or -1) or spin (values of +½, or -½), and in specifying the energy states of electrons in atomic orbitals (each of which is characterized by a unique set of four quantum numbers).

quantum requirement (in photosynthesis) the inverse of **quantum yield** (def. 2); i.e. the number of quanta of light energy required to bring about the release of one molecule of dioxygen. The theoretical value is eight, but the value observed in an experimental system is generally somewhat higher, depending on conditions.

quantum theory see **quantum**.

quantum yield 1 abbr.: *Q*; (in a luminescent system) the probability of luminescence occurring in given conditions, expressed by the ratio of the number of photons, i.e. quanta of light, emitted by the luminescing species to the number absorbed. Compare **quantum efficiency**. The measurement of *Q* requires the counting of photons, because

$$Q = \text{photons emitted} / \text{photons absorbed}.$$

It requires highly specialized instrumentation, but a less rigorous method may be employed in which the fluorescence of a fluor is compared with that of a fluor of known *Q*. Then

$$Q_x = [I_x Q_s A_s] / [I_s A_x]$$

where *Q_s* is the quantum yield of the standard, *I_s* and *I_x* are the fluorescence intensities of the sample and standard and *A_s* and *A_x* are the percentage of absorption of each solution at the exciting wavelength. 2 (in photosynthesis) the inverse of **quantum requirement**; i.e. the fractional number of molecules of carbon dioxide reduced per photon absorbed.

quartz a colourless, often transparent, glassy form of silica, SiO₂, used in optical devices and cuvettes because of its transparency to near-ultraviolet radiation. It is an antiferromagnetic material.

Bản đang đọc: một quyển sách được tặng bởi HUỠNG NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



quaternary

the right- and left-handed forms rotating the plane of polarized light in opposite directions.

quaternary 1 consisting of four (items, parts, etc.); by or in fours. 2 (*in chemistry*) consisting of or containing an atom bound to four atoms or groups. 3 of the fourth order; fourth in a series or hierarchy.

quaternary ammonium compound any compound that can be regarded as derived from ammonium hydroxide or an ammonium salt by replacement of all four hydrogen atoms of the NH_4^+ ion by organic groups. Certain compounds of this class – ones in which one of the organic groups is a long-chain (C_8 – C_{18}) alkyl group and the other three are shorter-chain alkyl or other groups – have the properties of cationic detergents and are powerful antimicrobial agents. They are bacteriostatic at low concentrations and bactericidal at higher concentrations, being generally more active against Gram-positive than Gram-negative organisms. They display relatively low toxicity to higher animals and humans, and hence are widely used as antiseptics and disinfectants.

quaternary structure the fourth order of complexity of structural organization exhibited by protein, nucleic acid, and nucleoprotein molecules. It refers to the arrangement in space of the subunits of a multimeric macromolecule and the ensemble of its intersubunit contacts and interactions without regard to the internal geometry of the subunits. Hence it is possessed only when the molecule in question is made up of at least two (identical or nonidentical) subunits that are (at least potentially) separable, i.e. are not linked by covalent bonds.

Quellung reaction or **Neufeld Quellung reaction** or **pneumococcus capsule swelling reaction** the swelling of the capsule observed when specific antibody is mixed with suitable bacterial cells (e.g. pneumococci). It is probably due to deposition of antibody on the outside of the capsule, making the latter clearly visible. [From the German, *Quellung*, swelling; described by Ferdinand Neufeld (1869–1945), German bacteriologist, in 1902.]

quench correction (*in liquid-scintillation counting*) a correction to be applied to an observed count rate to allow for losses resulting from **quenching** (def. 2).

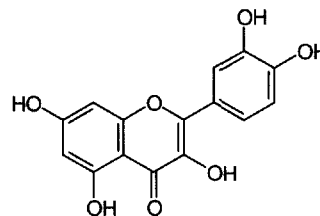
quenched flow *see* **rapid-reaction kinetics**.

quencher a substance capable of reducing or destroying luminescence through deactivation of an excited chemical species. In biological systems this usually occurs by chemical reaction (e.g. complex formation), by exchange of energy on collision, or by resonance energy transfer.

quenching 1 the reduction or destruction of luminescence, especially fluorescence, of a sample by addition or inclusion of a **quencher**. Fluorescence quenching is the basis of a useful technique for studying the binding of small molecules to proteins or other macromolecules. 2 the reduction by whatever means of the efficiency of energy transfer from beta particles to the photomultiplier(s) in a liquid-scintillation counter. Such means include: (1) **dilution quenching** – reduction of the probability of scintillation occurring through dilution of the scintillator by the sample; (2) **chemical** (or **impurity**) **quenching** – absorption by a component of the sample of some of the energy of the beta particles without emitting photons, or (without fluorescing and in competition with the primary fluor) of some of the photons emitted by excited solvent molecules; and (3) **colour quenching** – absorption by a coloured component of the sample of some of the photons emitted by the secondary fluor. 3 the termination of a chemical or enzymic reaction by addition of another reagent or sudden change of the conditions.

quercetin 3,3',4',5,7-pentahydroxyflavone; a member of the **flavonoids**, or **flavones**, widely distributed in plants, often as **glycosides**; it is the aglycon of **quercitrin**, **rutin**, and other materials. Known in folk medicine for its anti-inflammatory properties, it inhibits the **lipooxygenase** pathway, has some antiplatelet and antithrombotic activity *in vivo*, inhibits mast-cell degranulation, and possesses anti-asthmatic activity. It inhibits many

enzymes, including protein kinases, and also inhibits DNA, RNA, and protein synthesis. A number of its actions may reflect its property as a nucleoside antagonist.



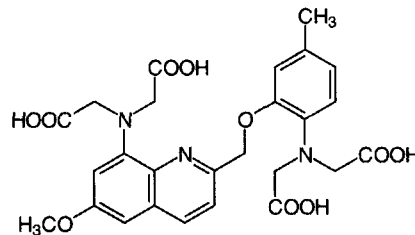
quercitrin quercetin-3-L-rhamnoside. *See also* **quercetin**.

query language very high level computer language used in interrogating databases.

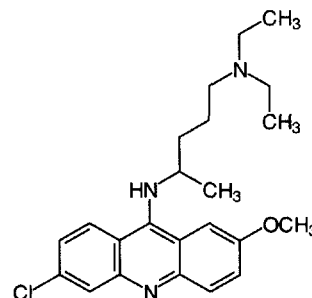
queuosine *symbol*: Q; any one of a series of nucleosides found in tRNA and having an additional pentenyl ring added via an NH group to the methyl group of 7-methylguanosine. The pentenyl ring may carry other substituents.

Qui *symbol* for a residue (or sometimes a molecule) of the deoxyaldohexose **quinovose**.

Quin-2 *abbr. for* 2-[(2-amino-5-methylphenoxy)methyl]-6-methoxy-8-aminoquinoline-*N,N,N',N'*-tetraacetic acid; a molecule functionally related to the Ca^{2+} -chelating agent **EGTA**, but containing fluorescent groups. The quantum yield of fluorescence is greatly enhanced on binding Ca^{2+} , leading to its use as a quantitative indicator of Ca^{2+} concentration. It is normally used as its tetraacetoxyster, which readily crosses the plasma membrane and is then hydrolysed within the cell by esterases to Quin-2, the polar nature of which causes it to be retained as an intracellular indicator. *See also* **Fura-2**, **INDO-1**.



quinacrine or **mepacrine** or **atebrine** 6-chloro-9-(4-diethylamino-1-methylbutylamino)-2-methoxyacridine; a derivative of **acridine**, formerly used as an antimalarial and anthelmintic. It fluoresces powerfully in UV light and can be used as a fluorochrome to label DNA in chromosomes.



quinacrine mustard a compound derived by chloro-substitution of the terminal methyls of **quinacrine** ethylamino groups, and possessing properties of an alkylating agent; this enhances its value as a DNA-labelling agent.